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DATA SHEET

NON-PERFORATED DRAIN

PRODUCT DESCRIPTION : Non-perforated flexible single wall pipe, with corrugated interior and exterior walls for drainage application use

FUNCTION : Gravity flow of water towards an outlet

MANUFACTURING STANDARD: BNQ 3624-115 Type 1

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 38 mm (1.5 in) to 250 mm (10 in)
Length: 3 m (9.8 ft) to 1,200 m (3,937 ft)
Strength in compression: 210 kPa
Manning: 0.015 to 0.017
Maximum backfill depth: 3 m (9.8 ft)

AVAILABLE COUPLERS : Soil tight: see technical data table



Non perforated DRAIN

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Length		Manning	HDPE soil tight connectors		
mm	in	mm	in	mm	in	m	ft	n	IC	DBS	SC
38	1.5	38	1.5	50	2.0	250	820.2	0.015			
50	2	50.5	2.0	60	2.4	250	820.2	0.015			
75	3	73	2.9	84	3.3	125	410.1	0.015		X	
100	4	99	3.9	116	4.6	3/15/30/45/75/600/1,200	9.8/49.2/98.4/147.6/246.1/1,968.1/3,937	0.015	X	X	
150	6	150	5.9	176	6.9	30/3	98.4/9.8	0.016	X	X	X
200	8	198	7.8	235	9.3	6/12	19.68/39.4	0.016		X	X
250	10	250	9.8	295	11.6	6/12	19.68/39.4	0.017		X	X

Note 1: Values in the table are approximate and may change without notice.

LEGEND
IC: internal coupler
DBS: double bell snap
SC: split coupler

APPLICATION : Drainage collector and outlet

ACCESSORIES : Consult the Drain Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of drain accessories.

SPECIFICATIONS

NON-PERFORATED DRAIN

SCOPE

These specifications apply to all non-perforated 38 mm (1.5 in) to 250 mm (10 in) diameter drain pipes used for gravity flow of water.

PIPE REQUIREMENTS

Pipes shall be manufactured with corrugated interior and exterior walls.

- Pipes 75 mm (3 in) to 250 mm (10 in) shall be certified as per standard BNQ 3624-115, Type 1, and shall have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 75 mm (3 in) to 250 mm (10 in): double bell snap (safety catch) to ensure proper positioning on pipes.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

PERFORATED DRAIN OR PERFORATED AND FILTERED DRAIN

PRODUCT DESCRIPTION : Perforated flexible single wall pipe, with corrugated interior and exterior walls for drainage application use

FUNCTION : Used to collect surface runoff and control the ground water level

MANUFACTURING STANDARD : BNQ 3624-115 Type 2, Type 3*

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter : 38 mm (1.5 in) to 250 mm (10 in)
Length : 3 m (9.8 ft) to 1,200 m (3,937 ft)
Strength in compression: 210 kPa
Manning : 0.015 to 0.017
Filter : Needle punched nonwoven TXC-10 geotextile, FOS 100 µm
Needle punched nonwoven TXC-250 geotextile, FOS 250 µm
Woven, FOS 450 µm
Maximum backfill depth : 3 m (9.8 ft)

AVAILABLE COUPLERS : Soil tight: see technical data table



TYPE 2 Perforated DRAIN



TYPE 3 (MEGA 3) Perforated DRAIN



TYPE 2 Perforated DRAIN filtered with TXC-10



TYPE 3 (MEGA 3) Perforated DRAIN filtered with TXC-250

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Length		Manning	Type		Filter			Soil tight connectors HDPE		
mm	in	mm	in	mm	in	m	ft	n	2	3	FOS 100 µm	FOS 250 µm	FOS 450 µm	IC	DCC	SC
38	1.5	38	1.5	50	2.0	250	820.2	0.015	X		X					
50	2	50.5	2.0	60	2.4	250	820.2	0.015	X		X					
75	3	73	2.9	80	3.3	30	98.4	0.015	X				X		X	
100	4	99	3.9	116	4.6	3/15/30/45/75/ 600/1,200	9.8/49.2/98.4/147.6/ 246.1/1,968.1/3,937	0.015	X	X	X	X	X	X	X	
150	6	150	5.9	176	6.9	30/300	98.4/984.3	0.016	X	X	X	X	X	X	X	X
200	8	198	7.8	235	9.3	6/120	9.8/393.7	0.016	X	X	X	X	X		X	X
250	10	250	9.8	295	11.6	6/120	9.8/393.7	0.017	X		X				X	X

Note 1: Values in the table are approximate and may change without notice.

LEGEND

IC: internal coupler

DBS: double bell snap

SC: split coupler

*The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

DATA SHEET

PERFORATED DRAIN OR PERFORATED AND FILTERED DRAIN (CONT.)

FILTER PROPERTIES TABLE

	Properties	Test method	FOS 100 µm	FOS 250 µm	FOS 450 µm
Mechanical	Tensile strength (N)	CAN-148.1 - No 7.3	95.0	200.0	n/a
	Elongation at break (%)	CAN-148.1 - No 7.3	Min.: 65; Max.: 105	Min.: 65; Max.: 100	n/a
Hydraulic	Permeability (cm/s)	CAN-148.1 - No 4	0.15	0.46	0.4
	Permittivity (s ⁻¹)	CAN-148.1 - No 4	2.0	8.0	> 2.4
	Filtration opening size (FOS, µm)	CAN-148.1 - No 10	110	250	450
	Wettability (cm)	CAN-148.1 - No 26.3	< 1.0	< 1.0	n/a
	Flow rate (l/s/m ²)	ASTM D4491	131	270	238
Construction	Raw material		Polyester	Polypropylene	Polyester
	Manufacturing process		Non-woven needle punched	Non-woven needle punched	Knit

APPLICATIONS : Trench drain
Foundation and basement drainage
Agricultural subsurface drainage
Turf and recreation drainage
Drainage collector and outlet

OPTION : Specific perforations are available upon request (minimum volume required).

ACCESSORIES : Consult the Drain Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of drain accessories.

SPECIFICATIONS

PERFORATED DRAIN TYPE 2

SCOPE

These specifications apply to all 100 mm (4 in) to 250 mm (10 in) diameter Type 2 perforated drain pipes used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be manufactured with corrugated interior and exterior walls.

- Pipes shall be certified as per standard BNQ 3624-115* and have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

- Perforations shall be Type 2, in compliance with standard BNQ 3624-115. Total perforation area shall be at least 32 cm² per meter (4.96 in²) per foot of length.

JOINT PERFORMANCE

- 75 mm (3 in) to 250 mm (10 in) : double bell snap (safety catch) to ensure proper positioning on pipes.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solenoproducts.com for installation recommendations.

* The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

SPECIFICATIONS

PERFORATED DRAIN TYPE 3

SCOPE

These specifications apply to all 100 mm (4 in) to 200 mm (8 in) diameter Type 3 perforated drain pipes used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be manufactured with corrugated interior and exterior walls.

- Pipes shall be certified as per standard BNQ 3624-115 and have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

- Perforations shall be Type 3, in compliance with standard BNQ 3624-115*. Total perforation area shall be at least 32 cm² per meter (4.96 in² per foot) of length.

JOINT PERFORMANCE

- Double bell snap (safety catch) to ensure proper positioning on pipes.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

* The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

SPECIFICATIONS

PERFORATED AND FILTERED DRAIN TYPE 2

SCOPE

These specifications apply to all 100 mm (4 in) to 250 mm (10 in) diameter Type 2 perforated drain pipes, covered with a geotextile, used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be manufactured with corrugated interior and exterior walls.

- Pipes shall be certified as per standard BNQ 3624-115 and have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

- Perforations shall be Type 2, in compliance with standard BNQ 3624-115*. Total perforation area shall be at least 32 cm² per meter (4.96 in² per foot) of length.

FILTERS

- The filter shall be made from a TXC-10, FOS 100µm or Routex III-IV**, FOS 110µm, in compliance with standard BNQ GCTTG 3001, geotextile with UV protection from Soleno Textiles or factory-installed woven polyester

JOINT PERFORMANCE

- 75 mm (3 in) to 250 mm (10 in): double bell snap (safety catch) to ensure proper positioning on pipes.
- The same geotextile shall cover both the assemblies and the pipe.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solenoproducts.com for installation recommendations.

* The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

** Routex III-IV is also compliant with the OPSS 1840 Class 1 standard.

SPECIFICATIONS

PERFORATED AND FILTERED DRAIN TYPE 3

SCOPE

These specifications apply to all 100 mm (4 in) to 200 mm (8 in) diameter Type 3 perforated drain pipes covered with a geotextile, used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be manufactured with corrugated interior and exterior walls.

- Pipes shall be certified as per standard BNQ 3624-115* and have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

- Perforations shall be Type 3, in compliance with standard BNQ 3624-115. Total perforation area shall be at least 130 cm² per meter (20.1 in² per foot) of length.

FILTERS

- The filter shall be made from a TXC-10, FOS 100µm geotextile, TXC-250, FOS 250µm geotextile, Routex III-IV**, BNQ GCTTG 3001, FOS 110µm geotextile with UV protection from Soleno Textiles or factory-installed woven geotextile, FOS 450 µm.

JOINT PERFORMANCE

- Double bell snap (safety catch) to ensure proper positioning on pipes.
- The same geotextile shall cover both the assemblies and the pipe.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solenoproducts.com for installation recommendations.

* The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

** Routex III-IV is also compliant with the OPSS 1840 Class 1 standard.

DATA SHEET

SOLIFLEX

PRODUCT DESCRIPTION : Non-perforated, perforated or perforated and filtered flexible dual wall pipe, with smooth interior and corrugated exterior walls for drainage application use

FUNCTION : Used to collect surface runoff and control the ground water level

MANUFACTURING STANDARD : BNQ 3624-115 Type 3

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Strength in compression: 210 kPa
Manning : 0.012
Perforations : Type 3
Filter : needle punched nonwoven TXC-250 geotextile, FOS 250 µm
Maximum backfill depth : 3 m (9.8 ft)



SOLIFLEX DRAIN

AVAILABLE COUPLERS : Soil tight: see technical data table

TECHNICAL DATA TABLE

Nom. diameter		Int. diameter		Ext. diameter		Length ⁽¹⁾		HDPE soil tight connectors		
mm	po	mm	po	mm	po	m	pi	IC	DBS	SC
100	4	99	3.9	116	4.6	45	1147.6	x	x	x

Note 1 : Maxi available, contact the Soleno representative in your region for more information.

Note 2 : Values in the table are approximate and may change without notice.

Note 3 : Type 3 perforations are recommended in the presence of iron ochre or for an installation with a stone backfill.

LEGEND

IC: internal coupler

DBS: double bell snap

SC: split coupler

APPLICATIONS :

Trench drain
Foundation and basement drainage
Agricultural subsurface drainage
Turf and recreation drainage
Drainage collector and outlet

ACCESSORIES : Consult the Drain Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of drain accessories.

SPECIFICATIONS

NON-PERFORATED SOLIFLEX

SCOPE

These specifications apply to all non-perforated 100 mm (4 in) diameter flexible dual wall pipes used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be manufactured with a smooth interior wall and a corrugated exterior wall.

- Pipes shall be certified as per standard BNQ 3624-115* and have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- Double bell snap (safety catch) to ensure proper positioning on pipes.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

SPECIFICATIONS

PERFORATED SOLIFLEX

SCOPE

These specifications apply to all 100 mm (4 in) diameter Type 3 perforated flexible dual wall pipes used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be manufactured with a smooth interior wall and a corrugated exterior wall.

- Pipes shall be certified as per standard BNQ 3624-115* and have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

- Perforations shall be Type 3, in compliance with standard BNQ 3624-115*. Total perforation area shall be at least 130 cm² per meter (20.1 in² per foot) of length.

JOINT PERFORMANCE

- Double bell snap (safety catch) to ensure proper positioning on pipes.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.soleno.com for installation recommendations.

* The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

SPECIFICATIONS

PERFORATED AND FILTERED SOLIFLEX

SCOPE

These specifications apply to all 100 mm (4 in) diameter Type 3 perforated flexible dual wall pipes covered with a geotextile, used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be manufactured with a smooth interior wall and a corrugated exterior wall.

- Pipes shall be certified as per standard BNQ 3624-115* and have a minimum stiffness of 210 kPa.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

- Perforations shall be Type 3, in compliance with standard BNQ 3624-115. Total perforation area shall be at least 130 cm² per meter (20.1 in² per foot) of length.

FILTERS

- The filter shall be made from a TXC-250, FOS 250µm geotextile with UV protection from Soleno Textiles.

JOINT PERFORMANCE

- Double bell snap (safety catch) to ensure proper positioning on pipes.
- The same geotextile shall cover both the assemblies and the pipe.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-115.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

* The Canadian General Standards Board has adopted in 1992 the BNQ Standard 3624-115 for implementation across the country (replacing the CGSB-41-GP-29-M76 standard).

DRAIN COUPLERS AND ACCESSORIES

COUPLERS



INTERNAL COUPLER

Diameter	Code
100 mm (4 in)	1M0040
150 mm (6 in)	1M0060N



DOUBLE BELL SNAP

Diameter	Code
75 mm (3 in)	1M0031
100 mm (4 in)	3MCC004CV
150 mm (6 in)	3MCC006C
200 mm (8 in)	3MCC008C
250 mm (10 in)	3MCC010C



SPLIT COUPLER

Diameter	Code
150 mm (6 in)	1M0061
200 mm (8 in)	1M0081
250 mm (10 in)	1M1010

ACCESSORIES



INTERNAL END CAP

Diameter	Code
75 mm (3 in)	1B0035
100 mm (4 in)	1B0045
150 mm (6 in)	1B0065



90-DEGREE ELBOW

Diameter	Code
100 mm (4 in)	1C0049



EXTERNAL REDUCER

Diameter	Code
150 mm X 100 mm (6 in x 4 in)	1R0064
200 mm X 150 mm (8 in x 6 in)	1R0086
250 mm X 200 mm (10 in x 8 in)	1R1008



EXTERNAL END CAP

Diameter	Code
100 mm (4 in)	2B100CV



NEW!
INTERNAL REDUCER

Diameter	Code
150 mm to 100 mm (6 in to 4 in)	1RI0064



TEE

Diameter	Code
75 mm (3 in)	1T0033
250 mm (10 in)	1T1010
300 mm (12 in)	1T1212

DRAIN COUPLERS AND ACCESSORIES (CONT.)



REDUCING TEE

Diameter	Code
100 mm X 100 mm x 75 mm (4 in X 4 in X 3 in)	1T0043
150 mm X 150 mm x 100 mm (6 in X 6 in X 4 in)	1T0064
200 mm X 150 mm x 100 mm (8 in X 6 in X 4 in)	1T8864



SUMP WELL COVER

Diameter	Code
500 mm (20 in)	1UCV025



TEE-WYE (Y)

Diameter	Code
100 mm (4 in)	1TY044
150 mm (6 in)	1TY066



DOWN SPOUT ADAPTER

Diameter	Code
WHITE 100 mm (4 in)	1G004
BLACK 100 mm (4 in)	1G004N



RADON SUMP BASIN

Dimension	Code
450 mm x 600 mm (18 in x 24 in)	1UCURADON



PVC SNAP ADAPTER

Diameter	Code
100 mm (4 in)	1ADRC04



SUMP WELL

Diameter	Code
500 mm X 600 mm (20 in x 24 in)	1UCU025



COVER FOR RADON SUMP BASIN

Diameter	Code
450 mm (18 in)	1UCVRADON



WYE (Y)

Diameter	Code
100 mm (4 in)	1DY004
150 mm (6 in)	1DY006

DRAIN COUPLERS AND ACCESSORIES (CONT.)



PLASTIC GRATE DISC

Diameter	Code
100 mm (4 in)	1TTG04D



STAINLESS STEEL INSIDE GRATE

Diameter	Code
300 mm (12 in)	1TTG12AI
375 mm (15 in)	1TTG15AI
450 mm (18 in)	1TTG18AI
600 mm (24 in)	1TTG24AI
750 mm (30 in)	1TTG30AI
900 mm (36 in)	1TTG36AI



VERTICAL DRAIN (Hickenbottom)

Diameter	Code
125 mm (5 in) LARGE HOLES	1VAG04
125 mm (5 in) SMALL HOLES	1VAP04
150 mm (6 in) LARGE HOLES	1VAG06
150 mm (6 in) SMALL HOLES	1VAP06
200 mm (8 in) LARGE HOLES	1VAG08
200 mm (8 in) SMALL HOLES	1VAP08
250 mm (10 in) LARGE HOLES	1VAG10
250 mm (10 in) SMALL HOLES	1VAP10



STAINLESS STEEL REINFORCED PLASTIC GRATE

Diameter	Code
100 mm (4 in)	1TTG04AG
150 mm (6 in)	1TTG06AG



STEEL GRATE

Diameter	Code
100 mm (4 in)	1TTG04A
150 mm (6 in)	1TTG06A
200 mm (8 in)	1TTG08A



VERTICAL DRAIN TEE

Diameter	Code
125 mm (5 in) - TEE	1VAT04
150 mm (6 in) - TEE	1VAT06
200 mm (8 in) - TEE	1VAT08
250 mm (10 in) - TEE	1VAT10



STAINLESS STEEL FORK GRATE

Diameter	Code
100 mm (4 in)	1TTG04AF
150 mm (6 in)	1TTG06AF
200 mm (8 in)	1TTG08AF
250 mm (10 in)	1TTG10AF



HDPE GRID

Diameter	Code
300 mm (12 in)	33GP012
375 mm (15 in)	33GP015
450 mm (18 in)	33GP018



VERTICAL DRAIN REDUCER

Diameter	Code
150 mm (6 in)	1VAR06

DRAIN COUPLERS AND ACCESSORIES (CONT.)



DRAINAGE ACCESS CHIMNEY

Diameter	Code
100 mm (4 in)	830004



TAPE

Diameter	Code
BLACK EXTENSIBLE TAPE	1ZRUBAS
52 mm x 35.65 m (2 in x 108 ft)	
WHITE TAPE	1ZRUBAW
48 mm X 50 m (2 in x 164 ft)	



PLASTIC GRATE WITH SMALL HOLES

Diameter	Code
100 mm (4 in)	1TTG04C
150 mm (6 in)	1TTG06C



FIBRO-DRAIN FLOOR DRAIN

Dimensions	Description	Code
300 mm x 300 mm (12 in x 12 in)	Complete drain (2 pcs.)	1ZD1212
355 mm x 405 mm (14 in x 16 in)	Drain (cover not included)	1ZD1416P
400 mm x 450 mm (16 in x 18 in)	Plastic cover for 1ZD1416P	1ZD1618C
450 mm x 600 mm (18 in x 24 in)	Drain (cover not included)	1ZD1824P
485 mm x 635 mm (19 in x 25 in)	Plastic cover for 1ZD1824P	1ZD1925C
485 mm x 635 mm (19 in x 25 in)	Steel cover for 1ZD1824P	1ZD1925CA
504 mm x 657 mm (20 in x 26 in)	Drain (cover not included)	1ZD2425P
554 mm x 707 mm (22 in x 28 in)	Plastic cover for 1ZD2425P	1ZD1925E



TUF-TITE FLOOR DRAIN

Dimensions	Description	Code
279 mm x 279 mm x 381 mm (11 in x 11 in x 15 in)	Complete drain (5 pcs.)	1ZD115
279 mm x 279 mm x 381 mm (11 in x 11 in x 15 in)	Cover for 1ZD115	1ZD115C
100 mm (4 in)	Drain O-ring (blue)	1ZDANB
100 mm (4 in)	Drain O-ring (yellow)	1ZDANJ
100 mm (4 in)	Cap (orange)	1ZDBOUO

DATA SHEET

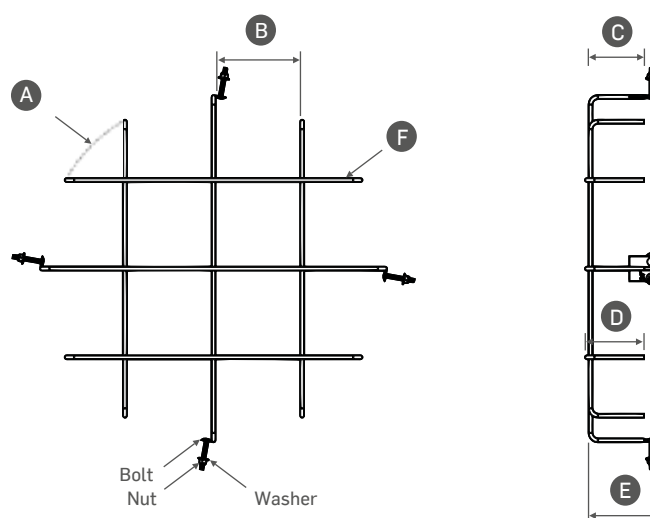
STAINLESS STEEL INSIDE GRATE

PRODUCT DESCRIPTION : stainless steel inside grate

FUNCTION : Allows to retain the detritus out of the pipe to which the grid is fixed

RAW MATERIAL : 304 stainless steel grate, nuts, washers and bolts

TECHNICAL DATA : Diameters : 300 mm (12 in) to 1 500 mm (60 in)



TECHNICAL DATA TABLE

Nominal diam. HDPE pipe		Ext. diam. A		B		C		D		E		Stem Diam. F		Bolts
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	Number
300	12	298	11.73	76	2.99	45	1.77	51	2.01	76	2.99	6.35	0.25	4
375	15	375	14.76	95	3.74	57	2.24	64	2.52	89	3.50	6.35	0.25	4
450	18	451	17.76	114	4.49	57	2.24	64	2.52	89	3.50	6.35	0.25	4
525	21	527	20.75	133	5.24	70	2.76	76	2.99	102	4.20	6.35	0.25	4
600	24	597	23.50	152	5.98	95	3.74	102	4.02	127	5.00	6.35	0.25	4
750	30	749	29.49	191	7.52	95	3.74	102	4.02	127	5.00	6.35	0.25	5
900	36	908	35.75	229	9.02	95	3.74	102	4.02	127	5.00	6.35	0.25	6
1050	42	1060	41.73	267	10.51	139	5.47	146	5.75	171	6.73	6.35	0.25	6
1200	48	1213	47.76	305	12.01	140	5.51	146	5.75	172	6.77	6.35	0.25	6
1500	60	1510	59.44	381	15.0	146	5.75	163	6.42	178	7.01	6.35	0.25	6

APPLICATION : Agricultural subsurface drainage

DATA SHEET

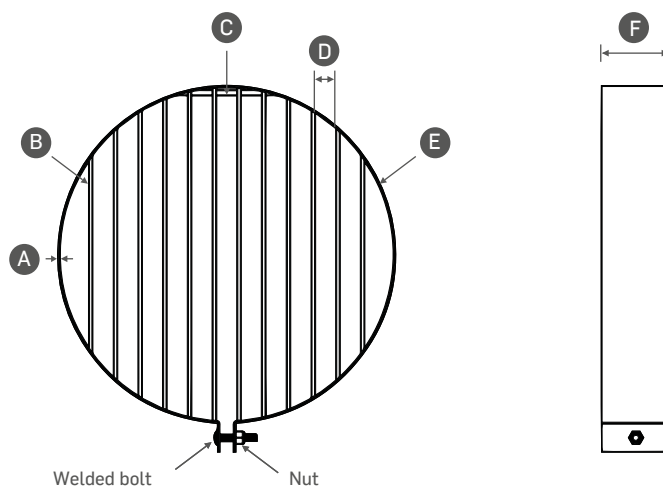
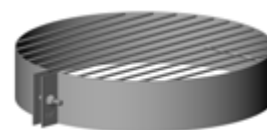
STAINLESS STEEL GRATE

PRODUCT DESCRIPTION : Stainless steel grate for HDPE pipes

FUNCTION : Allows to retain the detritus out of the pipe to which the grid is fixed

RAW MATERIAL : 304 stainless steel grate, nuts and bolts

TECHNICAL DATA : Diameters : 100 mm (4 in) to 250 mm (10 in)



TECHNICAL DATA TABLE

Nominal diam. HDPE pipe		A		Stem Diam. B		Stem Diam. C		D		E		F	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in
100	4	1.59	0.06	3.18	0.125	4.76	0.19	19	0.75	124	4.76	51	2.01
150	6	1.59	0.06	3.18	0.125	4.76	0.19	19	0.75	178	7.01	51	2.01
200	8	1.59	0.06	3.18	0.125	4.76	0.19	19	0.75	235	9.25	61	2.40
250	10	1.59	0.06	3.18	0.125	4.76	0.19	19	0.75	298	11.73	62	2.44

APPLICATION : Agricultural subsurface drainage

DATA SHEET

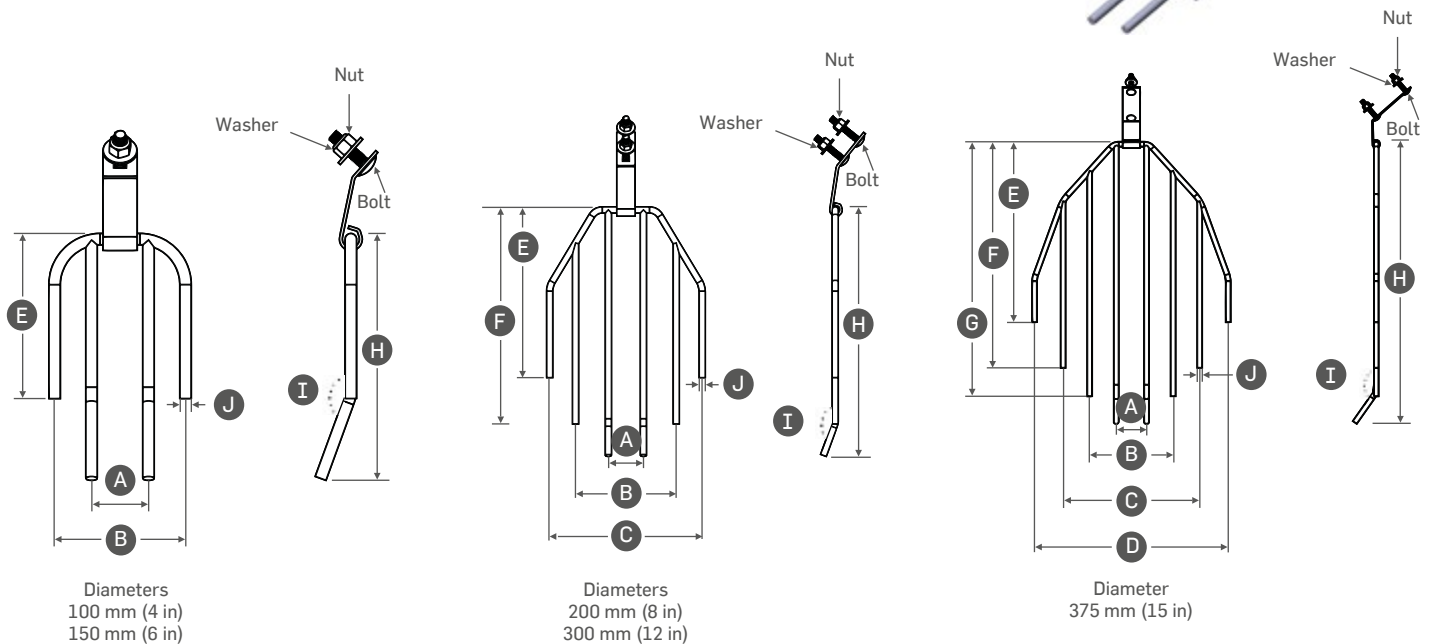
STAINLESS STEEL FORK GRATE

PRODUCT DESCRIPTION : Stainless steel fork grate

FUNCTION : Allows to retain the detritus out of the pipe to which the grid is fixed

RAW MATERIAL : 304 stainless steel grate, nuts, washers, hinge and bolts

TECHNICAL DATA : Diameters: 100 mm (4 in) to 375 mm (15 in)



TECHNICAL DATA TABLE

Nom. diam. HDPE pipe		A		B		C		D		E		F		G		H		I	Stem Diam. J	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	Degree	mm	in
100	4	32	1.26	73	2.87	N/A	N/A	N/A	N/A	92	3.62	N/A	N/A	N/A	N/A	138	5.43	160	6	0.24
150	6	36	1.42	102	4.02	N/A	N/A	N/A	N/A	137	5.39	N/A	N/A	N/A	N/A	192	7.56	160	6	0.24
200	8	37	1.46	105	4.13	159	6.26	N/A	N/A	178	7.01	226	8.90	N/A	N/A	260	10.24	160	6	0.24
250	10	45	1.77	124	4.88	191	7.52	N/A	N/A	235	9.25	298	11.73	N/A	N/A	343	13.50	155	6	0.24
300	12	51	2.01	152	5.98	254	10.00	N/A	N/A	247	9.72	337	13.27	N/A	N/A	410	16.14	145	8	0.31
375	15	51	2.01	140	5.51	229	9.02	324	12.76	302	11.89	378	14.88	426	16.77	472	18.58	145	8	0.31

APPLICATION : Agricultural subsurface drainage

DATA SHEET

NON-PERFORATED SOLFLO

PRODUCT DESCRIPTION : Non-perforated flexible single wall pipe, with corrugated interior and exterior walls for roadway use

FUNCTION : Gravity flow of water towards an outlet

MANUFACTURING STANDARDS : BNQ 3624-110 or AASHTO M294

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 100 mm (4 in) to 300 mm (12 in)
Length: 6 m (19.68 ft) to 1,200 m (3,937 ft)
Strength in compression: 210 kPa and 300 kPa
Manning: 0.015 to 0.019

AVAILABLE COUPLERS : Soil tight: see technical data table



Non-perforated SOLFLO

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Length		Manning	Strength	Backfill				Soil tight connectors		
										Minimum		Maximum		HDPE		
mm	in	mm	in	mm	in	m	ft	n	kPa	m	ft	m	ft	DBS	DBC	SC
100	4	100	3.9	118	4.6	30/600/1,200	98.4/1,968.5/3,937	0.015	300	0.4	1.3	12.0	39.4	X	X	
150	6	150	5.9	176	6.9	30/75/300	98.4/246.1/984.2	0.016	300	0.4	1.3	12.0	39.4	X		X
200	8	198	7.8	228	9.0	6/12	19.68/39.4	0.016	300	0.4	1.3	12.0	39.4	X		X
300	12	305	12.0	367	14.4	6	19.68	0.019	210*	0.4	1.3	10.0	32.8		X	X

Note 1: Values in the table are approximate and may change without notice.

Note 2: 100 mm (4 in) and 150 mm (6 in) Solflo: only 30 m lengths in inventory.

Note 3: 200 mm (8 in) and 300 mm (12 in) Solflo: only 6 m lengths in inventory.

Note 4: * certified according to standard M294.

LEGEND

DBS: double bell snap

DBC: double bell with clips

SC: split coupler

APPLICATIONS : Drainage collector and outlet
Surface evacuation

OPTION : Cord to ease the passage of the mandrill gauge.

ACCESSORIES : Consult the Solflo and Solflo Max Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of Solflo accessories.

SPECIFICATIONS

NON-PERFORATED SOLFLO

SCOPE

These specifications apply to all 100 mm (4 in) to 300 mm (12 in) diameter non-perforated Solflo pipes used for gravity flow of water.

PIPE REQUIREMENTS

Pipes shall be Solflo, manufactured with corrugated interior and exterior walls.

- Pipes shall be certified as per standard BNQ 3624-110 and have a minimum stiffness of 300 kPa for diameters between 100 mm (4 in) and 200 mm (8 in).
- Pipes shall be certified as per standard M294 and have a minimum of 210 kPa for 300 mm (12 in) diameters.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 100 mm (4 in) to 200 mm (8 in): double bell snap (safety catch) to ensure proper positioning on pipes.
- 300 mm (12 in): double bell with clips to ensure proper pipe insertion.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-110.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

PERFORATED OR PERFORATED AND FILTERED SOLFLO

PRODUCT DESCRIPTION : Perforated flexible single wall pipe, with corrugated interior and exterior walls for roadway use

FUNCTION : Used to collect surface runoff and control the ground water level

MANUFACTURING STANDARDS : BNQ 3624-110 or AASHTO M294

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 100 mm (4 in) to 300 mm (12 in)
Length: 6 m (19.68 ft) to 1,200 m (3,937 ft)
Strength in compression: 210 kPa and 300 kPa
Manning: 0.015 to 0.019
Filter: Needle punched nonwoven, Routex III-IV geotextile with 110-micron openings

AVAILABLE COUPLERS : Soil tight: see technical data table



Perforated unfiltered SOLFLO



SOLFLO perforated and filtered with Routex III-IV

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Length		Manning	Strength	Backfill				Soil tight connectors		
										Minimum	Maximum	Minimum	Maximum	HDPE		
mm	in	mm	in	mm	in	m	ft	n	kPa	m	ft	m	ft	DBS	DBC	SC
100	4	100	3.9	118	4.6	30/600/1,200	98.4/1,968.5/3,937	0.015	300	0.4	1.3	12.0	39.4	X	X	
150	6	150	5.9	176	6.9	30/75/300	98.4/246.1/984.2	0.016	300	0.4	1.3	12.0	39.4	X		X
200	8	198	7.8	228	9.0	6/12	19.68/39.4	0.016	300	0.4	1.3	12.0	39.4	X		X
300	12	305	12.0	367	14.4	6	19.68	0.019	210	0.4	1.3	10.0	32.8		X	X

Note 1: Values in the table are approximate and may change without notice.

Note 2: 100 mm (4 in) and 150 mm (6 in) Solflo: only 30 m (98.4 ft) lengths in inventory.

Note 3: 200 mm (8 in) and 300 mm (12 in) Solflo: only 6 m (19.68 ft) lengths in inventory.

Note 4: * certified according to standard M294.

LEGEND

DBS: double bell snap

DBC: double bell with clips

SC: split coupler

APPLICATIONS : Roadway drainage

Trench drain

Turf and recreation drainage

OPTIONS : Specific perforations (minimum volume required) and other filtering geotextiles are available upon request.

Cord to ease the passage of the mandrill gauge.

ACCESSORIES : Consult the Solflo and Solflo Max Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of Solflo accessories.

SPECIFICATIONS

PERFORATED SOLFLO

SCOPE

These specifications apply to all 100 mm (4 in) to 300 mm (12 in) diameter perforated Solflo pipes used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be Solflo, manufactured with corrugated interior and exterior walls.

- Pipes shall be certified as per standard BNQ 3624-110 and have a minimum stiffness of 300 kPa for diameters between 100 mm (4 in) and 200 mm (8 in).
- Pipes shall be certified as per standard M294 and have a minimum of 210 kPa for 300 mm (12 in) diameters.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

- Perforations shall be Type 2, in compliance with standard BNQ 3624-110. Total perforation area shall be at least 32 cm² per meter (1.5 in² per foot) of length.

JOINT PERFORMANCE

- 100 mm (4 in) to 200 mm (8 in): double bell snap (safety catch) to ensure proper positioning on pipes.
- 300 mm (12 in): double bell with clips to ensure proper pipe insertion.

ACCESSORIES

- Accessories will be manufactured according to specified requirements in the standard BNQ 3624-110.
- Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

NON-PERFORATED SOLFLO MAX

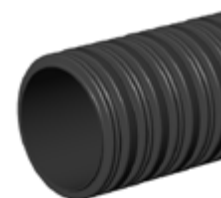
PRODUCT DESCRIPTION : Non-perforated rigid dual wall pipe, with smooth interior and corrugated exterior walls for roadway use

FUNCTION : Gravity flow of water in networks and water course piping

MANUFACTURING STANDARD: CSA B182.8, BNQ 3624-120 or AASHTO M294

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 100 mm (4 in) to 1,500 mm (60 in)
Length: 3.30 m (10.8 ft), 4 m (13.1 ft), 6 m (19.7 ft), 9 m (29.5 ft), 12 m (39.4 ft)
Strength in compression: see technical data table
Manning: 0.010 to 0.012



Non-perforated SOLFLO MAX

AVAILABLE COUPLERS : Watertight and soil tight: see technical data table

TECHNICAL DATA TABLES

Nom. dia.		Int. dia.		Ext. dia.		Manning	Strength		Minimum backfill		Maximum backfill			
mm	in	mm	in	mm	in	n	210 kPa	320 kPa	m	ft	210 kPa		320 kPa	
											m	ft	m	ft
100	4	101	4.0	120	4.7	0.010		X	0.3	1.0			20.0	65.6
150	6	150	5.9	176	6.9	0.010		X	0.3	1.0			19.0	62.3
200	8	205	8.1	236	9.3	0.010		X	0.3	1.0			18.0	59.1
250	10	251	9.9	296	11.7	0.010		X	0.3	1.0			18.0	59.1
300	12	303	11.9	363	14.3	0.010	X	X	0.3	1.0	16.0	52.5	17.0	55.8
375	15	381	15.0	449	17.7	0.010	X	X	0.3	1.0	17.0	55.8	17.0	55.8
450	18	457	18.0	543	21.4	0.010	X	X	0.3	1.0	17.0	55.8	17.0	55.8
525	21	535	21.1	640	25.2	0.012	X	X	0.3	1.0	16.0	52.5	16.0	52.5
600	24	603	23.8	722	28.4	0.012	X	X	0.3	1.0	15.0	49.2	15.0	49.2
750	30	756	29.8	890	35.0	0.012	X	X	0.3	1.0	13.0	42.7	15.0	49.2
900	36	912	35.9	1,046	41.2	0.012	X	X	0.3	1.0	13.0	42.7	15.0	49.2
1,050	42	1,067	42.0	1,225	48.2	0.012	145 kPa*		0.3	1.0	13.0 m (145 kPa)		42.7 ft (145 kPa)	
1,200	48	1,220	48.0	1,377	54.2	0.012	135 kPa*		0.3	1.0	13.0 m (135 kPa)		42.7 ft (135 kPa)	
1,500	60	1,512	59.5	1,684	66.3	0.012	105 kPa*		0.5	1.6	11.0 m (105 kPa)		36.1 ft (105 kPa)	

Note 1: Values in the table are approximate and may change without notice.
Note 2: * Certified AASHTO M294.

DATA SHEET

NON-PERFORATED SOLFLO MAX (CONT.)

AVAILABLE COUPLERS

Nom. dia.		Int. dia.		Ext. dia.		Soil tight connectors				Watertight connectors			
						HDPE				HDPE			
mm	in	mm	in	mm	in	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	101	4.0	120	4.7		X					X	
150	6	150	5.9	176	6.9		X		X			X	
200	8	205	8.1	236	9.3		X		X			X	
250	10	251	9.9	296	11.7		X		X			X	
300	12	303	11.9	363	14.3			X	X				X
375	15	381	15.0	449	17.7	X			X	X			
450	18	457	18.0	543	21.4	X			X	X			
525	21	535	21.1	640	25.2	X			X	X			
600	24	603	23.8	722	28.4	X			X	X			
750	30	756	29.8	890	35.0	X			X	X			
900	36	912	35.9	1,046	41.2	X			X	X			
1,050	42	1,067	42.0	1,225	48.2				X		X		
1,200	48	1,220	48.0	1,377	54.2				X		X		
1,500	60	1,512	59.5	1,684	66.3				X		X		

Note 1: Values in the table are approximate and may change without notice.

LENGTH 320 KPA

Nom. dia.		Type of coupler													
		PE		BC		DBS		DBC		BIGC		DBIG		DBIGC	
mm	in	m	ft	m	ft	m	ft	m	ft	m	ft	m	ft	m	ft
100	4	6	19.68			6	19.68					4/6	13.1/19.68		
150	6	6	19.68			6	19.68					4/6	13.1/19.68		
200	8	6	19.68			6	19.68					4/6	13.1/19.68		
250	10	6	19.68			6	19.68					4/6	13.1/19.68		
300	12	6/9	19.68/29.5					6	19.68					4/6	13.1/19.68
375	15	6/9	19.68/29.5	6	19.68					4/6	13.1/19.68				
450	18	6/9	19.68/29.5	6	19.68					4/6	13.1/19.68				
525	21	6/9	19.68/29.5	6	19.68					4/6	13.1/19.68				
600	24	6/9	19.68/29.5	6	19.68					4/6	13.1/19.68				
750	30	6/9	19.68/29.5	6	19.68					4/6	13.1/19.68				
900	36	6/9	19.68/29.5	6	19.68					4/6	13.1/19.68				

LENGTH 210 KPA

Nom. dia.		Type of coupler					
		PE		BC		DBC	
mm	in	m	ft	m	ft	m	ft
300	12	6/9/12	19.68/29.5/39.4			6	19.68
375	15	6/9/12	19.68/29.5/39.4	6	19.68		
450	18	6/9/12	19.68/29.5/39.4	6	19.68		
525	21	6/9/12	19.68/29.5/39.4	6	19.68		
600	24	6/9/12	19.68/29.5/39.4	6	19.68		
750	30	6/9/12	19.68/29.5/39.4	6	19.68		
900	36	6/9	19.68	6	19.68		

LENGTH AASHTO M294

Nom. dia.		Type of coupler			
		PE		BG	
mm	in	m	ft	m	ft
1,050	42	6	19.68	3.08	10.1
1,200	48	6	19.68	3.08	10.1
1,500	60	6	19.68	6	19.68

LEGEND
PE: plain end
BC: bell with clips
DBC: double bell with clips

DBIG: double bell with integrated gasket
SC: split coupler
DBIGC: double bell with integrated gasket with clips

DBS: double bell snap
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DATA SHEET

NON-PERFORATED SOLFLO MAX (CONT.)

APPLICATIONS : CONVEYING

Driveway culvert
HDPE culvert
Storm water sewer
Beaver Barrier
Ventilation
Ditch piping
Roadway drainage (outlet)
Drainage collector and outlet

STORAGE

Detention system

OPTIONS : Special lengths are available upon request

Beveled cut
Service entrance/catch basin connection
Fish baffles

ACCESSORIES : Consult the Solflo and Solflo Max Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of Solflo Max accessories.

SPECIFICATIONS

WATERTIGHT SOLFLO MAX

SCOPE

These specifications apply to all 100 mm (4 in) to 1,500 mm (60 in) diameter Solflo pipes used for drainage, water course piping and gravity flow storm water sewers.

PIPE REQUIREMENTS

Pipes shall be Solflo Max, manufactured with smooth interior and corrugated exterior walls.

- 100 mm (4 in) to 900 mm (36 in) diameter pipes shall be conformed to CSA B182.8 and shall be certified as per standard BNQ 3624-120, shall have minimum stiffness of 320 kPa and be 6 m (19.68 ft) long.
- 1,050 mm (42 in) and 1,200 mm (48 in) diameter pipes shall be certified as per standard AASHTO M294, shall have minimum stiffness of 145 kPa for 1,050 mm (42 in) diameters, 135 kPa for 1,200 mm (48 in) diameters and be 3.30 m (10.8 ft) long.
- 1,500 mm (60 in) diameter pipes shall be certified as per standard AASHTO M294, shall have minimum stiffness of 105 kPa and be 6 m (19.68 ft) long.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 100 mm (4 in) to 250 mm (10 in): double bell with integrated gasket. Double bells shall be certified as per standard CSA B182.8, Type 1 (watertight at 10.8 psi). The gasket shall be co-injected into the bell and shall overlap two corrugations.
- 300 mm (12 in): double bell with integrated gasket with clips to ensure proper pipe insertion. Double bell shall be conformed to CSA B182.8, Type 1 (Watertight at 10.8 psi). The gasket shall be co-injected into the bell and shall overlap two corrugations.
- 381 mm (15 in) to 900 mm (36 in): bell with integrated gasket with clips welded to the pipe to ensure proper pipe insertion. Double bell shall be conformed to CSA B182.8, Type 1 (Watertight at 10.8 psi). The gasket shall be co-injected into the bell.
- 1,050 mm (42 in) to 1,500 mm (60 in): bell with O-Ring gasket installed by the manufacturer onto the male end, covered by a protective film for shipping. The assembly shall be certified as per standard AASHTO M294.
- O-ring gaskets shall comply with standards ASTM F477 and BNQ 3624-120. Fused and/or glued gaskets will not be permitted.

ACCESSORIES

- 100 to 900 mm (4 to 36 in): accessories will be manufactured according to specified requirements in the standard BNQ 3624-120.
- 1,050 to 1,500 mm (42 to 60 in): accessories will be manufactured according to specified requirements in the standard AASHTO M294.
- **Connections, service connections:**
 - Service connections for 100 mm (4 in), 150 mm (6 in) and 200 mm (8 in) diameter PVC pipes shall be monolithic (factory welded), as required by standard BNQ 1809-300, and equipped with an integrated O-ring gasket.
 - Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth exterior wall and an end bell with integrated gasket, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded, both outside and inside, when this side is accessible.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.soleno.com for installation recommendations.

SPECIFICATIONS

SOIL TIGHT SOLFLO MAX

SCOPE

These specifications apply to all 100 mm (4 in) to 1,500 mm (60 in) diameter Solflo Max pipes used for drainage and water course piping.

PIPE REQUIREMENTS

Pipes shall be Solflo Max, manufactured with smooth interior and corrugated exterior walls.

- 100 mm (4 in) to 900 mm (36 in) diameter pipes shall be conformed to CSA B182.8 and shall be certified as per standard BNQ 3624-120; they shall have minimum stiffness of 320 kPa for 100 mm (4 in) to 250 mm (10 in) diameters, 210 kPa or 320 kPa for 300 mm (12 in) to 900 mm (36 in) diameters and be 6 m (19.68 ft) long.
- 1,050 mm (42 in) and 1,200 mm (48 in) diameter pipes shall be certified as per standard AASHTO M294; they shall have minimum stiffness of 145 kPa for 1,050 mm (42 in) diameters, 135 kPa for 1,200 mm (48 in) diameters and be 3.30 m (10.8 ft) long.
- 1,500 mm (60 in) diameter pipes shall be certified as per standard AASHTO M294; they shall have minimum stiffness of 105 kPa and be 6 m (19.68 ft) long.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 100 mm (4 in) to 300 mm (10 in): double bell snap (safety catch) to ensure proper positioning on pipes.
- 300 mm (12 in): double bell with clips to ensure proper pipe insertion.
- 375 mm (15 in) to 900 mm (36 in): bell with clips welded to the pipe to ensure proper pipe insertion.
- 1,050 mm (42 in) to 1,500 mm (60 in): bell with O-Ring gasket installed by the manufacturer onto the male end, covered by a protective film for shipping.

ACCESSORIES

- Accessories from 100 mm (4 in) to 900 mm (36 in) will be manufactured according to specified requirements in the standard BNQ 3624-120.
- Accessories from 1,050 mm (42 in) to 1,500 mm (60 in) will be manufactured according to specified requirements in the standard AASHTO M294.
- Every connection to a concrete structure shall be made with:
 - Service connections for 100 mm (4 in), 150 mm (6 in) and 200 mm (8 in) diameter PVC pipes shall be monolithic (factory welded), as required by standard BNQ 1809-300, and equipped with an integrated O-ring gasket.
 - Every connection has a concrete structure and shall be made with a monolithic HDPE adapter with smooth exterior wall and an end bell with integrated gasket, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded, both outside and inside, when this side is accessible.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

PERFORATED OR PERFORATED AND FILTERED SOLFLO MAX

PRODUCT DESCRIPTION : Perforated rigid dual wall pipe, with smooth interior and corrugated exterior walls for roadway use

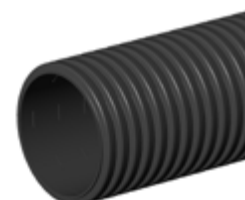
FUNCTION : Used to collect surface runoff and control the ground water level

MANUFACTURING STANDARD: BNQ 3624-120 or AASHTO M294

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 100 mm (4 in) to 1,500 mm (60 in)
Length: 3.30 m (10.8 ft), 6 m (19.68 ft), 9 m (29.5 ft), 12 m (39.4 ft)
Strength in compression: see technical data table
Manning: 0.010 to 0.012
Filter: Needle punched nonwoven geotextile
Routex III-IV with 110-micron openings

AVAILABLE COUPLERS : Soil tight: see technical data table



Perforated SOLFLO MAX



SOLFLO MAX perforated and filtered with Routex III-IV

TECHNICAL DATA TABLES

Nom. dia.		Int. dia.		Ext. dia.		Manning	Strength		Minimum backfill		Maximum backfill			
mm	in	mm	in	mm	in	n	210 kPa	320 kPa	m	ft	210 kPa		320 kPa	
											m	ft	m	ft
100	4	101	4.0	120	4.7	0.010		X	0.3	1.0			20.0	65.6
150	6	150	5.9	176	6.9	0.010		X	0.3	1.0			19.0	62.3
200	8	205	8.1	236	9.3	0.010		X	0.3	1.0			18.0	59.1
250	10	251	9.9	296	11.7	0.010		X	0.3	1.0			18.0	59.1
300	12	303	11.9	363	14.3	0.010	X	X	0.3	1.0	16.0	52.5	17.0	55.8
375	15	381	15.0	449	17.7	0.010	X	X	0.3	1.0	17.0	55.8	17.0	55.8
450	18	457	18.0	543	21.4	0.010	X	X	0.3	1.0	17.0	55.8	17.0	55.8
525	21	535	21.1	640	25.2	0.012	X	X	0.3	1.0	16.0	52.5	16.0	52.5
600	24	603	23.8	722	28.4	0.012	X	X	0.3	1.0	15.0	49.2	15.0	49.2
750	30	756	29.8	890	35.0	0.012	X	X	0.3	1.0	13.0	42.7	15.0	49.2
900	36	912	35.9	1,046	41.2	0.012	X	X	0.3	1.0	13.0	42.7	15.0	49.2
1,050	42	1,067	42.0	1,225	48.2	0.012	145 kPa*		0.3	1.0	13.0 m (145 kPa)		42.7 ft (145 kPa)	
1,200	48	1,220	48.0	1,377	54.2	0.012	135 kPa*		0.3	1.0	13.0 m (135 kPa)		42.7 ft (135 kPa)	
1,500	60	1,512	59.5	1,684	66.3	0.012	105 kPa*		0.5	1.6	11.0 m (105 kPa)		36.1 ft (105 kPa)	

Note 1: Values in the table are approximate and may change without notice.

Note 2: * Certified AASHTO M294.

DATA SHEET

PERFORATED OR PERFORATED AND FILTERED SOLFLO MAX (CONT.)

AVAILABLE COUPLERS

Nom. dia.		Int. dia.		Ext. dia.		Soil tight connectors				
						HDPE				
mm	in	mm	in	mm	in	BC	DBS	DBC	SC	BG
100	4	101	4.0	120	4.7		X			
150	6	150	5.9	176	6.9		X		X	
200	8	205	8.1	236	9.3		X		X	
250	10	251	9.9	296	11.7		X		X	
300	12	303	11.9	363	14.3			X	X	
375	15	381	15.0	449	17.7	X			X	
450	18	457	18.0	543	21.4	X			X	
525	21	535	21.1	640	25.2	X			X	
600	24	603	23.8	722	28.4	X			X	
750	30	756	29.8	890	35.0	X			X	
900	36	912	35.9	1,046	41.2	X			X	
1,050	42	1,067	42.0	1,225	48.2				X	X
1,200	48	1,220	48.0	1,377	54.2				X	X
1,500	60	1,512	59.5	1,684	66.3				X	X

Note 1: Values in the table are approximate and may change without notice.

LENGTH 320 KPA

Nom. dia.		Type of coupler							
		PE		BC		DBS		DBC	
		m	ft	m	ft	m	ft	m	ft
100	4	6	19.68			6	19.68		
150	6	6	19.68			6	19.68		
200	8	6	19.68			6	19.68		
250	10	6	19.68			6	19.68		
300	12	6/9	19.68/29.5					6	19.68
375	15	6/9	19.68/29.5	6	19.68				
450	18	6/9	19.68/29.5	6	19.68				
525	21	6/9	19.68/29.5	6	19.68				
600	24	6/9	19.68/29.5	6	19.68				
750	30	6/9	19.68/29.5	6	19.68				
900	36	6/9	19.68/29.5	6	19.68				

LENGTH 210 KPA

Nom. dia.		Type of coupler					
		PE		BC		DBC	
mm	in	m	ft	m	ft	m	ft
300	12	6/9/12	19.68/29.5/39.4			6	19.68
375	15	6/9/12	19.68/29.5/39.4	6	19.68		
450	18	6/9/12	19.68/29.5/39.4	6	19.68		
525	21	6/9/12	19.68/29.5/39.4	6	19.68		
600	24	6/9/12	19.68/29.5/39.4	6	19.68		
750	30	6/9/12	19.68/29.5/39.4	6	19.68		
900	36	6/9	19.68	6	19.68		

LENGTH AASHTO M294

Nom. dia.		Type of coupler			
		PE		BG	
mm	in	m	ft	m	ft
1,050	42	6	19.68	3.08	10.1
1,200	48	6	19.68	3.08	10.1
1,500	60	6	19.68	6	19.68

LEGEND

PE: plain end
SC: split coupler

DBC: double bell with clips
DBS: double bell snap

BC: bell with clips
BG: O-ring bell gasket

DATA SHEET

PERFORATED OR PERFORATED AND FILTERED SOLFLO MAX (CONT.)

APPLICATIONS : Trench drain
Roadway drainage
Turf and recreation drainage
Leachate collection
Ditch piping
Aerial and subsurface ventilation pipes
Retention system

OPTIONS : Special lengths, specific perforations and other filtering geotextiles are available upon request.

ACCESSORIES : Consult the Solflo and Solflo Max Couplers and Accessories section in this technical catalogue to obtain more information about our complete line of Solflo Max accessories.

SPECIFICATIONS

PERFORATED SOLFLO MAX

SCOPE

These specifications apply to all 100 mm (4 in) to 1,500 mm (60 in) diameter perforated Solflo Max pipes used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be Solflo Max, manufactured with smooth interior and corrugated exterior walls.

- 100 mm (4 in) to 900 mm (36 in) diameter pipes shall be certified as per standard BNQ 3624-120; they shall have a minimum stiffness of 320 kPa for diameters between 100 mm (4 in) and 250 mm (10 in), 210 kPa or 320 kPa for diameters between 300 mm (12 in) and 900 mm (36 in) and be 6 m (19.68 ft) long.
- 1,050 mm (42 in) and 1,200 mm (48 in) diameter pipes shall be certified as per standard AASHTO M294; they shall have minimum stiffness of 145 kPa for 1,050 mm (42 in) diameters, 135 kPa for 1,200 mm (48 in) diameters and be 3.30 m (10.8 ft) long.
- 1,500 mm (60 in) diameter pipes shall be certified as per standard AASHTO M294; they shall have a minimum stiffness of 105 kPa and be 6 m (19.68 ft) long.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS

Perforations shall be Type 2, in compliance with standard BNQ 3624-120. Total perforation area shall be at least 32 cm² per meter (1.5 in² per foot) of length.

JOINT PERFORMANCE

- 100 mm (4 in) to 250 mm (10 in): double bell snap (safety catch) to ensure proper positioning on pipes.
- 300 mm (12 in): double bell with clips to ensure proper pipe insertion.
- 375 mm (15 in) to 900 mm (36 in): bell with clips welded to the pipe to ensure proper pipe insertion.
- 1,050 mm (42 in) to 1,500 mm (60 in): bell with O-Ring gasket installed by the manufacturer onto the male end, covered by a protective film for shipping.

ACCESSORIES

- Accessories from 100 mm (4 in) to 900 mm (36 in) will be manufactured according to specified requirements in the standard BNQ 3624-120.
- Accessories from 1,050 mm (42 in) to 1,500 mm (60 in) will be manufactured according to specified requirements in the standard AASHTO M294.
- **Connections, service connections:**
 - Service connections for 100 mm (4 in), 150 mm (6 in) and 200 mm (8 in) diameter PVC pipes shall be monolithic (factory welded), as required by standard BNQ 1809-300, and equipped with an integrated O-ring gasket.
 - Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded, both outside and inside, when this side is accessible.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

SPECIFICATIONS

PERFORATED AND FILTERED SOLFLO MAX

SCOPE

These specifications apply to all 100 mm (4 in) to 1,500 mm (60 in) diameter perforated Solflo Max pipes covered with a geotextile, used to collect surface runoff and control the ground water level.

PIPE REQUIREMENTS

Pipes shall be Solflo Max, manufactured with smooth interior and corrugated exterior walls.

- 100 mm (4 in) to 900 mm (36 in) diameter pipes shall be certified as per standard BNQ 3624-120; they shall have minimum stiffness of 320 kPa for diameters between 100 mm (4 in) and 250 mm (10 in), 210 kPa or 320 kPa for diameters between 300 mm (12 in) and 900 mm (36 in) and be 6 m (19.68 ft) long.
- 1,050 mm (42 in) to 1,500 mm (60 in) diameter pipes shall be certified as per standard AASHTO M294, shall have minimum stiffness of 145 kPa for diameters of 1,050 mm (42 in), 135 kPa for diameters of 1,200 mm (48 in) and be 3.30 m (10.8 ft) long.
- 1,500 mm (60 in) diameter pipes shall be certified as per standard AASHTO M294, shall have minimum stiffness of 105 kPa and be 6 m (19.68 ft) long.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

PERFORATIONS AND FILTERS

Perforations shall be Type 2, in compliance with standard BNQ 3624-120. Total perforation area shall be at least 32 cm² per meter (1.5 in² per foot) of length. The filter shall be made from a Routex III-IV BNQ G.C.T.T.G 3001 geotextile with Soleno Textile factory-installed UV protection.

JOINT PERFORMANCE

- 100 mm (4 in) to 250 mm (10 in): double bell snap (safety catch) to ensure proper positioning on pipes.
- 300 mm (12 in): double bell with clips to ensure proper pipe insertion.
- 375 mm (15 in) to 900 mm (36 in): bell with clips welded to the pipe to ensure proper pipe insertion.
- 1,050 mm (42 in) to 1,500 mm (60 in): bell with O-Ring gasket installed by the manufacturer onto the male end, covered by a protective film for shipping.
- The same geotextile shall cover both the assemblies and the pipe.

ACCESSORIES

- Accessories from 100 mm (4 in) to 900 mm (36 in) will be manufactured according to specified requirements in the standard BNQ 3624-120.
- Accessories from 1,050 mm (42 in) to 1,500 mm (60 in) will be manufactured according to specified requirements in the standard AASHTO M294.
- **Connections, service connections:**
 - Service connections for 100 mm (4 in), 150 mm (6 in) and 200 mm (8 in) diameter PVC pipes shall be monolithic (factory welded), as required by standard BNQ 1809-300, and equipped with an integrated O-ring gasket.
 - Every connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth interior wall and an end bell, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Accessories shall be factory welded, both outside and inside, when accessible.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

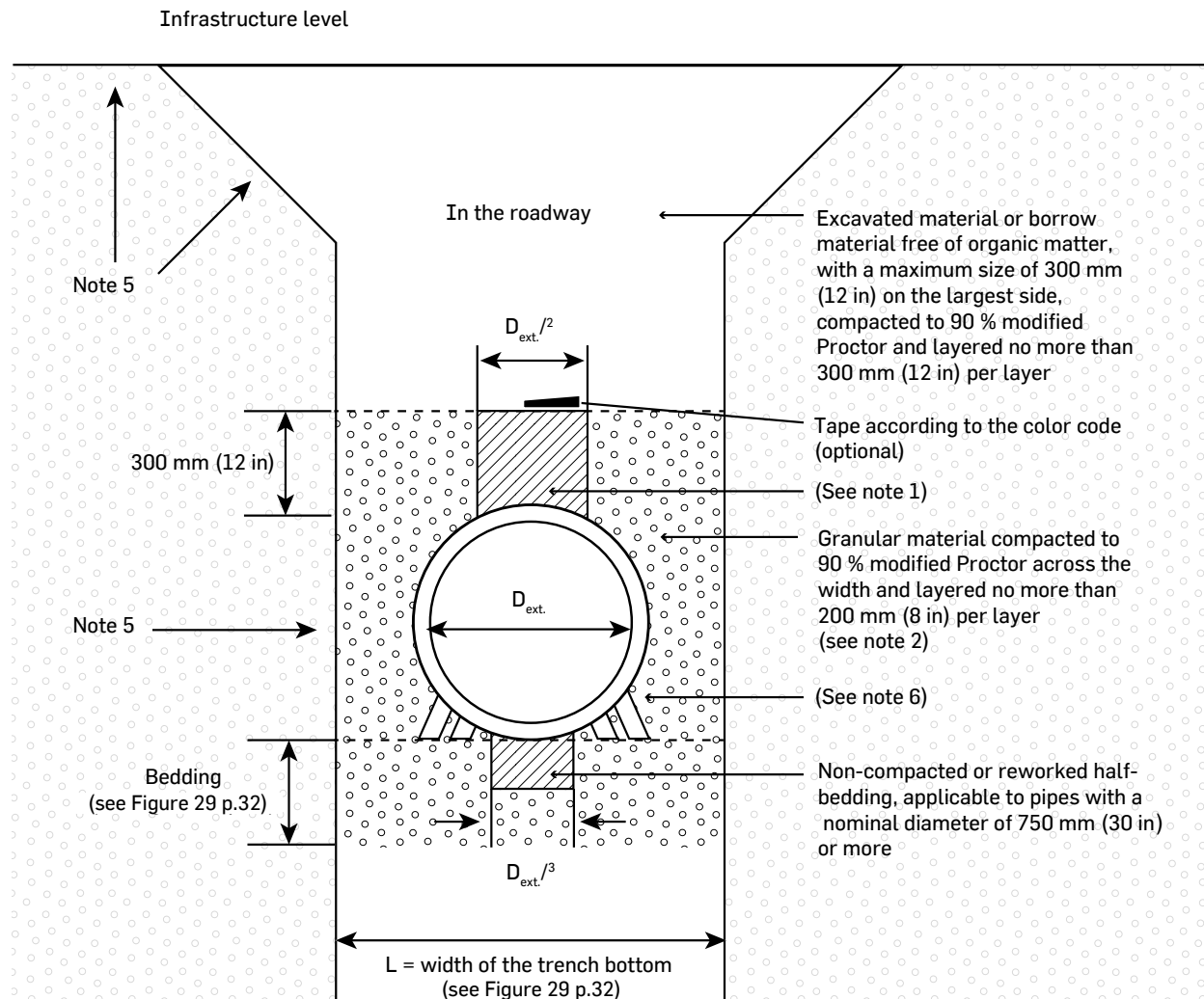
Installation shall be carried out in compliance with standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.soleno.com for installation recommendations.

INSTALLATION GUIDE

SOLFLO MAX HDPE PIPE INSTALLATION

The method of installation of SOLFLO MAX HDPE pipe, shown below, is based on the BNQ 1809-300 R2007 standard. However, depending on the location of the project, other regulations or standards may apply. We recommend that you comply with the regulation or standard of the city or province of reference. Otherwise, we recommend the application of the method shown here.

UNDER THE ROADWAY



Note 1 : The compaction equipment should never be driven in this area.

Note 2 : CG-14 granular material must be used, as required by Standard BNQ 2560-114-III.

Note 3 : $D_{ext.}$ is the outer diameter of the pipe.

Note 4 : For the first meter above the pipe, the only accepted compaction equipment are the groomer, the plate compactor and vibrating drum rollers. The total force applied from these equipments must not exceed 50,000 N.

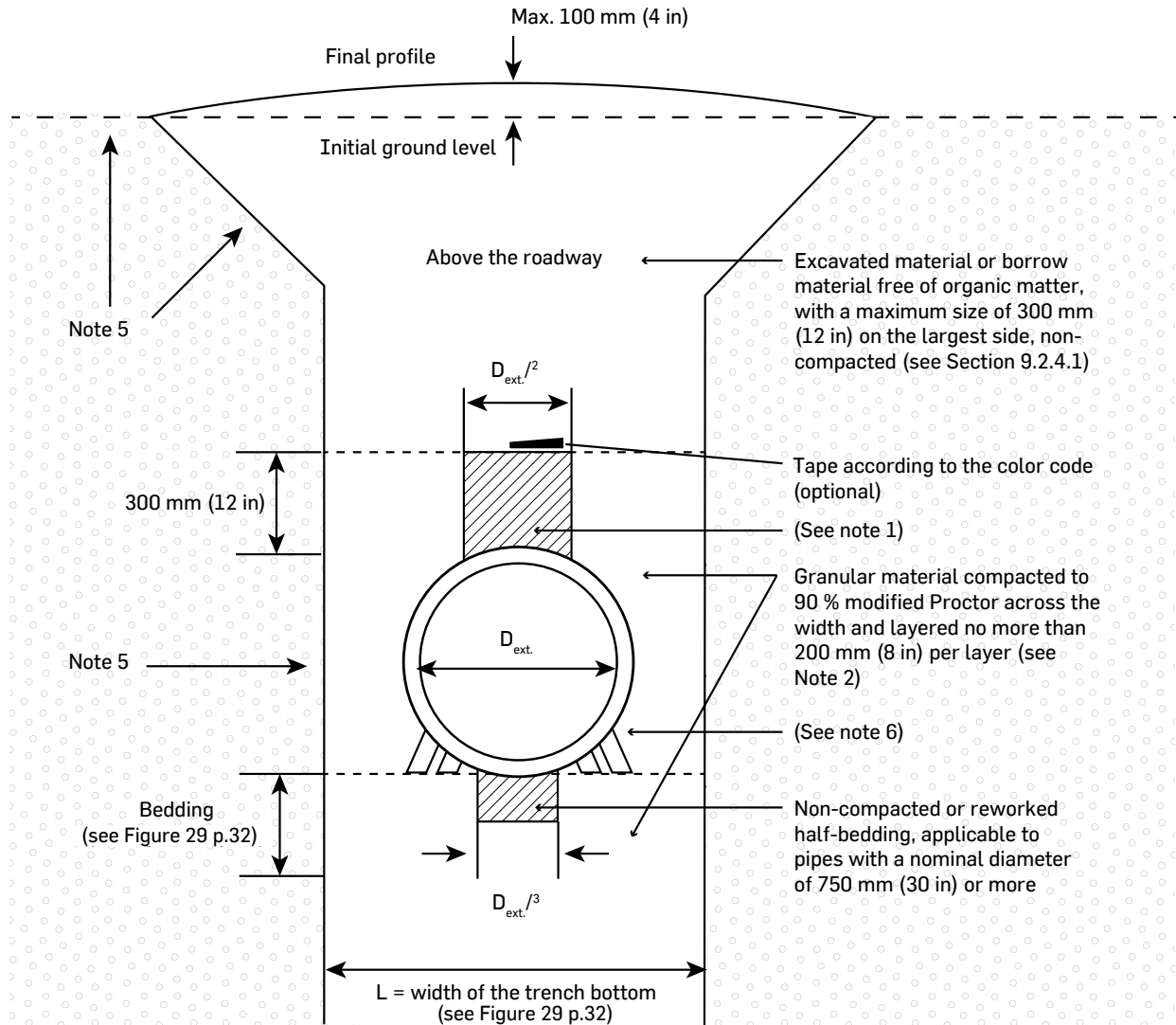
Note 5 : The slopes of the excavation are not restricted to only those slopes shown in the figure above. The excavation must meet the requirements of the *Safety Code for the Construction Industry*, particularly with regard to storage of equipment, the movement of vehicles around an excavation in the area, and slope stability.

Note 6 : It is important to place and firmly pack backfill material in this area to ensure adequate support to the pipe.

INSTALLATION GUIDE

SOLFLO MAX HDPE PIPE INSTALLATION

ABOVE THE ROADWAY



Note 1 : The compaction equipment should never be driven in this area.

Note 2 : CG-14 granular material must be used, as required by Standard BNQ 2560-114-III (see Section 6.5).

Note 3 : $D_{ext.}$ is the outer diameter of the pipe.

Note 4 : The top surface of backfill must not rise above 100 mm from the initial ground level.

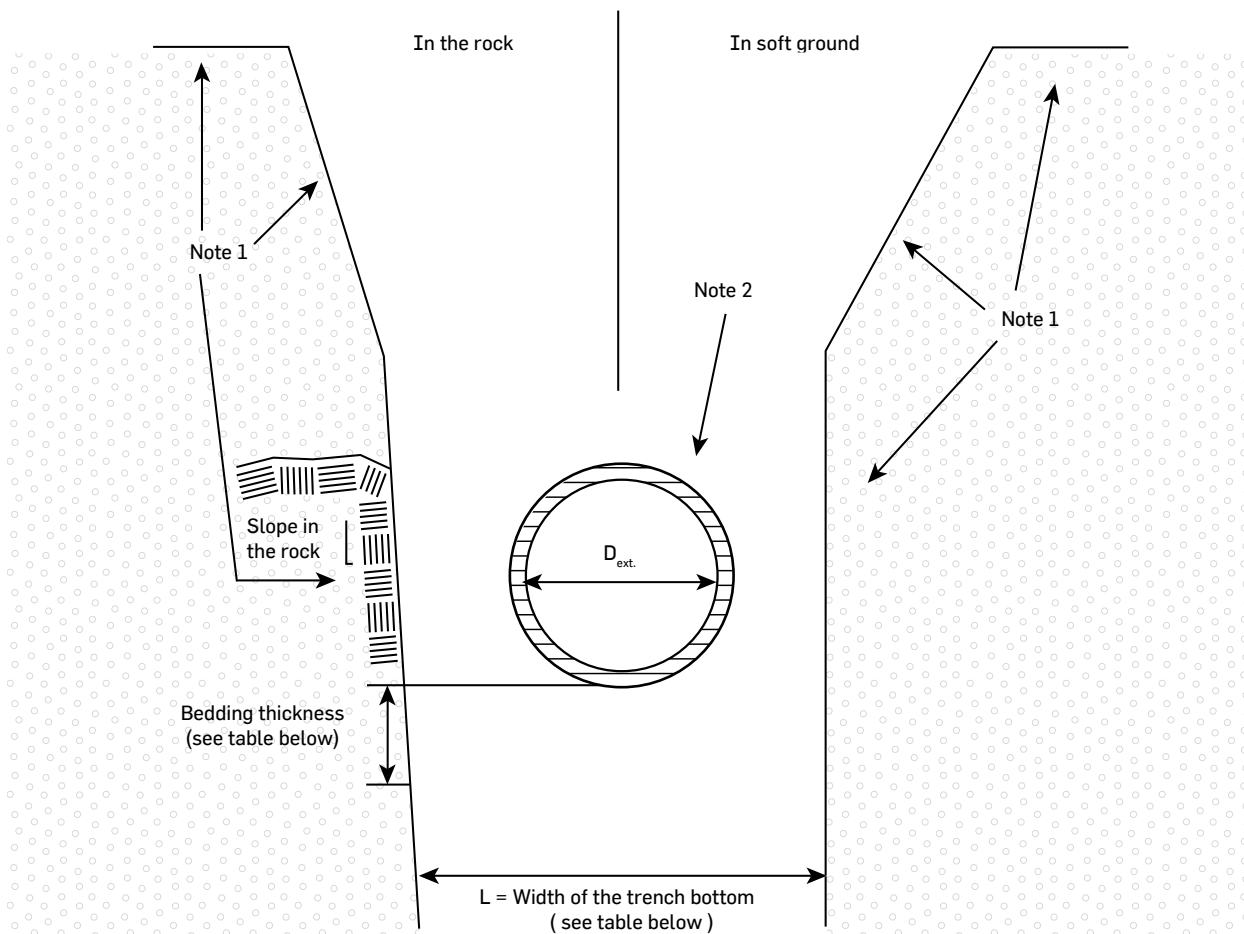
Note 5 : The slopes of the excavation are not restricted to only those slopes shown in the figure above. The excavation must meet the requirements of the *Safety Code for the Construction Industry*, particularly with regard to storage of equipment, the movement of vehicles around an excavation in the area, and slope stability.

Note 6 : It is important to place and firmly pack backfill material in this area to ensure adequate support to the pipe.

INSTALLATION GUIDE

SOLFLO MAX HDPE PIPE INSTALLATION

BEDDING AND TRENCH WIDTH FIGURE 29



Note 1 : The slopes of the excavation are not restricted to only those slopes shown in the figure above. The excavation must meet the requirements of the Safety Code for the

Construction Industry, particularly with regard to storage of equipment, the movement of vehicles around an excavation in the area, and slope stability.

Note 2 : The pipe must be positioned in the center of the trench. $D_{ext.}$ is the outer diameter of the pipe, expressed in millimeters.

Note 3 : The width of the bottom of the trench is the clear distance between the two faces of the shoring, if the latter is installed.

Nominal pipe diameter		Minimum thickness of the bedding in soil or in rock	Width at the bottom of the trench L
mm	po	mm	mm
300 and less	12 and less	150	$D_{ext.} + 600$
350 to 600	14 to 24	150	$D_{ext.} + 900$
750 to 1200	30 to 48	200	$D_{ext.} + 1200$
1350 to 1500	54 to 60	250	$D_{ext.} + 1200$
1800 and more	72 and more	300	a) $D_{ext.} + 1200$, if the trench is shored. b) $D_{ext.} + 900$, if the trench is not shored.

INSTALLATION GUIDE

SOLFLO MAX HDPE PIPE INSTALLATION

The particle size, physical properties and other characteristics of granular materials must comply with the requirements of Standard NQ-2560-114 III (see table below taken from this standard).

GRADING RANGE SPECIFICATIONS FOR THE BEDDING MATERIAL AND EMBEDMENT

Material	Sieve (mm)				Sieve (µm)		
	31.5	20	14	5	315	160	80
CG-14		100		35-100			0-10

Note 1: The new "CG-14" designation regroups the granulometric requirements for granular material used for the pipe bedding material and embedment, previously known by one or other of the following names: "0-¾ crushed stone" or "0-¾ gravel" and "Class A sand". The new designation is expressed as CG-14, in order to be consistent with the current nomenclature based on granular classes.

Note 2: The technical aspects of a project may require that more detailed specifications are to be described in the estimation of special technical clauses. Certain special technical conditions may cause the design engineer to specify a different grading range, especially if increased bearing capacity is sought.

SOLFLO AND SOLFLO MAX COUPLERS AND ACCESSORIES

COUPLERS



WATERTIGHT BELL
model shown double bell
with integrated gasket with clips

Type	Diameter	Code
DBIG	100 mm (4 in)	3MCC004GI
DBIG	150 mm (6 in)	3MCC006GI
DBIG	200 mm (8 in)	3MCC008GI
DBIG	250 mm (10 in)	3MCC010GI
DBIGC	300 mm (12 in)	3MCC012GI
BIGC	375 mm (15 in)	3MCC015GI
BIGC	450 mm (18 in)	3MCC018GI
BIGC	525 mm (21 in)	3MCC021GI
BIGC	600 mm (24 in)	3MCC024GI
BIGC	750 mm (30 in)	3MCC030GI
BIGC	900 mm (36 in)	3MCC036GI



SOIL TIGHT BELL
model shown double bell with clips

Type	Diameter	Code
DBS	100 mm (4 in)	3MCC004C
DBS	150 mm (6 in)	3MCC006C
DBS	200 mm (8 in)	3MCC008C
DBS	250 mm (10 in)	3MCC010C
DBC	300 mm (12 in)	3MCC012C
BC	375 mm (15 in)	3MCC015C
BC	450 mm (18 in)	3MCC018C
BC	525 mm (21 in)	3MCC021C
BC	600 mm (24 in)	3MCC024C
BC	750 mm (30 in)	3MCC030C
BC	900 mm (36 in)	3MCC036C



SPLIT COUPLER

Diameter	Code
300 mm (12 in)	2M300F
375 mm (15 in)	2M375F
450 mm (18 in)	2M450F
525 mm (21 in)	2M525F
600 mm (24 in)	2M600F
750 mm (30 in)	2M750F
900 mm (36 in)	2M900F
1,200 mm (48 in)	2M1200F

LEGEND

DBIG: double bell with integrated gasket
DBS: double bell snaps

DBIGC: double bell with integrated gasket with clips
DBC: double bell with clips

BIGC: bell with integrated gasket with clips
BC: bell with clips

ACCESSORIES



END CAP

Diameter	Code
100 mm (4 in)	2B100
150 mm (6 in)	2B150
200 mm (8 in)	2B200
250 mm (10 in)	2B250
300 mm (12 in)	2B300
375 mm (15 in)	2B375
450 mm (18 in)	2B450
525 mm (21 in)	2B525
600 mm (24 in)	2B600
750 mm (30 in)	2B750
900 mm (36 in)	2B900
1,200 mm (48 in)	2B1200



22.5-DEGREE ELBOW*

Diameter	Code
100 mm (4 in)	33C2204
150 mm (6 in)	33C2206
200 mm (8 in)	33C2208
250 mm (10 in)	33C2210
300 mm (12 in)	33C2212
375 mm (15 in)	33C2215
450 mm (18 in)	33C2218
525 mm (21 in)	33C2221
600 mm (24 in)	33C2224
750 mm (30 in)	33C2230
900 mm (36 in)	33C2236
1,050 mm (42 in)	33C2242
1,200 mm (48 in)	33C2248



45-DEGREE ELBOW*

Diameter	Code
100 mm (4 in)	33C4504
150 mm (6 in)	33C4506
200 mm (8 in)	33C4508
250 mm (10 in)	33C4510
300 mm (12 in)	33C4512
375 mm (15 in)	33C4515
450 mm (18 in)	33C4518
525 mm (21 in)	33C4521
600 mm (24 in)	33C4524
750 mm (30 in)	33C4530
900 mm (36 in)	33C4536
1,050 mm (42 in)	33C4542
1,200 mm (48 in)	33C4548

Note 1: Our Solflo and Solflo Max accessories are available with plain ends, watertight and soil tight couplers. Consult our price list to find out what diameters we offer.

*See products data sheet on following pages: 38 (22.5-DEGREE ELBOW) and 39 (45-DEGREE ELBOW).

SOLFLO AND SOLFLO MAX COUPLERS AND ACCESSORIES (CONT.)



90-DEGREE ELBOW*

Diameter	Code
100 mm (4 in)	33C9004
150 mm (6 in)	33C9006
200 mm (8 in)	33C9008
250 mm (10 in)	33C9010
300 mm (12 in)	33C9012
375 mm (15 in)	33C9015
450 mm (18 in)	33C9018
525 mm (21 in)	33C9021
600 mm (24 in)	33C9024
750 mm (30 in)	33C9030
900 mm (36 in)	33C9036
1,050 mm (42 in)	33C9042
1,200 mm (48 in)	33C9048



INLET

Diameter	Code
100 mm (4")	33ES04
150 mm (6")	33ES06
200 mm (8")	33ES08
250 mm (10")	33ES10



90-DEGREE ELBOW—3 SECTIONS*

Diameter	Code
100 mm (4 in)	33S9004
150 mm (6 in)	33S9006
200 mm (8 in)	33S9008
250 mm (10 in)	33S9010
300 mm (12 in)	33S9012
375 mm (15 in)	33S9015
450 mm (18 in)	33S9018
525 mm (21 in)	33S9021
600 mm (24 in)	33S9024
750 mm (30 in)	33S9030
900 mm (36 in)	33S9036
1,050 mm (42 in)	33S9042
1,200 mm (48 in)	33S9048



**UNIVERSAL SOL TEE
(Sol-Tee)**

Diameter	Code
100 mm (4 in)	33LS041248
150 mm (6 in)	33LS061548
200 mm (8 in)	33LS081848



**O-RING
(gasket)**

Diameter	Code
100 mm (4 in)	MPG004
150 mm (6 in)	MPG006
200 mm (8 in)	MPG008
250 mm (10 in)	MPG010
300 mm (12 in)	MPG012
375 mm (15 in)	MPG015
450 mm (18 in)	MPG018
525 mm (21 in)	MPG021W
600 mm (24 in)	MPG024W
750 mm (30 in)	MPG030W
900 mm (36 in)	MPG036W
1,050 mm (42 in)	MPG042WS
1,200 mm (48 in)	MPG048WS



**PVC TRANSITION O-RING
(gasket)**

Diameter	Code
100 mm (4 in)	MPGT04
150 mm (6 in)	MPGT06
200 mm (8 in)	MPGT08
250 mm (10 in)	MPGT10

Note 1: Our Solflo and Solflo Max accessories are available with plain ends, watertight and soil tight couplers.
Consult our price list to find out what diameters we offer.

*See products data sheet on following pages: 40 (90-DEGREE ELBOW) and 41 (90-DEGREE ELBOW—3 SECTIONS).

SOLFLO AND SOLFLO MAX COUPLERS AND ACCESSORIES (CONT.)



CROSS*

Diameter	Code
100 mm (4 in)	33X0404
150 mm (6 in)	33X0606
200 mm (8 in)	33X0808
250 mm (10 in)	33X1010
300 mm (12 in)	33X1212
375 mm (15 in)	33X1515
450 mm (18 in)	33X1818
525 mm (21 in)	33X2121
600 mm (24 in)	33X2424
750 mm (30 in)	33X3030
900 mm (36 in)	33X3636
1,050 mm (42 in)	33X4242
1,200 mm (48 in)	33X4848



CROSS-REDUCER*

Diameter	Code
150 mm x 150 mm x 100 mm (6 in x 6 in x 4 in)	33X0604
200 mm x 200 mm x 100 mm (8 in x 8 in x 4 in)	33X0804
250 mm x 250 mm x 100 mm (10 in x 10 in x 4 in)	33X1004
250 mm x 250 mm x 150 mm (10 in x 10 in x 6 in)	33X1006
250 mm x 250 mm x 200 mm (10 in x 10 in x 8 in)	33X1008
300 mm x 300 mm x 100 mm (12 in x 12 in x 4 in)	33X1204
300 mm x 300 mm x 150 mm (12 in x 12 in x 6 in)	33X1206
300 mm x 300 mm x 200 mm (12 in x 12 in x 8 in)	33X1208
300 mm x 300 mm x 250 mm (12 in x 12 in x 10 in)	33X1210
375 mm x 375 mm x 150 mm (15 in x 15 in x 6 in)	33X1506
375 mm x 375 mm x 200 mm (15 in x 15 in x 8 in)	33X1508
375 mm x 375 mm x 250 mm (15 in x 15 in x 10 in)	33X1510
375 mm x 375 mm x 300 mm (15 in x 15 in x 12 in)	33X1512
450 mm x 450 mm x 150 mm (18 in x 18 in x 6 in)	33X1806
450 mm x 450 mm x 200 mm (18 in x 18 in x 8 in)	33X1808
450 mm x 450 mm x 250 mm (18 in x 18 in x 10 in)	33X1810
450 mm x 450 mm x 300 mm (18 in x 18 in x 12 in)	33X1812
450 mm x 450 mm x 375 mm (18 in x 18 in x 15 in)	33X1815



TEE*

Diameter	Code
100 mm (4 in)	33T0404
150 mm (6 in)	33T0606
200 mm (8 in)	33T0808
250 mm (10 in)	33T1010
300 mm (12 in)	33T1212
375 mm (15 in)	33T1515
450 mm (18 in)	33T1818
525 mm (21 in)	33T2121
600 mm (24 in)	33T2424
750 mm (30 in)	33T3030
900 mm (36 in)	33T3636
1,050 mm (42 in)	33T4242
1,200 mm (48 in)	33T4848



REDUCER TEE*

Diameter	Code
200 mm x 200 mm x 100 mm (8 in x 8 in x 4 in)	33T0804
250 mm x 250 mm x 100 mm (10 in x 10 in x 4 in)	33T1004
250 mm x 250 mm x 150 mm (10 in x 10 in x 6 in)	33T1006
250 mm x 250 mm x 200 mm (10 in x 10 in x 8 in)	33T1008
300 mm x 300 mm x 100 mm (12 in x 12 in x 4 in)	33T1204
300 mm x 300 mm x 150 mm (12 in x 12 in x 6 in)	33T1206
300 mm x 300 mm x 200 mm (12 in x 12 in x 8 in)	33T1208
300 mm x 300 mm x 250 mm (12 in x 12 in x 10 in)	33T1210
375 mm x 375 mm x 150 mm (15 in x 15 in x 6 in)	33T1506
375 mm x 375 mm x 200 mm (15 in x 15 in x 8 in)	33T1508
375 mm x 375 mm x 250 mm (15 in x 15 in x 10 in)	33T1510
375 mm x 375 mm x 300 mm (15 in x 15 in x 12 in)	33T1512
450 mm x 450 mm x 150 mm (18 in x 18 in x 6 in)	33T1806
450 mm x 450 mm x 200 mm (18 in x 18 in x 8 in)	33T1808
450 mm x 450 mm x 250 mm (18 in x 18 in x 10 in)	33T1810
450 mm x 450 mm x 300 mm (18 in x 18 in x 12 in)	33T1812
450 mm x 450 mm x 375 mm (18 in x 18 in x 15 in)	33T1815

Note 1: Our Solflo and Solflo Max accessories are available with plain ends, watertight and soil tight couplers.

Consult our price list to find out what diameters we offer.

*See products data sheet on following pages: 42 (CROSS), 43 (CROSS REDUCER), 45 (TEE) and 46 (REDUCER TEE).

SOLFLO AND SOLFLO MAX COUPLERS AND ACCESSORIES (CONT.)



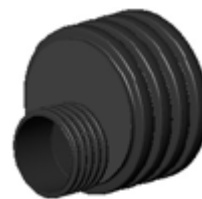
WYE (Y)*

Diameter	Code
100 mm (4 in)	33Y0404
150 mm (6 in)	33Y0606
200 mm (8 in)	33Y0808
250 mm (10 in)	33Y1010
300 mm (12 in)	33Y1212
375 mm (15 in)	33Y1515
450 mm (18 in)	33Y1818
525 mm (21 in)	33Y2121
600 mm (24 in)	33Y2424
750 mm (30 in)	33Y3030
900 mm (36 in)	33Y3636



WYE (Y) REDUCER*

Diameter	Code
150 mm x 150 mm x 100 mm (6 in x 6 in x 4 in)	33Y0604
200 mm x 200 mm x 100 mm (8 in x 8 in x 4 in)	33Y0804
250 mm x 250 mm x 100 mm (10 in x 10 in x 4 in)	33Y1004
250 mm x 250 mm x 150 mm (10 in x 10 in x 6 in)	33Y1006
250 mm x 250 mm x 200 mm (10 in x 10 in x 8 in)	33Y1008
300 mm x 300 mm x 100 mm (12 in x 12 in x 4 in)	33Y1204
300 mm x 300 mm x 150 mm (12 in x 12 in x 6 in)	33Y1206
300 mm x 300 mm x 200 mm (12 in x 12 in x 8 in)	33Y1208
300 mm x 300 mm x 250 mm (12 in x 12 in x 10 in)	33Y1210
375 mm x 375 mm x 150 mm (15 in x 15 in x 6 in)	33Y1506
375 mm x 375 mm x 200 mm (15 in x 15 in x 8 in)	33Y1508
375 mm x 375 mm x 300 mm (15 in x 15 in x 12 in)	33Y1512
450 mm x 450 mm x 150 mm (18 in x 18 in x 6 in)	33Y1806
450 mm x 450 mm x 200 mm (18 in x 18 in x 8 in)	33Y1808
450 mm x 450 mm x 300 mm (18 in x 18 in x 12 in)	33Y1812
450 mm x 450 mm x 375 mm (18 in x 18 in x 15 in)	33Y1815
600 mm x 600 mm x 300 mm (24 in x 24 in x 12 in)	33Y2412



REDUCER*

Diameter	Code
150 mm x 100 mm (6 in x 4 in)	33RED0604
200 mm x 100 mm (8 in x 4 in)	33RED0804
250 mm x 100 mm (10 in x 4 in)	33RED1004
250 mm x 150 mm (10 in x 6 in)	33RED1006
250 mm x 200 mm (10 in x 8 in)	33RED1008
300 mm x 100 mm (12 in x 4 in)	33RED1204
300 mm x 150 mm (12 in x 6 in)	33RED1206
300 mm x 200 mm (12 in x 8 in)	33RED1208
300 mm x 250 mm (12 in x 10 in)	33RED1210
375 mm x 150 mm (15 in x 6 in)	33RED1506
375 mm x 200 mm (15 in x 8 in)	33RED1508
375 mm x 250 mm (15 in x 10 in)	33RED1510
375 mm x 300 mm (15 in x 12 in)	33RED1512
450 mm x 150 mm (18 in x 6 in)	33RED1806
450 mm x 200 mm (18 in x 8 in)	33RED1808
450 mm x 250 mm (18 in x 10 in)	33RED1810
450 mm x 300 mm (18 in x 12 in)	33RED1812
450 mm x 375 mm (18 in x 15 in)	33RED1815
525 mm x 250 mm (21 in x 10 in)	33RED2110
525 mm x 300 mm (21 in x 12 in)	33RED2112
525 mm x 375 mm (21 in x 15 in)	33RED2115
525 mm x 450 mm (21 in x 18 in)	33RED2118
600 mm x 300 mm (24 in x 12 in)	33RED2412
600 mm x 375 mm (24 in x 15 in)	33RED2415
750 mm x 300 mm (30 in x 12 in)	33RED3012
750 mm x 375 mm (30 in x 15 in)	33RED3015
750 mm x 450 mm (30 in x 18 in)	33RED3018
750 mm x 525 mm (30 in x 21 in)	33RED3021
750 mm x 600 mm (30 in x 24 in)	33RED3024
900 mm x 450 mm (36 in x 18 in)	33RED3618
900 mm x 525 mm (36 in x 21 in)	33RED3621
900 mm x 600 mm (36 in x 24 in)	33RED3624
1,200 mm x 900 mm (48 in x 36 in)	33RED4836

Note 1: Our Solflo and Solflo Max accessories are available with plain ends, watertight and soil tight couplers. Consult our price list to find out what diameters we offer.

*See products data sheet on following pages: 48 (WYE), 49 (WYE REDUCER), 51 (REDUCER).

SOLFLO AND SOLFLO MAX COUPLERS AND ACCESSORIES

BEVEL



PIPE

Diameter	Length	kPa	Code
300 mm (12 in)	6 m (19.68 ft)	R210	BISE-211224
300 mm (12 in)	4 m (13.1 ft)	R320	BISE-331204
300 mm (12 in)	6 m (19.68 ft)	R320	BISE-331220
375 mm (15 in)	6 m (19.68 ft)	R210	BISE-211524
375 mm (15 in)	4 m (13.1 ft)	R320	BISE-331504
375 mm (15 in)	6 m (19.68 ft)	R320	BISE-331520
450 mm (18 in)	6 m (19.68 ft)	R210	BISE-211824
450 mm (18 in)	4 m (13.1 ft)	R320	BISE-331804
450 mm (18 in)	6 m (19.68 ft)	R320	BISE-331820
525 mm (21 in)	6 m (19.68 ft)	R210	BISE-212124
525 mm (21 in)	4 m (13.1 ft)	R320	BISE-332104
525 mm (21 in)	6 m (19.68 ft)	R320	BISE-332120
600 mm (24 in)	6 m (19.68 ft)	R210	BISE-212424
600 mm (24 in)	4 m (13.1 ft)	R320	BISE-332404
600 mm (24 in)	6 m (19.68 ft)	R320	BISE-332420
750 mm (30 in)	6 m (19.68 ft)	R210	BISE-213024
750 mm (30 in)	4 m (13.1 ft)	R320	BISE-333004
750 mm (30 in)	6 m (19.68 ft)	R320	BISE-333020
900 mm (36 in)	6 m (19.68 ft)	R210	BISE-213624
900 mm (36 in)	4 m (13.1 ft)	R320	BISE-333604
900 mm (36 in)	6 m (19.68 ft)	R320	BISE-333620

CUTTING

Diameter	Code
300 mm (12 in)	BISE-12-
375 mm (15 in)	BISE-15-
450 mm (18 in)	BISE-18-
525 mm (21 in)	BISE-21-
600 mm (24 in)	BISE-24-
750 mm (30 in)	BISE-30-
900 mm (36 in)	BISE-36-



BEVELED END

Diameter	Length	Code
300 mm (12 in)	1.5 m (4.9 ft)	33END12
375 mm (15 in)	1.5 m (4.9 ft)	33END15
450 mm (18 in)	1.5 m (4.9 ft)	33END18
525 mm (21 in)	1.5 m (4.9 ft)	33END21
600 mm (24 in)	1.6 m (5.2 ft)	33END24
750 mm (30 in)	1.7 m (5.6 ft)	33END30
900 mm (36 in)	1.8 m (5.9 ft)	33END36

Note 1: Our Solflo and Solflo Max accessories are available with plain ends, watertight and soil tight couplers. Consult our price list to find out what diameters we offer.

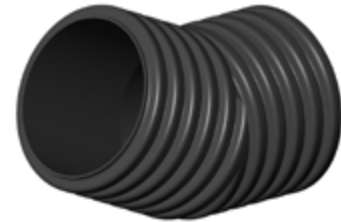
DATA SHEET

ELBOW 22.5°

PRODUCT DESCRIPTION: Factory-made elbow

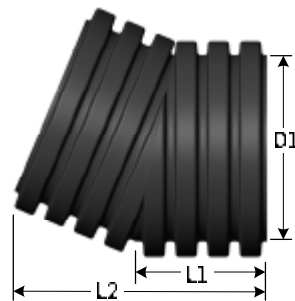
FUNCTION: Change direction of a pipe

MANUFACTURING STANDARDS: *100 mm (4 in) to 900 mm (36 in) diameter pipes:*
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294



RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	101	3.98	195	7.68	X		X		X			X	
150	6	135	5.31	259	10.20	X		X		X			X	
200	8	173	6.81	333	13.11	X		X		X			X	
250	10	226	8.90	434	17.09	X		X		X			X	
300	12	357	14.06	685	26.97	X			X	X				X
375	15	271	10.67	522	20.55	X	X			X	X			
450	18	325	12.80	612	24.09	X	X			X	X			
525	21	363	14.29	687	27.05	X	X			X	X			
600	24	473	18.62	890	35.04	X	X			X	X			
750	30	482	18.98	917	36.10	X	X			X	X			
900	36	498	19.61	959	37.76	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

ELBOW 45°

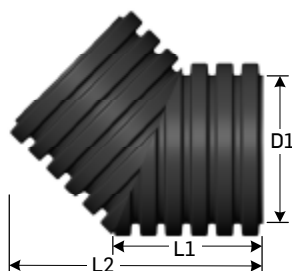
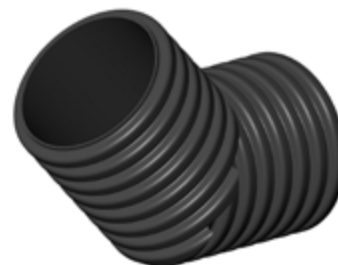
PRODUCT DESCRIPTION: Factory-made elbow

FUNCTION: Change direction of a pipe

MANUFACTURING STANDARDS: *100 mm (4 in) to 900 mm (36 in) diameter pipes:*
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	129	5.08	220	8.66	X		X		X			X	
150	6	135	5.31	259	10.20	X		X		X			X	
200	8	173	6.81	333	13.11	X		X		X			X	
250	10	226	8.90	434	17.09	X		X		X			X	
300	12	357	14.06	685	29.97	X			X	X				X
375	15	271	10.67	522	20.55	X	X			X	X			
450	18	325	12.80	612	24.09	X	X			X	X			
525	21	363	14.29	687	27.05	X	X			X	X			
600	24	473	18.62	890	35.04	X	X			X	X			
750	30	482	18.98	917	36.10	X	X			X	X			
900	36	498	19.61	959	37.76	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

ELBOW 90°

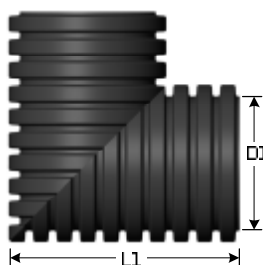
PRODUCT DESCRIPTION: Factory-made elbow

FUNCTION: Change direction of a pipe

MANUFACTURING STANDARDS: *100 mm (4 in) to 900 mm (36 in) diameter pipes:*
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Length L1		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	200	7.87	X		X		X			X	
150	6	276	10.87	X		X		X			X	
200	8	359	14.13	X		X		X			X	
250	10	465	18.31	X		X		X			X	
300	12	607	23.90	X			X	X				X
375	15	657	25.87	X	X			X	X			
450	18	746	29.37	X	X			X	X			
525	21	850	33.46	X	X			X	X			
600	24	1013	39.88	X	X			X	X			
750	30	1195	47.05	X	X			X	X			
900	36	1336	52.60	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

ELBOW 90° - 3 SECTIONS

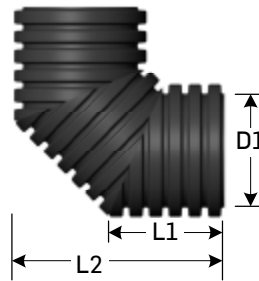
PRODUCT DESCRIPTION: Factory-made elbow

FUNCTION: Change direction of a pipe

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	129	5.08	204	8.03	X		X		X			X	
150	6	174	6.85	284	11.18	X		X		X			X	
200	8	224	8.82	370	14.57	X		X		X			X	
250	10	291	11.46	477	18.78	X		X		X			X	
300	12	392	15.43	618	24.33	X			X	X				X
375	15	388	15.28	678	26.69	X	X			X	X			
450	18	403	15.87	726	25.58	X	X			X	X			
525	21	510	20.08	914	35.98	X	X			X	X			
600	24	588	23.15	1037	40.83	X	X			X	X			
750	30	679	26.73	1232	48.50	X	X			X	X			
900	36	700	27.56	1323	52.09	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

CROSS

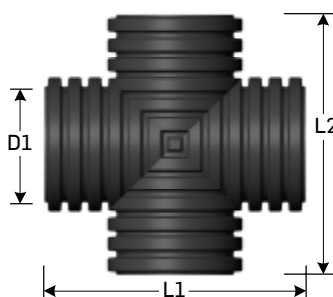
PRODUCT DESCRIPTION: Factory-made cross

FUNCTION: Junction between four connections

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	281	11.06	281	11.06	X		X		X			X	
150	6	376	14.80	376	14.80	X		X		X			X	
200	8	483	19.02	483	19.02	X		X		X			X	
250	10	632	24.88	632	24.88	X		X		X			X	
300	12	851	33.50	851	33.50	X			X	X				X
375	15	865	34.06	865	34.06	X	X			X	X			
450	18	949	37.36	949	37.36	X	X			X	X			
525	21	1060	41.73	1060	41.73	X	X			X	X			
600	24	1304	51.34	1304	51.34	X	X			X	X			
750	30	1501	59.09	1501	59.09	X	X			X	X			
900	36	1626	64.02	1626	64.02	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

CROSS (CONT'D)

TECHNICAL DATA TABLE - CGPF CROSS

Diameter D1		Length L1		Length L2		Watertight connectors HDPE
mm	in	mm	in	mm	in	DCGI-CGPF
100	4	281	11.06	149	5.87	X
150	6	376	14.80	218	8.62	X
200	8	483	19.02	279	10.98	X
250	10	632	24.88	632	24.88	X

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions, contact your Soleno representative.

LEGEND

DBIG: double bell with integrated gasket

CGPF : water tight PVC female

DATA SHEET

CROSS-REDUCER

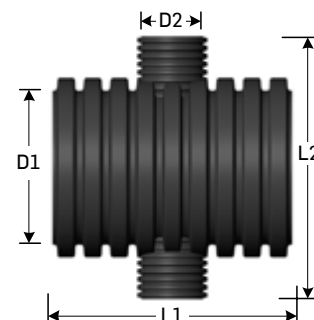
PRODUCT DESCRIPTION: Factory-made cross-reducer

FUNCTION: Junction between four connections (two different diameters)

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
150	6	100	4	337	13.27	349	13.74	X		X		X			X	
200	8	100	4	381	15.00	409	16.10	X		X		X			X	
200	8	150	6	432	17.00	444	17.48	X		X		X			X	
250	10	100	4	466	18.35	471	18.54	X		X		X			X	
250	10	150	6	532	20.94	505	19.88	X		X		X			X	
250	10	200	8	599	23.58	564	22.20	X		X		X			X	
300	12	100	4	650	25.59	537	21.14	X		X	X	X			X	X
300	12	150	6	701	27.60	572	22.52	X		X	X	X			X	X
300	12	200	8	751	29.57	630	24.80	X		X	X	X			X	X
300	12	250	10	801	31.54	713	28.07	X		X	X	X			X	X
375	15	100	4	532	20.94	623	24.53	X	X	X		X	X		X	
375	15	150	6	599	23.58	657	25.87	X	X	X		X	X		X	
375	15	200	8	665	26.18	716	28.19	X	X	X		X	X		X	
375	15	250	10	732	28.82	799	31.46	X	X	X		X	X		X	
375	15	300	12	799	31.46	975	38.39	X	X		X	X	X			X
450	18	100	4	543	21.38	716	28.19	X	X	X		X	X		X	
450	18	150	6	610	24.02	751	29.57	X	X	X		X	X		X	
450	18	200	8	678	26.69	810	31.89	X	X	X		X	X		X	
450	18	250	10	746	29.37	892	35.12	X	X	X		X	X		X	
450	18	300	12	814	32.05	1068	42.05	X	X		X	X	X			X
450	18	375	15	882	34.72	976	38.43	X	X			X	X			
525	21	100	4	606	23.86	814	32.05	X	X	X		X	X		X	
525	21	150	6	681	26.81	848	33.39	X	X	X		X	X		X	
525	21	200	8	757	29.80	907	35.71	X	X	X		X	X		X	
525	21	250	10	757	29.80	990	38.98	X	X	X		X	X		X	
525	21	300	12	833	32.80	1166	45.91	X	X		X	X	X			X
525	21	375	15	908	25.75	1073	42.24	X	X			X	X			
525	21	450	18	1060	41.73	1081	42.56	X	X			X	X			
600	24	100	4	803	31.61	896	35.28	X	X	X		X	X		X	
600	24	150	6	803	31.61	930	36.61	X	X	X		X	X		X	

DATA SHEET

CROSS-REDUCER (CONT'D)

TECHNICAL DATA TABLE (CONT'D)

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
600	24	200	8	903	35.55	989	38.94	X	X			X	X			
600	24	250	10	903	35.55	1072	42.20	X	X			X	X			
600	24	300	12	1003	39.49	1248	49.13	X	X		X	X	X			X
600	24	375	15	1104	43.46	1155	45.47	X	X			X	X			
600	24	450	18	1204	47.40	1163	45.79	X	X			X	X			
600	24	525	21	1304	51.34	1214	47.80	X	X			X	X			
750	30	100	4	801	31.54	1063	41.85	X	X	X		X	X		X	
750	30	150	6	801	31.54	1098	43.23	X	X	X		X	X		X	
750	30	200	8	901	35.47	1156	45.51	X	X	X		X	X		X	
750	30	250	10	901	35.47	1239	48.78	X	X	X		X	X		X	
750	30	300	12	1001	39.41	1415	55.71	X	X		X	X	X			X
750	30	375	15	1101	43.35	1322	52.05	X	X			X	X			
750	30	450	18	1201	47.28	1331	52.40	X	X			X	X			
750	30	525	21	1301	51.22	1382	54.41	X	X			X	X			
750	30	600	24	1401	55.16	1542	60.71	X	X			X	X			
900	36	100	4	813	32.01	1220	48.03	X	X	X		X	X		X	
900	36	150	6	813	32.01	1255	49.41	X	X	X		X	X		X	
900	36	200	8	914	35.98	1313	51.69	X	X	X		X	X		X	
900	36	250	10	914	35.98	1396	54.96	X	X	X		X	X		X	
900	36	300	12	1016	40.00	1572	61.89	X	X		X	X	X			X
900	36	375	15	1118	44.02	1479	58.23	X	X			X	X			
900	36	450	18	1219	47.99	1487	58.54	X	X			X	X			
900	36	525	21	1321	52.01	1538	60.55	X	X			X	X			
900	36	600	24	1422	55.98	1699	66.89	X	X			X	X			
900	36	750	30	1524	60.00	1697	66.81	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end

BC: bell with clips

DBS: double bell snap

DBC: double bell with clips

SC: split coupler

BIGC: bell with integrated gasket with clips

BG: O-ring bell gasket

DBIG: double bell with integrated gasket

DBIGC: double bell with integrated gasket with clips

DATA SHEET

CROSS-REDUCER (CONT'D)

TECHNICAL DATA TABLE - CGPF CROSS-REDUCER

Diameter D1		Diameter D2		Length L1		Length L2		Watertight connectors HDPE
mm	in	mm	in	mm	in	mm	in	DCGI-CGPF
150	6	100	4	337	13.27	217	8.54	X
200	8	100	4	381	15.00	277	10.91	X
200	8	150	6	432	17.01	285	11.22	X
250	10	100	4	466	18.35	339	13.35	X
250	10	150	6	532	20.94	347	13.66	X
250	10	200	8	599	23.58	361	14.21	X

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions, contact your Soleno representative.

LEGEND

DBIG: double bell with integrated gasket

CGPF : water tight PVC female

DATA SHEET

TEE

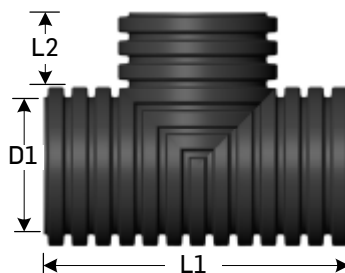
PRODUCT DESCRIPTION: Factory-made tee

FUNCTION: Splits a main line to a secondary line at 90°

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	281	11.06	80	3.15	X		X		X			X	
150	6	376	14.80	100	3.94	X		X		X			X	
200	8	483	19.02	123	4.84	X		X		X			X	
250	10	632	24.88	167	6.57	X		X		X			X	
300	12	851	33.50	244	9.61	X			X	X				X
375	15	865	34.06	208	8.19	X	X			X	X			
450	18	949	37.36	203	7.99	X	X			X	X			
525	21	1060	41.73	210	8.27	X	X			X	X			
600	24	1304	51.34	291	11.46	X	X			X	X			
750	30	1501	59.09	306	12.05	X	X			X	X			
900	36	1626	64.02	290	11.42	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

TEE (CONT'D)

TECHNICAL DATA TABLE - CGPF TEE

Diameter D1		Length L1		Length L2		Watertight connectors HDPE
mm	in	mm	in	mm	in	DCGI-CGPF
100	4	281	11.06	14	0.55	X
150	6	376	14.80	21	0.83	X
200	8	483	19.02	22	0.87	X
250	10	632	24.88	167	6.57	X

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions, contact your Soleno representative.

LEGEND

DBIG: double bell with integrated gasket

CGPF : water tight PVC female

DATA SHEET

REDUCER TEE

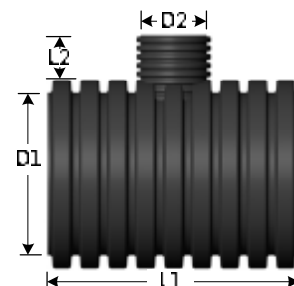
PRODUCT DESCRIPTION: Factory-made reducer tee

FUNCTION: Splits a main line to a secondary line at 90°

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
150	6	100	4	337	13.27	87	3.43	X		X		X			X	
200	8	100	4	381	15	87	3.43	X		X		X			X	
200	8	150	6	432	17	104	4.09	X		X		X			X	
250	10	100	4	466	18.35	87	3.43	X		X		X			X	
250	10	150	6	532	20.94	104	4.09	X		X		X			X	
250	10	200	8	599	23.58	133	5.24	X		X		X			X	
300	12	100	4	650	25.59	87	3.43	X		X	X	X			X	X
300	12	150	6	701	27.6	104	4.09	X		X	X	X			X	X
300	12	200	8	751	29.57	133	5.24	X		X	X	X			X	X
300	12	250	10	801	31.54	175	6.89	X		X	X	X			X	X
375	15	100	4	532	20.94	87	3.43	X	X	X		X	X		X	
375	15	150	6	599	23.58	104	4.09	X	X	X		X	X		X	
375	15	200	8	665	26.18	133	5.24	X	X	X		X	X		X	
375	15	250	10	732	28.82	175	6.89	X	X	X		X	X		X	
375	15	300	12	799	31.46	263	10.35	X	X		X	X	X			X
450	18	100	4	543	21.38	87	3.43	X	X	X		X	X		X	
450	18	150	6	610	24.02	104	4.09	X	X	X		X	X		X	
450	18	200	8	678	26.69	133	5.24	X	X	X		X	X		X	
450	18	250	10	746	29.37	175	6.89	X	X	X		X	X		X	
450	18	300	12	814	32.05	263	10.35	X	X		X	X	X			X
450	18	375	15	882	24.72	216	8.5	X	X			X	X			
525	21	100	4	606	23.86	87	3.43	X	X	X		X	X		X	
525	21	150	6	681	26.81	104	4.09	X	X	X		X	X		X	
525	21	200	8	757	29.8	133	5.24	X	X	X		X	X		X	
525	21	250	10	757	29.8	175	6.89	X	X	X		X	X		X	
525	21	300	12	833	32.8	263	10.35	X	X		X	X	X			X
525	21	375	15	908	35.75	216	8.5	X	X			X	X			
525	21	450	18	1060	41.73	220	8.66	X	X			X	X			
600	24	100	4	803	31.61	87	3.43	X	X	X		X	X		X	
600	24	150	6	803	31.61	104	4.09	X	X	X		X	X		X	

DATA SHEET

REDUCER TEE (CONT'D)

TECHNICAL DATA TABLE (CONT'D)

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
600	24	200	8	903	35.55	133	5.24	X	X			X	X			
600	24	250	10	903	35.55	175	6.89	X	X			X	X			
600	24	300	12	1003	39.49	263	10.35	X	X		X	X	X			X
600	24	375	15	1104	43.46	216	8.5	X	X			X	X			
600	24	450	18	1204	47.40	220	8.66	X	X			X	X			
600	24	525	21	1304	51.34	246	9.69	X	X			X	X			
750	30	100	4	801	31.54	87	3.43	X	X	X		X	X		X	
750	30	150	6	801	31.54	104	4.09	X	X	X		X	X		X	
750	30	200	8	901	35.47	133	5.24	X	X	X		X	X		X	
750	30	250	10	901	35.47	175	6.89	X	X	X		X	X		X	
750	30	300	12	1001	39.41	263	10.35	X	X		X	X	X			X
750	30	375	15	1101	43.35	216	8.5	X	X			X	X			
750	30	450	18	1201	47.28	220	8.66	X	X			X	X			
750	30	525	21	1301	51.22	246	9.69	X	X			X	X			
750	30	600	24	1401	55.16	326	12.83	X	X			X	X			
900	36	100	4	813	32	87	3.43	X	X	X		X	X		X	
900	36	150	6	813	32	104	4.09	X	X	X		X	X		X	
900	36	200	8	914	35.98	133	5.24	X	X	X		X	X		X	
900	36	250	10	914	35.98	175	6.89	X	X	X		X	X		X	
900	36	300	12	1016	40	263	10.35	X	X		X	X	X			X
900	36	375	15	1118	44.02	216	8.5	X	X			X	X			
900	36	450	18	1219	47.99	220	8.66	X	X			X	X			
900	36	525	21	1321	52	246	9.69	X	X			X	X			
900	36	600	24	1422	55.98	326	12.83	X	X			X	X			
900	36	730	30	1524	60	325	12.79	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end

BC: bell with clips

DBS: double bell snap

DBC: double bell with clips

SC: split coupler

BIGC: bell with integrated gasket with clips

BG: O-ring bell gasket

DBIG: double bell with integrated gasket

DBIGC: double bell with integrated gasket with clips

DATA SHEET

REDUCER TEE (CONT'D)

TECHNICAL DATA TABLE - CGPF REDUCER TEE

Diameter D1		Diameter D2		Length L1		Length L2		Watertight connectors HDPE
mm	in	mm	in	mm	in	mm	in	DCGI-CGPF
150	6	100	4	337	13.27	21	0.79	X
200	8	100	4	381	15.00	21	0.79	X
200	8	150	6	432	17.01	25	0.98	X
250	10	100	4	466	18.35	21	0.79	X
250	10	150	6	532	20.94	25	0.98	X
250	10	200	8	599	23.58	32	1.26	X

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions, contact your Soleno representative.

LEGEND

DBIG: double bell with integrated gasket

CGPF : water tight PVC female

DATA SHEET

WYE (Y)

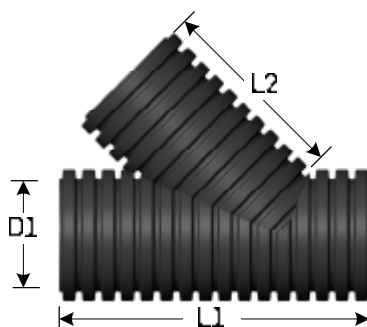
PRODUCT DESCRIPTION: Factory-made wye (y)

FUNCTION: Splits a main line to two same-diameter lines

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE

Diameter D1		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
100	4	330	12.99	201	7.91	X		X		X			X	
150	6	456	17.95	281	11.06	X		X		X			X	
200	8	584	22.99	361	14.21	X		X		X			X	
250	10	765	30.12	474	18.66	X		X		X			X	
300	12	851	33.50	608	23.94	X			X	X				X
375	15	1065	41.93	677	26.65	X	X			X	X			
450	18	1153	45.39	750	29.53	X	X			X	X			
525	21	1362	53.62	852	33.54	X	X			X	X			
600	24	1605	63.19	1018	40.08	X	X			X	X			
750	30	1901	74.84	1222	48.11	X	X			X	X			
900	36	2134	84.02	1434	56.46	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

WYE (Y) (CONT'D)

TECHNICAL DATA TABLE - CGPF WYE (Y)

Diameter D1		Length L1		Length L2		Watertight connectors HDPE
mm	in	mm	in	mm	in	DCGI-CGPF
100	4	330	12.99	135	5.31	X
150	6	456	17.98	202	7.95	X
200	8	584	22.99	259	10.20	X
250	10	765	30.12	474	18.66	X

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions, contact your Soleno representative.

LEGEND

DBIG: double bell with integrated gasket

CGPF : water tight PVC female

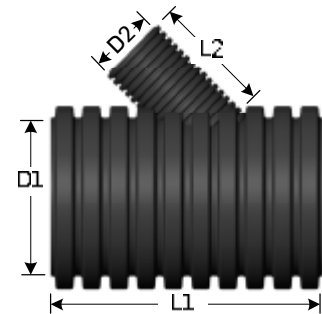
DATA SHEET

WYE (Y) REDUCER

PRODUCT DESCRIPTION: Factory-made wye (y) reducer

FUNCTION: Splits a main line to a secondary line

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:
Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294



RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting

TECHNICAL DATA TABLE

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
150	6	100	4	376	14.80	207	8.15	X		X		X			X	
200	8	100	4	432	17.01	207	8.15	X		X		X			X	
200	8	150	6	508	20.00	280	11.02	X		X		X			X	
250	10	100	4	532	20.94	207	8.15	X		X		X			X	
250	10	150	6	599	23.58	280	11.02	X		X		X			X	
250	10	200	8	699	27.52	369	14.53	X		X		X			X	
300	12	100	4	701	27.60	207	8.15	X		X	X	X			X	X
300	12	150	6	751	29.57	280	11.02	X		X	X	X			X	X
300	12	200	8	851	33.50	369	14.53	X		X	X	X			X	X
300	12	250	10	951	37.44	472	18.58	X		X	X	X			X	X
375	15	100	4	599	23.58	207	8.15	X	X	X		X	X		X	
375	15	150	6	665	26.18	280	11.02	X	X	X		X	X		X	
375	15	200	8	799	31.46	369	14.53	X	X	X		X	X		X	
375	15	250	10	865	34.06	472	18.58	X	X	X		X	X		X	
375	15	300	12	932	36.69	627	24.69	X	X		X	X	X			X
450	18	100	4	610	24.02	207	8.15	X	X	X		X	X		X	
450	18	150	6	678	26.69	280	11.02	X	X	X		X	X		X	
450	18	200	8	746	29.37	369	14.53	X	X	X		X	X		X	
450	18	250	10	882	34.72	472	18.58	X	X	X		X	X		X	
450	18	300	12	949	37.36	627	24.69	X	X		X	X	X			X
450	18	375	15	1085	42.72	666	26.22	X	X			X	X			
525	21	100	4	681	26.81	207	8.15	X	X	X		X	X		X	
525	21	150	6	747	29.41	280	11.02	X	X	X		X	X		X	
525	21	200	8	833	32.80	369	14.53	X	X	X		X	X		X	
525	21	250	10	908	35.75	472	18.58	X	X	X		X	X		X	
525	21	300	12	984	38.74	627	24.69	X	X		X	X	X			X
525	21	375	15	1135	44.69	666	26.22	X	X			X	X			
525	21	450	18	1287	50.77	764	30.08	X	X			X	X			
600	24	100	4	803	31.61	207	8.15	X	X	X		X	X		X	
600	24	150	6	903	35.55	280	11.02	X	X	X		X	X		X	

DATA SHEET

WYE (Y) REDUCER (CONT'D)

TECHNICAL DATA TABLE (CONT'D)

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE					Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC	
600	24	200	8	1003	39.49	369	14.53	X	X			X	X				
600	24	250	10	1104	43.46	472	18.58	X	X			X	X				
600	24	300	12	1204	47.40	627	24.69	X	X		X	X	X			X	
600	24	375	15	1304	51.34	666	26.22	X	X			X	X				
600	24	450	18	1405	55.31	764	30.08	X	X			X	X				
600	24	525	21	1605	63.19	886	34.88	X	X			X	X				
750	30	100	4	801	31.54	207	8.15	X	X	X		X	X		X		
750	30	150	6	901	35.47	280	11.02	X	X	X		X	X		X		
750	30	200	8	1001	39.41	369	14.53	X	X	X		X	X		X		
750	30	250	10	1101	43.35	472	18.58	X	X	X		X	X		X		
750	30	300	12	1201	47.28	627	24.69	X	X		X	X	X			X	
750	30	375	15	1301	51.22	666	26.22	X	X			X	X				
750	30	450	18	1401	55.16	764	30.08	X	X			X	X				
750	30	525	21	1601	63.03	886	34.88	X	X			X	X				
750	30	600	24	1701	66.97	1048	41.26	X	X			X	X				
900	36	100	4	813	32.01	207	8.15	X	X	X		X	X		X		
900	36	150	6	914	35.98	280	11.02	X	X	X		X	X		X		
900	36	200	8	1016	40.00	369	14.53	X	X	X		X	X		X		
900	36	250	10	1118	44.02	472	18.58	X	X	X		X	X		X		
900	36	300	12	1219	47.99	627	24.69	X	X		X	X	X			X	
900	36	375	15	1321	52.01	666	26.22	X	X			X	X				
900	36	450	18	1422	55.98	764	30.08	X	X			X	X				
900	36	525	21	1524	60.00	886	34.88	X	X			X	X				
900	36	600	24	1727	67.99	1048	41.26	X	X			X	X				
900	36	730	30	1930	75.98	1215	47.83	X	X			X	X				

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end
BC: bell with clips
DBS: double bell snap
DBC: double bell with clips

SC: split coupler
BIGC: bell with integrated gasket with clips
BG: O-ring bell gasket

DBIG: double bell with integrated gasket
DBIGC: double bell with integrated gasket with clips

DATA SHEET

WYE (Y) REDUCER (CONT'D)

TECHNICAL DATA TABLE - CGPF WYE (Y) REDUCER

Diameter D1		Diameter D2		Length L1		Length L2		Watertight connectors HDPE
mm	in	mm	in	mm	in	mm	in	DCGI-CGPF
150	6	100	4	376	14.80	141	5.55	X
200	8	100	4	432	17.01	141	5.55	X
200	8	150	6	508	20.00	201	7.91	X
250	10	100	4	532	20.94	141	5.55	X
250	10	150	6	599	23.58	201	7.91	X
250	10	200	8	699	27.52	268	10.55	X

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions, contact your Soleno representative.

LEGEND

DBIG: double bell with integrated gasket

CGPF : water tight PVC female

DATA SHEET

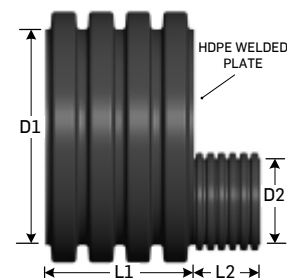
REDUCER

PRODUCT DESCRIPTION: Factory-made reducer

FUNCTION: Reduction in diameter of a pipe

MANUFACTURING STANDARDS: 100 mm (4 in) to 900 mm (36 in) diameter pipes:

Made from pipes that meet standards BNQ 3634-120 (R320). On special order, made from pipes that meet standards CSA B182.8, AASHTO M252 or AASTHO M294
1050 mm (42 po) to 1200 mm (48 po) diameter pipes:
Made from pipes that meet standards CSA B182.8 and AASHTO M294



RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting

TECHNICAL DATA TABLE

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
150	6	100	4	119	4.69	99	3.90	X		X		X			X	
200	8	100	4	152	5.98	99	3.90	X		X		X			X	
200	8	150	6	152	5.98	119	4.69	X		X		X			X	
250	10	100	4	200	7.87	99	3.90	X		X		X			X	
250	10	150	6	200	7.87	119	4.69	X		X		X			X	
250	10	200	8	200	7.87	152	5.98	X		X		X			X	
300	12	100	4	300	11.81	99	3.90	X		X	X	X			X	X
300	12	150	6	300	11.81	119	4.69	X		X	X	X			X	X
300	12	200	8	300	11.81	152	5.98	X		X	X	X			X	X
300	12	250	10	300	11.81	200	7.87	X		X	X	X			X	X
375	15	100	4	266	10.47	99	3.90	X	X	X		X	X		X	
375	15	150	6	266	10.47	119	4.69	X	X	X		X	X		X	
375	15	200	8	266	10.47	152	5.98	X	X	X		X	X		X	
375	15	250	10	266	10.47	200	7.87	X	X	X		X	X		X	
375	15	300	12	266	10.47	300	11.81	X	X		X	X	X			X
450	18	100	4	271	10.67	99	3.90	X	X	X		X	X		X	
450	18	150	6	271	10.67	119	4.69	X	X	X		X	X		X	
450	18	200	8	271	10.67	152	5.98	X	X	X		X	X		X	
450	18	250	10	271	10.67	200	7.87	X	X	X		X	X		X	
450	18	300	12	271	10.67	300	11.81	X	X		X	X	X			X
450	18	375	15	271	10.67	266	10.47	X	X			X	X			
525	21	100	4	303	11.93	99	3.90	X	X	X		X	X		X	
525	21	150	6	303	11.93	119	4.69	X	X	X		X	X		X	
525	21	200	8	303	11.93	152	5.98	X	X	X		X	X		X	
525	21	250	10	303	11.93	200	7.87	X	X	X		X	X		X	
525	21	300	12	303	11.93	300	11.81	X	X		X	X	X			X
525	21	375	15	303	11.93	266	10.47	X	X			X	X			
525	21	450	18	303	11.93	271	10.67	X	X			X	X			
600	24	100	4	401	15.79	99	3.90	X	X	X		X	X		X	
600	24	150	6	401	15.79	119	4.69	X	X	X		X	X		X	

DATA SHEET

REDUCER (CONT'D)

TECHNICAL DATA TABLE (CONT'D)

Diameter D1		Diameter D2		Length L1		Length L2		No connector	Soil tight connectors HDPE				Watertight connectors HDPE			
mm	in	mm	in	mm	in	mm	in	PE	BC	DBS	DBC	SC	BIGC	BG	DBIG	DBIGC
600	24	200	8	401	15.79	152	5.98	X	X			X	X			
600	24	250	10	401	15.79	200	7.87	X	X			X	X			
600	24	300	12	401	15.79	300	11.81	X	X		X	X	X			X
600	24	375	15	401	15.79	266	10.47	X	X			X	X			
600	24	450	18	401	15.79	271	10.67	X	X			X	X			
600	24	525	21	401	15.79	303	11.93	X	X			X	X			
750	30	100	4	400	15.75	99	3.90	X	X	X		X	X		X	
750	30	150	6	400	15.75	119	4.69	X	X	X		X	X		X	
750	30	200	8	400	15.75	152	5.98	X	X	X		X	X		X	
750	30	250	10	400	15.75	200	7.87	X	X	X		X	X		X	
750	30	300	12	400	15.75	300	11.81	X	X		X	X	X			X
750	30	375	15	400	15.75	266	10.47	X	X			X	X			
750	30	450	18	400	15.75	271	10.67	X	X			X	X			
750	30	525	21	400	15.75	303	11.93	X	X			X	X			
750	30	600	24	400	15.75	401	15.79	X	X			X	X			
900	36	100	4	406	15.98	99	3.90	X	X	X		X	X		X	
900	36	150	6	406	15.98	119	4.69	X	X	X		X	X		X	
900	36	200	8	406	15.98	152	5.98	X	X	X		X	X		X	
900	36	250	10	406	15.98	200	7.87	X	X	X		X	X		X	
900	36	300	12	406	15.98	300	11.81	X	X		X	X	X			X
900	36	375	15	406	15.98	266	10.47	X	X			X	X			
900	36	450	18	406	15.98	271	10.67	X	X			X	X			
900	36	525	21	406	15.98	303	11.93	X	X			X	X			
900	36	600	24	406	15.98	401	15.79	X	X			X	X			
900	36	750	30	406	15.98	400	15.75	X	X			X	X			

Note 1 : Dimensions shown above may vary slightly

Note 2 : For non-standard dimensions (1050 mm and 1200 mm (42 in and 48 in)), contact your Soleno representative.

LEGEND

PE: plain end

BC: bell with clips

DBS: double bell snap

DBC: double bell with clips

SC: split coupler

BIGC: bell with integrated gasket with clips

BG: O-ring bell gasket

DBIG: double bell with integrated gasket

DBIGC: double bell with integrated gasket with clips

BEVEL

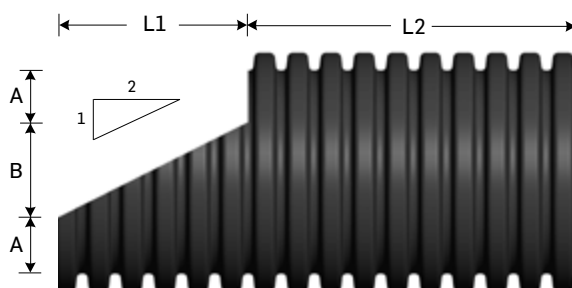
PRODUCT DESCRIPTION: Pipe end, cut at an angle

FUNCTION: Improves hydraulic efficiency of the culvert while adapting to the slope

MANUFACTURING STANDARDS: BNQ 3624-120, CSA B182.8, AASHTO 294

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

AVAILABLE FITTINGS: Fits any Solflo Max fitting



TECHNICAL DATA TABLE - STANDARDIZED BEVEL

Diameter		Length L1		Length L2		A		B		No connectors	Soil tight connectors HDPE			Watertight connectors HDPE	
mm	in	mm	in	mm	in	mm	in	mm	in	PE	BC	DBC	SC	BIGC	DBIGC
300	12	305	12	1000	39.37	76	2.99	152	5.98	X		X	X		X
375	15	381	15	1000	39.37	95	3.74	191	7.52	X	X		X	X	
450	18	457	18	1000	39.37	114	4.49	229	9.02	X	X		X	X	
525	21	533	21	1000	39.37	133	5.24	267	10.51	X	X		X	X	
600	24	610	24	1000	39.37	152	5.98	305	12.00	X	X		X	X	
750	30	762	30	1000	39.37	191	7.52	381	15.00	X	X		X	X	
900	36	914	36	1000	39.37	229	9.02	457	17.99	X	X		X	X	

Note 1 : These bevels meet the requirements of Volume III of the Quebec Ministry of Transportation

Note 2 : Other sizes are available. Contact your representative.

Note 3 : The bevel cut will be covered with a strip of high density polyethylene (HDPE).

LEGEND

PE: plain end

BC : bell with clips

DBC : double bell with clips

SC : split coupler

BIGC : double bell with integrated gasket

DBIGC : double bell with integrated gasket with clips

APPLICATIONS: culvert

TECHNICAL SHEET

ADAPTER FOR MANHOLE

PRODUCT DESCRIPTION: Molded or factory-made part

FUNCTION: Allows the connection of an HDPE pipe to a concrete manhole

RAW MATERIALS: Made of high density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA: Diameters from 100 mm (4 in) to 1200 mm (48 in)
Reference: see Annex C of standard BNQ 3624-120

AVAILABLE CONNECTIONS: The type of fittings available depend on the diameter and type of pipe considered for the manhole.



ADAPTER FOR MANHOLE
PVC DR35 Entry Type

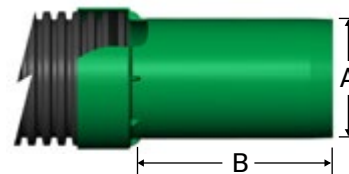


ADAPTER FOR MANHOLE
Soleno Entry Type (concrete)

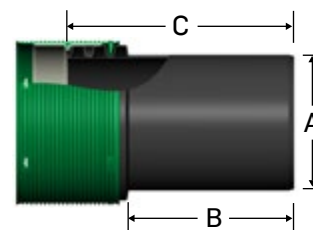
TECHNICAL DATA CHART

MANHOLE ADAPTER - PVC DR 35 ENTRY TYPE

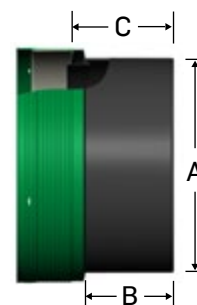
Nominal diameter of Solflo Max		Manhole entry type	Adapter outside diameter (A)		Length of the smooth portion (B)		Total length (C)	
mm	in		mm	in	mm	in	mm	in
100	4	PVC 100 mm	107	4.2	142	5.6	N/A	N/A
150	6	PVC 150 mm	159	6.3	180	7.1	N/A	N/A
200	8	PVC 200 mm	213	8.4	361	14.2	N/A	N/A
250	10	PVC 250 mm	267	10.5	452	17.8	N/A	N/A



Nominal diameter of Solflo Max		Manhole entry type	Adapter outside diameter (A)		Length of the smooth portion (B)		Total length (C)	
mm	in		mm	in	mm	in	mm	in
300	12	PVC 300 mm	317	12.5	381	15.0	531	20.9



Nominal diameter of Solflo Max		Manhole entry type	Adapter outside diameter (A)		Length of the smooth portion (B)		Total length (C)	
mm	in		mm	in	mm	in	mm	in
375	15	PVC 375 mm	389	15.3	381	15.0	448	17.6
450	18	PVC 450 mm	475	18.7	381	15.0	449	17.7
525	21	PVC 525 mm	559	22.0	381	15.0	457	18.0
600	24	PVC 600 mm	630	24.8	381	15.0	481	18.9



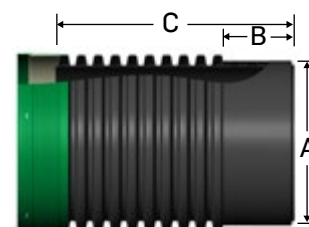
TECHNICAL SHEET

ADAPTER FOR MANHOLE (CONT'D)

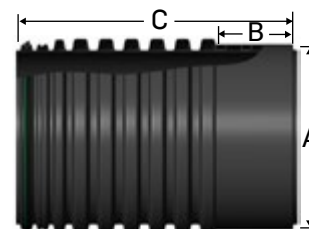
TECHNICAL DATA CHART

MANHOLE ADAPTER - SOLENO ENTRY TYPE (CONCRETE)

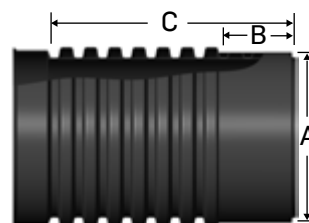
Nominal diameter of Solflo Max		Manhole entry type	Adapter outside diameter (A)		Length of the smooth portion (B)		Total length (C)	
mm	in		mm	in	mm	in	mm	in
525	21	Concrete 450 mm	624	24.6	383	15.1	1362	53.6
600	24	Concrete 525 mm	710	28.0	406	16.0	1405	55.3
750	30	Concrete 600 mm	805	31.7	401	15.8	1401	55.2
900	36	Concrete 750 mm	978	38.5	407	16.1	1422	56.0



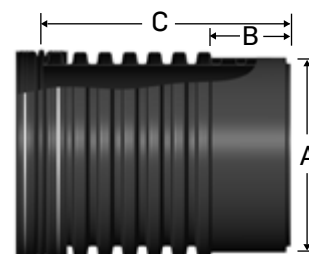
Nominal diameter of Solflo Max		Manhole entry type	Adapter outside diameter (A)		Length of the smooth portion (B)		Total length (C)	
mm	in		mm	in	mm	in	mm	in
1050 M	42 M	Concrete 900 mm	1156	45.5	446	17.6	1625	64.0
1200 M	48 M	Concrete 1050 mm	1334	52.5	451	17.8	1775	69.9



Nominal diameter of Solflo Max		Manhole entry type	Adapter outside diameter (A)		Length of the smooth portion (B)		Total length (C)	
mm	in		mm	in	mm	in	mm	in
1050 F	42 F	Concrete 900 mm	1156	45.5	446	17.6	1511	59.5
1200 F	48 F	Concrete 1050 mm	1334	52.5	451	17.8	1521	59.9



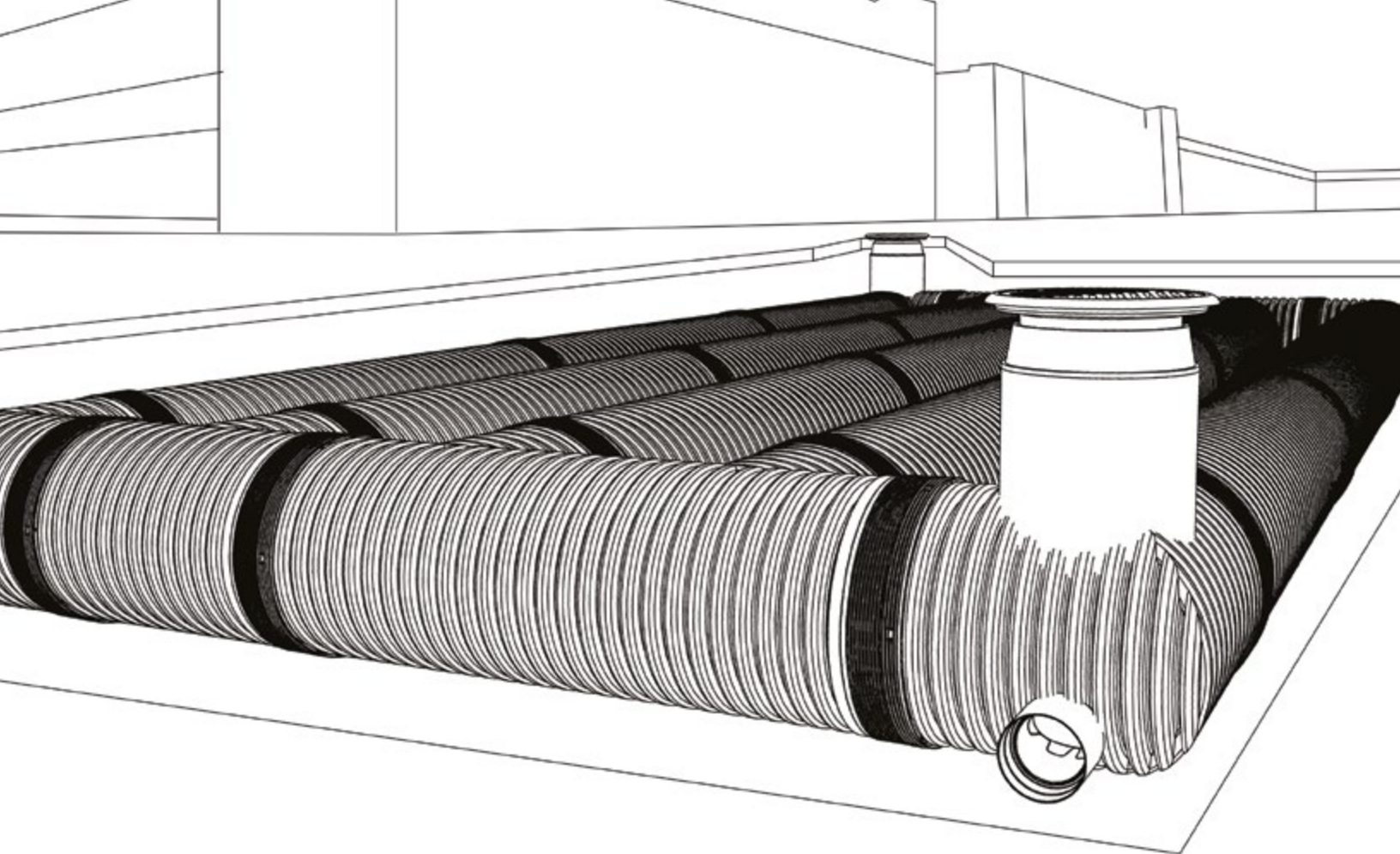
Nominal diameter of Solflo Max		Manhole entry type	Adapter outside diameter (A)		Length of the smooth portion (B)		Total length (C)	
mm	in		mm	in	mm	in	mm	in
1050 CS	42 CS	Concrete 900 mm	1156	45.5	446	17.6	1473	58.0
1200 CS	48 CS	Concrete 1050 mm	1334	52.5	451	17.8	1483	58.4



Legend: M = male outlet with O-ring seal;
F = female outlet;
CS = smooth outlet with collar sleeve; both male rings of the pipe to connect must be cut before installation.

APPLICATION: Storm sewer

OPTION: Special lengths and diameters available upon request.



DETENTION/RETENTION SOLFLO MAX SYSTEM

INSTALLATION GUIDE

INSTALLATION INSTRUCTIONS FOR A PIPE-BASED BASIN

STEPS

1. Before undertaking the work
2. Inspection of the materials
3. Identification of special parts
4. Site preparation
5. Assembly
6. Basin assembly
7. Backfilling
8. Connection
9. Finishing

STEP 1

BEFORE UNDERTAKING THE WORK

Contact your Soleno representative at least 48 hours before work begins.

STEP 2

INSPECTION OF THE MATERIALS

Upon receipt of the merchandise, make sure that all items specified in the shop drawing are delivered. Handle parts carefully.



STEP 3

IDENTIFICATION OF SPECIAL PARTS

Each special part (other than a pipe of standard length) is identified by a letter (on the last ring or on the access port). These identified items must be installed in accordance with the shop drawing.



STEP 4

PREPARE THE BEDDING

Level and compact the bedding at the bottom of the trench, following the specifications requirements (material, thickness and compaction).

STEP 5

ASSEMBLY

For ease of assembly, pipe grease can be used. Also, to facilitate insertion of the pipes, a waste pipe section or plywood panel can be used.



STEP 6A

BASIN ASSEMBLY

Begin assembly of the basin starting from the outlet (unless otherwise indicated on the shop drawings). Connect the discharge duct to the outlet collector. Complete assembly of the outlet collector.



STEP 6B

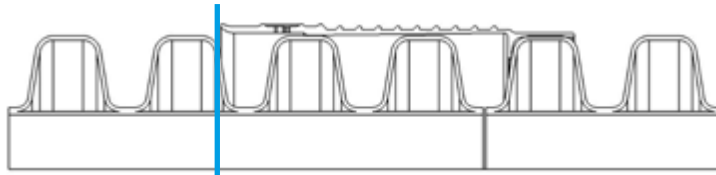
BASIN ASSEMBLY

Insert each length of the first section.

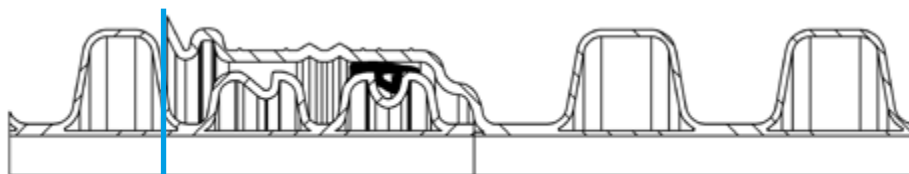


Make sure these pipes are fully inserted until the insertion line:

- diameter of 300 mm (12 in) to 900 mm (36 in)



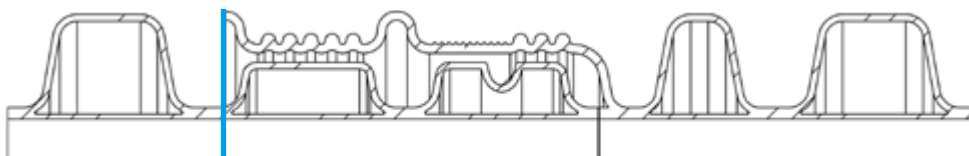
- diameter of 1050 mm (42 in)



- diameter of 1200 mm (48 in)



- diameter of 1500 mm (60 in)

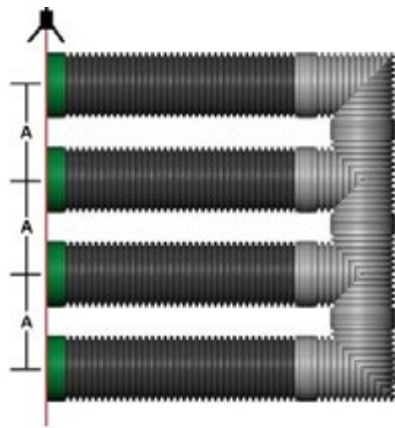


Do not hesitate to enter into the pipe to inspect the joints from inside and to make sure there remains no space.

STEP 6C

BASIN ASSEMBLY

Once the assembly of a completed section is done, check the parallelism between the rows. Also make sure that the rows have the same length.



STEP 6D

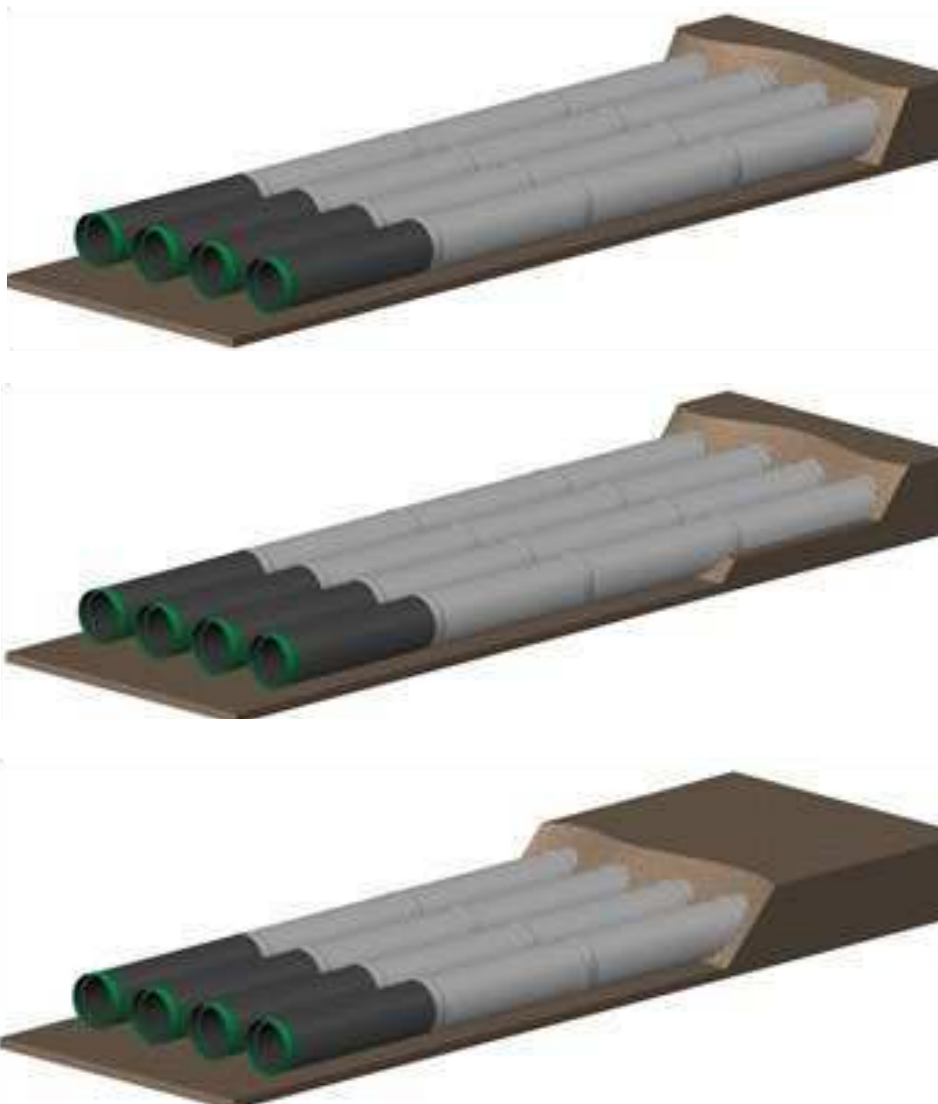
BASIN ASSEMBLY

Insert the following lengths, section by section, by making the checks described in step 6C after each section



STEP 7 BACKFILL

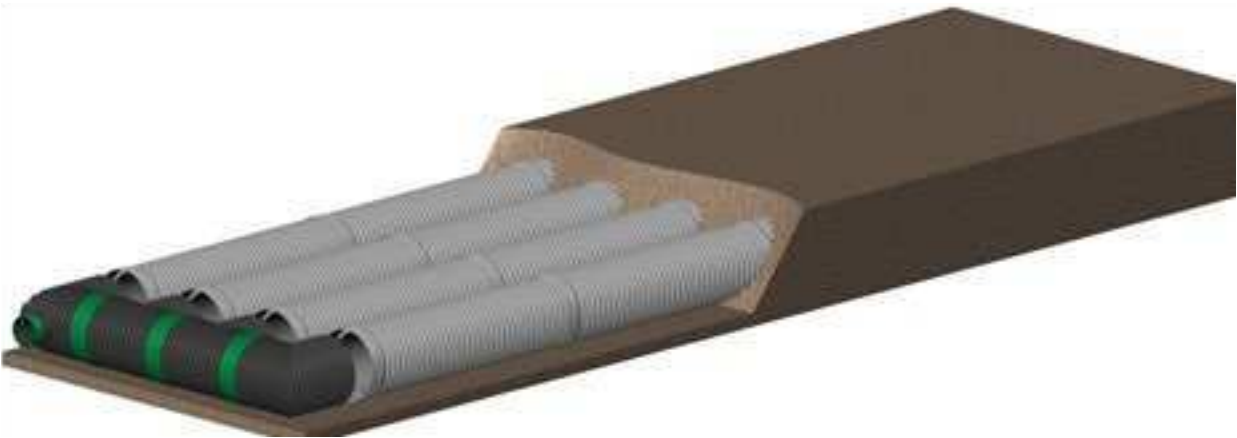
Backfill and compact progressively as required in the specifications, making sure to keep the last 3 sections open to allow final adjustments. During backfilling, ensure an equal fill height on each side of the pipe.



STEP 8

CONNECTION

If possible, pre-assemble the inlet collector and connect it to the pipes of the last section.



STEP 9

FINISHING

Complete the backfilling according to the specifications requirements. Make sure you have the minimum backfill height before circulating over the basin: 300 mm (12 in) for 1200 mm diameters (48 inches) or less and 500 mm (18 in) for the 1500 mm (60 in) diameter.

DATA SHEET

SLUICE PIPE

PRODUCT DESCRIPTION : Non-perforated dual wall half-pipe with anchors, with smooth interior and corrugated exterior walls

FUNCTION : Surface evacuation of surface runoff to prevent erosion of the in situ soil on steeply sloped land

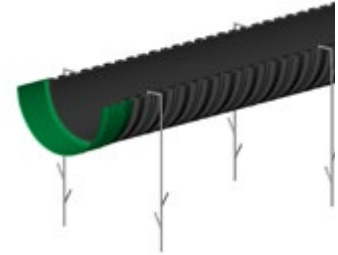
RAW MATERIALS : Pipe: Made from high-density polyethylene (HDPE)
ASTM D3350
Anchors: Steel

TECHNICAL DATA : Diameter: 450 mm (18 in) to 1,200 mm (48 in)
Standard length: 6 m (19.68 ft) for 450 mm (18 in) to 900 mm (36 in) sluice pipes, 3.30 m (10.8 ft) for 1,050 mm (42 in) and 1,200 mm (48 in) sluice pipes
Manning: 0.010 for 450 mm (18 in) and 0.012 for 525 mm (21 in) to 1,200 mm (48 in)

AVAILABLE COUPLER : Bell with clips

APPLICATIONS : Surface evacuation

OPTION : Deflector



SLUICE PIPE and anchors

SPECIFICATIONS

SLUICE PIPE

SCOPE

These specifications apply to all 450 mm (18 in) to 1,200 mm (48 in) diameter sluice pipes used for surface evacuation of surface runoff.

SLUICE PIPE REQUIREMENTS

Each sluice pipe is made from a half-section of Solflo Max pipe built with smooth interior and corrugated exterior walls, and steel anchors.

- 450 mm (18 in) to 900 mm (36 in) pipes shall be certified as per standard BNQ 3624-120; they shall have minimum stiffness of 320 kPa and be 6 m (19.68 ft) long.
- 1,050 mm (42 in) and 1,200 mm (48 in) diameter pipes shall be certified as per standard AASHTO M294; they shall have minimum stiffness of 145 kPa for 1,050 mm (42 in) diameters, 135 kPa for 1,200 mm (48 in) diameters and be 3.30 m (10.8 ft) long.

PRODUCT COMPONENTS

- The sluice pipe shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350-05.
- Anchors are manufactured from steel complying with standard ASTM A370.

JOINT PERFORMANCE

- Locking half-bell welded to the sluice pipe, allowing for proper insertion between sections with steel anchors, ensuring a stable pipe/soil interface.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact your local Soleno representative or visit our website at www.solen.com for installation recommendations.

DATA SHEET

SPELLWAY

PRODUCT DESCRIPTION : Custom-designed control structure with smooth interior and exterior walls

FUNCTION : For controlling the water level in an open-air retention system while regulating flow towards an outlet

RAW MATERIALS : Structure: Made from high-density polyethylene (HDPE) ASTM D3350
Accessories: 304 stainless steel or galvanized steel

TECHNICAL DATA : Diameter: 450 mm (18 in) to 3,300 mm (132 in)
Standard height: CUSTOM-MADE
Structural strength: CAN/CSA - S6 - 02 (CL-625)
Watertightness: ASTM D 3212/10.8 PSI

AVAILABLE COUPLERS : HDPE/bell with integrated gasket, bell with clips, Sol Tee
PVC/multi-axial outlet, Sol Tee
Concrete/male-female connection

OPENINGS : Sizing is based on retention volume and drought flows

APPLICATION : Water level control

OPTIONS : Special couplers are available upon request
Flow regulators
Debris grate
Riser
Ladder
Lift eyelet



SPELLWAY

SPECIFICATIONS

SPILLWAY

SCOPE

These specifications apply to all 450 mm (18 in) to 3,300 mm (132 in) diameter spillways used for controlling the water level in an open-air retention system while regulating flow.

SPILLWAY REQUIREMENTS

A spillway shall be made with smooth interior and exterior walls on its lower parts. The upper part of a spillway shall be made with a half-section with smooth interior and exterior walls, equipped with a 304 stainless steel or galvanized steel grate sized based on retention volume and drought flows.

PRODUCT COMPONENTS

- The spillway shall be made from a polyethylene resin that complies with properties classification PE 324420C as defined in standard ASTM D3350-05.
- The grate shall be made from 304 stainless steel or hot dip galvanized steel according to standard ASTM A123. The galvanizing thickness shall be at least 20 microns according to standard ASTM A153.

JOINT PERFORMANCE

- 100 mm (4 in) to 200 mm (8 in): Multiaxial gasket welded to the wall of the catch basin allowing insertion and retraction of the PVC pipe at various angles.
- 250 mm (10 in): Coupler welded to a catch basin equipped with a double bell with integrated gasket with transition to PVC. The gasket shall be co-injected into the double bell.
- 300 mm (12 in): Coupler welded to a catch basin equipped with a double bell with integrated gasket with clips to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the double bell.
- Diameter: 375 mm (18 in) to 900 mm (36 in) Bell with clips with integrated gasket welded to the catch basin to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the bell.
- Diameter: 1,050 mm (4 in) to 1,500 mm (60 in) HDPE bell welded to the spillway with gasket installed by the manufacturer onto the male end, covered by a protective film for shipping.
- Fused or glued gaskets will not be permitted.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation of the spillway will be made according to the installation requirements of the concrete catch basin specified in standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.soleno.com for installation recommendations.

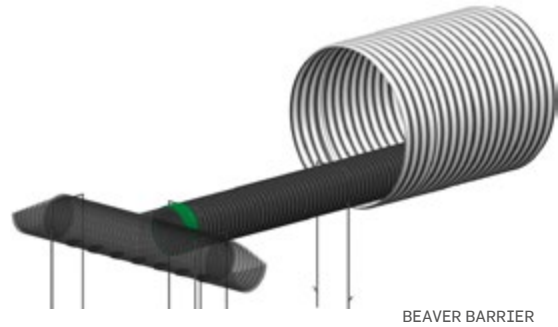
DATA SHEET

BEAVER BARRIER

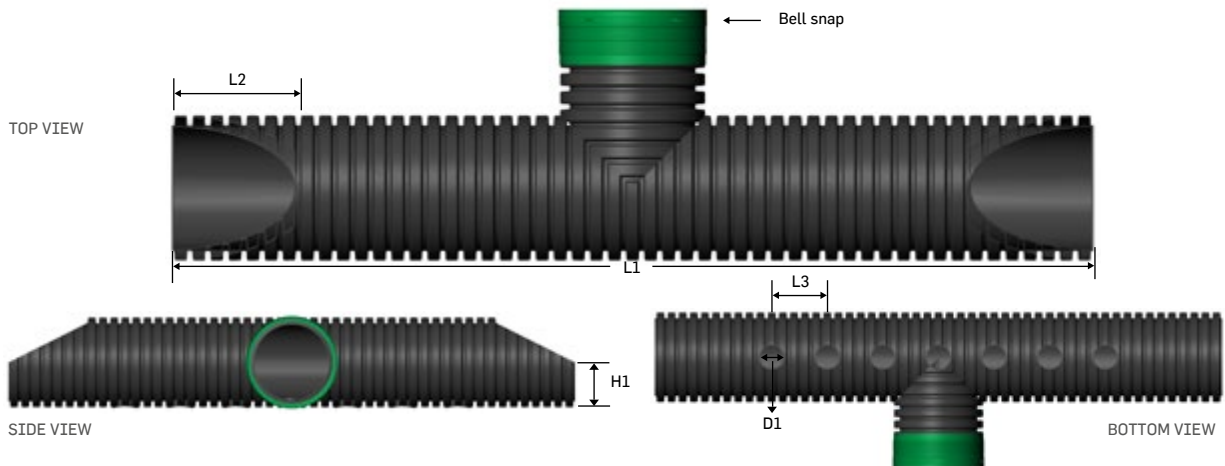
PRODUCT DESCRIPTION : Rigid dual wall, smooth interior wall and corrugated exterior wall, tee section with openings, directed down, attached to a length of Solflo Max.

FUNCTION: For conveying water towards an outlet in spite of beaver dams

RAW MATERIALS : Pipe: Made from ASTM D3350 high-density polyethylene (HDPE)
Anchors : 13 mm (0.51 in) steel re-bars, 1.52 m (5 ft) long, equipped with a fish hook at each end, folded and inserted between pipe corrugations



AVAILABLE COUPLERS : Double bell snap for 300 mm (12 in)
Bell with clips for 375 mm (15 in) and 450 mm (18 in)



TECHNICAL DATA TABLE

Nom. dia.		Length (L1)		Length (L2)		Length(L3)		Diameter (D1)		Height H1)		Openings
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	
300	12	3450	135	450	17.7	339	13.3	150	6.0	76.2	3.0	7
375	15	3450	135	466	18.3	339	13.3	150	6.0	95.2	3.7	7
450	18	3450	135	475	18.7	339	13.3	150	6.0	114.3	4.5	7

Length of transverse pipe: 9 m (29.5 ft)

DATA SHEET

PIPE WITH SMOOTH EXTERIOR WALL

PRODUCT DESCRIPTION : Pipe with smooth interior and exterior walls

FUNCTION : Pipe lining

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 450 mm (18 in) to 1,500 mm (60 in)
Length: 450 mm (18 in) thru 900 mm (36 in) is 3m (9.84 ft), 1,050 mm (42 in) and 1,200 mm (48 in) is 3.08m (10.1 ft)
1,500 mm (60 in) is 6m (19.68 ft)
Strength in compression: see technical data table
Manning: 0.010 to 0.012



PIPE with smooth exterior wall

AVAILABLE COUPLERS : Watertight: see technical data table

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Ext. dia. bell		Manning	Strength	Watertight connectors	
mm	in	mm	in	mm	in	mm	in	n	kPa	HDPE	
										BIGC	BG
450	18	457	18.0	552	21.1	562	22.1	0.010	320	X	X
525	21	535	21.1	650	25.6	663	26.1	0.012	320	X	X
600	24	603	23.8	746	29.4	762	30	0.012	320	X	X
750	30	756	29.8	899	35.4	917	36.1	0.012	320	X	X
900	36	912	35.9	1,056	41.6	1,077	42.4	0.012	320	X	X
1,050	42	1,067	42	1,234	48.6	1,234	48.6	0.012	145		X
1,200	48	1,220	48	1,387	54.6	1,387	54.6	0.012	135		X
1,500	60	1,511	59.5	1,678	66.1	1,678	66.1	0.012	105		X

Note 1: Values in the table are approximate and may change without notice.

LEGEND

BIGC: bell with integrated gasket with clips

BG: O-ring bell gasket

APPLICATIONS : HDPE culvert
Culvert lining

ACCESSORIES : Steel cross
Sacrificial pipe

Consult the Solflo and Solflo Max Couplers and Accessories section in this technical catalogue to obtain more information about steel crosses and sacrificial pipes.

SPECIFICATIONS

PIPE WITH SMOOTH EXTERIOR WALL

SCOPE

These specifications apply to all 450 mm (18 in) to 1,500 mm (60 in) diameter Solflo Max pipes with smooth exterior walls used to rehabilitate pipes by insertion.

PIPE REQUIREMENTS

Pipes shall be Solflo Max, manufactured with smooth interior and exterior walls.

- 450 mm (18 in) to 900 mm (36 in) pipes shall be conform to CSA B 182.8 and will be manufactured according to specified requirements as per standard BNQ 3624-120 and by adding a smooth exterior wall; they shall have minimum stiffness of 320 kPa and be 3 m (9.8 ft) long.
- 1,050 mm (42 in) to 1,500 mm (60 in) pipes shall be certified as per standard AASHTO M294; they shall have minimum stiffness of 145 kPa for 1,050 mm (42 in) diameters, 135 kPa for 1,200 mm (48 in) diameters, 105 kPa for 1,500 mm (60 in) diameters and be 3 m (9.8 ft) long for 1,050 mm (42 in) and 1,200 mm (48 in) and 6 m (19.68 ft) long for 1,500 mm (60 in).

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 334420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 450 mm (18 in) to 900 mm (36 in): bell with integrated gasket with clips welded to the pipe to ensure proper pipe insertion. The gasket shall be co-injected into the bell. The assembly shall be certified as per standard CSA B182.8.
- 1,050 mm (42 in) to 1,500 mm (60 in): bell with O-Ring gasket installed by the manufacturer onto the male end, covered by a protective film. The assembly shall be certified as per standard AASHTO M294.
- Fused or glued gaskets will not be permitted.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation of the pipe with smooth exterior wall will be made following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

WEHOLITE®

PRODUCT DESCRIPTION: Pipe with smooth interior and exterior walls

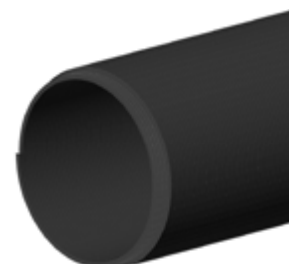
FUNCTION: HDPE culvert, culvert lining and storm water sewer

MANUFACTURING STANDARD: ASTM F894

RAW MATERIAL: Made of high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA: Diameters: 460 mm (18 in) to 3355 mm (132 in)
Lengths: 5 m (16.5 ft), 7.6 m (25 ft) and 15.2 m (50 ft)
Compression strength: see the technical data table
Manning: 0.01 n

AVAILABLE FITTINGS: Sediment-proof threaded joints
Threaded joints covered with Wehoseal to prevent any infiltration of grout during a rehabilitation project
Fused joints resistant to a pressure of 103 kPa (15 psi)



WEHOLITE



* For the purpose of transportation, check the overall length including the threaded end

TECHNICAL DATA TABLE

Nominal dia.		Int. dia.		Ext. dia. maximum		Streght	Mass		Maximum grout pressure		Backfill height				External threaded length	
mm	in	mm	in	mm	in	RSC	kg/m	lb/ft	kPa	lb/in ²	Minimum		Maximum		mm	in
460	18	460	18.1	528	20.8	160	26.9	18	79	11.5	0.4	1.3	16.76	55.0	89	3.5
495	19.5	500	19.6	577	22.7	160	28.4	19	52	7.5	0.5	1.6	16.76	55.0	105	4.1
530	21	530	20.8	615	24.2	160	31.3	21	51	7.4	0.55	1.8	16.76	55.0	105	4.1
610	24	610	24	701	27.6	160	35.8	24	59	8.6	0.55	1.8	16.76	55.0	121	4.8
690	27	690	27.1	787	31	160	40.3	27	53	7.7	0.55	1.8	16.76	55.0	136	5.4
760	30	760	29.9	871	34.3	160	44.8	30	48	7.0	0.61	2.0	16.76	55.0	151	5.9
840	33	840	33	967	38.1	160	49.3	33	50	7.3	0.64	2.1	16.76	55.0	151	5.9
910	36	910	35.8	1024	40.3	250	76.1	51	48	7.0	0.61	2.0	27.74	91.0	184	7.2
1020	40	1020	40.1	1140	44.9	250	89.6	60	43	6.2	0.64	2.1	24.38	80.0	207	8.1
1070	42	1070	42.1	1219	48	250	104.5	70	41	5.9	0.67	2.2	22.86	75.0	217	8.5
1220	48	1220	48	1376	54.2	250	119.4	80	37	5.4	0.7	2.3	27.43	90.0	235	9.3
1370	54	1370	53.9	1549	61	250	134.3	90	33	4.8	0.76	2.5	23.77	78.0	264	10.4

WEHOLITE® (CONT'D)

TECHNICAL DATA TABLE (CONT'D)

Nominal dia.		Int. dia.		Ext. dia. maximum		Strenght	Mass		Maximum grout pressure		Backfill height				External threaded length	
											Minimum		Maximum			
mm	in	mm	in	mm	in	RSC	kg/m	lb/ft	kPa	lb/in ²	m	ft	m	ft	mm	in
1520	60	1520	59.8	1722	67.8	250	149.3	100	30	4.4	0.82	2.7	22.25	73.0	293	11.5
1680	66	1680	66.1	1895	74.6	250	179.1	120	28	4.1	0.85	2.8	22.55	74.0	323	12.7
1830	72	1830	72	2067	81.4	250	223.9	150	26	3.8	0.91	3.0	22.55	74.0	352	13.9
1980	78	1980	77.9	2220	87.4	250	238.8	160	23	3.3	0.91	3.0	20.73	68.0	352	13.9
2130	84	2130	83.8	2393	94.2	250	283.6	190	22	3.2	0.94	3.1	21.03	69.0	381	15.0
2290	90	2290	90.1	2548	100.3	250	298.5	200	21	3.0	0.94	3.1	19.51	64.0	381	15.0
2440	96	2440	96	2718	107	250	343.3	230	19	2.8	0.98	3.2	19.81	65.0	410	16.1
2740	102	2590	101.9	2890	113.8	250	373.1	250	18	2.6	1	3.3	19.81	65.0	440	17.3
2740	108	2740	107.8	3045	119.9	250	403	270	17	2.5	1.04	3.4	18.9	62.0	440	17.3
3050	120	3050	120	3371	132.7	250	492.5	330	16	2.3	1.1	3.6	18.29	60.0	469	18.5
3355	132	3355	132	3716	146.3	250	552.2	370	16	2.3	1.19	3.9	17.37	57.0	528	20.8

Notes:

1. Backfill heights are based on the design principles in PPI's Engineering Handbook and take into account a Cl-625 (CSA) or H-25 (AASHTO) load.
2. This table assumes an aquifer level below the invert of the pipe. The calculations are performed according to the Deep Fill method.
3. The minimum backfill is based on the Shallow cover technical bulletin from KWH Pipe's Technical Services.
4. This table assumes the use of backfill soil with a modulus of 2000 psi, a density of 95% Standard Proctor.
5. For diameters from 2290 mm to 3350 mm, RSC Class 400 is available upon request. For this class, the minimum backfill is 0.910 m (3 ft) and the maximum backfill is 22.25 m (73 ft).

APPLICATIONS: Retention basin without groundwater replenishment, HDPE culvert and culvert rehabilitation

SPECIFICATIONS

WEHOLITE®

SCOPE

These specifications apply to all 450 mm (18 in) to 3,300 mm (132 in) diameter Weholite pipes used for pipe rehabilitation by insertion and for water course piping.

PIPE REQUIREMENTS

Pipes shall be Weholite, built with smooth interior and exterior walls.

- 450 mm (18 in) to 825 mm (33 in) pipes shall comply with standard ASTM F894-06; they shall have minimum stiffness of RSC 160 for diameters between 450 mm (18 in) and 750 mm (30 in), RSC 250 for diameters between 825 mm (33 in) and 3,300 mm (132 in) and shall be 7.62 m (25 ft) long.

RAW MATERIALS

Pipes shall be made from a polyethylene resin that complies with properties classification PE 334433C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

The pipe assembly is built with threaded joints resistant to water expansion during freeze-thaw cycles.

ACCESSORIES

- Accessories shall comply with standard ASTM F894-06.
- Accessories shall be factory welded, both outside and inside, when the inside is accessible.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

TURBULENCE

PRODUCT DESCRIPTION : Rigid, non-perforated single wall pipe, with corrugated interior and exterior wall, supplemented with 3 reinforcement strips for roadway use

FUNCTION : A structure placed in a stream, providing flow conditions similar to a creek, to allow the free flow of water and the free movement of aquatic fauna. HDPE culverts are mainly used in woodland, in mining and rural environments. They allow the crossing of a stream without disturbing the flora and fauna of an ecosystem.

RAW MATERIAL: Made from high-density polyethylene (HDPE) ASTM D3350

AVAILABLE COUPLERS : Soil tight: see technical data table

APPLICATION: HDPE culverts



TECHNICAL DATA TABLE

Nom. dia.		Int. dia. ⁽²⁾		Ext. dia.		Length		Manning Strength		Backfill				No connector	Soil tight connectors HDPE	
mm	in	mm	in	mm	in	m	ft	n	kPa	Minimum		Maximum		PE	BC	SC
										m	pi	m	pi			
450	18.0	461	18.1	543	21.4	6.0	19.7	0.021	210	0.4	1.3	10.0	32.8	X	X	X
600	24.0	605	23.8	722	28.4	6.0	19.7	0.022	210	0.4	1.3	8.0	26.2	X	X	X

Note 1: Values in the table are approximate and may change without notice

Note 2 : The inner diameter measurement does not include the thickness of the reinforcement strips

LEGEND

PE: plain end

BC: bell with clips

SC: split coupler

DATA SHEET

STEEL PIPE

PRODUCT DESCRIPTION : Corrugated galvanized steel pipes (CSP)

FUNCTION : Water course piping

MANUFACTURING STANDARDS : CSA G401

RAW MATERIALS : Galvanized steel CSA G401, ASTM A929

TECHNICAL DATA : Diameter: 150 mm (6 in) to 3,600 mm (144 in)
Length: 3 m (9.8 ft) to 18 m (59 ft)
Manning: 0.013 to 0.025



Steel PIPES

AVAILABLE COUPLER : Soil tight: see technical data table

TECHNICAL DATA TABLE

Nom. dia. (mm) / Corrugation contour (mm)			Weight (kg/m) / Pipe thickness (mm)				Manning	Minimum backfill (mm)		Maximum backfill (m) / Pipe thickness (mm)				Soil tight connector SC	
38 x 6.5	68 x 13	125 x 26	1.6	2	2.8	3.5		CL-625	E-80	1.6	2	2.8	3.5	No. of parts	No. of corrugations
150			7				0.013	300	300	70				1	3
200			10				0.013	300	300	70				1	3
250			12				0.013	300	300	70				1	3
	300		14	18			0.013	300	300	70	91			1	3
	375		18	22			0.013	300	300	57	73			1	3
	400		19	24			0.014	300	300	53	68			1	3
	450		21	27			0.014	300	300	48	68			1	3
	500		24	30			0.015	300	300	42	54			1	3
	600		28	35	49		0.016	300	300	35	45	66		1	3
	700		33	41	57	67	0.016	300	300	30	39	57	69	1	5
	750		36	44	63	73	0.016	300	300	27	35	52	63	1	5
	800		37	47	65	90	0.017	300	300	26	34	50	60	1	5
	900		42	53	73	100	0.018	300	300	23	30	44	56	1	5
	1.000			58	81	120	0.019	300	300		27	40	50	1	5
	1.200			70	97	125	0.022	300	300		23	33	42	1	5
		1.400		83	116	150	0.022	300	500		20	29	35	2	5
		1.500		89	124	160	0.022	300	500		20	27	33	2	5
		1.600		95	132	179	0.023	300	500		18	25	31	2	5
		1.800			148	205	0.024	300	500			22	28	2	9
		2.000			165	225	0.025	300	500			20	25	2	9
		2.200			181	245	0.025	300	700			18	23	2	9
		2.400			197	276	0.025	500	700			17	21	2	9
		2.700				306	0.025	500	700				18	3	9
		3.000				336	0.025	500	1.000				16	3	9
		3.300*				367	0.025	500	1.000				14**	3	9
		3.600*				401	0.025	700	1.000				12**	3	9

Note 1: Values in the table are approximate and may change without notice.

LEGEND

* As of May 1, 2016, these pipes aren't certified by the BNQ.

** for specific uses only, as the recommended flexibility limit has been exceeded

CL-625 (CAN CSA-06-06)

E-80 (AREMA Cooper)

SC: split coupler

DATA SHEET

STEEL PIPE (CONT.)

APPLICATIONS : Driveway culvert
Steel culvert
Ventilation
Drainage outlet

OPTIONS : Special lengths are available upon request
Lift eyelet
Arch pipes
Aluminum or polymer cover
Fish baffles
Perforation

COUPLERS AND BEVELED CUTS : Consult the Steel Couplers and Beveled cuts section in this catalogue to obtain more technical information about our complete line of accessories.

SPECIFICATIONS

STEEL PIPE

SCOPE

These specifications apply to all 300 mm (12 in) to 3,600 mm (144 in) diameter corrugated galvanized steel pipes (CSP) with locked seam used for drainage, water course piping and gravity flow storm water sewers.

PIPE REQUIREMENTS

Pipes shall be built with corrugated galvanized steel, and shall comply with standard CSA G401. Corrugations shall be spiral-shaped, except at the ends, where they shall be re-rolled, to allow the use of corrugated split couplers.

RAW MATERIALS

The steel used for these pipes is cold rolled and hot dip galvanized according to standard ASTM A929. The total weight of the zinc coating, for both faces, shall be at least 610 g/m² (2.013 oz/ft²) (commonly known as Z610), measured by the three-edge test.

JOINT PERFORMANCE

Split couplers shall have a corrugated collar, be made from the same material as the main pipe and shall comply with standard CSA G401. A Soleno Textiles TX-80 BNQ G.C.T.T.G 3001 geotextile covers the assembly.

ACCESSORIES

Couplers are factory welded. All welds shall be continuous, brushed, then painted with zinc according to standard CAN/CGSB-1.181-92.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

TECHNICAL DATA SHEET

ENVIRO-SPAN

PRODUCT DESCRIPTION : Modular arch culvert patented system

FUNCTION : Allows the crossing of a stream without disturbing its bed or banks and without disturbing the flora and fauna in its ecosystem

RAW MATERIAL : Made of Metton polymer

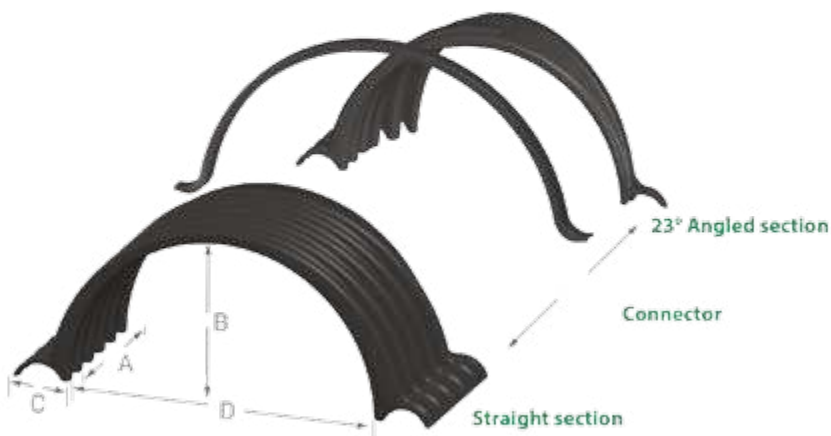
TECHNICAL INFORMATION : Manning: 0.025



ENVIRO-SPAN

DIMENSIONAL TABLE

Description	Length (A)	Height (B)	Bankseat width (C)	Span (D)	Weight
Straight span section	1070 mm (42 in)	1120 mm (44 in)	305 mm (12 in)	2150 mm (85 in)	141 lbs (64 kg)
Section angled at 23°	1130 mm (45 in)	1120 mm (44 in)	305 mm (12 in)	2150 mm (85 in)	95 lbs (43 kg)
Connector	97 mm (3.8 in)	1016 mm (40 in)	N/A	2438 mm (96 in)	10 lbs (4.5 kg)



HEIGHT ACCORDING TO BACKFILL

Traffic loads	Min height mm (in)	Max height mm (in)
H-20, H-25	600 (24)	35.05 (115)
H-30, L100, L150	900 (36)	35.05 (115)

INSTALLATION GUIDE

ENVIRO-SPAN

This guide is based on engineering principles and good practice for the installation of arch culverts. Stream crossings are highly regulated in many areas. Check with your local regulatory agencies before installation to ensure that the ENVIRO-SPAN arch culvert is authorized for your specific crossing.

Please read this installation guide before starting work.
Supervision by an engineer is recommended.



Preparation:

Make a detailed site plan for the access path based on:

- an initial site assessment;
- the stream hydrology;
- an assessment of the ground stability;
- other factors relating to the site.

Choose the location of the access path so that the wetted perimeter of the waterway sits entirely between the bankseats, with a width not exceeding 2100 mm (6.9 ft). For example, for a stream having a normal width of 1.070 mm (3.5 ft), we recommend choosing a 2150 mm (7 ft) ENVIRO-SPAN culvert.

Hydrological calculations should be performed to determine the peak flow carried by runoff from the upstream watershed, to ensure that the structure is properly sized for each application.

Warning:

- a. Do not travel on the ENVIRO-SPAN arch culvert with a tracked vehicle before backfilling at least 600 mm (24 in) above the arch.
- b. Do not travel on the ENVIRO-SPAN arch culvert with a wheeled vehicle before the minimum recommended backfill height above the arch (see data sheet).



Installation:

1. Select two straight timber logs which will serve as bankseat footings. Plastic pipes with rigid walls (Sclairpipe) and steel pipe (foundation piles) can also be used. The log dimensions are as follows:

Log diameter	200 mm (8 in) 300 mm (12 in)
Log length	Length of the structure + 1.200 mm (48 in)

The footings used to support the ENVIRO-SPAN arch culvert system can be installed without anchoring on ground slopes of 2% or less. When the slope exceeds 2%, it is recommended to anchor the logs with steel anchors at each end of the logs, and at intervals of 2.5 m (8 ft), if needed.

2. Place the logs parallel to the stream bed with a centre to centre spacing of 2,455 mm (96.65 in) corresponding to the width of the ENVIRO-SPAN arch culvert.



Note: If more than one log is necessary to achieve the required length, stagger the log joints from both sides of the structure to ensure its stability.

INSTALLATION GUIDE

ENVIRO-SPAN

3. Installing the sections:

- a. Install the first section at 600 mm (24 in) from one end of the log, positioning the culvert bankseats on the log footings.
- b. Place the second section next to the first one and place the connector on top of the two ENVIRO-SPAN sections so as to overlap the end rings of the two sections. No additional accessories are required.
- c. Continue adding sections up to the length specified in the design drawing.



4. Riprap:

For installations with seasonal flooding and during assembly, it is recommended to place inside the sections, one after the other,

- a. riprap along the bankseats
- b. a geogrid on the banks before placing riprap.



5. Coverage with a geotextile:

After assembly is completed, it is important to cover the entire structure with a TX-F type geotextile. The geotextile should cover the system and have an excess length of 300 mm (12 in) on each side of the ENVIRO-SPAN arch structure.



6. Backfilling:

- a. Backfilling can be done using local soil. Any stone exceeding 150 mm (6 in) in diameter must be removed.
- b. Backfilling should be laid in 300 mm layers (12 in) spread simultaneously and equally on each side of the structure.
- c. Each layer of backfill should be compacted using a vibrating plate or power shovel.
- d. Refer to the data sheet for the minimum and maximum fill heights.
- e. Take care not to displace the connectors when backfilling.



DATA SHEET

CATCH BASIN WITH SMOOTH EXTERIOR WALL

PRODUCT DESCRIPTION : Smooth interior and exterior walls for roadway and non-roadway use

FUNCTION : For collecting surface runoff and recovering sediments

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 450 mm (18 in) to 1,500 mm (60 in)
Standard height: 2 m (6.5 ft)
Structural strength: CAN/CSA - S6 - 02 (CL-625)
Watertightness: ASTM D 3212/10.8 PSI

AVAILABLE COUPLERS : Watertight: see technical data table
The type of coupler available is based on the diameter of the connecting pipe.



CATCH BASIN with smooth exterior wall and MG outlet, rectangular cast iron adjustable frame and grate

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Weight		Frames and grates	
mm	in	mm	in	mm	in	kg/m	lbs/ft	Non-roadway	Roadway
450	18	457	18.0	552	21.8	37.9	25.4	S series cast iron	None
600	24	603	23.8	746	29.4	68.3	45.8	S series cast iron	Adjustable cast iron
750	30	756	29.8	899	35.4	110.9	74.4	S series cast iron	Adjustable cast iron
900	36	912	35.9	1,056	41.6	127.5	85.5	S series cast iron	Adjustable cast iron
1,050	42	1,067	42.0	1,234	48.6	154.0	103.3	S series cast iron	Adjustable cast iron
1,200	48	1,220	48.0	1,387	54.6	166.0	111.3	S series cast iron	Adjustable cast iron
1,500	60	1,511	59.5	1,678	66.1	169.0	113.6	S series cast iron	Adjustable cast iron

Note 1: Values in the table are approximate and may change without notice.

Note 2: Weight does not include frames and grates.

Note 3: Contact your castings supplier for exact dimensions of available frames and grates.

Note 4: 450 mm (18 in), 1,050 mm (42 in), 1,200 mm (48 in) and 1,500 mm (60 in) catch basins with smooth exterior wall are available by special order.

DATA SHEET

CATCH BASIN WITH SMOOTH EXTERIOR WALL (CONT.)

AVAILABLE COUPLERS/DIAMETER OF THE CONNECTING PIPE

Nom. dia. connecting pipe		Watertight connectors					
		HDPE			PVC		
mm	in	DBIG	DBIGC	BIGC	S	MG	S
100	4	X			X	X	X
150	6	X			X	X	X
200	8	X			X	X	X
250	10	X					
300	12		X				
375	15			X			
450	18			X			
525	21			X			
600	24			X			
750	30			X			
900	36			X			

LEGEND

DBIG: double bell with integrated gasket
 DBIGC: double bell with integrated gasket with clips
 BIGC: bell with integrated gasket with clips
 MG: multiaxial gasket
 S: Sol Tee

APPLICATIONS : Surface drainage
 Percolation well
 Leachate collection

OPTIONS : Other diameters and special couplers are available upon request
 Riser
 Lift eyelet
 Special frames and grates
 Flow regulators
 Hook or eyelet for hooking system

SPECIFICATIONS

CATCH BASIN WITH SMOOTH EXTERIOR WALL

SCOPE

These specifications apply to all 450 mm (18 in) to 1,500 mm (60 in) diameter catch basins with smooth exterior walls used to collect surface runoff.

PIPE REQUIREMENTS

The catch basin shall be made with smooth interior and exterior walls, manufactured from a single part and meet CL-625 highway loading requirements as defined in standard CAN/CSA-S6-06.

PRODUCT COMPONENTS

- The catch basin shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350-05.

JOINT PERFORMANCE

- 100 mm (4 in) to 200 mm (8 in): multi-axial gasket welded to the wall of the catch basin allowing insertion and retraction of the PVC pipe at various angles.
- 250 mm (10 in): coupler welded to a catch basin equipped with a double bell with integrated gasket with transition to PVC. The gasket shall be co-injected into the double bell.
- 300 mm (12 in): coupler welded to the catch basin equipped with a double bell with integrated gasket with clips to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the double bell.
- 375 mm (15 in) to 900 mm (36 in): bell with integrated gasket with clips welded to the pipe to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the bell.
- Fused or glued gaskets will not be permitted.

FRAMES AND GRATES

- For roadway use, frames and grates shall be made of cast iron and adjustable, allowing transfer of loads to the surrounding soil for 600 mm (24 in) to 1,500 mm (60 in) diameter catch basins.
- For non-roadway use, frames and grates shall be made from S series cast iron, for 450 mm (18 in) to 1,500 mm (36 in) diameter catch basins.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation of the catch basin with smooth exterior wall will be made according to the installation requirements of the concrete catch basin specified in standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

MANHOLE WITH SMOOTH EXTERIOR WALL

PRODUCT DESCRIPTION : Smooth interior and exterior walls for roadway and non-roadway use

FUNCTION : Changes in direction and elevation, inspection and cleaning of buried networks (storm water, electrical, telephone, valve chamber, etc.)

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 600 mm (24 in) to 3,300 mm (132 in)
Standard height: CUSTOM-MADE
Structural strength: CAN/CSA - S6 - 02 (CL-625)
Watertightness: ASTM D 3212/10.8 PSI

AVAILABLE COUPLERS : Watertight and soil tight: see technical data table
The type of coupler available is based on the diameter of the connecting pipe.



MANHOLE with smooth exterior wall and BIGC outlets (bell with integrated gasket with clips), adjustable cast iron grate and cover

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Weight		Frames and grates	
mm	in	mm	in	mm	in	kg/m	lbs/ft	Non-roadway	Roadway
600	24	603	23.8	746	29.4	68.3	45.8	S series cast iron	Adjustable cast iron
750	30	756	29.8	899	35.4	110.9	74.4	S series cast iron	Adjustable cast iron
900	36	912	35.9	1,056	41.6	127.5	85.5	S series cast iron	Adjustable cast iron
1,050	42	1,067	42.0	1,234	48.6	154.0	103.3	S series cast iron	Adjustable cast iron
1,200	48	1,220	48.0	1,387	54.6	166.0	111.3	S series cast iron	Adjustable cast iron
1,350	54	1,370	54.0	1,549	61.0	169.0	113.6	S series cast iron	Adjustable cast iron
1,500	60	1,520	60.0	1,722	67.8	181.0	121.6	S series cast iron	Adjustable cast iron
1,650	66	1,680	66.0	1,895	74.6	206.0	138.4	S series cast iron	Adjustable cast iron
1,800	72	1,830	72.0	2,067	81.4	264.0	177.4	S series cast iron	Adjustable cast iron
1,950	78	1,980	78.0	2,220	87.4	284.0	190.8	S series cast iron	Adjustable cast iron
2,100	84	2,130	84.0	2,393	94.2	351.0	229.8	S series cast iron	Adjustable cast iron
2,250	90	2,290	90.0	2,548	100.3	408.0	274.2	S series cast iron	Adjustable cast iron
2,400	96	2,440	96.0	2,718	107.0	460.0	309.1	S series cast iron	Adjustable cast iron
2,250	102	2,590	102.0	2,890	113.8	520.0	349.4	S series cast iron	Adjustable cast iron
2,700	108	2,740	108.0	3,045	119.9	601.0	403.9	S series cast iron	Adjustable cast iron
3,000	120	3,050	120.0	3,371	132.7	730.0	490.6	S series cast iron	Adjustable cast iron
3,300	132	3,355	132.0	3,716	146.3	886.0	595.4	S series cast iron	Adjustable cast iron

Note 1: Couplers of 100 mm (4 in) to 250 mm (10 in) diameter are available for special cases.

Note 2: Contact your castings supplier for exact dimensions of available grates and covers.

Note 3: Weight does not include grates and covers.

Note 4: For the number of inlets and outlets from the manhole, see reference document "Minimum angle between two pipes".

Note 5: 600 mm (24 in), 750 mm (30 in), 1,350 mm (54 in), 1,650 mm (66 in) and larger manholes with smooth exterior walls are available by special order.

DATA SHEET

MANHOLE WITH SMOOTH EXTERIOR WALL (CONT.)

AVAILABLE COUPLERS/DIAMETER OF THE CONNECTING PIPE

Nom. dia. connecting pipe		Watertight connectors			Soil tight connectors	
		HDPE			HDPE	Concrete
mm	in	DBIGC	BIGC	BG	BC/PE	Other
300	12	X			X	X
375	15		X		X	X
450	18		X		X	X
525	21		X		X	X
600	24		X		X	X
750	30		X		X	X
900	36		X		X	X
1,050	42			X		X
1,200	48			X		X
1,500	60			X		X

Note 1: For the number of inlets and outlets from the manhole, see reference document "Minimum angle between two pipes".

LEGEND

DBIGC: double bell with integrated gasket with clips

BIGC: bell with integrated gasket with clips

BG: O-ring bell gasket

BC/PE: bell with clips/plain end

APPLICATIONS : Storm water sewer
Mechanical inspection room (electrical, fibre optic, etc.)
Valve chamber

OPTIONS : Special couplers are available upon request
Riser
Lift eyelet
Special grates and covers
Safety platform
Ladder

SPECIFICATIONS

MANHOLE WITH SMOOTH EXTERIOR WALL

SCOPE

These specifications apply to all 600 mm (24 in) to 3300 mm (132 in) diameter manholes with smooth interior and exterior walls used for changes in direction and elevation, inspection and cleaning of storm water sewer networks and retention system pipes.

MANHOLE REQUIREMENTS

The manhole shall be made with smooth interior and exterior walls, manufactured from a single part and meet CL-625 highway loading requirements as defined in standard CAN/CSA-S6

PRODUCT COMPONENTS

- The manhole shall be made from high-density polyethylene (HDPE) resin that complies with standard ASTM D3350.

JOINT PERFORMANCE

- 300 mm (12 in): coupler welded to the manhole equipped with a double bell with integrated gasket with clips to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the double bell.
- 375 mm (15 in) to 900 mm (36 in): bell with integrated gasket with clips welded to the manhole to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the bell.
- 1,050 mm (42 in) to 1,500 mm (60 in): HDPE bell welded to the manhole with gasket installed by the manufacturer onto the male end, covered by a protective film for shipping.
- Fused and/or glued gaskets will not be permitted.

GRATES AND COVERS

- For roadway use, frames and covers shall be made of cast iron and adjustable, allowing transfer of loads to the surrounding soil.
- For non-roadway use, grates and covers shall be made from S series cast iron.

ACCESSORIES

Ladder made of galvanized steel or stainless steel (upon request).

Anchors made of stainless steel.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation of the manhole with smooth exterior wall will be made according to the installation requirements of the concrete manhole specified in standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solenio.com for installation recommendations.

DATA SHEET

CORRUGATED MANHOLE/CATCH BASIN

PRODUCT DESCRIPTION : Smooth interior and corrugated exterior walls for non-roadway use

FUNCTION : Used to collect surface runoff, for changes in direction and elevation, inspection and cleaning of storm sewer networks.

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 300 mm (12 in) to 1,200 mm (48 in)
Standard height: 1.3 m (4.3 ft)
Structural strength: non-roadway

AVAILABLE COUPLERS : Watertight and soil tight: see technical data table
The type of coupler available is based on the diameter of the connecting pipe.



Corrugated MANHOLE/CATCH BASIN with BIGC outlets (bell with integrated gasket with clips) and cover made from S series cast iron

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Weight		Strength	Frames and grates
mm	in	mm	in	mm	in	kg/m	lbs/ft	kPa	
300	12	303	11.9	363	14.3	9.5	6.4	320	HDPE and S series cast iron
375	15	381	15.0	449	17.7	12.0	8.0	320	HDPE and S series cast iron
450	18	457	18.0	543	21.4	16.5	11.1	320	S series cast iron
600	24	603	23.8	722	28.4	25.8	17.3	320	S series cast iron
750	30	756	29.8	890	35.0	38.0	25.5	320	S series cast iron
900	36	912	35.9	1,046	41.2	60.8	40.8	320	S series cast iron
1,050	42	1,067	42.0	1,225	48.2	70.5	47.8	145*	S series cast iron
1,200	48	1,220	48.0	1,377	54.2	75.5	50.6	135*	S series cast iron

Note 1: Values in the table are approximate and may change without notice.

Note 2: Weight does not include frames and grates.

Note 3: * Certified AASHTO M294.

Note 4: Contact your castings supplier for exact dimensions of available cast iron frames and grates.

DATA SHEET

CORRUGATED MANHOLE/CATCH BASIN (CONT.)

AVAILABLE COUPLERS/DIAMETER OF THE CONNECTING PIPE

Nom. dia. connecting pipe		Watertight connectors							Soil tight connectors	
		HDPE					PVC		HDPE	Concrete
mm	in	DBIG	DBIGC	BIGC	BG	S	MG	S	BC/PE	Other
100	4	X				X	X	X	X	X
150	6	X				X	X	X	X	X
200	8	X				X	X	X	X	X
250	10	X							X	X
300	12		X						X	X
375	15			X					X	X
450	18			X					X	X
525	21			X					X	X
600	24			X					X	X
750	30			X					X	X
900	36			X					X	X
1,050	42				X					X
1,200	48				X					X

LEGEND

DBIG: double bell with integrated gasket

DBIGC: double bell with integrated gasket with clips

BIGC: bell with integrated gasket with clips

BG: O-ring bell gasket

MG: multi-axial gasket

S: Sol Tee

BC/PE: bell with clips/plain end

APPLICATIONS : Surface drainage
Leachate collection

OPTIONS : Flow regulator
Ladder
Hooking system with hook or eyelet
Frames and covers
Special couplers are available upon request

SPECIFICATIONS

CORRUGATED MANHOLE/CATCH BASIN

SCOPE

These specifications apply to all 300 mm (12 in) to 1,200 mm (48 in) diameter corrugated manholes/catch basins used for collecting non-roadway surface runoff.

MANHOLE/CATCH BASIN REQUIREMENTS

The manhole/catch basin shall be made with smooth interior and corrugated exterior walls and manufactured from a single part.

PRODUCT COMPONENTS

- The manhole/catch basin shall be made from a polyethylene resin that complies with properties classification PE 334420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 100 mm (4 in) to 200 mm (8 in): multi-axial gasket welded to the wall of the catch basin allowing insertion and retraction of the PVC pipe at various angles.
- 250 mm (10 in): coupler welded to a catch basin equipped with a double bell with integrated gasket with transition to PVC. The gasket shall be co-injected into the double bell.
- 300 mm (12 in): coupler welded to the catch basin equipped with a double bell with integrated gasket with clips to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the double bell.
- 375 mm (15 in) to 900 mm (36 in): bell with integrated gasket with clips welded to the pipe to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the bell.
- 1,050 mm (42 in) to 1,200 mm (48 in): HDPE bell welded to the manhole with gasket installed by the manufacturer onto the male end, covered by a protective film for shipping.
- Fused or glued gaskets will not be permitted.

FRAMES, GRATES AND COVERS

Frames, grates and covers shall be made from S series cast iron.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation of the corrugated manhole/catch basin will be made according to the installation requirements of the concrete manhole/catch basin specified in standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solenio.com for installation recommendations.

DATA SHEET

LINEAR CATCH BASIN

PRODUCT DESCRIPTION : Continuous horizontal grate assembly on a non-perforated rigid dual wall pipe with smooth interior and corrugated exterior walls.

FUNCTION : For perpendicular collection of surface runoff in the direction of flow across the full width of the drainage surface. This product can be factory-adjusted to tailor it to the site requirements.

MANUFACTURING STANDARDS : MTQ DN 010 Tome II

RAW MATERIALS : Pipe: Made from high-density polyethylene (HDPE) ASTM D3350
Grate and funnel: Galvanized steel grate and stainless steel funnel

TECHNICAL DATA : Diameter: 300 mm (12 in) to 600 mm (24 in)
Length: 4 m (13.1 ft)
Strength in compression: 320 kPa
Manning: 0.010 to 0.012

AVAILABLE COUPLERS : Watertight: see technical data table.



LINEAR CATCH BASIN with bell with clips and integrated gasket

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Weight		Manning	Backfill	Watertight connectors	
										HDPE	
mm	in	mm	in	mm	in	kg/m	lbs/ft	n	Concrete	DBIGC	BIGC
300	12	303	11.9	363	14.3	40.0	27.0	0.010	X	X	
375	15	381	15.0	449	17.7	42.0	28.2	0.010	X		X
450	18	457	18.0	543	21.4	45.3	30.5	0.010	X		X
525	21	535	21.1	640	25.2	49.5	33.3	0.012	X		X
600	24	603	23.8	722	28.4	53.5	35.9	0.012	X		X

Note 1: Values in the table are approximate and may change without notice.

LEGEND

DBIGC: double bell with integrated gasket with clips.

BIGC: bell with integrated gasket with clips.

SC: split coupler

APPLICATION : Linear collection

OPTIONS : Other diameters and special couplers are available upon request
Variable width grate, from 50 mm (1.9 in) to 150 mm (5.9 in)
Access shafts
Cap

SPECIFICATIONS

LINEAR CATCH BASIN

SCOPE

These specifications apply to all 300 mm (12 in) to 600 mm (24 in) diameter linear catch basins used to collect surface runoff.

LINEAR CATCH BASIN REQUIREMENTS

Each linear catch basin is composed of a galvanized steel grate supported by a stainless steel funnel, which is attached to a Solflo Max pipe with smooth interior and corrugated exterior walls.

- Grate: at least 175 mm (6.9 in) wide with maximum openings of 30 mm (1.2 in) per 100 mm (4 in). Can support CL-625 loads.
- Funnel: attached to the pipe with stainless steel screws and wedged in heated HDPE to keep everything in place.
- Pipe: shall be certified according to standard BNQ 3624-120, shall have a minimum stiffness of 320 kPa and be 4 m (13.1 ft) long.

RAW MATERIALS

- Grate: manufactured from steel complying with standard ASTM A36 and hot dip galvanized according to standard ASTM A123 and/or CSA G164 M92. The galvanized coating weight shall be at least 260 g/m² (0.858 oz/ft²).
- Funnel: manufactured from 304 stainless steel with 16 gauge (1.29 mm) minimum thickness and shall comply with Stainless steel standard AISI 304 (UNS S30400).
- Pipe: shall be made from a polyethylene resin that complies with properties classification PE 324420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 300 mm (12 in): double bell with integrated gasket with clips to ensure proper pipe insertion. Double bells shall be certified as per standard CSA B182.8, Type 1 (watertight at 10.8 psi). The gasket shall be co-injected into the bell and shall overlap two corrugations.
- 375 mm to 600 mm (15 in to 24 in): bell with integrated gasket with clips welded to the pipe to ensure proper pipe insertion. The gasket shall be co-injected into the bell.
- Fused and/or glued gaskets will not be permitted.

ACCESSORIES

- **Connections, service connections:**
Service connections for 100 mm (4 in), 150 mm (6 in) and 200 mm (8 in) diameter PVC pipes shall be monolithic (factory welded), as required by standard BNQ 1809-300, and equipped with an integrated O-ring gasket.
Each connection to a concrete structure shall be made with a monolithic HDPE adapter with smooth exterior wall and an end bell with integrated gasket, as required by standard BNQ 1809-300. PVC and urethane adapters will not be permitted.
- Caps and access shafts

Each series of catch basins shall be equipped with ____ cap(s) and ____ HDPE access chimney(s) to ensure proper water flow, maintenance and cleaning.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation of the linear catch basin will be made following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solenoproducts.com for installation recommendations.

DATA SHEET

TREATMENT CATCH BASIN

PRODUCT DESCRIPTION : Smooth interior and exterior walls for roadway and non-roadway use

FUNCTION : Structure allowing pre-treatment of surface runoff at source by preventing dispersion of floating debris into the storm sewer network

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350 Separator, HDPE internal mechanism and air vent, variable height, welded by continuous fusion.

TECHNICAL DATA : Diameter: 600 mm (24 in), 750 mm (30 in) and 900 mm (36 in)
Standard height: 2 m (6.5 ft)
Structural strength: CAN/CSA - S6 - 02 (CL-625)
Watertightness: ASTM D 3212/10.8 PSI

AVAILABLE COUPLERS : Watertight: see technical data table
The type of coupler available is based on the diameter of the connecting pipe.



Treatment CATCH BASIN with cast iron adjustable frame and grate

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Weight		Frames and grates		Depth of sump					
										300 mm (12 in)		600 mm (24 in)		900 mm (36 in)	
mm	in	mm	in	mm	in	kg/m	lbs/ft	Non-roadway	Roadway	Sediment vol. available					
										m³	ft³	m³	ft³	m³	ft³
600	24	603	23.8	746	29.4	68.3	45.8	S series cast iron	Adjustable cast iron	0.09	3.14	0.18	6.28	0.27	9.42
750	30	756	29.8	899	35.4	110.9	74.4	S series cast iron	Adjustable cast iron	0.13	4.83	0.26	9.66	0.39	14.49
900	36	912	35.9	1056	41.6	127.5	85.5	S series cast iron	Adjustable cast iron	0.20	7.03	0.4	14.06	0.60	21.09

Note 1: Values in the table are approximate and may change without notice.

Note 2: Weight does not include frames and grates.

Note 3: Contact your castings supplier for exact dimensions of available frames and grates.

DATA SHEET

TREATMENT CATCH BASIN (CONT.)

AVAILABLE COUPLERS/DIAMETER OF THE CONNECTING PIPE

Nom. dia. connecting pipe		Watertight connectors					
		HDPE			PVC		
mm	in	DBIG	DBIGC	BIGC	S	MG	S
100	4	X			X	X	X
150	6	X			X	X	X
200	8	X			X	X	X
250	10	X					
300	12		X				
375	15			X			
450	18			X			
525	21			X			
600	24			X			
750	30			X			
900	36			X			

LEGEND

DBIG: double bell with integrated gasket

DBIGC: double bell with integrated gasket with clips

BIGC: bell with integrated gasket with clips

MG: multiaxial gasket

S: Sol Tee



DATA SHEET

TREATMENT CATCH BASIN (CONT.)

APPLICATION : Control of sediments and floating debris

OPTIONS : Other diameters and special couplers are available upon request

Riser

Lift eyelet

Special frames and covers

Flow regulators

SPECIFICATIONS

TREATMENT CATCH BASIN

SCOPE

These specifications apply to all 600 mm (24 in) to 900 mm (36 in) diameter source treatment catch basins used for pretreating surface runoff.

TREATMENT CATCH BASIN REQUIREMENTS

The catch basin shall be made with smooth interior and exterior walls, manufactured from a single part and meet CL-625 highway loading requirements as defined in standard CAN/CSA-S6-06. The catch basin shall have a ____ mm ____ in deep sump and available sedimentation volume of ____ m³ ____ ft³.

PRODUCT COMPONENTS

- The catch basin shall be made from high-density polyethylene (HDPE) resin that complies with standards ASTM D3350.

JOINT PERFORMANCE

- 100 mm (4 in) à 250 mm (10 in) : coupler 90° welded to air vent penetrating the catch basin. The outer section of the coupler shall be equipped with a double bell with integrated gasket with transition to PVC. The gasket shall be co-injected into the double bell.
- 300 mm (12 in) : coupler 90° welded to air vent penetrating the catch basin. The outer section of the coupler shall be equipped with a double bell with clips to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the double bell.
- 375 mm (15 in) à 900 mm (36 in) : coupler 90° welded to air vent penetrating the catch basin. The outer section of the coupler shall be equipped with a double bell with clips to ensure proper insertion of the HDPE pipe. The gasket shall be co-injected into the bell.
- Fused and/or glued gaskets will not be permitted.

FRAMES AND GRATES

- For roadway use, frames and grates shall be made of cast iron and adjustable, allowing transfer of loads to the surrounding soil.
- For non-roadway use, grates and covers shall be made from S series cast iron.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation of the treatment catch basin will be made according to the installation requirements of the concrete catch basin specified in standard BNQ 1809-300, and following Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

PERCOLATION CATCH BASIN

PRODUCT DESCRIPTION : Smooth interior and exterior walls on the upper section, perforated corrugated exterior walls on the lower section (for roadway and non roadway use)

FUNCTION : For collecting surface runoff and replenishing groundwater

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 450 mm (18 in) to 900 mm (36 in)
Total standard height: 2 m (6.5 ft)
Height of the perforated lower section: 1 m (3.3 in)
Perforation diameter: 9 mm (0.3 in)
Minimum number of perforation rows:
8 per section (see technical drawing)
Structural strength: CAN/CSA - S6 - 06 (CL-625)

AVAILABLE COUPLERS : Watertight: see technical data table
The type of coupler available is based on the diameter of the connecting pipe.



Percolation CATCH BASIN with smooth exterior wall, adjustable cast iron frame and grate

TECHNICAL DATA TABLE

Nom. dia.		Int. dia.		Ext. dia.		Weight (smooth wall)		Frames and grates	
mm	in	mm	in	mm	in	kg/m	lbs/ft	Non-roadway	Roadway
450	18	457	18.0	552	21.8	37.9	25.4	S series cast iron	None
525	21	535	21.1	650	25.6	48.8	32.7	S series cast iron	None
600	24	603	23.8	746	29.4	68.3	45.8	S series cast iron	Adjustable cast iron
750	30	756	29.8	899	35.4	110.9	74.4	S series cast iron	Adjustable cast iron
900	36	912	35.9	1056	41.6	127.5	85.5	S series cast iron	Adjustable cast iron

Note 1: Values in the table are approximate and may change without notice.

Note 2: Weight does not include frames and grates.

Note 3: Contact your castings supplier for exact dimensions of available frames and grates.

DATA SHEET

PERCOLATION CATCH BASIN (CONT.)

AVAILABLE COUPLERS/DIAMETER OF THE CONNECTING PIPE

Nom. dia. connecting pipe		Watertight connectors					
		HDPE			PVC		
mm	in	DBIG	DBIGC	BIGC	S	MG	S
100	4	X			X	X	X
150	6	X			X	X	X
200	8	X			X	X	X
250	10	X					
300	12		X				
375	15			X			
450	18			X			
525	21			X			
600	24			X			
750	30			X			
900	36			X			

LEGEND

DBIG: double bell with integrated gasket
 DBIGC: double bell with integrated gasket with clips
 BIGC: bell with integrated gasket with clips
 MG: multiaxial gasket
 S: Sol Tee

OPTIONS : Other diameters and special couplers are available upon request

Riser
 Lift eyelet
 Special frames and covers
 Flow regulator

APPLICATIONS : Percolation well

SPECIFICATIONS

PERCOLATION CATCH BASIN

SCOPE

These specifications apply to all 450 mm (18 in) to 900 mm (36 in) diameter percolation catch basins used to collect surface runoff and replenish ground water.

PIPE REQUIREMENTS

The upper section of the catch basin shall be made with smooth interior and exterior walls and the lower section shall be made with smooth interior and corrugated exterior walls. The lower section shall be perforated according to percolation requirements. It shall be manufactured from a single part based on the specified height and shall meet CL-625 highway loading requirements as defined in standard CAN/CSA-S6-06.

PRODUCT COMPONENTS

- The catch basin shall be made from a polyethylene resin that complies with properties classification PE 334420C, as defined in standard ASTM D3350.

JOINT PERFORMANCE

- 100 mm (4 in) to 200 mm (8 in): integrated gasket co-injected into the catch basin wall allowing insertion and retraction of the pipe at various angles.
- 250 mm (10 in): coupler welded to a catch basin equipped with a double bell with integrated gasket. The gasket shall be co-injected into the double bell.
- 300 mm (12 in): coupler welded to a catch basin equipped with a double bell with integrated gasket with clips to ensure proper pipe insertion. The gasket shall be co-injected into the double bell.
- 375 mm (15 in) to 900 mm (36 in): bell with integrated gasket with clips welded to the pipe to ensure proper pipe insertion. The gasket shall be co-injected into the bell.
- Fused and/or glued gaskets will not be permitted.

FRAMES AND GRATES

- For roadway use, frames and grates shall be made of cast iron and adjustable, allowing transfer of loads to the surrounding soil for 600 mm (24 in) to 900 mm (36 in) diameter catch basins.
- For non-roadway use, frames and grates shall be made from S series cast iron, for 450 mm (18 in) to 900 mm (36 in) diameter catch basins.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and/or drawings.

INSTALLATION

Installation of the percolation catch basin will be made according to the installation requirements of the concrete catch basin specified in standard 1809-300, and following Soleno's recommendations. Contact your local Soleno representative or visit our website at www.solen.com for installation recommendations.

DATA SHEET

STANDARD OR CUSTOM-MADE RISER FOR MANHOLE OR CATCH BASIN WITH SMOOTH EXTERIOR WALL

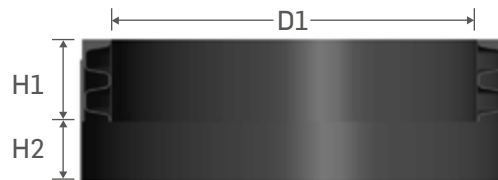
PRODUCT DESCRIPTION: Factory-made riser

FUNCTION: To increase the height of a manhole or a catch basin

RAW MATERIAL: Made of high-density polyethylene (HDPE) ASTM D3350



RISER



TECHNICAL DATA TABLE - 2 OR 3-RING RISER

Nominal diameter	2 rings				3 rings			
D1	H1		H2		H1		H2	
	mm	in	mm	in	mm	in	mm	in
450 mm (18 in)	136	5.35	89	3.50	203	7.99	89	3.50
525 mm (21 in)	151	5.94	101	3.98	227	8.94	101	3.98
600 mm (24 in)	201	7.91	138	5.43	301	11.85	138	5.43
750 mm (30 in)	200	7.87	137	5.39	300	11.81	137	5.39
900 mm (36 in)	203	7.99	140	5.51	305	12.00	140	5.51

DATA SHEET

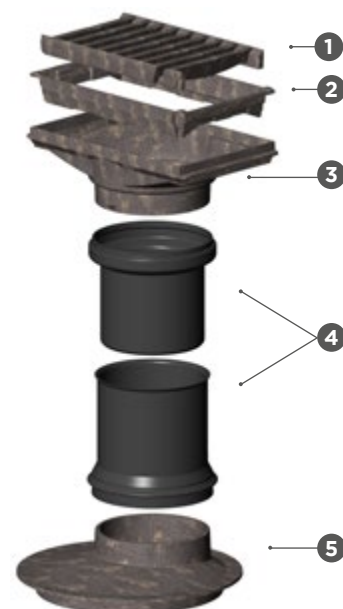
PUISTAR

PRODUCT DESCRIPTION: Adjustable collection system for in road catch basin

FUNCTION: Increased stormwater inflow using an adjustable head on 3 axes:
height, angle and rotation, preventing the infiltration of water between the catch basin and surrounding backfill.

RAW MATERIALS: High density polyethylene (HDPE) ASTM D3350
Ductile iron ASTM A536, grade 65-45-12
Grey cast iron ASTM A48, Class 30

TECHNICAL DATA: Maximum vertical adjustment of 180 mm (7 in)
Tilt as a function of the slope of the roadway up to 8 degrees (14% slope)
Possible head rotation on 360°
Diameters: 600 mm (24 in), 750 mm (30 in), 900 mm (36 in)
Supplemental final adjustment of 50 or 60 mm (1.96 or 2.36 in)



TECHNICAL DATA TABLE - DUCTILE IRON GRATE (1)

Length		Width		Height		Bar spacing (top of the grate)		Mass	
mm	in	mm	in	mm	in	mm	in	kg	lbs
613	24.13	370	14.5	63.5	2.50	4.52	1.78	42	93

TECHNICAL DATA TABLE - DUCTILE IRON SEAT FRAME (2)

Length (interior)		Width (interior)		Seat frame height		Overall height		Mass	
mm	in	mm	in	mm	in	mm	in	kg	lbs
640	25.2	318	12.5	40	1.57	87	3.4	13.2	29
640	25.2	318	12.5	50	1.97	87	3.4	14.3	31.5
640	25.2	318	12.5	60	2.36	87	3.4	15.4	34

TECHNICAL DATA TABLE - DUCTILE IRON COLLECTOR FRAME (3)

Length (interior)		Width (interior)		Overall height		Mass	
mm	in	mm	in	mm	in	kg	lbs
638.3	25.13	369.21	14.54	213	8.38	34	75

DATA SHEET

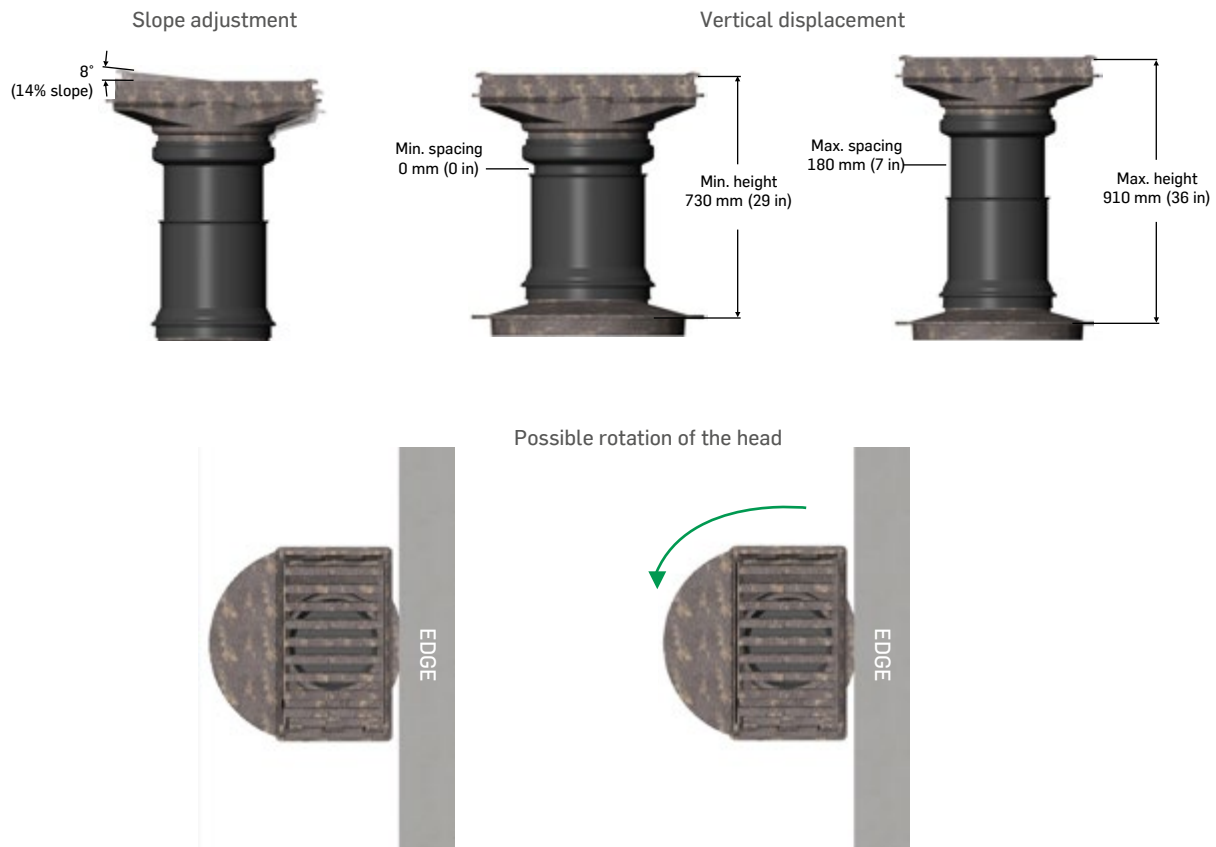
PUISTAR (CONT'D)

TECHNICAL DATA TABLE - HIGH-DENSITY POLYETHYLENE (HDPE) FLANGE (4)

Diameter		Minimum height		Maximum height		Mass	
mm	in	mm	in	mm	in	kg	lbs
377	14.8	495.3	19.5	673.1	26.5	6.0	13.0

TECHNICAL DATA TABLE - GREY IRON HEAD (5)

Diameter		Overall height		Mass	
mm	in	mm	in	kg	lbs
600	24	121	4.76	34.4	76
750	30	121	4.76	54.4	120
900	36	121	4.76	63	139



SPECIFICATIONS

PUISTAR

SCOPE

These specifications apply to all 600 mm (24 in) to 900 mm (36 in) adjustable catch basin systems used to collect surface runoff.

RAW MATERIALS

- The grate, the seat frame and the collector frame shall be made of ductile iron according to ASTM A536, Grade 65-45-12 standard.
- The catch basin head is made of grey cast iron according to ASTM A48, Class 30 standard.
- The flange shall be made of high-density polyethylene (HDPE) certified to ASTM D3350 standard.

PERFORMANCE OF THE GRATE AND THE COLLECTOR FRAME

- The bars of the grate are inclined 12 degrees optimizing the collection of water runoff and its entry into the catch basin.
- The grate is provided with a lock to prevent removal without tools.
- The collector frame shall be expanded to prevent the infiltration of water between the catch basin and surrounding backfill. Consequently, the water will be conveyed under the grate and toward the flange, preventing freeze-thaw problems.
- Seat frames of ductile iron are available for a final adjustment following the elevation of the road surface at 50 or 60 mm (1.96 or 2.36 in).

PERFORMANCE OF THE ASSEMBLY

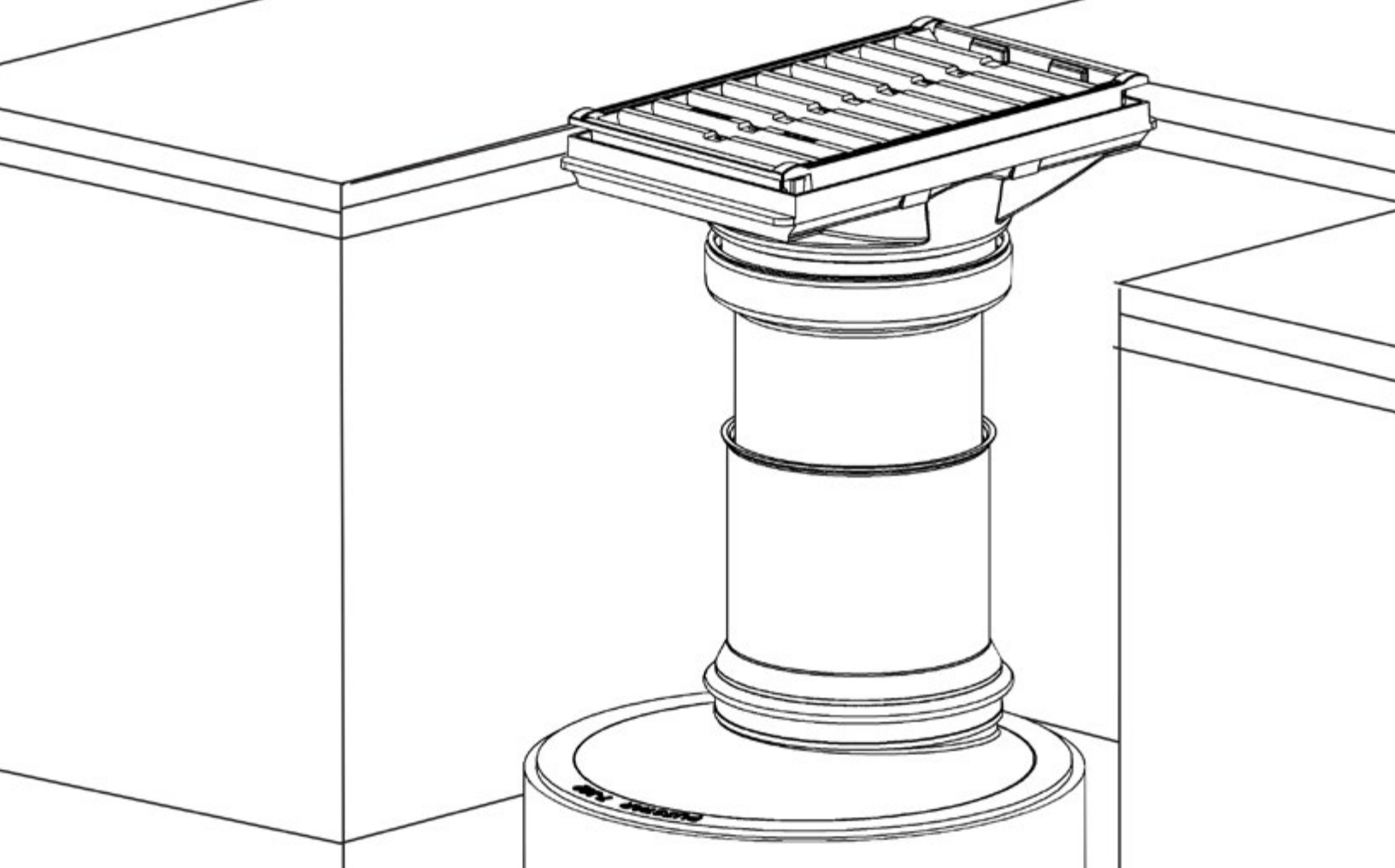
- The system shall be able to withstand the CL-625 highway loading requirements as defined in standard CAN/CSA-S6.
- Tolerances between the individual parts shall permit adjustment of the final slope of the grate between 0 and 8 degrees (14 % slope).
- The system shall allow a maximum vertical adjustment of 180 mm (7 in).
- The catch basin head must be adjustable through 360 degrees to allow proper positioning of the grate relative to the edge.
- The grate and its frame are separated from the head of the catch basin by the adjustable flange, the impact loads are redistributed in the soil and not directly on the catch basin.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and plans.

INSTALLATION

Installation of the puistar shall be carried out in compliance with Soleno's installation guide. Contact your local Soleno representative or visit our website at www.solen.com for installation recommendations.



PUISTAR®

INSTALLATION GUIDE

INSTALLATION INSTRUCTIONS FOR THE PUISTAR® SYSTEM

STEPS

1. Before undertaking the work
2. Merchandise reception
3. HDPE catch basin installation
4. Rehabilitation of an existant catch basin
5. PUISTAR® installation
6. Removing the grate

STEP 1

BEFORE UNDERTAKING THE WORK

In case of discrepancy between the instructions contained in this guide and those contained in the plans and specifications, please contact your Soleno sales representative.

Contact your Soleno sales representative at least 48 hours before work begins. A visit from your authorized Soleno representative is recommended after receipt of the materials on site or/and before the work begins.

STEP 2

MERCHANDISE RECEPTION

Upon receipt of the materials, ensure that all items required on the slip are delivered and in good condition. Please notify immediately your Soleno representative in case of damage.



STEP 3

HDPE CATCH BASIN INSTALLATION

(For a rehabilitation, skip to step 4)

- Excavate to allow the installation of the catch basin as well as the connecting pipe.
- If required, install a trench box.
- Clear the top of the main storm water pipe and proceed to the drilling of this pipe. Make the catch basin lead connection to the storm sewer line with the appropriate universal sol tee connection.
- Install the universal sol tee as well as the necessary pipeline to get to the catch basin location.

STEP 3

HDPE CATCH BASIN INSTALLATION (CONT'D)

(For a rehabilitation, skip to step 4)

- Install the catch basin at the required location.
- Insert the pipe inside the catch basin.
- Proceed to the backfilling and the compaction according to the plans and specifications sheet and conform to the local norms in effect.
- If required, install the hooking system.
- Go to step 5.

STEP 4

REHABILITATION OF AN EXISTANT CATCH BASIN

- Proceed to the paving cut in the needed rehabilitated zone.
- Excavate the backfill material to remove the broken section of the catch basin.
- Remove the defective elements and other accessories (frame, grate, head, and riser) to free the top of the catch basin.
- Ensure that the catch basin is structurally sound, stable and that the hooking system (if required) is properly placed.
- If a spigot lip is present over the catch basin, remove with the help of a concrete saw.

STEP 5

PUISTAR INSTALLATION



STEP 5

PUISTAR INSTALLATION (CONT'D)

- Lay the flange-iron head assembly (1) on top of the catch basin.
- Properly align the iron head opening with the grate location, position the flange at the required distance between the grate and the edge of the concrete or of the sidewalk.
- Insert the collector frame-flange assembly (2) in the flange-iron head assembly (1). Incline the collector frame to the same grade of the street profile pavement.
- Proceed to the backfilling with the help of compactable materials, with successive layers of 200 mm (8 in). The compaction needs to reach 90% of the M.P. on all of the width of the backfilling. Adjust the elevation and slop of the collector frame during the backfilling procedures.
- Adjust the height of the collector frame as to obtain the following dimensions:



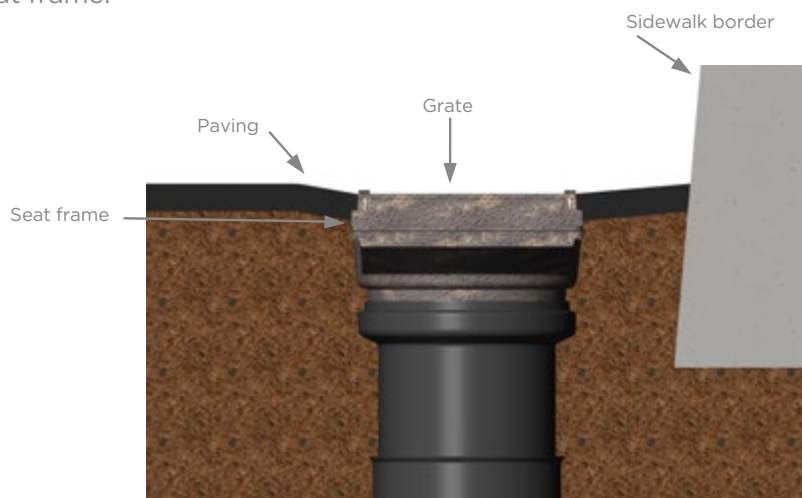
- Clean the exterior of the collector frame.
- Insert the grate-seat frame assembly (3) inside the collector frame. In the case that the grate is installed on a unidirectional slope, ensure the positioning of the grate to optimize the water flow/drainage.



STEP 5

PUISTAR INSTALLATION (CONT'D)

- Perform a fast pass with the compaction equipment on the grate to properly sit the collector frame into the backfilling.
- Temporarily cover the grate with a cardboard or geotextile (to protect the grate and prevent the infiltration of granular material).
- Proceed to the paving by taking care of aligning the top of the paving and the border of the seat frame.

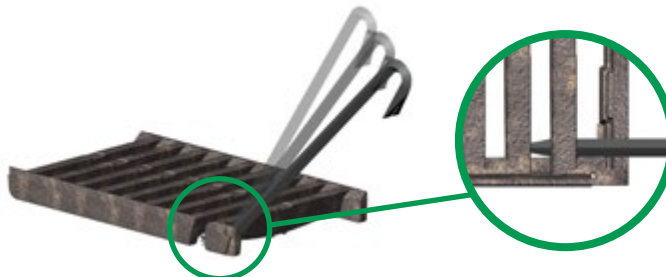


- Remove the cardboard or geotextile protection.

STEP 6

REMOVING THE GRATE

- Insert a lever or a crowbar in the first section of the grate, as shown in the picture below, at the side of the opening.
- Push down, as shown, in order to pry the first bar from the seat frame. This action will release and lift the grate up.
- To replace, simply insert one end of the grate from the bar ends into the seat frame and firmly push down the other end to fully insert the grate in place.



DATA SHEET

CAST IRON FRAME AND GRATE

PRODUCT DESCRIPTION: Cast iron frame and grate

FUNCTION: For collecting surface runoff for conveying in a non-roadway manhole

MANUFACTURING STANDARDS: ASTM A536

RAW MATERIAL: Ductile iron

TECHNICAL DATA: Diameter : 300 mm (12 in) to 900 mm (36 in)

AVAILABLE:



450 mm (18 in)



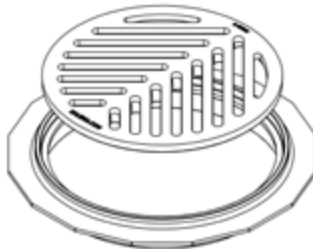
300 mm (12 in)



375 mm (15 in)



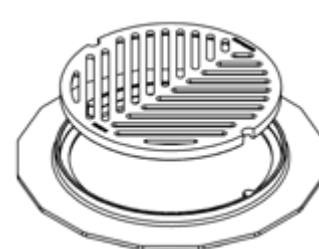
450 mm (18 in)



600 mm (24 in)

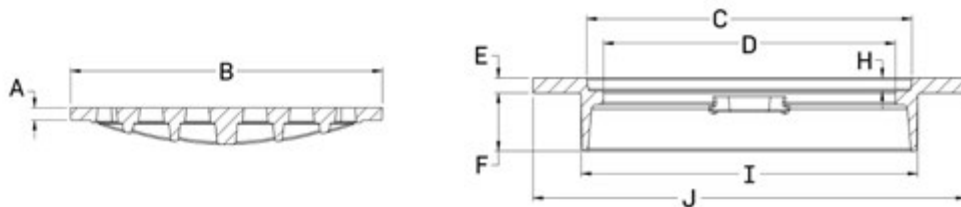


750 mm (30 in)



900 mm (36 in)

TECHNICAL DATA TABLE 300 mm (12 in) DIAMETER

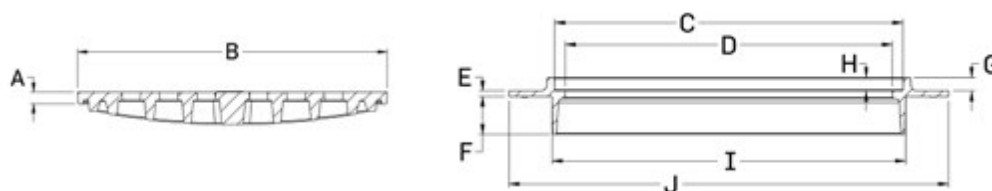


A		B		C		D		E		F		H		I		J	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in
11	0.43	276	10.87	279	10.98	253	9.96	13	0.51	50	1.958	11	0.43	292	11.50	382	15.04

DATA SHEET

CAST IRON FRAME AND GRATE (CONT'D)

TECHNICAL DATA TABLE 375 mm (15 in) TO 900 mm (36 in) DIAMETER



Diameter		A		B		C		D		E		F		G		H		I		J	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in
375	15	25	0.98	375	14.76	381	15.00	348	13.70	10	0.39	65	2.56	25	0.98	25	0.98	365	14.37	459	18.07
450	18	25	0.98	452	17.79	458	18.03	419	16.49	10	0.39	70	2.76	25	0.98	25	0.98	439	17.28	561	22.09
600	24	22	0.87	572	22.52	578	22.76	543	21.38	9	0.35	63	2.48	22	0.87	22	0.87	587	23.11	744	29.29
750	30	41	1.61	751	29.57	757	29.80	705	27.76	15	0.59	63	2.48	41	1.61	41	1.61	735	28.94	907	35.71
900	36	41	1.61	751	29.57	757	29.80	690	27.17	15	0.59	63	2.48	41	1.61	41	1.61	883	34.76	1057	41.61

TECHNICAL DATA TABLE

Diameter		Frame weight		Grate weight	
mm	in	kg	lbs	kg	lbs
300	12	4.50	9.9	7.05	15.5
375	15	12.50	27.5	13.50	29.8
450	18	19.80	43.7	17.08	39.2
600	24	24.10	53.1	34.80	76.7
750	30	43.80	96.6	67.60	149.0
900	36	71.40	157.4	67.60	149.0

MINIMUM ANGLE BETWEEN CONNECTIONS FOR MANHOLES OR CATCH BASINS

CONNECTION DIAMETER		MANHOLE OR CATCH BASIN DIAMETER									
INLET	OUTLET	300 mm (12 in.)	375 mm (15 in.)	450 mm (18 in.)	525 mm (21 in.)	600 mm (24 in.)	750 mm (30 in.)	900 mm (36 in.)	1050 mm (42 in.)	1200 mm (48 in.)	1500 mm (60 in.)
100 mm (4 in.)	100 mm (4 in.)	70	55	45	35	35	30	20	25	20	15
150 mm (6 in.)	100 mm (4 in.)	80	60	50	40	40	30	25	25	20	15
150 mm (6 in.)	150 mm (6 in.)	90	70	60	50	45	35	30	30	25	20
200 mm (8 in.)	100 mm (4 in.)	90	70	60	50	45	35	30	25	25	20
200 mm (8 in.)	150 mm (6 in.)	105	80	65	55	50	40	35	30	25	20
200 mm (8 in.)	200 mm (8 in.)	115	90	70	60	60	45	40	35	30	25
250 mm (10 in.)	100 mm (4 in.)	105	80	70	55	50	40	35	30	30	25
250 mm (10 in.)	150 mm (6 in.)	120	90	75	60	55	45	40	30	30	25
250 mm (10 in.)	200 mm (8 in.)	130	95	85	70	60	50	45	35	35	25
250 mm (10 in.)	250 mm (10 in.)	145	105	90	75	70	55	45	40	35	30
300 mm (12 in.)	100 mm (4 in.)	135	90	70	60	60	45	40	30	30	25
300 mm (12 in.)	150 mm (6 in.)	145	100	80	65	65	50	40	35	30	25
300 mm (12 in.)	200 mm (8 in.)	155	110	90	70	70	55	45	35	35	30
300 mm (12 in.)	250 mm (10 in.)	165	125	100	80	75	55	50	40	40	30
300 mm (12 in.)	300 mm (12 in.)	180	140	105	85	80	65	55	45	40	35
375 mm (15 in.)	100 mm (4 in.)		125	90	75	65	50	45	40	35	25
375 mm (15 in.)	150 mm (6 in.)		135	105	80	70	55	45	40	35	30
375 mm (15 in.)	200 mm (8 in.)		145	110	85	75	60	50	45	40	30
375 mm (15 in.)	250 mm (10 in.)		155	120	90	85	65	55	50	45	30
375 mm (15 in.)	300 mm (12 in.)		165	125	100	90	70	60	50	45	35
375 mm (15 in.)	375 mm (15 in.)		180	140	115	100	80	65	55	50	40
450 mm (18 in.)	100 mm (4 in.)			115	100	80	55	50	40	40	30
450 mm (18 in.)	150 mm (6 in.)			125	105	85	60	55	45	40	35
450 mm (18 in.)	200 mm (8 in.)			135	110	90	65	60	50	45	35
450 mm (18 in.)	250 mm (10 in.)			140	115	95	75	65	50	45	40
450 mm (18 in.)	300 mm (12 in.)			150	125	100	80	70	55	50	40
450 mm (18 in.)	375 mm (15 in.)			165	140	110	85	75	60	55	45
450 mm (18 in.)	450 mm (18 in.)			180	150	125	90	80	65	60	50
525 mm (21 in.)	100 mm (4 in.)				120	95	65	60	50	40	35
525 mm (21 in.)	150 mm (6 in.)				115	95	70	60	50	45	35
525 mm (21 in.)	200 mm (8 in.)				125	100	75	65	55	45	40
525 mm (21 in.)	250 mm (10 in.)				135	110	80	70	60	50	45
525 mm (21 in.)	300 mm (12 in.)				145	115	85	75	65	55	45
525 mm (21 in.)	375 mm (15 in.)				155	130	90	80	70	55	50
525 mm (21 in.)	450 mm (18 in.)				165	140	100	90	75	60	50
525 mm (21 in.)	525 mm (21 in.)				180	150	105	95	80	65	55
600 mm (24 in.)	100 mm (4 in.)					105	80	60	55	45	40
600 mm (24 in.)	150 mm (6 in.)					110	80	65	60	50	40
600 mm (24 in.)	200 mm (8 in.)					120	85	70	60	50	45
600 mm (24 in.)	250 mm (10 in.)					125	90	70	60	55	45
600 mm (24 in.)	300 mm (12 in.)					135	95	75	65	55	50
600 mm (24 in.)	375 mm (15 in.)					145	100	80	70	60	55
600 mm (24 in.)	450 mm (18 in.)					160	110	90	75	65	55
600 mm (24 in.)	525 mm (21 in.)					170	120	95	80	70	60
600 mm (24 in.)	600 mm (24 in.)					180	130	100	85	75	65
750 mm (30 in.)	100 mm (4 in.)						105	75	65	60	45
750 mm (30 in.)	150 mm (6 in.)						110	80	70	60	50
750 mm (30 in.)	200 mm (8 in.)						115	80	70	65	50
750 mm (30 in.)	250 mm (10 in.)						120	85	75	65	55
750 mm (30 in.)	300 mm (12 in.)						125	90	80	70	55
750 mm (30 in.)	375 mm (15 in.)						130	95	85	75	60
750 mm (30 in.)	450 mm (18 in.)						140	105	90	80	60
750 mm (30 in.)	525 mm (21 in.)						150	115	95	85	65
750 mm (30 in.)	600 mm (24 in.)						160	120	95	90	70
750 mm (30 in.)	750 mm (30 in.)						180	135	110	100	75
900 mm (36 in.)	100 mm (4 in.)							100	80	70	55
900 mm (36 in.)	150 mm (6 in.)							105	80	75	55

TABLE CONTINUES ON NEXT PAGE

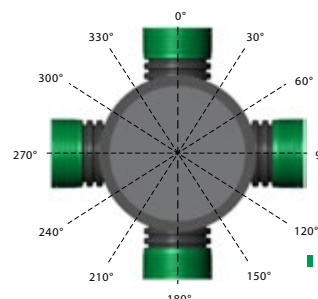
MINIMUM ANGLE BETWEEN CONNECTIONS FOR MANHOLES OR CATCH BASINS

CONNECTION DIAMETER		MANHOLE OR CATCH BASIN DIAMETER									
INLET	OUTLET	300 mm (12 in.)	375 mm (15 in.)	450 mm (18 in.)	525 mm (21 in.)	600 mm (24 in.)	750 mm (30 in.)	900 mm (36 in.)	1050 mm (42 in.)	1200 mm (48 in.)	1500 mm (60 in.)
900 mm (36 in.)	200 mm (8 in.)							110	85	75	60
900 mm (36 in.)	250 mm (10 in.)							115	85	80	60
900 mm (36 in.)	300 mm (12 in.)							120	90	80	65
900 mm (36 in.)	375 mm (15 in.)							125	95	85	70
900 mm (36 in.)	450 mm (18 in.)							135	105	90	75
900 mm (36 in.)	525 mm (21 in.)							140	110	95	75
900 mm (36 in.)	600 mm (24 in.)							150	115	100	80
900 mm (36 in.)	750 mm (30 in.)							165	130	110	85
900 mm (36 in.)	900 mm (36 in.)							180	140	125	95
1050 mm (42 in.)	100 mm (4 in.)								105	85	65
1050 mm (42 in.)	150 mm (6 in.)								105	85	65
1050 mm (42 in.)	200 mm (8 in.)								110	90	70
1050 mm (42 in.)	250 mm (10 in.)								115	90	70
1050 mm (42 in.)	300 mm (12 in.)								115	95	75
1050 mm (42 in.)	375 mm (15 in.)								120	100	75
1050 mm (42 in.)	450 mm (18 in.)								125	105	80
1050 mm (42 in.)	525 mm (21 in.)								130	110	85
1050 mm (42 in.)	600 mm (24 in.)								135	115	90
1050 mm (42 in.)	750 mm (30 in.)								150	125	95
1050 mm (42 in.)	900 mm (36 in.)								165	140	100
1050 mm (42 in.)	1050 mm (42 in.)								180	150	110
1200 mm (48 in.)	100 mm (4 in.)									105	75
1200 mm (48 in.)	150 mm (6 in.)									105	75
1200 mm (48 in.)	200 mm (8 in.)									110	80
1200 mm (48 in.)	250 mm (10 in.)									110	80
1200 mm (48 in.)	300 mm (12 in.)									115	85
1200 mm (48 in.)	375 mm (15 in.)									115	85
1200 mm (48 in.)	450 mm (18 in.)									120	90
1200 mm (48 in.)	525 mm (21 in.)									125	95
1200 mm (48 in.)	600 mm (24 in.)									130	100
1200 mm (48 in.)	750 mm (30 in.)									140	105
1200 mm (48 in.)	900 mm (36 in.)									150	115
1200 mm (48 in.)	1050 mm (42 in.)									165	125
1200 mm (48 in.)	1200 mm (48 in.)									180	135
1500 mm (60 in.)	100 mm (4 in.)										100
1500 mm (60 in.)	150 mm (6 in.)										100
1500 mm (60 in.)	200 mm (8 in.)										105
1500 mm (60 in.)	250 mm (10 in.)										105
1500 mm (60 in.)	300 mm (12 in.)										110
1500 mm (60 in.)	375 mm (15 in.)										115
1500 mm (60 in.)	450 mm (18 in.)										120
1500 mm (60 in.)	525 mm (21 in.)										120
1500 mm (60 in.)	600 mm (24 in.)										125
1500 mm (60 in.)	750 mm (30 in.)										135
1500 mm (60 in.)	900 mm (36 in.)										140
1500 mm (60 in.)	1050 mm (42 in.)										150
1500 mm (60 in.)	1200 mm (48 in.)										165
1500 mm (60 in.)	1500 mm (60 in.)										180

Note 1: The configurations indicated in this document assume that the pipes are at the same level. If the pipes are at different levels, please contact your Technical Service Engineer.

Note 2: To get the minimum angles between the connections for manholes and catch basins with a diameter greater than 1500 mm (60 inches), please contact your sales representative or Technical Service Engineer.

CONNECTION ANGLES :



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DATA SHEET

AQUA-SWIRL®

PRODUCT DESCRIPTION : Hydrodynamic separator

FUNCTION : System that maximizes removal of Total Suspended Solids (TSS), oils and floating debris from surface runoff before it is conveyed to an outlet.

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 750 mm (30 in) to 3300 mm (132 in)
Higher flow rates are custom manufactured
Height: The height of the unit and the stack are variable
Size of particles to control: from coarse silt to very coarse sand (60 microns and over)
Installation: networked with or parallel to the storm water sewer system

AVAILABLE COUPLERS : Adapts to all types of HDPE, PVC, concrete, steel and CSP pipes



AQUA-SWIRL

TECHNICAL DATA TABLE

Model	Nom. dia. of the chamber		Ext. dia. of the chamber		Height of the chamber		Weight		Maximum nom. diam. of the pipe connection				Oils and floating debris storage capacity		Sediment storage capacity	
	mm	ft	mm	in	mm	in	kg	lb	In parallel		Networked		liters	gallons	m ³	ft ³
AS-2	750	2.5	871	34.3	1524	60	141	311	200	8	300	12	140	37	0.3	10
AS-3	1050	3.5	1219	48	2642	104	464	1024	250	10	525	21	416	110	0.6	20
AS-4	1350	4.5	1549	61	2642	104	686	1512	300	12	600	24	719	190	0.9	32
AS-5	1500	5	1722	67.8	2642	104	816	1799	300	12	750	30	1022	270	1.3	45
AS-6	1800	6	2067	81.4	2642-2794	104-110	1108-1142	2443-2518	300	12	900	36	1476	390	1.8	65
AS-7	2100	7	2393	94.2	2896-2946	114-116	1467-1482	3235-3267	375	15	900	36	2044	540	2.6	90
AS-8	2400	8	2718	107	2896-3099	114-122	1770-1841	3901-4058	375	15	1200	48	2687	710	3.3	115
AS-9	2700	9	3045	119.9	2896-3251	114-128	2172-2315	4788-5103	450	18	1200	48	3444	910	4.1	145
AS-10	3000	10	3371	132.7	2896-3251	114-128	2523-2701	5563-5103	525	21	1500	60	4277	1130	5.1	180
AS-11	3300	11	3716	146.3	2896-3251	114-128	3277-3526	7226-7774	525	21	1500	60	5383	1422	6.3	222
AS-XX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX

Note 1: Values in the table are approximate and may change without notice.

AS-XX: Custom made unit.

* Contact your Soleno representative to know the treatment throughput for the unit, according to local regulations.

APPLICATION : Control of TSS, oils and floating debris by hydrodynamic separation

OPTION : Factory-welded bell with integrated gasket with clips

SPECIFICATIONS

AQUA-SWIRL®

SCOPE

These specifications apply to the performance, materials and manufacture of models AS-2 to AS-11 Aqua-Swirl storm water treatment tanks supplied by Soleno.

TREATMENT UNIT REQUIREMENTS

The Aqua-Swirl storm water treatment unit shall be supplied by Soleno Inc. and shall comply with these materials and performance specifications based on specified flow and storage capacity.

RAW MATERIALS

The storm water treatment unit shall be made from high-density polyethylene (HDPE) resin.

- **Physical properties of the resin**
 - The filtration system shall be made from high-density polyethylene (HDPE) resin that complies with standards ASTM D3350.
- **Chemical properties of the resin**
 - HDPE resists corrosion, as it is not an electrical conductor.
 - Biological resistance is unaffected by microbial organisms.
 - It can be used in applications with pH values ranging from 1.5 to 14 due to its inert molecular structure.

The storm water treatment unit may be refused if it does not meet the requirements of these specifications.

PERFORMANCE

- The storm water treatment unit must include a networked ____ mm (____ in) interior diameter vortex hydrodynamic treatment chamber for treating the water entering the unit. A tangential inlet pipe shall induce a whirlpool which will deposit solid sediments at the bottom of the treatment chamber and prevent these sediments from being resuspended. An arched deflector in the chamber shall prevent oils, grease, and floating debris from exiting the unit while facilitating the chamber's vortex action.
- The storm water treatment unit shall include a ____ m³ (____ ft³) sediment storage reservoir and may also contain ____ litres (____ gallons) of oils, grease, and floating debris. The storm water treatment unit shall capture the oils, grease, waste and floating debris and ____ % of Total Suspended Solids (TSS) entering the chamber (annually).
- Size of particles to control shall be from coarse silt to very coarse sand (60 microns and over).
- The storm water treatment unit shall include a 750 mm (30 in) interior diameter access shaft, located on the upper part of the chamber, to provide safe access for routine inspection and maintenance.

SPECIFICATIONS

AQUA-SWIRL® (CONT.)

MANUFACTURING THE TREATMENT UNIT

- The storm water treatment unit shall be made from HDPE pipes that comply with standard ASTM F894 or ASTM F714, based on the diameters required.
- The thickness of the plate at the bottom of the treatment unit shall be determined based on standard ASTM F1759 requirements. Calculations shall be provided to substantiate this thickness.
- Inlets and outlets are HDPE extrusion-welded inside and outside the structure.
- The deflector shall be made from HDPE and welded to the inside of the treatment unit with 180-degree couplers at each end.
- A lifting cable shall be attached to the base of the processing unit to avoid unwanted stress on some critical elements during loading, unloading and handling of the unit.
- The upper section of the treatment unit shall be built in compliance with site requirements and confirmed by shop drawings. When the backfill is deep, the upper section shall be reinforced to resist the loads encountered. For AS-5 to AS-11 devices, in roadway use or when the backfill exceeds 2.44 m (8 pi), the treatment unit shall be covered with a structural concrete slab to distribute the load over the entire surface of the unit. The cast iron frame and cover must also be able to support these loads.

Buoyancy calculations (Archimedes' principle) may be provided upon request. Standard installation drawings for concrete anti-buoyancy structures may also be provided. The contractor shall provide the anti-buoyancy structure.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

The Contractor shall contact a representative of the manufacturer at least 48 hours before the start of the installation.

Installation shall be carried out in compliance with manufacturer recommendations.

- **Excavation and bedding**
The trench and bedding shall meet the requirements of standard BNQ 1809-300, article 9, Excavation and Backfill. The storm water treatment unit shall be installed on stable bedding composed of CG-14 granular material, compacted to 95 % modified Proctor density.
The contractor is responsible for all safety precautions during treatment unit installation.
- **Backfill**
Backfill material shall be composed of CG-14 granular material, compacted to 90 % modified Proctor density. The backfill and bedding shall be free of debris. Backfill shall meet the requirements of standard BNQ 1809-300, Article 9, Excavation and Backfill. Backfill shall extend at least 1.0 m (3.3 ft) beyond the perimeter of the unit, from below the bedding to the full height of the unit, and laterally up to the undisturbed part of the soil.
- **Connecting inlet and outlet pipes**
Flexible couplers with stainless steel tightening collars shall be used to connect unit inlets and outlets to the pipes. When connecting to a Solflo Max pipe, a bell with integrated gasket with clips shall be factory-welded to the treatment unit.

SPECIFICATIONS

AQUA-SWIRL® (CONT.)

SHARED RESPONSIBILITY

- **Solenio's responsibilities**
Solenio is responsible for delivering the storm water treatment system to the site. The system includes the treatment unit with debris deflector, inlet and outlet connections, lifting cable, 750 mm (30 in) interior diameter access chimneys up to the finish grade level, with temporary cover(s), cast iron frame(s) and cover(s).
- **Contractor's responsibility**
The contractor shall prepare the site for system installation. This includes, but is not limited to, temporary diversion of the water flow, excavation, cutting and removing existing pipe (if applicable), supplying new pipe, bedding, backfill and compacting, etc. The contractor is responsible for providing suitable lifting equipment for unloading the treatment unit from the delivery truck. The contractor shall provide any concrete that may be required for the anti-buoyancy structure, anchors, slabs, ring clamps, etc., and connection adapters required. The contractor is responsible for cutting on site, if required, and installing access chimneys to the finish grade level. The contractor is responsible for watertightness of connections to the treatment unit's inlet and outlet pipes, backfill, labor, tools and any other equipment that may be required.

APPROVAL DOCUMENTS

Shop drawings showing all system dimensions shall be provided to the contractor, who must submit them to the consulting engineer for approval prior to manufacture.

QUALITY ASSURANCE AND INSPECTION

- **Materials**
The quality of materials used, the manufacturing process and the entire unit are subject to inspection by the consulting engineer, at any time during the course of the project (at the manufacturing plant or on site). If it does not meet all the requirements of these specifications, the treatment unit may be refused, even if parts of the unit were previously accepted at the factory. All parts refused after delivery to the site shall be marked for identification purposes and removed from the site. Any damaged part of the unit that cannot be repaired on site shall be refused. If this part is already installed, it must be repaired to the satisfaction of the consulting engineer, if accepted, or completely replaced at the contractor's expense.
- **Inspection**
All parts of the unit shall be inspected for overall appearance, dimensions, general condition, etc.
- **Defects**
Structural defects may be repaired (subject to acceptance by the consulting engineer) after the manufacturer has demonstrated the durability and strength of the proposed repair. The consulting engineer shall carefully verify the quality of the repairs carried out before final acceptance of all the components.

INSPECTION AND MAINTENANCE

The manufacturer shall include in its price a quality assurance program that shall include a series of four (4) inspections at intervals of six (6) months, each accompanied by a detailed report on the nature and quantity of pollutants accumulated in the units, and the recommendations to undertake cleaning when required. In addition, at the end of the work, the manufacturer shall provide, in two (2) copies, an inspection and maintenance manual for each installed equipment.

DATA SHEET

STORM WATER SETTLING AND SEPARATION UNIT

PRODUCT DESCRIPTION : Horizontal structure equipped with internal deflectors and access chimneys for inspection and maintenance

FUNCTION : Separates suspended sediments from storm water and stores floating oils and debris in a cell.

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350

TECHNICAL DATA : Diameter: 750 mm (30 in) to 3,300 mm (132 in)
Size of particles to control: from very fine sand to very coarse sand (75 microns and over)
Length: 6 m (19.68 ft) to 15 m (49.2 ft)
Installation: networked with or parallel to the storm sewer system



STORM WATER SETTLING AND SEPARATION UNIT

AVAILABLE COUPLERS : Adapts to all types of HDPE, PVC, concrete, steel and CSP pipes.

TECHNICAL DATA TABLES

Unit diameter		Length		Nom. dia. entry		Nom. dia. outlet		Nom. dia. access chimney	Treatment rate		Sediment volume		Oil volume		Particle size	
mm	in	m	ft	mm	in	mm	in		l/s	g/s	m ³	ft ³	m ³	ft ³	mm	SIEVE
750	30	6	20	200	8	200	8	750 mm (30 in)	33	8,7	1,5	52	0,4	13,4	0,106	140
750	30	12	40	200	8	200	8		56	14,8	2,7	94,1	1,2	42,3	0,106	140
750	30	6	20	200	8	200	8		15	4	1,5	52	0,4	13,4	0,075	200
750	30	12	40	200	8	200	8		36	9,5	2,7	94,1	1,2	42,3	0,075	200
900	36	6	20	250	10	250	10		42	11,1	2,1	75	0,6	20	0,106	140
900	36	12	40	250	10	250	10		67	17,7	3,8	135,8	1,8	62,2	0,106	140
900	36	6	20	250	10	250	10		20	5,3	2,1	75	0,6	20	0,075	200
900	36	12	40	250	10	250	10		45	11,9	3,8	135,8	1,8	62,2	0,075	200
1050	42	6	20	300	12	300	12		49	13	2,9	101,5	0,8	26,9	0,106	140
1050	42	12	40	300	12	300	12		104	27,5	5,2	183,7	2,4	85,2	0,106	140
1050	42	6	20	300	12	300	12		24	6,3	2,9	101,5	0,8	26,9	0,075	200
1050	42	12	40	300	12	300	12		52	13,7	5,2	183,7	2,4	85,2	0,075	200
1200	48	6	20	300	12	300	12		64	17	4	140	1,1	38	0,106	140
1200	48	12	40	300	12	300	12		112	29,6	7,2	253	3,4	119	0,106	140
1200	48	6	20	300	12	300	12		32	8,5	4	140	1,1	38	0,075	200
1200	48	12	40	300	12	300	12		68	18	7,2	253	3,4	119	0,075	200
1370	54	7,2	24	375	15	375	15		78	20,6	5,6	198	2,7	93,9	0,106	140
1370	54	7,2	24	375	15	375	15		39	10,3	5,6	198	2,7	93,9	0,075	200
1500	60	6	20	375	15	375	15		84	22,2	6,6	234	1,7	59	0,106	140
1500	60	12	40	375	15	375	15		176	46,5	11	389	5,2	185	0,106	140
1500	60	6	20	375	15	375	15		42	11,1	6,6	234	1,7	59	0,075	200
1500	60	12	40	375	15	375	15		88	23,2	11	389	5,2	185	0,075	200

Note : Contact your Soleno representative to obtain the properties of the units with diameters not shown in this table.

SPECIFICATIONS

STORM WATER SETTLING AND SEPARATION UNIT

SCOPE

These specifications apply to the performance, materials and manufacture of Soleno's storm water settling and separation units.

SETTLING UNIT REQUIREMENTS

The storm water treatment unit shall be supplied by Soleno Inc. and shall comply with these materials and performance specifications based on specified flow and storage capacity.

RAW MATERIALS

The storm water settling unit shall be manufactured from high-density polyethylene (HDPE) resin.

- HDPE resin: the treatment unit must be made of high-density resin, with high molecular weight (HMW). Resin shall meet the criteria of standard ASTM D3350-02 with minimum cell classification 345420C.
- **Physical properties of the resin**
 - The filtration system shall be made from high-density polyethylene (HDPE) resin that complies with standards ASTM D3350.
- **Chemical properties of the resin**
 - HDPE resists corrosion, as it is not an electrical conductor.
 - Biological resistance is unaffected by microbial organisms.
 - It can be used in applications with pH values ranging from 1.5 to 14 due to its inert molecular structure.

The storm water treatment unit may be refused if it does not meet the requirements of these specifications.

PERFORMANCE

- The storm water settling unit shall include a _____ mm _____ in interior diameter and _____ mm _____ in long sedimentation chamber for treating the water entering the unit, as well as a _____ mm long oil separation chamber.
- The storm water settling unit shall include a _____ m³ _____ ft³ sediment storage reservoir and may also contain _____ m³ _____ ft³ of hydrocarbons. The storm water treatment unit shall capture the oils, grease, waste and floating debris and 80 % of Total Suspended Solids (TSS) entering the chamber (annually).
- An inlet pipe with deflector shall eliminate water turbulence to allow solid sediments to be deposited at the bottom of the sedimentation chamber to prevent these sediments from being resuspended.
- Size of particles to control: from very fine sand to very coarse sand (100 microns and over)
- Deflectors inside each chamber shall separate and facilitate solid sedimentation and oil floatation in their respective chambers. An outlet section with a 90° elbow shall prevent liquid oils and floating solids from exiting the unit.
- The storm water settling unit shall include two 750 mm (30 in) interior diameter access shafts with triple walls, located at the top of each chamber, to provide safe access for routine inspection and maintenance.

SPECIFICATIONS

STORM WATER SETTLING AND SEPARATION UNIT (CONT'D)

MANUFACTURING SETTLING UNITS

- The storm water settling unit shall be made from HDPE pipes that comply with standards BNQ 3624-120, AASHTO M294 or ASTM F894, based on the diameters required.
- Inlets and outlets are HDPE extrusion-welded inside and outside the structure.
- Deflectors shall be made from HDPE and extrusion welded to the inside of the settling unit.
- HDPE lift eyelets may be attached to the outside of the settling unit so as to prevent undesirable stress on critical elements during loading, unloading and maintenance of the unit.
- The storm water settling unit shall be built in compliance with site requirements, confirmed by shop drawings and, if necessary, approved by the engineer in charge of the project. In a roadway installation, the settling unit shall be equipped with one or more adjustable grate(s) and cast iron cover(s) complying with standard BNQ 1809-300.

Buoyancy calculations (Archimedes' principle) may be provided upon request. The consulting engineer in charge of the project is responsible for final approval of the anti-buoyancy structure. The contractor shall provide the anti-buoyancy structure.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations.

- **Excavation and bedding**
The trench and bedding shall meet the requirements of standard BNQ 1809-300, article 9, Excavation and Backfill. The storm water settling unit shall be installed on stable bedding composed of CG-14 granular material (standard BNQ 2560-114-III, article 6.5), with a minimum depth of 300 mm (12 in), compacted to 95 % modified Proctor density. The contractor is responsible for all safety precautions during settling unit installation.
- **Backfill**
Backfill material shall be composed of CG-14 granular material compacted to 90 % modified Proctor density. The backfill and bedding shall be free of debris. Backfill shall meet the requirements of standard BNQ 1809-300, Article 9, Excavation and Backfill. Backfill shall extend at least 1.0 m (3.3 ft) beyond the perimeter of the unit, from below the bedding to the full height of the unit, and laterally up to the undisturbed part of the soil.
- **Connecting inlet and outlet pipes**
Bell couplings with integrated gaskets shall be installed to connect unit inlets and outlets to the pipes.

SHARED RESPONSIBILITY

- **Soleno's responsibilities**
Soleno shall deliver the storm water settling system to the site. The system includes the settling unit with debris deflector, inlet and outlet connections, 750 mm (30 in) interior diameter access chimneys with smooth exterior walls, up to the finish grade level.
- **Contractor's responsibility**
The contractor shall prepare the site for system installation, including, but is not limited to, temporary diversion of the water flow, excavation, cutting and removing existing pipe, supplying new pipe, bedding, backfill and compacting, etc. The contractor shall also provide suitable lifting equipment for unloading the settling unit from the delivery truck. The contractor shall provide any concrete that may be required for the anti-buoyancy structure, anchors, slabs, collars, etc., and any belts and connection adapters required. The contractor is also responsible for cutting on site, if required, and installing access chimneys to the grade level indicated on approved drawings. The contractor is also responsible for watertightness of connections from the settling unit to the inlet and outlet pipes, backfill, labor, tools and any other equipment that may be required.

APPROVAL DOCUMENTS

Shop drawings showing all system dimensions shall be provided to the contractor, who must submit them to the consulting engineer for approval prior to manufacture.

SPECIFICATIONS

STORM WATER SETTLING AND SEPARATION UNIT (CONT'D)

QUALITY ASSURANCE AND INSPECTION

- **Materials**

The quality of materials used, the manufacturing process and the entire unit are subject to inspection by the consulting engineer, at any time during the course of the project (at the manufacturing plant or on site). If it does not meet the requirements of these specifications, the settling unit may be refused, even if parts of the unit were previously accepted at the factory. All parts refused after delivery to the site shall be marked for identification purposes and removed from the site. Any damaged part of the unit that cannot be repaired on site shall be refused. If this part is already installed, it must be repaired to the satisfaction of the consulting engineer, if accepted, or completely replaced at the contractor's expense.

- **Inspection**

All parts of the unit shall be inspected for overall appearance, dimensions, general condition, etc.

- **Defects**

Structural defects may be repaired (with the approval of the consulting engineer) after the manufacturer has demonstrated the durability and strength of the proposed repair. The consulting engineer shall carefully verify the quality of the repairs carried out before final acceptance of all the components.

DATA SHEET

AQUA-FILTER

PRODUCT DESCRIPTION : Treatment train, including a hydrodynamic separator and a filtration treatment unit.

FUNCTION : System that maximizes the removal of Total Suspended Solids (TSS), nutrients, heavy metals and hydrocarbons from surface runoff before it is conveyed to an outlet. The filtration treatment unit is always preceded by an AQUA-SWIRL hydrodynamic separator.

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350.

FILTER MEDIA : Variable based on the pollutants to be treated
Perlite, Zeolite, activated charcoal, compost and other

TECHNICAL DATA : Hydrodynamic separator: see AQUA-SWIRL data sheet
Filtration treatment unit
Size of particles to control: from clay to medium silt (20 microns and over)
Diameter: 1,800 mm (72 in)
Variable length: see technical data table.
Installation: networked with or parallel to the storm water sewer system
Access chimney diameters : 750 mm (30 in) and 900 mm (36 in)



AQUA-FILTER

AVAILABLE COUPLERS : Adapts to all types of HDPE, PVC, concrete, steel and CSP pipes.

TECHNICAL DATA TABLE

Model	Length of the filtration treatment tank		Approx. total length including the AQUA-SWIRL		Maximum nom. diam.		Filtration rate	
	m	ft	m	ft	mm	in	litres/s	gallons/s
AF-X.1	2.9	9.6	4.9	16	100	4	14	0.5
AF-X.2	3.7	12.0	5.5	18	150	6	28	1.0
AF-X.3	4.4	14.3	6.4	21	200	8	43	1.5
AF-X.4	5.1	16.6	7.3	24	250	10	57	2.0
AF-X.5	5.7	18.7	8.5	28	300	12	71	2.5
AF-X.6	6.4	21.0	9.4	31	300	12	85	3.0
AF-X.7	7.2	23.6	10.4	34	300	12	99	3.5
AF-X.8	7.9	25.9	11.0	36	300	12	113	4.0
AF-X.9	8.6	28.2	11.6	38	300	12	127	4.5
AF-X.10	9.3	30.5	12.2	40	300	12	141	5.0
AF-X.11	10.0	32.8	12.8	42	350	14	155	5.5
AF-X.12	10.9	35.6	13.7	45	350	14	169	6.0
AF-XX	XXX	XXX	XXX	XXX	XXX	XXX	169+	6.0+

Note 1: Values in the table are approximate and may change without notice.
AF-XX: Custom-made.

APPLICATIONS : Control of TSS, nutrients, heavy metals and hydrocarbons

SPECIFICATIONS

AQUA-FILTER

SCOPE

These specifications apply to the performance, materials and manufacture of Aqua-Filter storm water filtration units supplied by Soleno.

FILTRATION UNIT REQUIREMENTS

The Aqua-Filter storm water filtration unit shall be supplied by Soleno Inc. and shall comply with these materials and performance specifications based on specified flow and pollutants to be treated.

RAW MATERIALS

The Aqua-Filter storm water filtration system, including the Aqua-Swirl treatment unit, shall be made from high-density polyethylene (HDPE) resin.

- **Physical properties of the resin**
 - The filtration system shall be made from high-density polyethylene (HDPE) resin that complies with standards ASTM D3350.
- **Chemical properties of the resin**
 - HDPE resists corrosion, as it is not an electrical conductor.
 - Biological resistance is unaffected by microbial organisms.
 - It can be used in applications with pH values ranging from 1.5 to 14 due to its inert molecular structure.

PERFORMANCE

- The Aqua-Filter storm water filtration system must include a ____ mm ____ in interior diameter Aqua-Swirl vortex hydrodynamic separator for pretreating the water entering the system. A tangential inlet pipe shall induce a whirlpool which will deposit solid sediments at the bottom of the treatment chamber and prevent these sediments from being resuspended. An arched deflector in the chamber shall prevent oils, grease, and floating debris from exiting the Aqua-Swirl unit.
- The Aqua-Filter storm water filtration system shall include a ____ mm ____ in interior diameter filtration chamber with a filtering bed at least ____ m ____ ft long, containing ____ filter cells. A distribution system shall equally distribute low flows across the filter cells. A derivative of an overflow spillway shall allow flows greater than the design capacity of the filtering bed to sidestep the bed and exit the filtration chamber. Filter media shall be made from natural materials capable of removing up to 80 % of total suspended solids (TSS), 95 % of total petroleum hydrocarbons (TPH), 80 % of phosphorus and 90 % of zinc, on a net annual basis.
- The Aqua-Filter™ storm water filtration system shall include a ____ m³ ____ ft³ sediment storage reservoir and may also contain ____ litres ____ gallons of oils, grease, and floating debris. The system shall be designed for ____ m³/s ____ ft³/s peak flow and ____ m³/s ____ ft³/s water quality filtration rate. Size of particles to control: from clay to medium silt (20 microns and over).
- The treatment unit shall include between one (1) and four (4) 750 mm (30 in) interior diameter access chimneys, including one with a ladder to provide access to all maintenance zones.

SPECIFICATIONS

AQUA-FILTER (CONT'D)

MANUFACTURING THE TREATMENT Unit

- The storm water treatment unit shall be made from HDPE pipes that comply with standards ASTM F894 or ASTM F714, based on the diameters required.
- The thickness of the plate at the bottom of the treatment unit shall be determined based on standard ASTM F1759 requirements. Calculations shall be provided to substantiate this thickness.
- Inlets and outlets are HDPE extrusion-welded inside and outside the structure.
- The deflector shall be made from HDPE and welded to the inside of the treatment unit with 180-degree couplers at each end.
- HDPE lift eyelets may be attached to the outside of the treatment unit so as to prevent undesirable stress on critical elements during loading, unloading and maintenance of the unit. Lift eyelets are an integral part of the unit and shall be HDPE extrusion-welded.
- The upper section of the treatment unit shall be built in compliance with site requirements and confirmed by shop drawings. When the backfill is deep, the upper section shall be reinforced to resist the loads encountered. In roadway use, the treatment unit shall be covered with a concrete slab to distribute the load over the entire surface of the unit. The cast iron frame and cover must also be able to support these loads. The consulting engineer shall approve the concrete slab design. Buoyancy calculations (Archimedes' principle) may be provided upon request. Standard installation drawings for concrete anti-buoyancy structures may also be provided. The consulting engineer in charge of the project is responsible for final approval of the anti-buoyancy structure. The contractor shall provide the anti-buoyancy structure.

JOINT PERFORMANCE

- The filtration chamber of the Aqua-Filter system shall be made from HDPE pipes that comply with standards ASTM F894.
- Inlets and outlets are HDPE extrusion-welded inside and outside the structure.
- All filtration chamber outlet couplers with diameters exceeding 100 mm (4 in) shall be fusion welded end to end, electrofusion welded or held together with clamps. Threaded transition couplers may also be used for smaller diameter pipes.
- The ends of the filtration chamber shall be built in compliance with site requirements and confirmed by shop drawings. If an air test is required, hermetic covers are required to seal all openings.
- In roadway installations, when required by the consulting engineer, the filtration chamber shall be covered with a concrete slab to distribute the load over the entire unit surface. A cast iron frame and cover able to support these loads shall also be required for each access chimney. The consulting engineer shall approve the concrete slab design. The consulting engineer's calculations shall be included in the bid.
- As needed, the contractor shall provide anti-buoyancy or anti-collapsing anchors with suitable clamps or attachment devices.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations.

■ **Excavation and bedding**

The trench and bedding shall meet the requirements of standard BNQ 1809-300, article 9, Excavation and Backfill. The HDPE Aqua-Filter system shall be installed on stable bedding composed of CG-14 granular material (standard BNQ 2560-114-III, article 6.5), at least 300 mm (12 in) deep, compacted to 95 % modified Proctor density. To ensure adequate support, any additional bedding shall be uniformly compacted under the lower section of the filtration chamber up to the pipe axis. The contractor is responsible for all safety precautions during Aqua-Filter system installation. 28888

SPECIFICATIONS

AQUA-FILTER (CONT'D)

INSTALLATION (CONT.)

- **Backfill**
Backfill material shall be composed of CG-14 granular material, with a minimum depth of 300 mm (12 in), compacted to 90 % modified Proctor density. The backfill and bedding shall be free of debris. Backfill shall meet the requirements of standard BNQ 1809-300, Article 9, Excavation and Backfill. Backfill shall extend at least 1.0 m (3.3 ft) beyond the perimeter of the Aqua-Filter system, from below the bedding to the full height of the system, and extend laterally up to the undisturbed part of the soil.
- **Connecting inlet and outlet pipes**
Flexible couplers with stainless steel tightening collars shall be used to connect treatment unit inlets and outlets to the pipes. When connecting to a Solflo Max pipe, a bell with integrated gasket with clips shall be factory-welded to the filtration and treatment unit.

SHARED RESPONSIBILITY

- **Solenio's responsibilities**
Solenio shall deliver the the Aqua-Filter system to the site. This system includes the Aqua-Swirl unit with all its internal components and surface access chimney, the filtration chamber with all its internal components and its surface access chimneys, filter media, connections between the Aqua-Swirl unit and the filtration chamber, and cast iron frames and covers.
- **Contractor's responsibility**
The contractor shall prepare the site for system installation. This includes, but is not limited to, temporary diversion of the water flow, excavation, cutting and removing existing pipe, supplying new pipe, bedding, backfill and compacting, etc. The contractor is responsible for providing suitable lifting equipment for unloading the system from the delivery truck. The contractor shall provide any concrete that may be required for the anti-buoyancy structure, anchors, slabs, collars, etc., and any belts and connection adapters required. The contractor is responsible for cutting on site, if required, and installing access chimneys to the finish grade level. The contractor is responsible for watertightness of connections to the inlet and outlet pipes of the filtration and treatment unit, backfill, labor, tools and any other equipment that may be required.

APPROVAL DOCUMENTS

Shop drawings showing all system dimensions shall be provided to the contractor, who must submit them to the consulting engineer for approval prior to manufacture.

QUALITY ASSURANCE AND INSPECTION

- **Materials**
The quality of materials used, the manufacturing process and the entire unit are subject to inspection by the consulting engineer, at any time during the course of the project (at the manufacturing plant or on site). If it does not meet all the requirements of these specifications, the treatment unit may be refused, even if parts of the unit were previously accepted at the factory. All parts refused after delivery to the site shall be marked for identification purposes and removed from the site. Any damaged part of the unit that cannot be repaired on site shall be refused. If this part is already installed, it must be repaired to the satisfaction of the consulting engineer, if accepted, or completely replaced at the contractor's expense.
- **Inspection**
All parts of the unit shall be inspected for overall appearance, dimensions, general condition, etc.
- **Defects**
Structural defects may be repaired, subject to acceptance by the consulting engineer, after the manufacturer has demonstrated the durability and strength of the proposed repair. The consulting engineer shall carefully verify the quality of the repairs carried out before final acceptance of all the components.

TECHNICAL DATA SHEET

HYDROSTOR™

PRODUCT DESCRIPTION: Semi-circular retention and infiltration chambers

APPLICATION: Underground storage of runoff water with or without groundwater replenishment. Possible evacuation to an outlet

MANUFACTURING STANDARDS: ASTM F2418, ASTM F2787, CSA B184.2

RAW MATERIAL: ASTM D4101 polypropylene



CHAMBER TECHNICAL DATA TABLE

Features	HS75		HS180	
				
Length	2212 mm	87.1 in	2253 mm	88.7 in
Width	1295 mm	51 in	1976 mm	77.8 in
Height	754 mm	29.7 in	1156 mm	45.5 in
Installed length	2157 mm	84.9 in	2166 mm	85.3 in
Volume	1.31 m ³	46.4 ft ³	3.22 m ³	113.6 ft ³
Installed volume (min.)	2.12 m ³	75 ft ³	5.1 m ³	180 ft ³
Weight	70 lbs	31.8 kg	127 lbs	57.6 kg
Minimum backfill height	457 mm	18 in	597 mm	23.5 in
Maximum backfill height	2.44 m	8 ft	2.44 m	8 ft

END CAP TECHNICAL DATA TABLE

Features	HS75		HS180	
				
Length	559 mm	22 in	665 mm	26.2 in
Width	1227 mm	48.3 in	1941 mm	76.4 in
Height	747 mm	29.4 in	1133 mm	44.6 in
Installed length	518 mm	20.4 in	551 mm	21.7 in
Volume	0.16 m ³	5.8 ft ³	0.54 m ³	19 ft ³
Installed volume (min.)	0.61 m ³	21.5 ft ³	1.36 m ³	48 ft ³
Weight	16 lbs	7.26 kg	65 lbs	29.48 kg

TECHNICAL DATA SHEET

HYDROSTOR™ (cont'd)

STORAGE VOLUMES

	HS75		HS75		HS180		HS180	
								
Bedding								
150 mm (6 in)	2.12 m³	75.0 ft³	0.61 m³	21.5 ft³	Minimum bedding of 229 mm (9 in)			
229 mm (9 in)	2.22 m³	78.4 ft³	0.65 m³	23.0 ft³	5.10 m³	180.0 ft³	1.36 m³	48.0 ft³
300 mm (12 in)	2.31 m³	81.7 ft³	0.68 m³	24.0 ft³	5.23 m³	184.7 ft³	1.41 m³	49.8 ft³
375 mm (15 in)	2.41 m³	85.1 ft³	0.72 m³	25.4 ft³	5.37 m³	189.6 ft³	1.46 m³	51.6 ft³
450 mm (18 in)	2.50 m³	88.4 ft³	0.75 m³	26.5 ft³	5.51 m³	194.6 ft³	1.51 m³	53.3 ft³

MINIMUM EXCAVATION VOLUMES AND WEIGHTS

	HS75		HS75		HS180		HS180	
								
Bedding								
150 mm (6 in)	4.2 m ³	148.3 ft ³	1.5 m ³	53.0 ft ³	Minimum bedding of 229 mm (9 in)			
229 mm (9 in)	4.5 m ³	158.9 ft ³	1.6 m ³	56.5 ft ³	9.1 m ³	321.4 ft ³	2.6 m ³	92.2 ft ³
300 mm (12 in)	4.7 m ³	166.0 ft ³	1.7 m ³	60.0 ft ³	9.4 m ³	332.0 ft ³	2.7 m ³	96.4 ft ³
375 mm (15 in)	5.0 m ³	176.6 ft ³	1.8 m ³	63.6 ft ³	9.8 m ³	346.1 ft ³	2.8 m ³	100.5 ft ³
450 mm (18 in)	5.2 m ³	183.6 ft ³	1.9 m ³	67.1 ft ³	10.1 m ³	356.7 ft ³	2.9 m ³	104.7 ft ³

Based on a minimum backfill of 450 mm (18 in) for HS75 chambers and 600 mm (24 in) for HS180 chambers, spacing between chambers of 150 mm (6 in) for HS75 chambers and 200 mm (8 inches) for HS180 chambers and a spacing of 300 mm (12 in) for HS75 chambers and 150 mm (6 in) for HS180 between the end caps and the trench. Voids in the stone: 40%. If the backfill is bigger, the volume of excavation will have to be adjusted as a consequence.

TECHNICAL DATA SHEET

HYDROSTOR™

MINIMUM VOLUMES OF CLEAN STONE

	HS75		HS75		HS180		HS180	
								
Bedding								
150 mm (6 in)	2.0 m³	3.4 m.t.	1.0 m³	1.7 m.t.	Minimum bedding of 229 mm (9 in)			
229 mm (9 in)	2.3 m³	3.9 m.t.	1.1 m³	1.9 m.t.	4.5 m³	7.5 m.t.	2.1 m³	3.5 m.t.
300 mm (12 in)	2.5 m³	4.2 m.t.	1.2 m³	2.0 m.t.	4.9 m³	8.2 m.t.	2.2 m³	3.7 m.t.
375 mm (15 in)	2.8 m³	4.7 m.t.	1.3 m³	2.2 m.t.	5.2 m³	8.7 m.t.	2.3 m³	3.9 m.t.
450 mm (18 in)	3.0 m³	5.0 m.t.	1.3 m³	2.4 m.t.	5.6 m³	9.4 m.t.	2.4 m³	4.1 m.t.

Based on a minimum backfill with clean stone of 150 mm (6 in) for HS75 chambers and 292 mm (11.5 in) for HS180 chambers, spacing between chambers of 150 mm (6 in) for HS75 chambers and 200 mm (8 in) for HS180 chambers and a spacing of 300 mm (12 in) for HS75 chambers and 150 mm (6 in) for HS180 chambers between the end caps and the trench. Voids in the stone: 40%

APPLICATIONS: Retention system
Detention system

OPTIONS: An available pretreatment system is recommended

Access port
Diffuser
Collector
Flow regulators
Control and/or bypass manholes

SPECIFICATIONS

HYDROSTOR™

SCOPE

This specification applies to all HydroStor™ chambers supplied by Soleno. These chambers can be used to recharge the water table or to temporarily retain water and return it to the storm sewer when the rain event is over.

REQUIREMENTS FOR CHAMBERS

HydroStor™ chambers form a continuous arch with an open bottom, which helps maximize water infiltration. Each chamber shall feature four lifting handles, making it easy to move and install.

- Each HS75 HydroStor™ chamber shall have 14 annular exterior corrugations.
- Each HS180 HydroStor™ chamber shall have 8 annular exterior corrugations.

End caps shall be curved and have adequate structural capacity to allow traffic overload under CSA CL-625 and AASHTO H-20 standards. End cap may be cut at any invert elevation to receive a pipe at any elevation.

Chambers and end caps shall meet or exceed the following standards : CSA B184.2, ASTM F2418 and ASTM F2787.

RAW MATERIAL

Chambers shall be injection molded and constructed of polypropylene resin resistant to environmental stress cracking (ESCR) and with ability to maintain adequate stiffness through the construction and service life of the chamber. The end caps shall be thermoformed or injection molded from polyethylene resin. The chamber material shall meet or exceed the requirements of designation PP0330B99945 as defined in ASTM D4101.

The end caps shall be made of a polypropylene resin or polyethylene and molded by thermoforming or by injection.

JOINT PERFORMANCE

- Joining of chambers shall be accomplished by overlapping the corrugations of longitudinally adjacent chambers along lengths of chambers. Chambers and joints shall be installed in the direction stamped in the valley of the corrugation.
- End caps are designed to fit over top of a corrugation on either end of the chamber.

CHAMBER STORAGE CAPACITIES

Each HS180 chamber shall contain a minimum of 1.43 m³ of water per linear meter (15 ft³ per linear foot), not including the amount of water contained in the clean stone. Each HS75 chamber must contain a minimum of 0.59 m³ of water per linear meter (6.4 ft³ per linear foot) not including the amount of water contained in the clean stone.

STRUCTURAL CAPACITY OF CHAMBERS

The system must be able to withstand a traffic load under CSA CL-625 or AASHTO H-20.

- For chamber HS180, the minimum cover height shall be 595 mm (23.5 in) (measured from the top of the chamber to the bottom of the pavement layer) and the maximum height shall be 2400 mm (8 ft) (measured from the top of the chamber to the top the pavement layer).
- For chamber HS75, the minimum cover height shall be 450 mm (18 in) (measured from the top of the chamber to the bottom of the pavement layer) and the maximum height shall be 2400 mm (8 ft) (measured from the top of the chamber to the top the pavement layer).

RECOVERY OF SEDIMENT, OILS, GREASES AND FLOATING DEBRIS

A HydroStor™ pretreatment system shall be designed to remove the sediment, oils and floating debris upstream from the chamber system inlet. It shall include a diffuser with baffle, an elbow at each outlet and two smooth outer wall access shafts 750 mm (30 in) in diameter. Maintenance of the system will not require any work in confined spaces. Each access shaft will be provided with a lid and adjustable cast iron frame. If required, a diffuser shall be installed at the end of the pretreatment system to reach the additional rows of chambers. The system will be sized according to the expected inflow.

STABILIZATION GEOGRID

A HydroStor™ geogrid used to distribute the load at the bottom of the chambers and to prevent the stone displacement caused by the water flow shall be installed underneath all chambers.

SEPARATION GEOTEXTILE

A Soleno TX-90 geotextile compliant with BNQ G.C.T.T.G. 3001 shall be installed on the perimeter of the excavation and on top of the system in order to isolate clean stone from the existing soil and/or road or parking foundation.

SPECIFICATIONS

HYDROSTOR™ (CONT'D)

INSPECTION AND MAINTENANCE

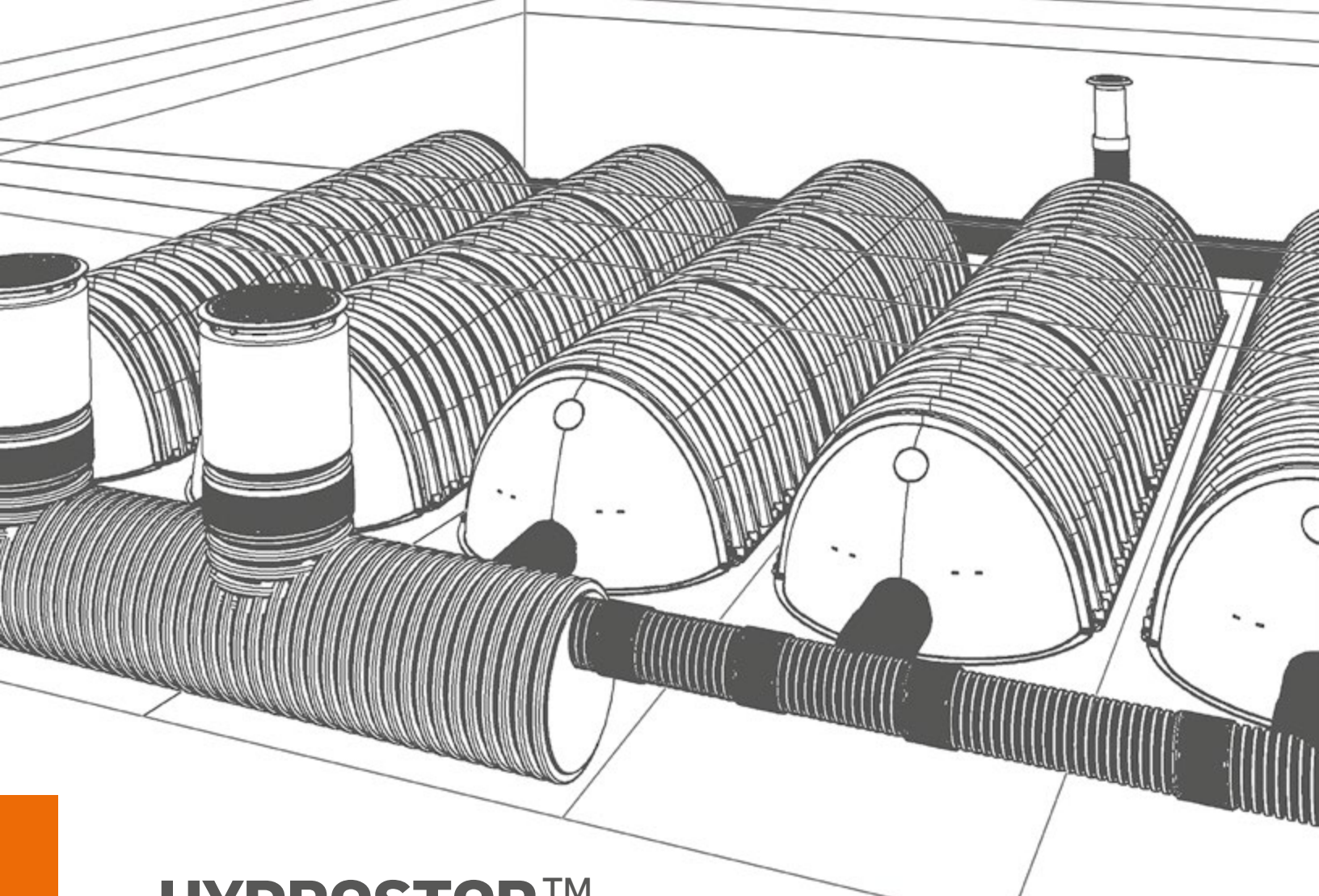
The manufacturer shall include in its price a quality assurance program that shall include a series of four (4) inspections at intervals of six (6) months, each accompanied by a detailed report on the nature and quantity of pollutants accumulated in the units and the recommendations to undertake cleaning when required. In addition, at the end of the work, the manufacturer shall provide, in two (2) copies, an inspection and maintenance manual for each installed equipment.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall be consistent as shown on the bid documentation and plans.

INSTALLATION

The installation shall be performed in accordance with the Soleno Installation Guide and must contain all the components specified in the plans, specifications and instructions provided by Soleno. Contact your local Soleno representative or visit our website at www.solenio.com to view installation recommendations.



HYDROSTOR™

INSTALLATION GUIDE

INSTALLATION INSTRUCTIONS FOR THE HYDROSTOR SYSTEM

STEPS

1. Before undertaking the work
2. Receiving the material
3. Handling and storage
4. Preparing the trench
5. Placing the geotextile
6. Preparing the bedding
7. Installing the pretreatment system (if required)
8. Installing the geogrid
9. Handling the chambers
10. Connecting the end caps
11. Installing the chambers
12. Installing the access ports (if required)
13. Backfilling
14. Initial backfill
15. Final backfill

STEP 1

BEFORE UNDERTAKING THE WORK

In case of discrepancy between the instructions contained in this guide and those contained in the plans and specifications, please contact your Soleno representative.

Contact your Soleno representative at least 48 hours before work begins. A visit from your authorized Soleno representative is recommended after receipt of the materials on site or before work begins.

STEP 2

RECEIVING THE MATERIAL

Upon receipt of the materials, ensure that all items required on the shop drawing are delivered and in good condition. Please notify immediately your Soleno representative in case of damage.



CHAMBER



END CAP



PRETREATMENT
HYDROSTOR
(unique to our system)



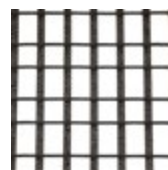
DIFFUSER



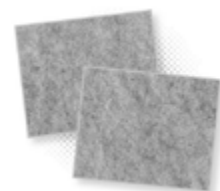
COLLECTOR



ACCESS PORT



GEOGRID
HYDROSTOR



GEOTEXTILE TX-90

STEP 3

HANDLING AND STORAGE

Unloading of the chambers must be done with a lift truck fitted with forks measuring a minimum of 1.8 m (6 ft) in length.

We recommend that you keep the chambers and end caps on their respective pallets until use.

Chamber	Number of chambers per pallet	Weight of a full pallet
HS75	32	1134 kg (2500 lbs)
HS180	17	1089 kg (2400 lbs)

STEP 4

PREPARING THE TRENCH

- Proceed with the basin localization and excavation according to the approved shop drawings and engineering plans. Make sure the bottom of the excavation is dry before proceeding further.
- Level the bottom of the basin.
If required, unroll the drain on the downstream side, on the clean stone.
An spacing of 150 mm (6 in) is required between the drain and the chambers.



STEP 5

PLACING THE GEOTEXTILE

Install the geotextile TX-90* vertically on the walls and add 300 mm (12 in) at the bottom of the excavation (do not install a geotextile at the bottom of the excavation).
Provide an overlap of 300 mm (12 in).
Provide a surplus to fold back over the stone at the end of the work.



*The warranty applies only if all the components specified in the HYDROSTOR system are installed. No substitute part will be accepted.

STEP 6

PREPARING THE BEDDING

Level and compact the washed clean stone at the bottom of the basin with a vibrating plate. The stone must be 20 mm in size (3/4 inch net), crushed and washed. Refer to the shop drawing and engineering plan for the required thickness of the stone layer.

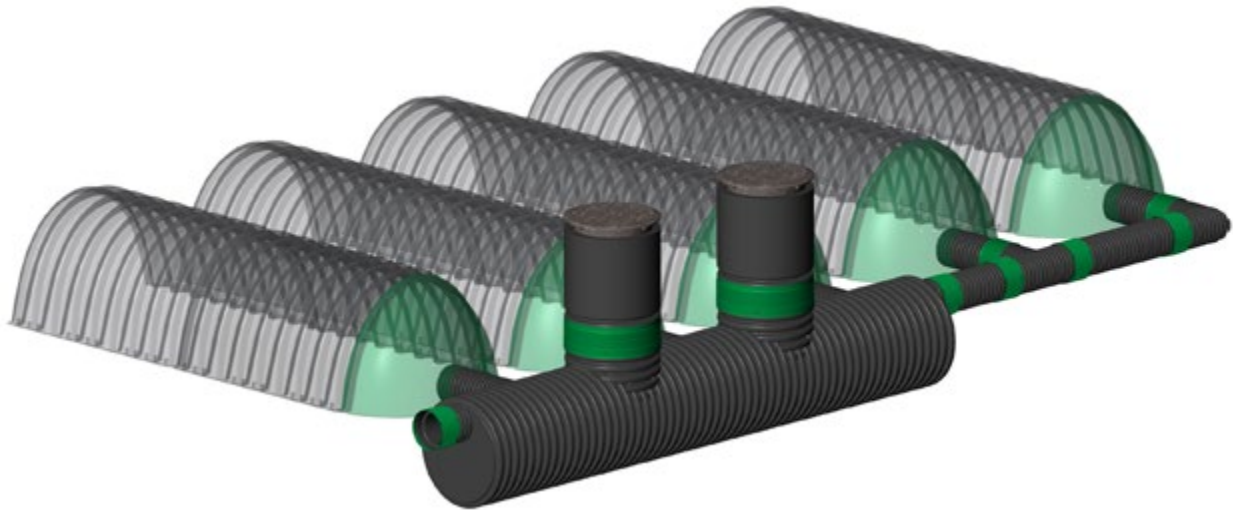
Chamber	Minimum bedding thickness
HS75	150 mm (6 in)
HS180	229 mm (9 in)



STEP 7

INSTALLING A PRETREATMENT SYSTEM (IF REQUIRED)

Install the HYDROSTOR pretreatment system by connecting the inlet pipe and the outlet diffuser, as appropriate.



STEP 8

INSTALLING THE GEOGRID

Unroll the HYDROSTOR stabilization geogrid on clean stone, transversely to the chambers, providing an overlap of 300 mm (12 in).



*The warranty applies only if all the components specified in the HYDROSTOR system are installed. No substitute part will be accepted.

STEP 9

HANDLING THE CHAMBERS

To remove the chambers from the pallet, carefully cut the retaining straps around the chambers. Use the side handles to lift and remove the chambers. Two persons (one on each side) are required for this step.

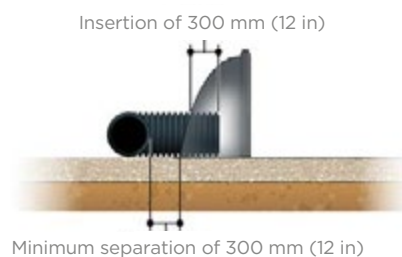


WARNING : The chambers must never be moved by tipping them. This approach could cause product breakage, personal injury or even death.

STEP 10

CONNECTING THE END CAPS

- Align the end cap of each row with the corresponding inlet pipe, taking care to position the parts at the right elevations.
- Drill the end cap at the required diameter and location, and then insert the pipe according to the figure below:



Provide a geotextile to prevent stone infiltration.

STEP 11

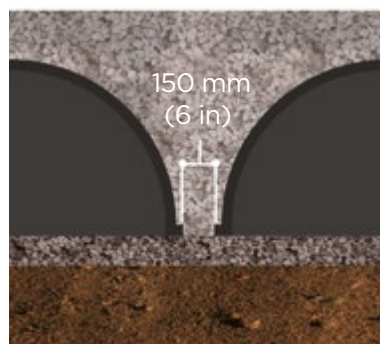
INSTALLING THE CHAMBERS

Installation of the chambers must always start from the UPSTREAM side.
Orient chambers as per arrows.

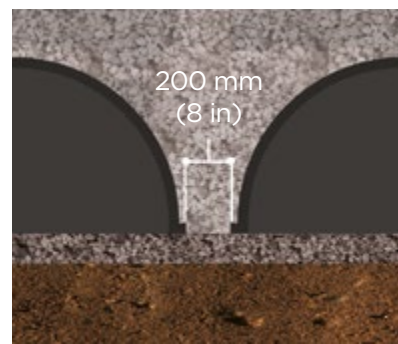


- Install the end cap on the first chamber ring.
- Secure the end caps to the chambers with four evenly-spaced screws.
- Observe the required spacings shown in the table below, throughout the installation process.

Chamber	Distance between chambers	Distance between chambers, end caps and excavation
HS75	150 mm (6 in)	300 mm (12 in)
HS180	200 mm (8 in)	



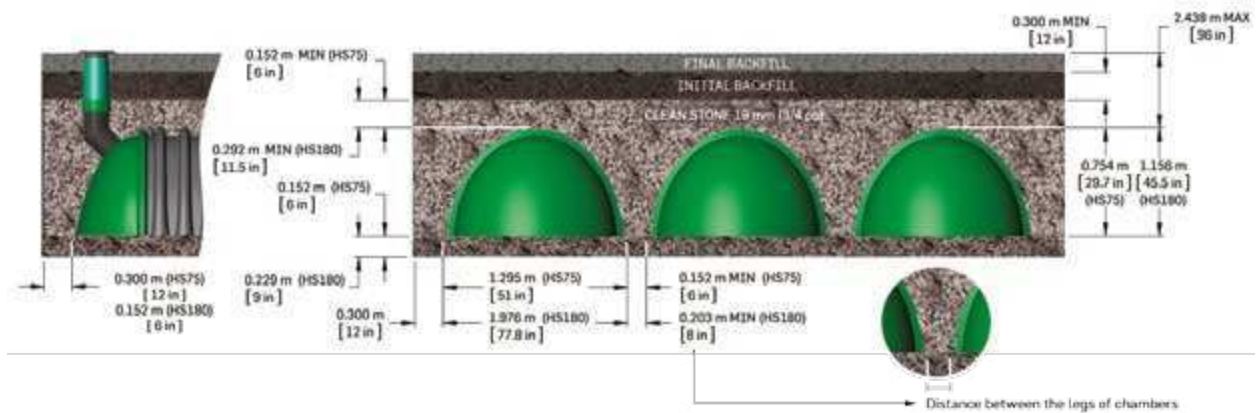
HS75



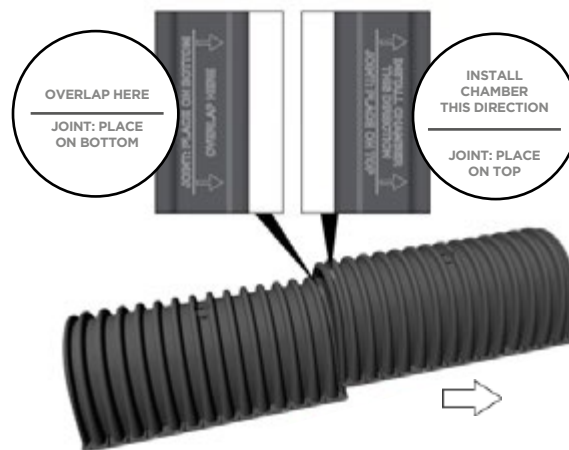
HS180

STEP 11

INSTALLING THE CHAMBERS (CONT'D)



Spacers are recommended to maintain the spacing during installation. Continue with the chamber installation, always observing the direction of the arrows. Follow the overlap indicated at the chamber extremities.

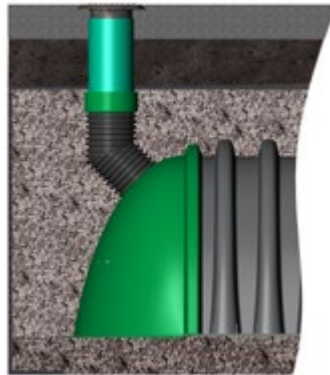


Use end caps to close each row, following the instructions at the beginning of this step. When required, install the outlet pipe or collector.

STEP 12

INSTALLING THE ACCESS PORTS (IF REQUIRED)

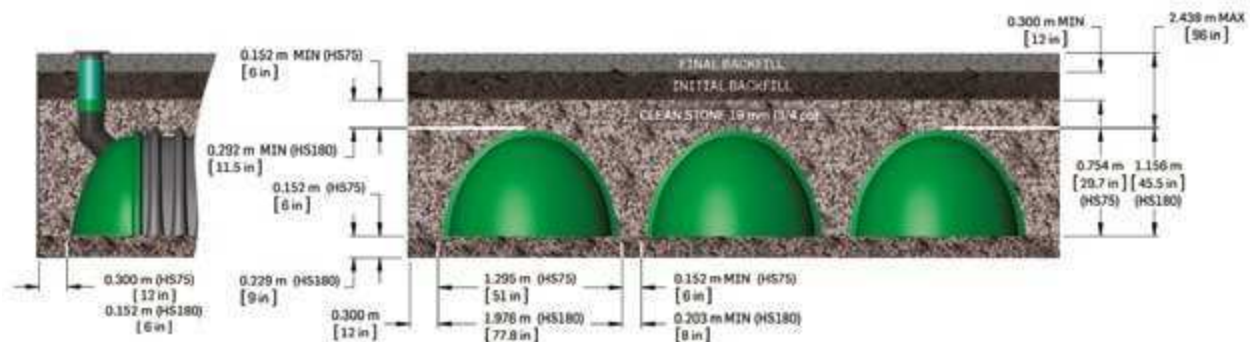
These access ports are recommended for inspection and maintenance, when no pretreatment system is to be installed.



STEP 13

BACKFILLING

Backfill is critical to ensure the sustainability of the HYDROSTOR system.



STEP 13

BACKFILLING (CONT'D)

The stone used for backfilling the chambers and their periphery should be 20 mm in size (3/4 in net), crushed and washed. **No equipment is to circulate above the chambers before reaching the minimum height of initial backfill (as shown in the previous figure).**

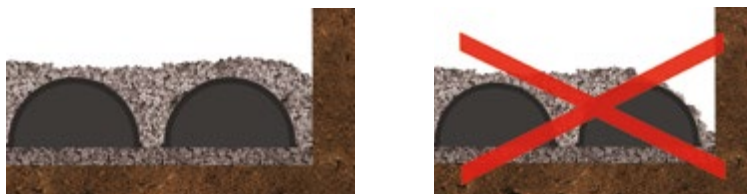
An excavator or stone conveyor system can be used to deposit the stone on the chambers.



- Carefully deposit the stone on top of the chambers, in such a way as to prevent displacement of the chambers and to ensure adequate spacing.
- Backfill the pre-treatment system, the diffuser and the collector in the same way.
- During backfilling, ensure that the stone height difference on each side of the chamber does not exceed 300 mm (12 in).



Complete the backfilling until a minimum of 150 mm (6 in) of stone covers HS 75 chambers and 290 mm (11.5 in) for the HS 180 chambers.



STEP 13

BACKFILLING (CONT'D)

Complete backfilling in the trench periphery, following the same method.
Cover the top of the clean stone completely with the geotextile, providing a overlap of 300 mm (12 in).



STEP 14

INITIAL BACKFILL

Install a layer of compactable granular material at a minimum thickness of 300 mm (12 in), compacted at 90% Modified Proctor. The compaction equipment must circulate parallel to the chambers.

STEP 15

FINAL BACKFILL

i) UNDER ROADWAY (with pavement)

Use the material specified and backfill to required elevations.

Once the paving is completed, road traffic (loads standards CL-625 or H20) is allowed.

The maximum height between the top of the chambers and the top of the pavement should not exceed 2400 mm (8 ft).

ii) OFF ROADWAY (without pavement)

Use the material specified and backfill to required elevations.

A minimum of 150 mm (6 in) is required before circulation can be allowed.

The maximum height between the top of the chambers and the top of the pavement should not exceed 2400 mm (8 ft).

DATA SHEET

STORMCHAMBER®

PRODUCT DESCRIPTION : Semi-circular retention and infiltration chamber

FUNCTION : Subsurface storage of surface runoff with or without ground water replenishment and possible conveyance to an outlet

RAW MATERIALS : Made from high-density polyethylene (HDPE) ASTM D3350



STORMCHAMBER

CHAMBER TECHNICAL DATA TABLE

Features	StormChamber	
Width	1524 mm	60 in
Height	864 mm	34 in
Length (intermediate chamber)	2489 mm	98 in
Installed length (intermediate chamber)	2311 mm	91 in
Volume	2.12 m ³	75 ft ³
Installed volume (min.)	3.27 m ³	115 ft ³
Weight	50 kg	110 lb
Minimum backfill height	450 mm	18 in
Maximum backfill height	4.88 m	16 ft

SYSTEM TECHNICAL DATA TABLE

Bedding		Storage volume		Min. excavation volume		Min. volume of clean stone	
mm	in	m ³	ft ³	m ³	ft ³	m ³	ft ³
150	6	3.27	115	5.93	210	2.60	92
225	9	3.40	120	6.24	220	2.90	102
300	12	3.53	125	6.54	231	3.21	113
375	15	3.66	129	6.85	242	3.51	124
450	18	3.79	134	7.15	253	3.81	135

Based on a minimum backfill of 450 mm (18 in), spacing between chambers of 230 mm (9 in) and spacing on the perimeter of 300 mm (12 in). Voids in the stone: 40%.

APPLICATIONS : Detention system
Retention system

OPTIONS : An available pretreatment system is recommended
Maintenance and observation shaft
Flow regulator
Sediment trap
Multi-stage installation

SPECIFICATIONS

STORMCHAMBER® (CONT.)

SCOPE

These specifications apply to all StormChamber retention chambers supplied by Soleno.

CHAMBER REQUIREMENTS

To prevent any points of failure, the chamber shall be manufactured from a single part. The chamber shall be 864 mm (2.8 ft) high by 1,524 mm (5.0 ft) wide by 2,616 mm (8.6 ft) long. The useful length of the end chambers shall be 2,464 mm (8.1 ft) and that of the intermediate chambers shall be 2,311 mm (7.6 ft). Each chamber shall include 14 corrugations and a deflector wall at least 300 mm (12 in) high to reduce water turbulence.

Each chamber shall include a horizontal receptacle making it possible to add a 250 mm (10 in) diameter PVC DR35 cleaning shaft. Each chamber shall also include a vertical receptacle making it possible to connect Solflo Max, 320 kPa 200 mm (8 in) diameter interconnection or water intake pipes.

RAW MATERIALS

Chambers shall be made from dual wall (orange outer, black inner) high-density polyethylene resin.

PERFORATIONS

Perforation diameter shall be 12.7 mm (0.5 in). Each ring shall have 4 perforations (2 per side), except for the first interlocked ring. Each chamber shall therefore have 48 perforations.

CHAMBER STORAGE CAPACITY

Each chamber shall hold at least 2.12 m³ (74.8 ft³) of water, not including the water held in the clean stone.

CHAMBER STRUCTURAL CAPACITY

The system shall be able to withstand double the CL-625 highway load, for backfill depths from 450 mm (18 in) to 9.14 m (30 ft).

SEDIMENT RECOVERY

The inlet row shall have at least 2 sediment recovery SedimenTraps made from high-density polyethylene with 250 mm (10 in) diameter PVC DR35 vertical cleaning shafts. Each cleaning shaft shall be equipped with an adjustable cast iron frame and cover with identification.

GEOGRID

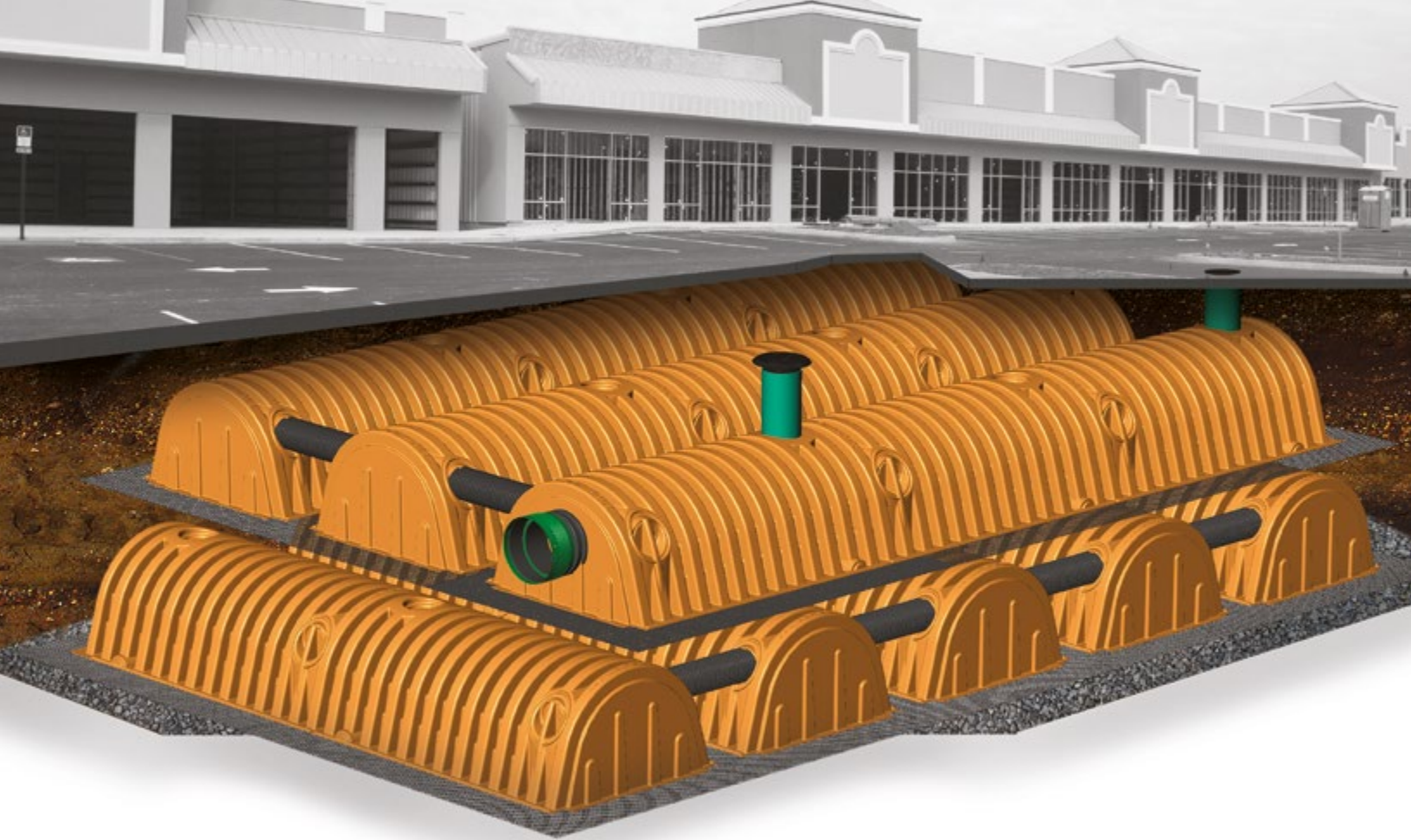
A geogrid shall be installed under all the chambers. To prevent any rock movement, an additional heavy-duty geogrid shall be installed under every chamber that has an external water uptake. The same geogrid shall be installed under every cleaning shaft, to prevent stones from being sucked into the cleaning hose.

GEOTEXTILES

A Soleno TX-90 BNQ G.C.T.T.G. 3001 geotextile shall be installed on the excavation perimeter and the top of the system, to insulate the clean stone of the in situ soil and/or road/parking lot foundation.

INSTALLATION

Installation shall be carried out in compliance with manufacturer recommendations. Contact your local Soleno representative or visit our website at www.soleno.com for installation recommendations.



STORMCHAMBER® RETENTION CHAMBERS

INSTALLATION GUIDE

INSTALLATION INSTRUCTIONS FOR STORMCHAMBER® RETENTION CHAMBERS

STEPS

1. Before undertaking the work
2. Inspection of the materials
3. Basin preparation
4. Geotextile installation
5. Excavation
6. Bedding
7. Installation of the geogrid
8. Installation of high-strength geogrid
9. Chamber preparation
10. Chamber installation
11. Clearances
12. Stabilization
13. Cutting
14. Hole cutting
15. Installing the maintenance shaft
16. Backfilling
17. Coverage
18. Backfilling
19. Finishing

STEP 1

BEFORE UNDERTAKING THE WORK

In case of discrepancy between the instructions contained in this guide and those contained in the plans and specifications, please contact your Soleno representative.

Contact your Soleno representative at least 48 hours before work begins. A visit from your authorized Soleno representative is recommended after receipt of the materials on site or before work begins.

STEP 2

INSPECTION OF THE MATERIALS

Upon receipt of the materials, ensure that all items specified in the shop drawing are delivered.



CHAMBER



GEOGRIDS

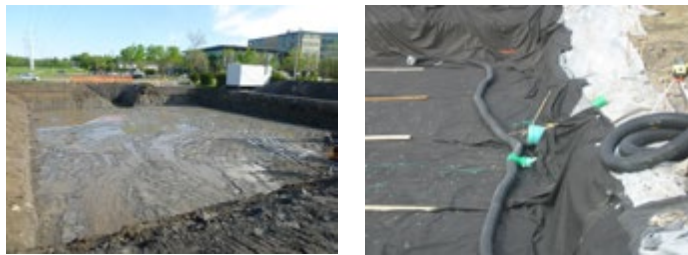


CAST IRON FRAMES AND
COVERS - 250 mm (10 in)

STEP 3

BASIN PREPARATION

- Proceed with the installation and excavation of the basin.
- Level the bottom of the basin.
- Unroll the drain on the downstream side over the clean stone (if required).



STEP 4

GEOTEXTILE INSTALLATION

- Install the TX-90* geotextile vertically on the walls and add 300 mm (12 in) to cover the bottom of the excavation.
- Provide an overlap of 300 mm (12 in) at the joints.
- Provide a surplus to fold back over the riprap at the end of the work.



* The warranty applies only if all the components specified in the Stormchamber system are installed. No substitute part will be accepted.

STEP 5

EXCAVATION

Excavate the exact location of the sediment collectors (if required in the plan) providing 150 mm (6 in) of clean stone underneath and around the collector. The top of the collector will be level with the chamber invert.



STEP 6

BEDDING

Level and compact 150 mm (6 in) (or specified thickness) of 20 mm washed clean stone (0.75 in) at the bottom of the basin.



STEP 7

INSTALLATION OF THE GEOGRID

Unroll the stabilization geogrid* on clean stone, transversely to the chambers, providing an overlap of 300 mm (12 in).



* The warranty applies only if all the components specified in the Stormchamber system are installed. No substitute part will be accepted.

STEP 8

INSTALLATION OF HIGH-STRENGTH GEOGRID

- Lay the high-strength geogrid underneath the inlet chambers and the chambers with a sediment collector.
- Cut the geogrids above the sediment collectors (temporarily cover the hole to prevent any risk of falling matter).



STEP 9

CHAMBER PREPARATION

- An iron bar or a piece of wood can be used to separate the stacked chambers on the pallets.
- Remove the 100 mm (4 in) small strip on the intermediate and end-of-line chambers.



STEP 10

CHAMBER INSTALLATION

Start the installation of Stormchamber chambers from the upstream side (the starting chambers are labeled “START”).



STEP 11

CLEARANCES

Provide a gap of 300 mm (12 in) on the perimeter of the chambers and 230 mm (9 in) between rows (measured at the base, at the chamber ends).



STEP 12

STABILIZATION

Stabilize the chambers by screwing together the base of two joined chambers.



STEP 13

CUTTING

Cut out openings at the chamber ends (full wall), taking into account the planned elevations, in order to connect the inlet and outlet pipes. Provide a geotextile at the openings to prevent the infiltration of clean stone.



STEP 14

HOLE CUTTING

Where indicated in the shop drawing, cut the lateral receptacles and insert a 200 mm (8 in) HDPE Solflomax pipe at a length of 1500 mm (60 in) between two rows of chambers. Provide a geotextile at the openings to prevent the infiltration of clean stone.



STEP 15

INSTALLING THE MAINTENANCE WELL

Above the sediment collectors, cut the top receptacle. Insert the bell outlet part of a 250 mm (10 in) PVC pipe or 250 mm (10 in) PVC adapter with integrated gasket. Insert the 250 mm (10 in) PVC pipe in the bell.



STEP 16

BACKFILLING

Backfill with 20 mm (0.75 in) washed clean stone around and between the rows of chambers and add 150 mm (6 in) (or specified thickness) on top of the chambers. Begin by filling the upper and central part of the chambers to prevent their displacement.



STEP 17

COVERAGE

Completely cover the top of the clean stone with the geotextile, providing a overlap of 300 mm (12 in).



STEP 18

BACKFILLING

Proceed to backfill to a minimum of 300 mm (12 in) of compactable granular material.



STEP 19

FINISHING

Once the final elevation is achieved, install the adjustable frames and lids on the maintenance shafts.



DATA SHEET

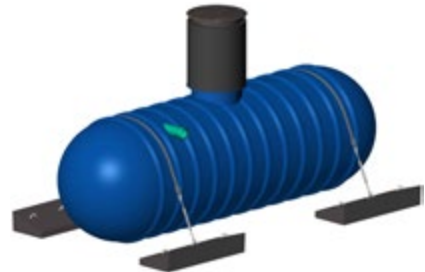
NEMO TANK

PRODUCT DESCRIPTION: Tubular retention tank

FUNCTION: Underground storage of rainwater

MANUFACTURING STANDARDS: CSA B66, ANSI/AWWA D20

RAW MATERIAL: Fiberglass



NEMO TANK 1800 mm x 3467 mm long, complete with manhole 750 mm in diameter, 150-mm PVC outlet, straps, turnbuckles and concrete blocks with anchors

TECHNICAL DATA TABLE TANK 1800 mm (6 ft) IN DIAMETER

Volume		Length		Weight		Anchoring (straps)
Litres	US Gallons	mm	ft	kg	lbs	
7600	2000	3467	11.37	318	700	2
9500	2500	4216	13.83	390	860	2
11500	3000	5080	16.66	463	1020	2
13000	3500	5664	18.58	540	1190	2
15000	4000	6502	21.33	617	1360	3
17000	4500	7112	23.33	660	1455	3
19000	5000	7823	25.67	703	1550	3
21000	5500	8560	28.08	771	1700	4
23000	6000	9271	30.42	835	1840	4

TECHNICAL DATA TABLE TANK 2400 mm (8 ft) IN DIAMETER

Volume		Length		Weight		Anchoring (straps)
Litres	US Gallons	mm	ft	kg	lbs	
18731	4948	5076	16.65	753	1661	2
20538	5425	5495	18.03	810	1787	2
22345	5903	5914	19.40	868	1914	2
24152	6380	6333	20.78	926	2041	2
25959	6858	6752	22.15	983	2168	4
27767	7335	7171	23.53	1041	2294	4
29574	7812	7590	24.90	1098	2421	4
31381	8290	8009	26.28	1156	2548	4
33188	8767	8429	27.65	1213	2675	4
34995	9245	8848	29.03	1271	2801	4
36802	9722	9267	30.41	1328	2928	4
38610	10200	9686	31.78	1386	3055	4
40417	10677	10105	33.15	1443	3182	4
42224	11154	10524	34.53	1501	3308	4

DATA SHEET

NEMO TANK (CONT'D)

TECHNICAL DATA TABLE TANK 2400 mm (8 ft) DIAMETER (CONT'D)

Volume		Length		Weight		Anchoring (straps)
Litres	US Gallons	mm	ft	kg	lbs	
44031	11632	10943	35.90	1558	3435	4
45838	12109	11362	37.28	1616	3562	4
47646	12587	11781	38.65	1673	3689	6
50083	13230	12340	40.49	1775	3914	6
51890	13708	12759	41.86	1833	4040	6
53697	14185	13178	43.23	1890	4167	6
55504	14663	13597	44.61	1948	4294	6
57311	15140	14016	45.98	2005	4421	6
59119	15617	14435	47.36	2063	4547	6
60926	16095	14855	48.74	2120	4674	6
62733	16572	15274	50.11	2178	4801	8
65170	17216	15833	51.95	2280	5026	8
66977	17693	16252	53.32	2337	5153	8
68784	18171	16671	54.69	2395	5279	8
70591	18648	17090	56.07	2452	5406	8
72399	19126	17509	57.44	2510	5533	8
74206	19603	17928	58.82	2567	5660	8
76013	20080	18347	60.19	2625	5786	8

TECHNICAL DATA TABLE TANK 3000 mm (10 ft) IN DIAMETER

Volume		Length		Weight		Anchoring (straps)
Litres	US Gallons	mm	ft	kg	lbs	
39133	10338	6370	20.90	1283	2829	3
42210	11151	6790	22.28	1367	3013	3
45228	11948	7209	23.65	1451	3198	4
48364	12776	7628	25.03	1534	3382	4
51442	13589	8047	26.40	1618	3566	4
54519	14402	8466	27.78	1701	3751	4
57596	15215	8885	29.15	1785	3935	4
60673	16028	9304	30.52	1869	4119	5
63750	16841	9724	31.90	1952	4304	5
66828	17654	10142	33.28	2036	4488	5
69905	18467	10562	34.65	2119	4672	5
72982	19280	10981	36.03	2203	4856	6
76059	20093	11400	37.40	2287	5041	6
79136	20905	11819	38.78	2370	5225	6
82214	21718	12238	40.15	2454	5409	6
85847	22678	12733	41.77	2614	5762	6

DATA SHEET

NEMO TANK (CONT'D)

TECHNICAL DATA TABLE TANK 3000 mm (10 ft) DIAMETER (CONT'D)

Volume		Length		Weight		Anchoring (straps)
Litres	US Gallons	mm	ft	kg	lbs	
88924	23491	13152	43.15	2697	5946	7
92001	24304	13571	44.52	2781	6163	7
95079	25117	13990	45.90	2865	6315	7
98156	25930	14410	47.28	2948	6499	7
101233	26743	14829	48.65	3032	6683	8
104310	27556	15248	50.03	3115	6868	8
107387	28369	15667	51.40	3199	7052	8
110465	29181	16086	52.78	3283	7236	8
113542	29994	16505	54.15	3366	7421	8
117175	30954	17000	55.77	3526	7773	8
120252	31767	17419	57.15	3610	7957	9
123330	32580	17839	58.53	3683	8142	9
126407	33393	18258	59.90	3777	8326	9
129484	34206	18677	61.28	3860	8510	9
132561	35019	19096	62.65	3944	8695	9
135638	35832	19515	64.02	4028	8879	9
138716	36645	19934	65.40	4111	9063	9
141793	37457	20353	73.34	4195	9248	9
144603	38200	20772	68.15	4278	9432	10
148503	39230	21268	69.78	4438	9784	10
151301	39969	21687	71.15	4522	9969	10
157735	41669	22525	73.90	4689	10337	10

BACKFILL HEIGHTS

Features	Roadway		Non-Roadway		Maximum burial depth	
	mm	in	mm	in	mm	in
1800 mm (6 ft)	1200	48	300	12	2100	82
2400 mm (8 ft)	1200	48	300	12	2100	82
3000 mm (10 ft)	1200	48	300	12	2100	82

When a concrete slab must be installed above the system, it should at least 300 mm (12 in) in thickness.

DATA SHEET

NEMO TANK (CONT'D)

APPLICATIONS: Detention system
Pumping station

OPTIONAL ACCESSORIES: Vent 100 mm (4 in)
Manhole with Nemo adapter 750 mm (30 in)
Outlet 100 mm (4 in)
Outlet 150 mm (6 in)
Outlet 200 mm (8 in)
Outlet 250 mm (10 in)
Outlet 300 mm (12 in)
Nemo strap/turnbuckle 1800 mm (6 ft)
Nemo strap/turnbuckle 2400 mm (8 ft)
Nemo strap/turnbuckle 3000 mm (10 ft)
Concrete block anchors 544 kg (1200 lbs)
Adjustable frame and cover

SPECIFICATIONS

NEMO TANK

SCOPE

These specifications apply to all NEMO tanks supplied by Soleno. These tanks can be used to retain water permanently or temporarily, i.e. returning water to the storm sewer when the rain event is over.

REQUIREMENTS FOR TANKS

The tanks shall comply with approved shop drawings and shall include the following accessories:

- 750 mm (30 in) access shaft with a smooth exterior wall, designed to receive CSA CL-625 and AASHTO H-20 traffic loads.
- Cast iron adjustable frame and cover.
- Factory welded, PVC SDR 35 inlet and outlet.
- Lifting rings (2) on top of the tank. The rings are designed to raise 4 times the mass of the empty tank.

RAW MATERIALS

The tanks shall be molded and constructed of a single wall covered with a reinforced fiber glass resin.

The reinforcements shall be covered with the same material.

The fiberglass shall comply with the CSA B66 standard.

TANK STRUCTURAL CAPACITY

The system shall be able to withstand CSA CL-25 and AASHTO H-20 traffic loads and the pressure of the water table, with a safety factor of 3:1.

Backfill height

- Roadway
 - Minimum 1200 mm (48 in) - Maximum 2 100 mm (82 in)
- Non-roadway
 - Minimum 300 mm (12 in) - Maximum 2 100 mm (82 in)

WATERTIGHTNESS

Each tank is tested at the factory under an internal pressure of 35 kPa (5 psi).

ANCHORING

When required, the tanks shall be equipped with precast anchors weights which are designed to be installed on each side of the tank at the bottom of the excavation.

The attachment points for the anchor weights shall be made of galvanized steel or stainless steel with a minimum of 20 mm (0.75 in) in diameter.

Fiberglass straps hold the reservoirs in place and are attached to the anchor weights.

Each strap supports 89 kN (20 000 lbf) in traction and has a minimum width of 50 mm (2 in). The safety factor is 2:1.

Turnbuckles shall be manufactured of galvanized steel or stainless steel with a minimum capacity of 23 kN (52 000 lbf). The safety factor is 5:1.

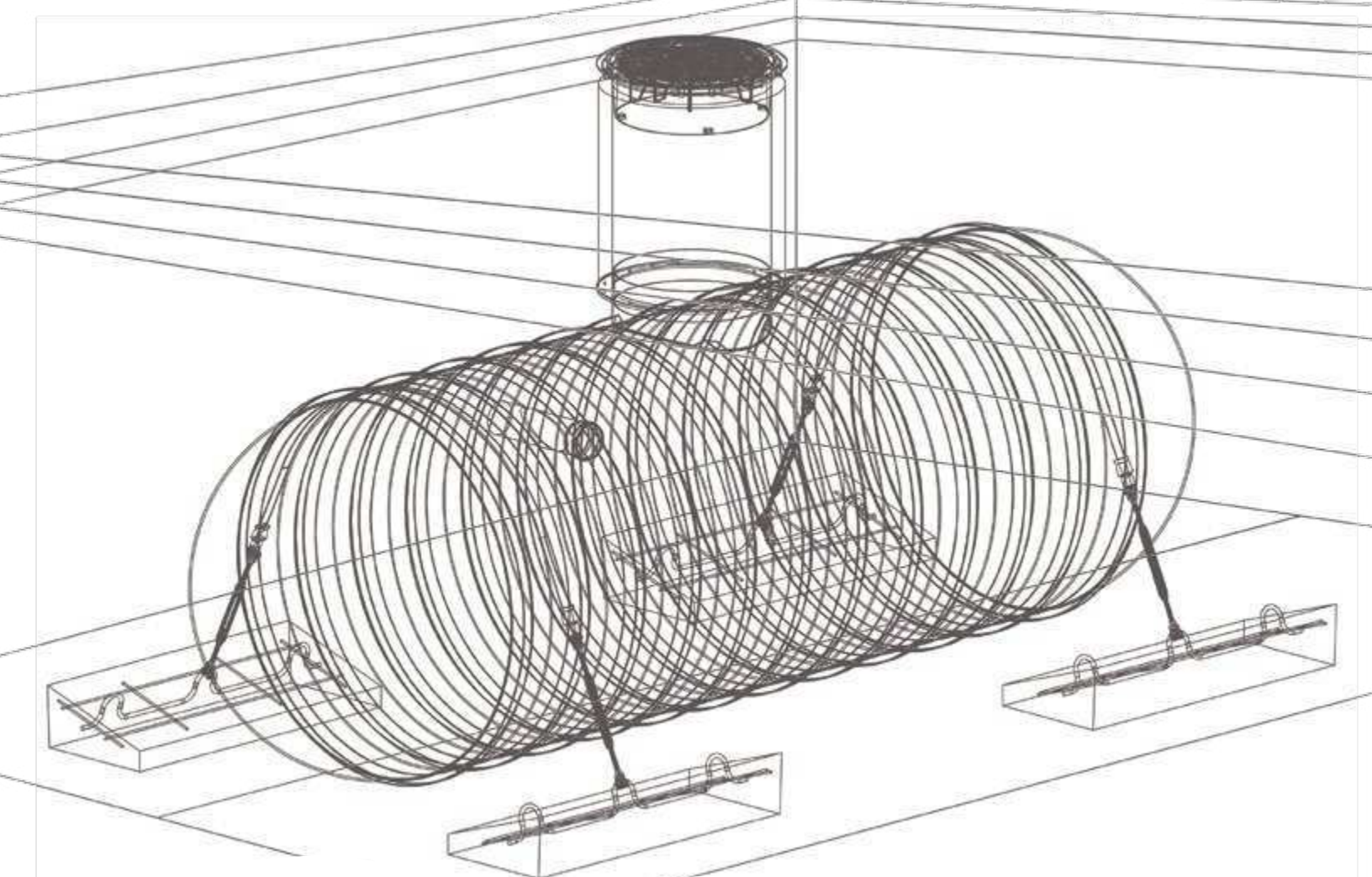
No chain or nylon strap is allowed.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

The installation will be carried out with the Soleno Installation Guide. Contact your local Soleno representative or visit our website at www.solen.com for installation recommendations.



NEMO TANK

INSTALLATION GUIDE

INSTALLATION INSTRUCTIONS FOR THE NEMO TANK

STAGE

1. Before undertaking the work
2. Receiving the materials
3. Handling and storage
4. Tank layout
5. Installing the tanks

STAGE 1

BEFORE UNDERTAKING THE WORK

In case of discrepancy between the instructions contained in this guide and those contained in the plans and specifications, contact your Soleno representative.

Contact your Soleno representative at least 48 hours before the work begins. The visit from your authorized Soleno representative is recommended after receipt of the materials on site or before the work begins.

STAGE 2

RECEIVING THE MATERIALS

Upon receipt of the materials, ensure that all the items required on the shop drawing are delivered and in good condition. Please notify immediately your Soleno representative in case of damage.

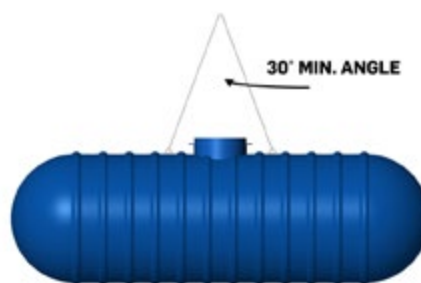
STAGE 3

HANDLING AND STORAGE

Ensure that all the lifting equipment used on the site is adapted to handle this load. Refer to the tank data sheet to find out its specific weight and select the appropriate equipment for its handling.

Select a level solid surface to deposit the tank. Ensure the surface is free of rocks, debris or other objects that could damage the tank.

When lifting the tank, comply with the angle shown below with the lifting straps or chains:



STAGE 3

HANDLING AND STORAGE (CONT'D)

Use the lifting rings to unload and install the tank.

Use a guide rope to move the tank (as needed) about its lifting axis.

When handling a tank equipped with plumbing accessories located on the bottom, take additional precautions not to damage them.

When the tank is stored temporarily on the site before its installation, anchor it solidly in place to prevent it from rolling.

Use ropes and wedges to stabilize the tank, as needed.

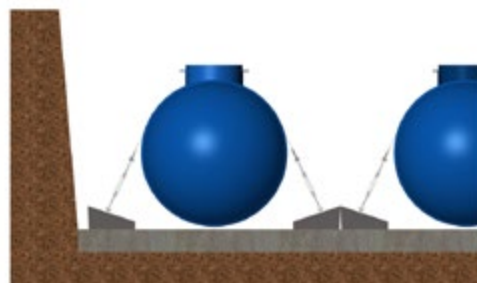


STAGE 4

TANK LAYOUT

Lay out the tanks according to the approved shop drawings and engineering plans.

The dimensions specified in this section must be considered as minimal and must be increased, as needed, to adapt to the conditions of the land (anchor weights or slabs)



STAGE 4

TANK LAYOUT (CONT'D)

Provide for enough clearance between tanks to allow a dummy exit for the anchor weights.

TANK SPACING

	Stable soil conditions	Unstable soil conditions*
Distance between tanks	The widest of 450 mm (18 in) or double the width of the anchor weight	The widest of 450 mm (18 in) or double the width of the anchor weight
Distance between the tanks and the trench	The widest of 450 mm (18 in) or double the width of the anchor weight	1200 mm (48 in)

*Unstable soil: peat bog, sand pit, landfill site, saturated soil

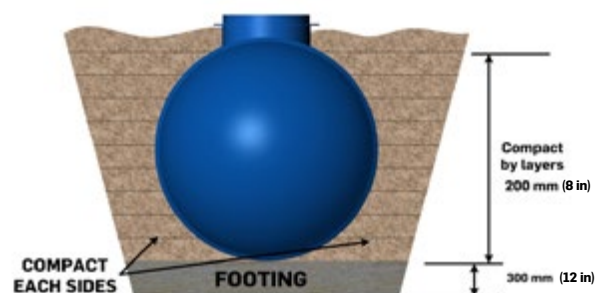
STAGE 5

INSTALLATION

The excavation bottom must be stable and undisturbed.

If water is present at the bottom of the excavation, keep it dry by pumping water out of the trench.

Prepare a base 300 mm (12 in) thick, composed of naturally rounded, clean gravel, ranging in diameter between 3 mm (0.125 in) and 19 mm (0.75 in). An alternative material composed of crushed stone or gravel, ranging in diameter between 3 mm (0.125 in) and 13 mm (0.5 in) may also be used. The base must be smooth and level.



STAGE 5

INSTALLATION (CONT'D)

Deposit the tank on the base. Never install a tank directly on a concrete slab, on beams or cradles, or on the natural soil in place. Ensure the tank is perfectly horizontal and that the minimum distances are respected between the tanks and the excavation walls.

If the tank requires anchoring, install the necessary hardware at this time, according to the approved shop drawings.

Deposit the anchor beams at the bottom of the excavation, parallel to the tank. The beams must not be positioned directly on the tanks.

Tie-down straps

Exclusively use tie-down straps supplied by Soleno to anchor the tank.

The positioning of the tie-down straps on each tank is identified by the arrow symbol ►◄ or by highlights on the reinforcements.

The straps must be used at each location identified for this purpose.

Never position the straps between the tank reinforcements. Non-compliance with the appropriate positioning of the straps can damage the tank.

Also distribute the tensile load by tightening all the tie-down straps uniformly until they are strongly tensioned without deforming the tank.

Measure the inner diameter of the tank horizontally and vertically to ensure there is no deformation after tightening of the straps.

The connection between the tie-down strap and the anchor point must be established by means of the following parts: tempered steel turnbuckle, looped cable or a combination of the two methods.



If and only if the installation is done in saturated soil, partially fill the tank with water to maintain it in place during backfilling. The water level in the tank must never exceed the level of the excavation by more than 300 mm (12 in) until final backfilling.

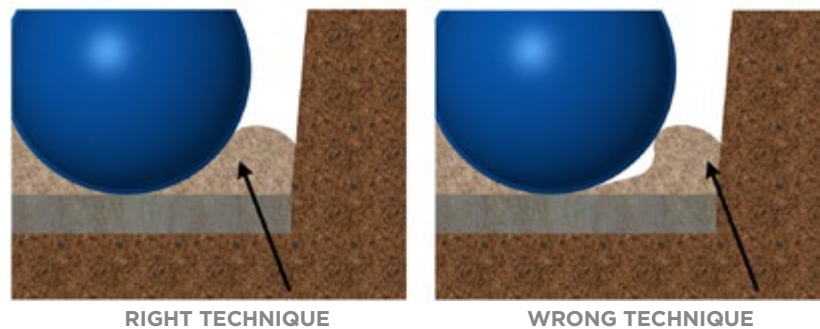
STAGE 5

INSTALLATION (CONT'D)

Deposit uniformly around the tank a 300 mm (12 in) layer of backfill material of the same composition as the base.

With the assistance of a long enough non-metal piece (for example, a 2' X 4" wooden plank), compact the fill material under the tank so that it is supported uniformly over its entire surface. Ensure that no area remains free.

Avoid banging the tank during backfilling.



Continue backfilling in layers of 200 mm (8 in) and continue compacting the filling material under the sides of the tank and between the reinforcements to eliminate voids and ensure the necessary maintenance.

Continue for each filling layer, at least up to mid-height of the tank.

Compact each layer to 90% of M.P.

Complete the backfilling to the top of the tank with the same filling material.

After installing the piping, continue backfilling to 300 mm (12 in) above the tank.

DATA SHEET

DUROMAXX®

PRODUCT DESCRIPTION: HDPE pipe with smooth interior wall and an exterior wall reinforced with encapsulated galvanized steel

FUNCTION: Large diameters HDPE culvert, culvert lining, storm water sewer and sanitary sewer and detention system

MANUFACTURING STANDARD: CSA B182.14, CSA B182.15, ASTM F2562 and AASHTO MP-20

RAW MATERIAL: Made of high-density polyethylene (HDPE) ASTM D3350 and galvanized steel ASTM A653/ASTM A653A, 230 MPa

TECHNICAL DATA: Diameters: 750 mm (30 in) to 3000 mm (120 in)
Lengths: 4.3 m (14 ft) and 14.6 m (48 ft)
Compression strength: see the technical data table
Manning: 0.012 n

AVAILABLE FITTINGS: Impervious to particles, low pressure (20.7 kPa or 3 psi), high performance (103.4 kPa or 15 psi), on-site welded (207 kPa or 30 psi), Electrofused (207 kPa or 30 psi)



TECHNICAL DATA TABLE

Nominal dia.		Int. dia.		Ext. dia.		Ext. Dia. (bell)		Mass*		Backfill height				Soil tight connectors		Watertight connectors			
										Minimum		Maximum		C	LP	HP	W	EF	
mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	m	ft	m	ft						
750	30	749	29.5	785	30.9	864	34.0	23	16	0.3	1.0	15	50	X	X	X			
900	36	900	35.4	942	37.1	1014	39.9	31	21	0.3	1.0	15	50	X	X	X	X	X	
1050	42	1049	41.3	1097	43.2	1163	45.8	39	27	0.3	1.0	15	50	X	X	X	X	X	
1200	48	1199	47.2	1257	49.5	1328	52.3	43	29	0.3	1.0	9	30	X	X	X	X	X	
1350	54	1351	53.2	1410	55.5	1478	58.2	52	35	0.3	1.0	9	30	X	X	X	X	X	
1500	60	1501	59.1	1560	61.4	1628	64.1	62	42	0.3	1.0	9	30	X	X	X	X	X	
1650	66	1651	65.0	1722	67.8	1819	71.6	85	57	0.5	1.5	9	30	X	X	X	X	X	
1800	72	1800	70.9	1882	74.1	1971	77.6	98	66	0.5	1.5	9	30	X	X	X	X	X	
2100	84	2100	82.7	2182	85.9	2258	88.9	114	76	0.6	2.0	9	30	X	X		X	X	
2400	96	2400	94.5	2497	98.3	N/A	N/A	129	87	0.6	2.0	9	30	X			X	X	
2750	108	2743	108.0	2827	111.3	N/A	N/A	148	100	0.8	2.5	7.6	25	X			X	X	
3000	120	3000	118.1	3097	121.9	N/A	N/A	162	109	0.9	3.0	7.6	25	X			X	X	

* Approximate mass : may vary depending on the length or type of connection used

LEGEND

C : Galvanized coupler and soil tight gasket
LP : Low pressure bell and gasket (20.7 kPa or 3 psi)
HP : High pressure bell and gasket (103.4 kPa or 15 psi)
W : On-site welding (207 kPa or 30 psi)
EF : Electrofused coupler (207 kPa or 30 psi)

APPLICATIONS: Storm water sewer
Sanitary sewer
Detention system
Large diameters HDPE culvert
Culvert lining
Reservoir
Irrigation pipe

SPECIFICATIONS

DUROMAXX®

SCOPE

These specifications apply to the DuroMaxx® pipes supplied by Soleno. These pipes can be used in a storm sewer, sanitary sewer, for retention and/or groundwater infiltration systems, drainage, as tanks or for the conveyance of industrial waste. They are available in diameters ranging from 750 mm (30 in) to 3000 mm (120 in).

PIPE REQUIREMENTS

The pipe shall be the DuroMaxx® type and shall consist of a smooth interior wall of reinforced polyethylene with a continuous exterior wall containing high-strength galvanized steel. The steel shall be completely encapsulated within the helical polyethylene profile. This profile includes a merged continuous joint. The pipe manufacture shall be compliant with standards ASTM F2562 and AASHTO MP-20.

- For storm sewer applications, the pipe shall comply with standard CSA B182.14.
- For sanitary sewer applications, the pipe shall comply with standard CSA B182.15.
- The Manning coefficient shall be 0.012.

RAW MATERIALS

- Polyethylene: virgin resin, high density, resistant to pressure, complies with standard ASTM D3350.
- High-strength galvanized steel: a minimum yield strength of 550 MPa with a minimum zinc coating weight of 183 g/m².

JOINT PERFORMANCE

- *Impervious to particles connection, M-type (750 to 3000 mm (30 to 120 in))*: smooth coupling sleeve made of aluminized type 2 steel, complete with an elastomer gasket. This sleeve shall include iron angles that allow its bolting.
- *Low-pressure bell with integrated gasket, LP-type (750 to 2100 mm (30 to 84 in))*: male end with a leak-proof gasket made of elastomer that is inserted into a bell made of polyethylene (according to the above polyethylene description). This joint has been laboratory tested to 20.7 kPa (3 psi), when tested in accordance with standard ASTM D3212.
- *High-pressure bell with integrated gasket, HP-type (750 to 2100 mm (30 to 84 in))*: male end with a leak-proof gasket made of elastomer that is inserted into a bell made of polyethylene (according to the above polyethylene description). The male end and the bell shall be reinforced with encapsulated steel. This joint has been laboratory tested to 103.4 kPa (15 psi), when tested in accordance with standard ASTM D3212.
- *S-type on-site welded connection (900 to 3000 mm (36 to 120 in))*: smooth coupling sleeve made of aluminized type 2 steel complete with an elastomer gasket for maintaining the smooth ends in place. This sleeve shall include iron angles that allow its bolting. Inside the pipes, there is a polyethylene smooth sleeve that is welded on-site. This joint has been laboratory tested to 207 kPa (30 psi), when tested in accordance with standard ASTM D3212.
- *On-site electrofused joint, EF-type (900 to 3000 mm (36 to 120 in))*: smooth coupling sleeve made of aluminized type 2 steel complete with an elastomeric gasket for maintaining the smooth ends in place. This sleeve shall include iron angles that allow its bolting. Inside the pipes, there is a polyethylene smooth sleeve that is welded on-site. This fitting shall be resistant to an on-site pressure test of 207 kPa (30 psi).

ACCESSORIES

The accessories and the connections manufactured and supplied by Soleno are designed to provide the same structural integrity and the same resistance to leaking as the pipes.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

The installation shall be performed according to the Soleno Installation Guide and shall contain all the components specified in the plans and specifications. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

ORIFICE FLOW REGULATORS

PRODUCT DESCRIPTION : Device shaped like a coupler or plate regulator.

FUNCTION : Controls water flow

RAW MATERIALS : Made from polyvinyl chloride (PVC)

TECHNICAL DATA : Diameter: 200 mm (8 in), 250 mm (10 in), 300 mm (12 in), 375 mm (15 in) and 450 mm (18 in)
Flow rate: 8 to 120 m³/s (282.7 to 4,240.32 ft³/s)

AVAILABLE COUPLERS : Adapts to all types of pipes (PVC, HDPE, concrete, TILE)



Connector FLOW REGULATOR



Plate REGULATOR

TECHNICAL DATA TABLES

CONNECTOR FLOW REGULATOR

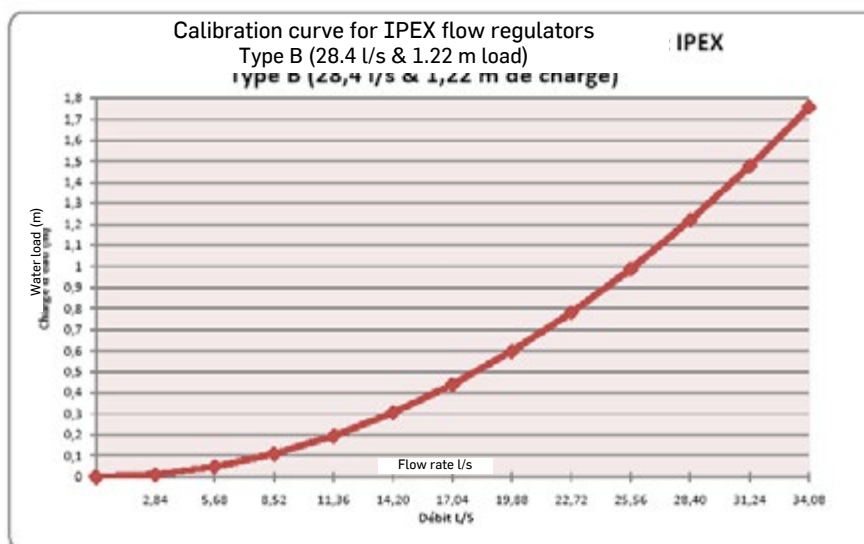
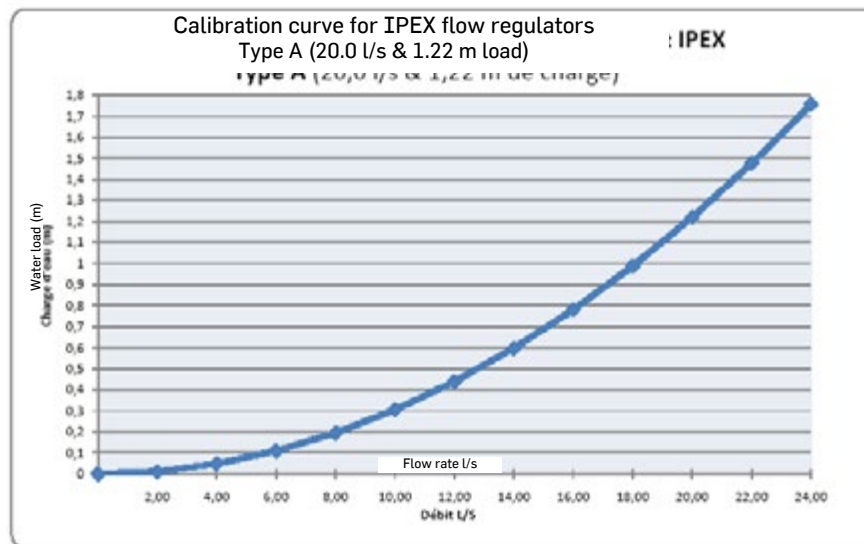
Nom. dia. of the pipe		Type				
		A	B	C	D	F
mm	in					
200	8	X	X			
250	10	X	X	X	X	
300	12	X	X	X	X	X
375	15	X	X	X	X	X
450	18	X	X	X	X	X
525	21	X	X	X	X	X
600	24	X	X	X	X	X

PLATE FLOW REGULATOR

Nom. dia. of the pipe		Type				
		A	B	C	D	F
mm	in					
200	8	X	X	X	X	X
250	10	X	X	X	X	X
300	12	X	X	X	X	X
375	15	X	X	X	X	X
450	18	X	X	X	X	X
525	21	X	X	X	X	X
600	24	X	X	X	X	X

DATA SHEET

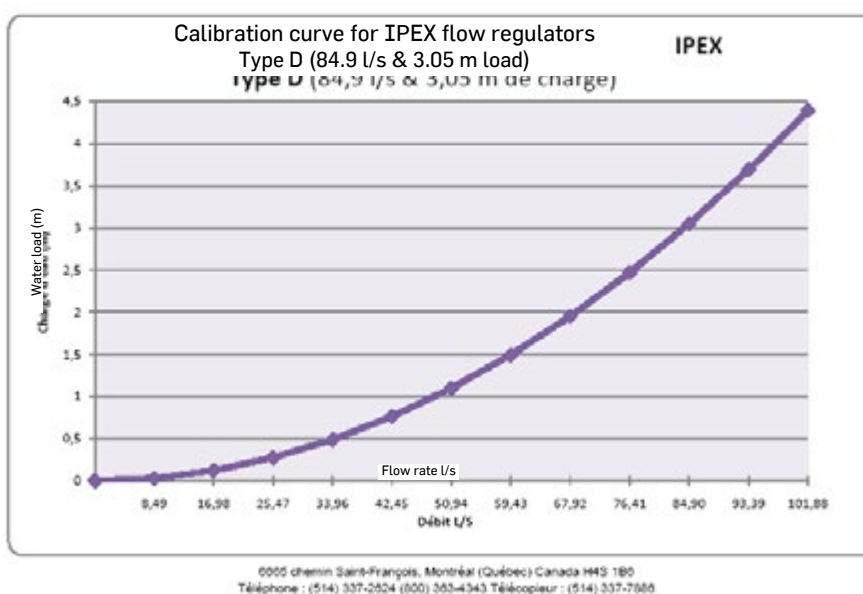
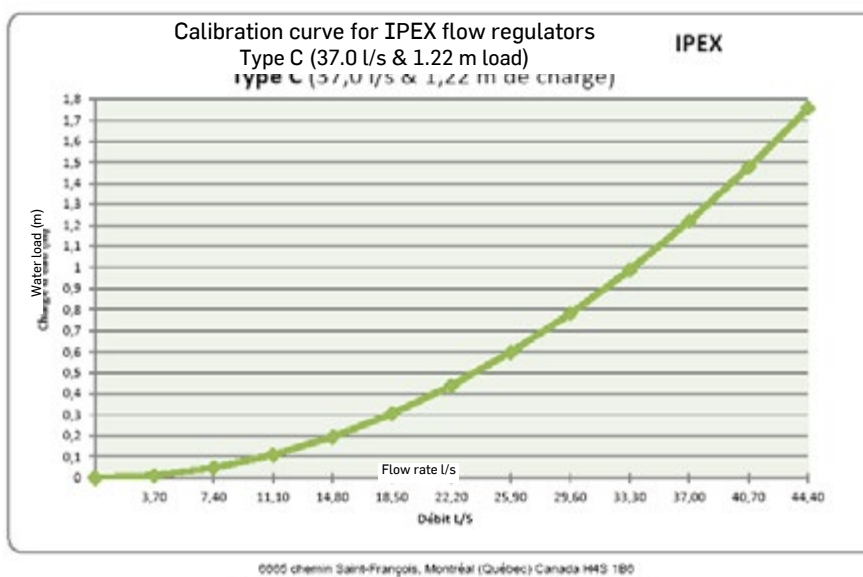
ORIFICE FLOW REGULATOR (CONT.)



6000 chemin Saint-François, Montréal (Québec) Canada H4S 1B9
Téléphone : (514) 337-2624 (800) 363-4343 Télécopieur : (514) 337-7666

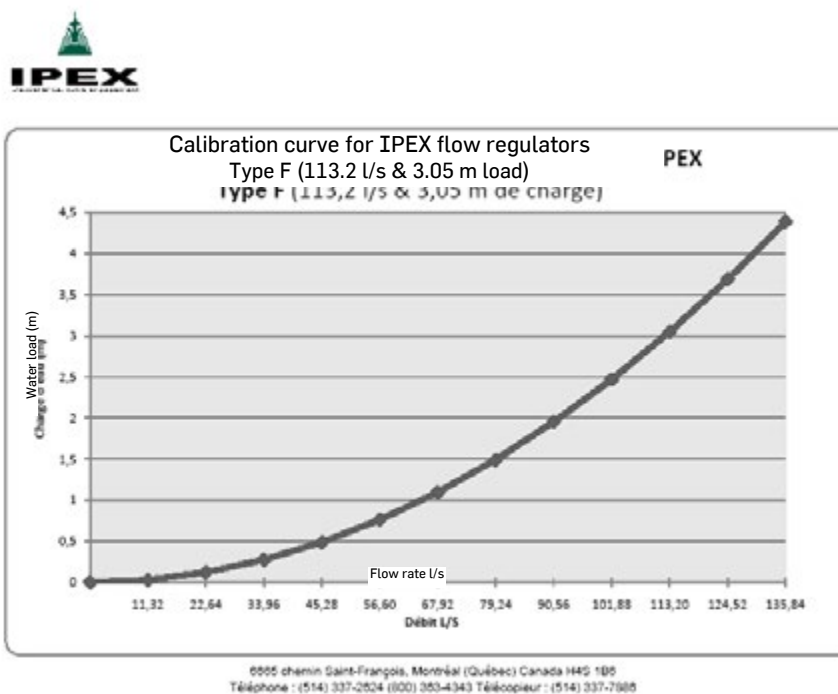
DATA SHEET

ORIFICE FLOW REGULATOR (CONT.)



DATA SHEET

ORIFICE FLOW REGULATOR (CONT.)



OPTION : Sizes of orifice based on specific water flows and loads are available upon request.

SPECIFICATIONS

CONNECTOR FLOW REGULATOR

SCOPE

These specifications apply to all 200 mm (8 in) to 600 mm (24 in) connector flow regulators used to regulate water flow.

FLOW REGULATOR REQUIREMENTS

- The regulator shall be short, slightly conical, inserted into the catch basin/manhole outlet pipe and kept in position by friction and hydrostatic pressure.
- The opening of the flow regulator's water passage shall be diamond-shaped with a keyed channel in its lower part to prevent the opening from getting clogged by debris during dry periods, and to generate a vortex for self-cleaning the pipe during heavy rain.
- When it is in position, the part of the flow regulator with holes will be even with the inner surface of the manhole/catch basin.
- The size of the opening shall be determined based on allowable flow and water load according to suitable calibration curves.

RAW MATERIALS

The flow regulator shall be made from polyvinyl chloride (PVC).

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

SPECIFICATIONS

PLATE FLOW REGULATOR

SCOPE

These specifications apply to all 200 mm (8 in) to 600 mm (24 in) plate flow regulators used to regulate water flow.

FLOW REGULATOR REQUIREMENTS

- The flow regulator shall be shaped like a plate and shall be kept in position by a grooved frame facilitating its installation and removal.
- The frame shall be located above the outlet pipe and bolted to the interior wall of the manhole or catch basin.
- The opening of the flow regulator's water passage shall be diamond-shaped with a keyed channel in its lower part to prevent the opening from getting clogged by debris during dry periods, and to generate a vortex for self-cleaning the pipe during heavy rain.
- The size of the opening shall be determined based on allowable flow and water load according to suitable calibration curves.

RAW MATERIALS

The flow regulator and frame shall be made from polyvinyl chloride (PVC).

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall comply with tender documents and drawings.

INSTALLATION

Installation shall be carried out in compliance with Soleno's recommendations. Contact the Soleno representative in your region or visit our website at www.solen.com for installation recommendations.

DATA SHEET

FILCOTEN ROBUST

PRODUCT DESCRIPTION : Drainage channel for residential applications and light loads (pedestrian zones and bike paths) equipped with a galvanized steel grate

FUNCTION : Perpendicular collection accross the full width of a drainage surface such as a parking deck, street or sloped residential driveway

APPLICABLE STANDARD : EN 1433 (A, 15KN)

RAW MATERIAL : Channel : fiberglass-reinforced composite concrete
Grate : galvanized steel

TECHNICAL DATA : Width : 100 mm (4 in)
Height : 97 mm (3.8 in)
Section length : 1000 mm (39.4 in)

AVAILABLE ACCESSORIES : Side connection well
Channel end plate (with or without outlet)
Vertical channel outlet

CONNECTION : Filcoten adapter compatible with drains, Solflo or Solflo Max



TECHNICAL DATA CHART

Model	Internal dimensions				Overall dimensions				Channel weight	Roughness (int. surface)	Bedding and bank sloping	Galvanized grate						
	Width		Height		Width		Height					Capacity			Openings			
	mm	in	mm	in	mm	in	mm	in				kN	kPa	PSI	mm x mm	in x in		
100 x 97	100	4	85	3.3	122	4.8	97	3.8	8.1	5.4	35	0.0014	Concrete	A15	480	70	85 x 8	3.3 x 0.3

APPLICATION : Linear collection

SPECIFICATIONS

FILCOTEN ROBUST

SCOPE

This estimate applies to all Filcoten Robust channel, used to collect and discharge storm water.

CHANNEL REQUIREMENTS

Each channel consists of a galvanized steel grate attached on a channel made of fiberglass-reinforced composite concrete.

- **Grate :** the grate has 8 x 85 mm openings (0.3 in x 3.3 in). It can support a load of 15 kN (class A, in accordance with EN 1433 standard).
- **Channel :**
 - The channel can be installed within the asphalt, concrete or paving stones.
 - Each channel length can be connected to a discharge outlet.
 - Manufacture of the channel should require no oily or greasy stripping agent, in order to maximize adhesion between the channel and the concrete filling. Such a precaution will eliminate the risk of water infiltration between the fill concrete and the body of the channel.
 - The outer upper part of the channel is perfectly smooth, which allows a perfect fit between the channel and the finishing material.
 - Each section has a net length of 1.0 m (39.4 in).
 - The minimum impact resistance shall be 20 MPa/mm².
 - The channel is certified A+ by the IBR Institute, which ensures that it emits no substance harmful to human health and can be installed indoors.

RAW MATERIAL

- **Grate :**
 - The pre-galvanized grate is in accordance with EN 10142, 10214 and 10215 standards.
- **Channel :**
 - The channel is made of composite fiber-reinforced concrete, with a minimum compressive strength of 80 MPa, in accordance with EN-206-1 standard.
 - The composite concrete must be certified in accordance with CE EN 1433 standard to -20 °C as well as 40 °C, which provides excellent resistance to freeze/thaw cycles, as well as to de-icing salts.
 - The structure of the channel must be impervious to UV rays.
 - The channel must be Class A1 nonflammable, in accordance with DIN 4102 standard.102.

TRACEABILITY AND IDENTIFICATION

Each section includes product identification, a bar code, the direction of flow and the date of manufacture.

ACCESSORIES

- The side connection well includes an outlet with integrated seal, which prevents the leakage of fine particles.
- The silt bucket is installed in its upper part, which prevents blocking of the outlet.
- Each section includes a preformed outlet opening adapted to receive the drain pipe, which is mechanically fastened to the channel.

DIMENSIONS AND QUANTITIES

Dimensions and quantities must be consistent as shown on the bid documentation and plans.

INSTALLATION

Installation will be performed in accordance with manufacturer's recommendations. Contact your local Soleno representative or visit our website at www.soleno.com to read our recommendations for installation.

DATA SHEET

FILCOTEN INFRA WITH 4 mm GALVANIZED STEEL ANGLE BRACKETS

PRODUCT DESCRIPTION : Drainage channel for commercial or industrial applications, forklifts, public areas and residential developments with heavy traffic, light or heavy traffic loads, main roads and highways (channels are not installed under areas of dynamic traffic loads). Equipped with galvanized steel grate or grating, polyamide or cast iron grate.

FUNCTION : Perpendicular collection across the full width of a drainage surface and routing to a discharge structure

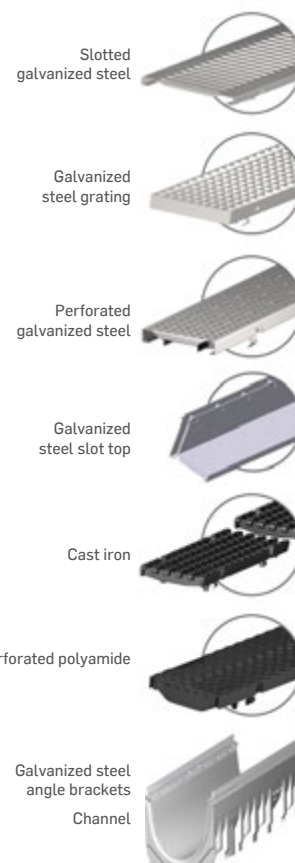
MANUFACTURING STANDARDS : EN 1433 (A, 15 kN / B, 125 kN / C, 250 kN, D, 400 kN, E, 600 kN), CSA S6-06 (CL-625)

RAW MATERIAL : Channel: fiber-reinforced composite concrete
Angle brackets: galvanized steel
Grates: galvanized steel, perforated polyamide or cast iron

AVAILABLE ACCESSORIES : Side connection well
Channel end plate (with or without outlet)
Vertical channel outlet
Vandal-proof locking system

CONNECTION : See Accessories - Filcoten Infra with 4 mm galvanized steel angle brackets Data Sheet

APPLICATION : Linear collection



TECHNICAL DATA TABLE 100 mm (4 in) WIDTH CHANNELS

Model	Internal dimensions				Overall dimensions								Impact resistance		Roughness (int. surface)		Density	
	Width		Height		Width		Height											
	mm	in	mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	MPa-mm	psi-in	µm	in	g/cm³	lb/ft³
100 x 142	100	4	100	4.0	138	5.4	142	5.6	1000	39.4	18.3	12.3	20	114.2	30	0.0012	2.40	150
100 x 195	100	4	150	6.0	138	5.4	195	7.7	1000	39.4	24.9	16.7	20	114.2	30	0.0012	2.40	150
100 x 245	100	4	200	8.0	138	5.4	245	9.6	1000	39.4	29.9	20.1	20	114.2	30	0.0012	2.40	150

DATA SHEET

FILCOTEN INFRA WITH 4 mm GALVANIZED STEEL ANGLE BRACKETS (CONT'D)

TECHNICAL DATA TABLE GRATES FOR 100 mm (4 in) WIDTH CHANNELS

Description	Capacity			Length ⁽¹⁾		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Slotted galvanized steel grate	A15	480	70	1000	39.4	8 x 80	0.3 x 3.2	n/a	n/a	1.5	1.0
Galvanized steel grating	B125	4000	580	1000	39.4	30 x 10	1.8 x 0.4	n/a	n/a	2.8	1.9
ADA* perforated galvanized steel grate	C250	8000	1160	1000	39.4	n/a	n/a	Ø 6.0	Ø 0.25	3.3	2.2
GALVANIZED STELL SLOT TOP	C250	8000	1160	1000	39.4	1000 x 12.5	39.4 x 0.5	n/a	n/a	5.4	3.6
ADA* cast iron grate	C250	8000	1160	500	19.7	29 x 6	1.2 x 0.3	n/a	n/a	7.2	4.8
Cast iron grate	D400	12800	1856	500	19.7	29 x 13	1.1 x 0.5	n/a	n/a	6.8	4.6
ADA* perforated polyamide grate	C250	8000	1160	500	19.7	n/a	n/a	Ø 7.3	Ø 0.29	2.2	1.5

Note 1 : For grates of 500 mm (19.7 in), forecast 2 grates per channel.

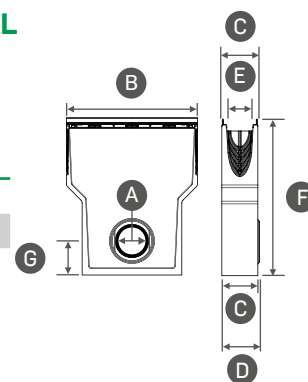
ADA : Conform to *Americans with disabilities Act* (www.ada.gov)

* : Heel proof grate

TECHNICAL DATA TABLE 100 mm (4 in) WIDTH SIDE CONNECTION WELL

Nominal diam. of the outlet (A)		Length (B)		Width (C)		Max. width (including the outlet) (D)		Nominal width (E)		Height (F)		Distance (outlet center to the bottom of the well) (G)		Weight	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
100	4	500	19.7	138	5.4	145	5.7	100	4	600	23.6	130	5.1	32.4	71.4

Note : An adapter is required for connection to a 100 mm (4 in) HDPE pipe and no adapter is required for connection to a 100 mm (4 in) PVC pipe.



TECHNICAL DATA TABLE GRATES FOR 100 mm (4 in) WIDTH SIDE CONNECTION WELL

Description	Capacity			Length		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Slotted galvanized steel grate	A15	480	70	500	19.7	8 x 80	0.3 x 3.2	n/a	n/a	1.6	1.1
Galvanized steel grating	B125	4000	580	500	19.7	30 x 10	1.8 x 0.4	n/a	n/a	2.8	1.9
ADA* perforated galvanized steel grate	C250	8000	1160	500	19.7	n/a	n/a	Ø 6.0	Ø 0.25	3.4	2.3
GALVANIZED STELL SLOT TOP	C250	8000	1160	500	19.7	1000 X 12.5	39.4 X 0.5	n/a	n/a	10	6.7
Galvanized steel maintenance unit	C250	8000	1160	500	19.7	500 X 12.5	19.9 X 0.5	n/a	n/a	10	6.7
ADA* cast iron grate	C250	8000	1160	500	19.7	29 X 6	1.2 X 0.3	n/a	n/a	7.2	4.8
Cast iron grate	D400	12800	1856	500	19.7	29 X 13	1.2 X 0.6	n/a	n/a	6.8	4.6
ADA* perforated polyamide grate	C250	8000	1160	500	19.7	n/a	n/a	Ø 7.3	Ø 0.29	2.2	1.5

ADA : Conform to *Americans with disabilities Act* (www.ada.gov)

* : Heel proof grate

DATA SHEET

FILCOTEN INFRA WITH 4 mm GALVANIZED STEEL ANGLE BRACKETS (CONT'D)

TECHNICAL DATA TABLE 150 mm (6 in) WIDTH CHANNELS

Model	Internal dimensions				Overall dimensions				Length		Weight		Impact resistance		Roughness (int. surface)		Density	
	Width		Height		Width		Height											
	mm	in	mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	MPa-mm	psi-in	µm	in	g/cm³	lb/ft³
100 x 210	150	6	165	6.5	188	7.4	210	8.3	1000	39.4	30.8	20.7	20	114.2	30	0.0012	2.40	150

TECHNICAL DATA TABLE GRATES FOR 150 mm (6 in) WIDTH CHANNELS

Description	Capacity			Length ⁽¹⁾		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Galvanized steel grating	B125	4000	580	1000	39.4	30 x 10	1.8 x 0.4	n/a	n/a	4.6	3.1
ADA* cast iron grate	D400	12800	1856	500	19.7	29 x 13	1.2 x 0.6	n/a	n/a	10.4	7.0

Note 1 : For grates of 500 mm (19.7 in), forecast 2 grates per channel.

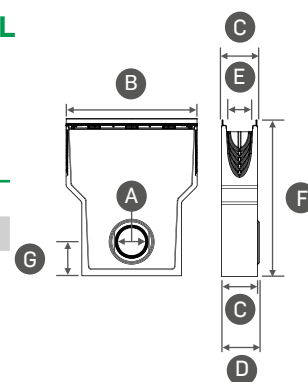
ADA : Conform to *Americans with disabilities Act* (www.ada.gov)

* : Heel proof grate

TECHNICAL DATA TABLE 150 mm (6 in) WIDTH SIDE CONNECTION WELL

Nominal diam. of the outlet (A)		Length (B)		Width (C)		Max. width (including the outlet) (D)		Nominal width (E)		Height (F)		Distance (outlet center to the bottom of the well) (G)		Weight	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
150	6	500	19.7	188	7.4	200	7.9	150	6	690	27.2	180	7.1	44.5	98.1

Note : An adapter is required for connection to a 150 mm (6 in) HDPE pipe and no adapter is required for connection to a 150 mm (6 in) PVC pipe.



TECHNICAL DATA TABLE GRATES FOR 150 mm (6 in) WIDTH SIDE CONNECTION WELL

Description	Capacity			Length		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Galvanized steel grating	B125	4000	580	500	19.7	30 x 10	1.8 x 0.4	n/a	n/a	4.8	3.2
ADA* cast iron grate	D400	12800	1856	500	19.7	29 x 13	1.2 x 0.6	n/a	n/a	10.4	3.2

ADA : Conform to *Americans with disabilities Act* (www.ada.gov)

* : Heel proof grate

DATA SHEET

FILCOTEN INFRA WITH 4 mm GALVANIZED STEEL ANGLE BRACKETS (CONT'D)

TECHNICAL DATA TABLE 200 mm (8 in) WIDTH CHANNELS

Model	Internal dimensions				Overall dimensions				Length		Weight		Impact resistance		Roughness (int. surface)		Density	
	Width		Height		Width		Height											
	mm	in	mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	MPa-mm	psi-in	µm	in	g/cm³	lb/ft³
200 x 265	200	8	220	8.7	238	9.4	265	10.4	1000	39.4	44	29.5	20	114.2	30	0.0012	2.40	150

TECHNICAL DATA TABLE GRATES FOR 200 mm (8 in) WIDTH CHANNELS

Description	Capacity			Length ⁽¹⁾		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Galvanized steel grating	B125	4000	580	1000	39.4	30 x 10	1.8 x 0.4	n/a	n/a	6.0	4.0
ADA* cast iron grate	D400	12800	1856	500	19.7	29 x 13	1.2 x 0.6	n/a	n/a	15.4	10.3

Note 1 : For grates of 500 mm (19.7 in), forecast 2 grates per channel.

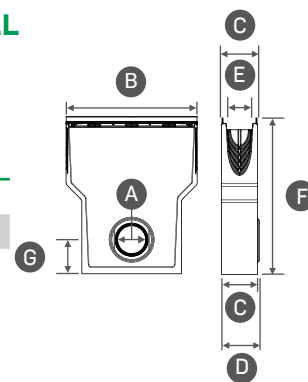
ADA : Conform to *Americans with disabilities Act* (www.ada.gov)

* : Heel proof grate

TECHNICAL DATA TABLE 200 mm (8 in) WIDTH SIDE CONNECTION WELL

Nominal diam. of the outlet (A)		Length (B)		Width (C)		Max. width (including the outlet) (D)		Nominal width (E)		Height (F)		Distance (outlet center to the bottom of the well) (G)		Weight	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
200	8	500	19.7	238	9.4	265	10.4	150	6	690	27.2	135	5.3	48	105.8

Note : An adapter is required for connection to a 200 mm (8 in) HDPE pipe and no adapter is required for connection to a 200 mm (8 in) PVC pipe.



TECHNICAL DATA TABLE GRATES FOR 200 mm (8 in) WIDTH SIDE CONNECTION WELL

Description	Capacity			Length		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Galvanized steel grating	B125	4000	580	500	19.7	30 x 10	1.8 x 0.4	n/a	n/a	6.2	4.2
ADA* cast iron grate	D400	12800	1856	500	19.7	29 x 13	1.2 x 0.6	n/a	n/a	15.4	10.3

ADA : Conform to *Americans with disabilities Act* (www.ada.gov)

* : Heel proof grate

SPECIFICATION

FILCOTEN INFRA WITH 4 mm GALVANIZED STEEL ANGLE BRACKETS

SCOPE

These specifications apply to all Filcoten Infra channels with 4 mm (0.16 in) galvanized steel angle brackets, used to collect and discharge storm water.

CHANNEL REQUIREMENTS

Each channel consists of a grate attached within galvanized steel angle brackets which are fitted on a channel made of fiber-reinforced composite concrete.

■ Grates :

- *Slotted galvanized steel grate 100 mm (4 in)* : the grate has 8 x 85 mm (0.3 in x 0.3 in) openings. It can support a load of 15 kN (3 372 lb) (class A, according to EN 1433 standard).
- *Galvanized steel grating 100 mm (4 in), 150 mm (6 in) and 200 mm (8 in)* : the grating has 10 x 30 mm (0.4 x 1.2 in) openings. It can support a load of 125 kN (28 100 lb) (class B, according to EN 1433 standard).
- *ADA perforated galvanized steel grate 100 mm (4 in)* : the grate has openings 6 mm (0.2 in) in diameter. It can support a load of 250 kN (56 200 lb) (class A, according to EN 1433 standard). Heelproof grate.
- *GALVANIZED STEEL SLOT TOP 100 mm (4 in)* : the grate has a slot of 12.5 x 500 mm (0.5 x 19.9 in). It can support a load of 250 kN (56 200 lb) (class A, according to EN 1433 standard).
- *Galvanized steel maintenance unit* : the maintenance unit has a slot of 12.5 x 500 mm (0.5 x 19.9 in). It can support a load of 250 kN (56 200 lb) (class A, according to EN 1433 standard).
- *Cast iron grate 100 mm (4 in)* : the grate has 13 x 29 mm (0.5 x 1.2 in) openings. It can support a load of 400 kN (89 923 lb) (class D according to EN 1433).
- *ADA cast iron grate 100 mm (4 in)* : the grate has 6 x 29 mm (0.3 x 1.2 in) openings. It can support a load of 250 kN (56 200 lb) (class C according to EN 1433). It is equipped with 4 pressure fasteners, to prevent lifting. Heelproof grate.
- *ADA cast iron grate 150 mm (6 in) and 200 mm (8 in)* : the grate has 13 x 29 mm (0.5 x 1.2 in) openings. It can support a load of 400 kN (89 923 lb) (class D according to EN 1433). It is equipped with 4 pressure fasteners, to prevent lifting. Heelproof grate.
- *ADA perforated polyamide grate 100 mm (4 in)* : the grate has openings 7.3 mm (0.29 in) in diameter. It can support a load of 250 kN (56 200 lb) (class C according to EN 1433). It is equipped with 4 pressure fasteners, to prevent lifting. Heelproof grate.

■ Angle brackets :

- The side angle brackets are fitted into the upper part of the channel, are part of the channel body and have a minimum thickness of 4 mm (0.16 in).
- In total, the angle brackets have 8 anchoring stops per channel section.

■ Channel :

- The channel can be installed within the asphalt, concrete or paving stones.
- Manufacture of the channel should require no oily or greasy stripping agent, in order to maximize adhesion between the channel and the concrete filling. Such a precaution will eliminate the risk of water infiltration between the fill concrete and the body of the channel.
- The outer upper part of the channel is perfectly smooth, which allows a perfect fit between the channel and the finishing material.
- The guide stops at the channel ends allow proper alignment.
- The minimum wall thickness of the channel shall be 19 mm (0.75 in).
- Each section has a net length of 1 m (39.4 in).
- The minimum impact resistance is 20 MPa-mm (114 psi-in).
- The channel is certified A+ by the IBR Institute, which ensures that it emits no substance harmful to human health and can be installed indoors.

SPECIFICATIONS

FILCOTEN INFRA WITH 4 mm GALVANIZED STEEL ANGLE BRACKETS (CONT'D)

RAW MATERIAL

- **Grates :**
 - The slotted galvanized steel grate and the ADA perforated galvanized steel grate are pre-galvanized in accordance with EN 10142, 10214 and 10215 standards.
 - The galvanized steel grating is hot galvanized in accordance with the ISO 1461 standard, with a minimum mass per unit area of 275 g/m² (0.00039 psi).
 - The cast iron grate and the ADA cast iron grate are made of ductile iron in accordance with EN-GJS-500-7 standard.
 - The ADA perforated polyamide grate is made of polyamide with a fiberglass honeycomb structure. It is resistant to UV rays, frost and heat.
- **Angles brackets :**
 - The brackets are made from pre-galvanized steel, in accordance with EN 10142, 10214 and 10215 standards.
- **Channel :**
 - The channel is made of composite fiber-reinforced concrete, with a minimum compressive strength of 80 MPa (11 600 psi), in accordance with EN-206-1 standard.
 - The composite concrete must be certified in accordance with CE EN 1433 standard to -20 °C (-4°F) as well as 40 °C (104 °F), which provides excellent resistance to freeze/thaw cycles as well as de-icing salts.
 - The structure of the channel must be resistant to UV.
 - The channel must be Class A1 non-flammable, in accordance with DIN 4102 standard.

TRACEABILITY AND IDENTIFICATION

Each section includes product identification, a bar code, the direction of flow and the date of manufacture.

ACCESSORIES

- The side connection well includes an outlet with integrated seal, which prevents the leakage of fine particles.
- The silt bucket is installed in its upper part, which prevents blocking of the outlet.
- No mounting accessory internal to the channel that would impede the surface flow is allowed.
- Optional: Vandal-proof lock.
- Channel end plate without outlet.
- Channel end plate with outlet which allows the longitudinal connection of a PVC or HDPE pipe
- Vertical channel outlet.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall be consistent with the bid documentation and plans.

INSTALLATION

Installation will be performed in accordance with manufacturer's recommendations. Contact your local Soleno representative or visit our website at www.soleno.com to read our recommendations for installation.

LEGEND

ADA: Americans with disabilities Act

DATA SHEET

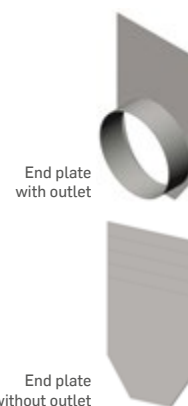
ACCESSORIES FOR FILCOTEN INFRA WITH 4 mm GALVANIZED STEEL ANGLE BRACKETS

TECHNICAL DATA TABLE CHANNEL END PLATES

For channel (width)		With outlet	Width		Outlet diameter	
mm	in		mm	in	mm	in
100	4	X	100	4	100	4
100	4		100	4	n/a	n/a
150	6	X	150	6	150	6
150	6		150	6	n/a	n/a
200	8	X	200	8	200	8
200	8		200	8	n/a	n/a

Note 1 : An adapter is required for connection to a 100 mm (4 in) to 200 mm (8 in) HDPE pipe and no adapter is required for connection to a PVC pipe.

Note 2 : The PVC connection must be made from the female end of the pipe.



TECHNICAL DATA TABLE VERTICAL CHANNEL OUTLET

For channel (width)		Outlet diameter	
mm	in	mm	in
100 ⁽¹⁾	4	100	4
150 ⁽²⁾	6	150	6
200 ⁽³⁾	8	200	8

Note 1 : No adapter is required for connection to a 100 mm (4 in) PVC and/or PEHD pipe.

Note 2 : An adapter is required for connection to a 100 mm (4 in) HDPE pipe and no adapter is required for connection to a 100 mm (4 in) PVC.

Note 3 : An adapter is required for connection to a 150 mm (6 in) HDPE pipe and no adapter is required for connection to a 150 mm (6 in) PVC.

Note 4 : The PVC connection must be made from the female end of the pipe.



VANDAL-PROOF LOCK (optional)



Note 1 : Compatible with galvanized steel grating, perforated galvanized steel grate and cast iron grate (optional).

Note 2 : Two units may be used per grate, but only one is required.

DATA SHEET

FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS

PRODUCT DESCRIPTION : Drainage channel for commercial or industrial applications, forklifts, public areas and residential developments with heavy traffic, light or heavy traffic loads, main roads and highways (channels are not installed under areas of dynamic traffic loads). Equipped with galvanized steel grate or grating, polyamide or cast iron grate.

FUNCTION : Perpendicular collection across the full width of a drainage surface and routing to a discharge structure

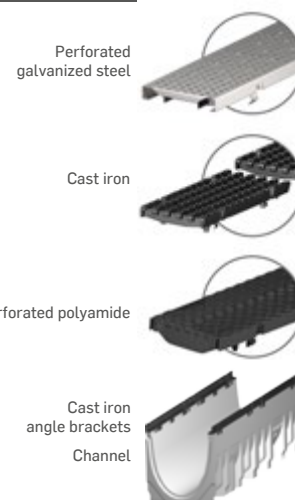
MANUFACTURING STANDARDS : EN 1433 (A, 15 kN / B, 125 kN / C, 250 kN, D, 400 kN, E, 600 kN), CSA S6-06 (CL-625)

RAW MATERIAL : Channel: fiber-reinforced composite concrete
Angle brackets: cast iron
Grates: galvanized steel, perforated polyamide or cast iron

AVAILABLE ACCESSORIES : Side connection well
Channel end plate (with or without outlet)
Vertical channel outlet
Vandal-proof locking system
Riser for side connection well

CONNECTION : See Accessories - Filcoten Infra with 5 mm cast iron angle brackets Data Sheet

APPLICATION : Linear collection



TECHNICAL DATA TABLE 100 mm (4 in) WIDTH CHANNELS

Model	Internal dimensions				Overall dimensions				Length		Weight		Impact resistance		Roughness (int. surface)		Density	
	Width		Height		Width		Height											
	mm	in	mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	MPa-mm	psi-in	µm	in	g/cm³	lb/ft³
100 x 195	100	4	150	5.9	138	5.4	195	7.7	1000	39.4	26.1	17.5	20	114.2	30	0.0012	2.40	150
100 x 245	100	4	200	8.0	138	5.4	245	9.6	1000	39.4	31.2	20.9	20	114.2	30	0.0012	2.40	150

DATA SHEET

FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS (CONT'D)

TECHNICAL DATA TABLE GRATES FOR 100 mm (4 in) WIDTH CHANNELS

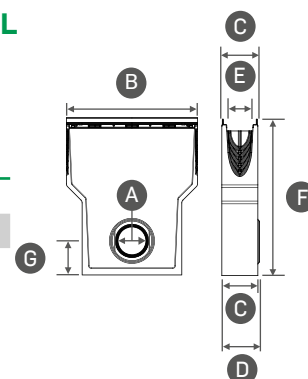
Description	Capacity			Length ⁽¹⁾		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
ADA* perforated galvanized steel grate	C250	8000	1160	1000	39.4	n/a	n/a	Ø 6.0	Ø 0.25	3.3	2.2
Cast iron grate ⁽²⁾	E600	16000	2321	500	39.4	22 x 13	0.9 x 0.5	n/a	n/a	8.4	5.6
ADA* perforated polyamide grate	C250	8000	1160	500	19.7	n/a	n/a	Ø 7.3	Ø 0.29	2.2	1.5

Note 1 : For grates of 500 mm (19.7 in), forecast 2 grates per channel.
 Note 2 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.
 ADA : Conform to *Americans with disabilities Act* (www.ada.gov)
 * : Heel proof grate

TECHNICAL DATA TABLE 100 mm (4 in) WIDTH SIDE CONNECTION WELL

Nominal diam. of the outlet (A)		Length (B)		Width (C)		Max. width (including the outlet) (D)		Nominal width (E)		Height (F)		Distance (outlet center to the bottom of the well) (G)		Weight	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
100	4	500	19.7	138	5.4	145	5.7	100	4	600	24.0	130	5.1	32.9	72.5

Note : An adapter is required for connection to a 100 mm (4 in) HDPE pipe and no adapter is required for connection to a 100 mm (4 in) PVC pipe.



TECHNICAL DATA TABLE GRATES FOR 100 mm (4 in) WIDTH SIDE CONNECTION WELL

Description	Capacity			Length		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
ADA* perforated galvanized steel grate	C250	8000	1160	500	19.7	n/a	n/a	Ø 6.0	Ø 0.25	1.7	3.7
Cast iron grate ⁽¹⁾	E600	16000	2321	500	39.4	22 x 13	0.9 x 0.5	n/a	n/a	8.4	5.6
ADA* perforated polyamide grate	C250	8000	1160	500	19.7	n/a	n/a	Ø 7.3	Ø 0.29	2.2	1.5

Note 1 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.
 ADA : Conform to *Americans with disabilities Act* (www.ada.gov)
 * : Heel proof grate

DATA SHEET

FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS (CONT'D)

TECHNICAL DATA TABLE 150 mm (6 in) WIDTH CHANNELS

Model	Internal dimensions				Overall dimensions				Length		Weight		Impact resistance		Roughness (int. surface)		Density	
	Width		Height		Width		Height											
	mm	in	mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	MPa-mm	psi-in	µm	in	g/cm³	lb/ft³
100 x 210	150	6	165	6.5	188	7.4	210	8.3	1000	39.4	31.6	21.2	20	114.2	30	0.0012	2.40	150

TECHNICAL DATA TABLE GRATES FOR 150 mm (6 in) WIDTH CHANNELS

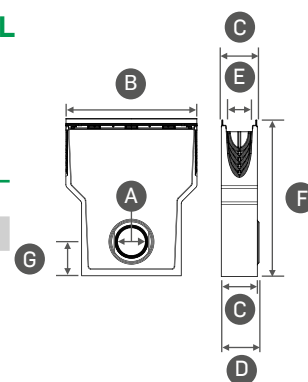
Description	Capacity			Length ⁽¹⁾		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Cast iron grate ⁽²⁾	E600	16000	2321	500	19.7	22 x 13	0.9 x 0.5	n/a	n/a	14,4	9,7

Note 1 : For grates of 500 mm (19.7 in), forecast 2 grates per channel.
Note 2 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.

TECHNICAL DATA TABLE 150 mm (6 in) WIDTH SIDE CONNECTION WELL

Nominal diam. of the outlet (A)		Length (B)		Width (C)		Max. width (including the outlet) (D)		Nominal width (E)		Height (F)		Distance (outlet center to the bottom of the well) (G)		Weight	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
150	6	500	19.7	188	7.4	200	7.9	150	6	690	27.2	180	7.1	45	99.2

Note : An adapter is required for connection to a 150 mm (6 in) HDPE pipe and no adapter is required for connection to a 150 mm (6 in) PVC pipe.



TECHNICAL DATA TABLE GRATES FOR 150 mm (6 in) WIDTH SIDE CONNECTION WELL

Description	Capacity			Length		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Cast iron grate ⁽¹⁾	E600	16000	2321	500	19.7	22 x 13	0.9 x 0.5	n/a	n/a	14.4	9.7

Note 1 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.

DATA SHEET

FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS (CONT'D)

TECHNICAL DATA TABLE 200 mm (8 in) WIDTH CHANNELS

Model	Internal dimensions				Overall dimensions				Length		Weight		Impact resistance		Roughness (int. surface)		Density	
	Width		Height		Width		Height											
	mm	in	mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	MPa-mm	psi-in	µm	in	g/cm³	lb/ft³
100 x 265	200	8	220	8.7	138	5.4	265	10.4	1000	39.4	44.8	30.1	20	114.2	30	0.0012	2.40	150

TECHNICAL DATA TABLE GRATES FOR 200 mm (8 in) WIDTH CHANNELS

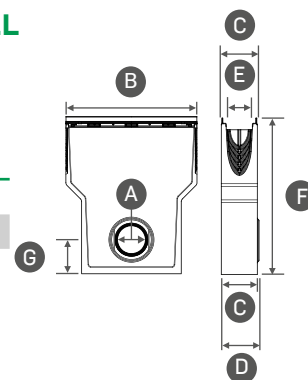
Description	Capacity			Length ⁽¹⁾		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Grille de fonte à barres ⁽²⁾	E600	16000	2321	500	19.7	22 x 13	0.9 x 0.51	n/a	n/a	17.6	11.8

Note 1 : For grates of 500 mm (19.7 in), forecast 2 grates per channel.
Note 2 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.

TECHNICAL DATA TABLE 200 mm (8 in) WIDTH SIDE CONNECTION WELL

Nominal diam. of the outlet (A)		Length (B)		Width (C)		Max. width (including the outlet) (D)		Nominal width (E)		Height (F)		Distance (outlet center to the bottom of the well) (G)		Weight	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
200	8	500	19.7	238	9.4	265	10.4	200	8	790	31.1	135	5.3	48.4	106.7

Note : An adapter is required for connection to a 200 mm (8 in) HDPE pipe and no adapter is required for connection to a 200 mm (8 in) PVC pipe.



TECHNICAL DATA TABLE GRATES FOR 200 mm (8 in) WIDTH SIDE CONNECTION WELL

Description	Capacity			Length		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Grille de fonte à barres ⁽¹⁾	E600	16000	2321	500	19.7	22 x 13	0.9 x 0.51	n/a	n/a	17.6	11.8

Note 1 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.

DATA SHEET

FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS (CONT'D)

TECHNICAL DATA TABLE 300 mm (12 in) WIDTH CHANNELS

Model	Internal dimensions				Overall dimensions				Length		Weight		Impact resistance		Roughness (int. surface)		Density	
	Width		Height		Width		Height											
	mm	in	mm	in	mm	in	mm	in	mm	in	kg/m	lb/ft	MPa-mm	psi-in	µm	in	g/cm³	lb/ft³
100 x 360	300	12	300	12	360	14.2	360	14.2	1000	39.4	77.3	51.9	20	114.2	30	0.0012	2.40	150

TECHNICAL DATA TABLE GRATES FOR 300 mm (12 in) WIDTH CHANNELS

Description	Capacity			Length ⁽¹⁾		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Cast iron grate ⁽²⁾	E600	16000	2321	500	19.7	29 x 13	1.1 x 0.5	n/a	n/a	35.8	24.0

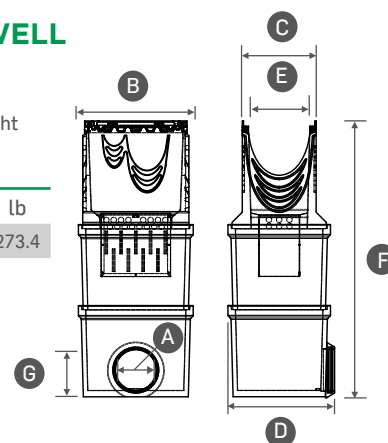
Note 1 : For grates of 500 mm (19.7 in), forecast 2 grates per channel.

Note 2 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.

TECHNICAL DATA TABLE 300 mm (12 in) WIDTH SIDE CONNECTION WELL

Nominal diam. of the outlet (A)		Length (B)		Width (C)		Max. width (including the outlet) (D)		Nominal width (E)		Height (F)		Distance (outlet center to the bottom of the well) (G)		Weight	
mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	kg	lb
300	12	560	22.0	360	14.2	495	19.5	300	12	950	37.4	130	5.1	124	273.4

Note : An adapter is required for connection to a 200 mm (8 in) HDPE pipe and no adapter is required for connection to a 200 mm (8 in) PVC pipe.



TECHNICAL DATA TABLE GRATES FOR 300 mm (12 in) WIDTH SIDE CONNECTION WELL

Description	Capacity			Length		Openings		Dia. of openings		Grate weight	
	kN	kPa	psi	mm	in	mm x mm	in x in	mm	in	kg/m	lb/ft
Cast iron grate ⁽¹⁾	E600	16000	2321	500	19.7	29 x 13	1.1 x 0.5	n/a	n/a	35.8	24.0

Note 1 : Cast iron grate to be bolted, 4 nuts and bolts required per 0.5 meter.

SPECIFICATIONS

FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS

SCOPE

These specifications apply to all Filcoten Infra channels with 5 mm (0.2 in) cast iron angle brackets, used to collect and discharge storm water.

CHANNEL REQUIREMENTS

Each channel consists of a grate attached within cast iron angle brackets which are fitted on a channel made of fiber-reinforced composite concrete.

- **Grates :**
 - *ADA perforated galvanized steel grate 100 mm (4 in)* : the grate has openings 6 mm (0.2 in) in diameter. It can support a load of 250 kN (56 200 lb) (class A, according to EN 1433 standard). Heelproof grate.
 - *Cast iron grate 100 mm (4 in), 150 mm (6 in) and 200 mm (8 in)* : the grate has 13 x 22 mm (0.5 x 0.9 in) openings. It can support a load of 600 kN (134 885 lb) (class E according to EN 1433).
 - *Cast iron grate 300 mm (12 in)* : the grate has 13 x 29 mm (0.5 x 1.2 in) openings. It can support a load of 600 kN (134 885 lb) (class E according to EN 1433).
 - *ADA perforated polyamide grate 100 mm (4 in)*: the grate has openings 7.3 mm (0.29 in) in diameter. It can support a load of 250 kN (56 200 lb) (class C according to EN 1433). It is equipped with 4 pressure fasteners, to prevent lifting. Heelproof grate.
- **Angle brackets :**
 - The side angle brackets are fitted into the upper part of the channel, are part of the channel body and have a minimum thickness of 5 mm (0.2 in).
 - In total, the angle brackets have 8 anchoring stops per channel section.
- **Channel :**
 - The channel can be installed within the asphalt, concrete or paving stones.
 - Manufacture of the channel should require no oily or greasy stripping agent, in order to maximize adhesion between the channel and the concrete filling. Such a precaution will eliminate the risk of water infiltration between the fill concrete and the body of the channel.
 - The outer upper part of the channel is perfectly smooth, which allows a perfect fit between the channel and the finishing material.
 - The guide stops at the channel ends allow proper alignment.
 - The minimum wall thickness of the channel shall be 19 mm (0.75 in).
 - Each section has a net length of 1 m (39.4 in).
 - The minimum impact resistance is 20 MPa-mm (114 psi-in).
 - The channel is certified A+ by the IBR Institute, which ensures that it emits no substance harmful to human health and can be installed indoors.

SPECIFICATIONS

FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS (CONT'D)

RAW MATERIAL

- **Grate:**
 - The slotted galvanized steel grate and the ADA perforated galvanized steel grate are pre-galvanized in accordance with EN 10142, 10214 and 10215 standards.
 - The galvanized steel grating is hot galvanized in accordance with the ISO 1461 standard, with a minimum basis mass per unit area of 275 g/m² (0.00039 psi).
 - The cast iron grate and ADA cast iron grate are made of ductile iron in accordance with EN-GJS-500-7 standard.
 - The ADA perforated polyamide grate is made of polyamide with a fiberglass honeycomb structure. It is resistant to UV rays, frost and heat.
- **Angle brackets :**
 - The brackets are made from ductile cast iron, in accordance with the EN-GJS-500-7 standard.
- **Channel :**
 - The channel is made of composite fiber-reinforced concrete, with a minimum compressive strength of 80 MPa (11 600 psi), in accordance with EN-206-1 standard.
 - The composite concrete must be certified in accordance with CE EN 1433 standard to -20 °C (-4°F) as well as 40 °C (104 °F), which provides excellent resistance to freeze/thaw cycles as well as de-icing salts.
 - The structure of the channel must be resistant to UV.
 - The channel must be Class A1 non-flammable, in accordance with DIN 4102 standard.

TRACEABILITY AND IDENTIFICATION

Each section includes product identification, a bar code, the direction of flow and the date of manufacture.

ACCESSORIES

- The side connection well includes an outlet with integrated seal, which prevents the leakage of fine particles.
- The silt bucket is installed in its upper part, which prevents blocking of the outlet.
- No mounting accessory internal to the channel that would impede the surface flow is allowed.
- Optional: Vandal-proof lock.
- Channel end plate without outlet.
- Channel end plate with outlet which allows the longitudinal connection of a PVC or HDPE pipe
- Vertical channel outlet.
- Riser for 300 mm (12 in) side connection well.

DIMENSIONS AND QUANTITIES

Dimensions and quantities shall be consistent with the bid documentation and plans.

INSTALLATION

Installation will be performed in accordance with manufacturer's recommendations. Contact your local Soleno representative or visit our website at www.soleno.com to read our recommendations for installation.

LEGEND

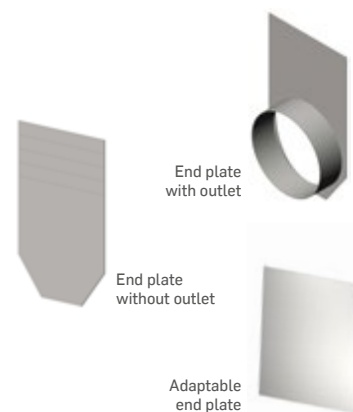
ADA: Americans with disabilities Act

DATA SHEET

ACCESSORIES FOR FILCOTEN INFRA WITH 5 mm CAST IRON ANGLE BRACKETS

TECHNICAL DATA TABLE CHANNEL END PLATES

For channel (width)		With outlet	Width		Outlet diameter	
mm	in		mm	in	mm	in
100	4	X	100	4	100	4
100	4		100	4	n/a	n/a
150	6	X	150	6	150	6
150	6		150	6	n/a	n/a
200	8	X	200	8	200	8
200	8		200	8	n/a	n/a
300	12	X	300	12	200	8
300	12		300	12	n/a	n/a
300	12	See Note 1	300	12	n/a	n/a



Note 1 : Adapter for connection to 100 mm (4 in) to 200 mm (8 in) channel.

Note 2 : An adapter is required for connection to a HDPE pipe of 100 mm (4 in) to 200 mm (8 in) and no adapter is required for connection to a PVC pipe.

Note 3 : The PVC connection must be made from the female end of the pipe.

Note 4 : The adaptable end plate allows passage from a 300 mm (12 in) channel to a 100 mm (4 in) or 150 mm (6 in) channel.

TECHNICAL DATA TABLE VERTICAL CHANNEL OUTLET

For channel (width)		Outlet diameter	
mm	in	mm	in
100 ⁽¹⁾	4	100	4
150 ⁽²⁾	6	150	6
200 ⁽³⁾	8	200	8
300 ⁽⁴⁾	12	200	8



Model for channel of
100, 150 and 200 mm
(4, 6 and 8 in)

Note 1 : No adapter is required for connection to a 100 mm (4 in) PVC and/or PEHD pipe.

Note 2 : An adapter is required for connection to a 100 mm (4 in) HDPE pipe and no adapter is required for connection to a 100 mm (4 in) PVC.

Note 3 : An adapter is required for connection to a 150 mm (6 in) HDPE pipe and no adapter is required for connection to a 150 mm (6 in) PVC.

Note 4 : An adapter is required for connection to a 200 mm (8 in) HDPE pipe and no adapter is required for connection to a 200 mm (8 in) PVC.

Note 5 : The PVC connection must be made from the female end of the pipe.

VANDAL-PROOF LOCK (optional)



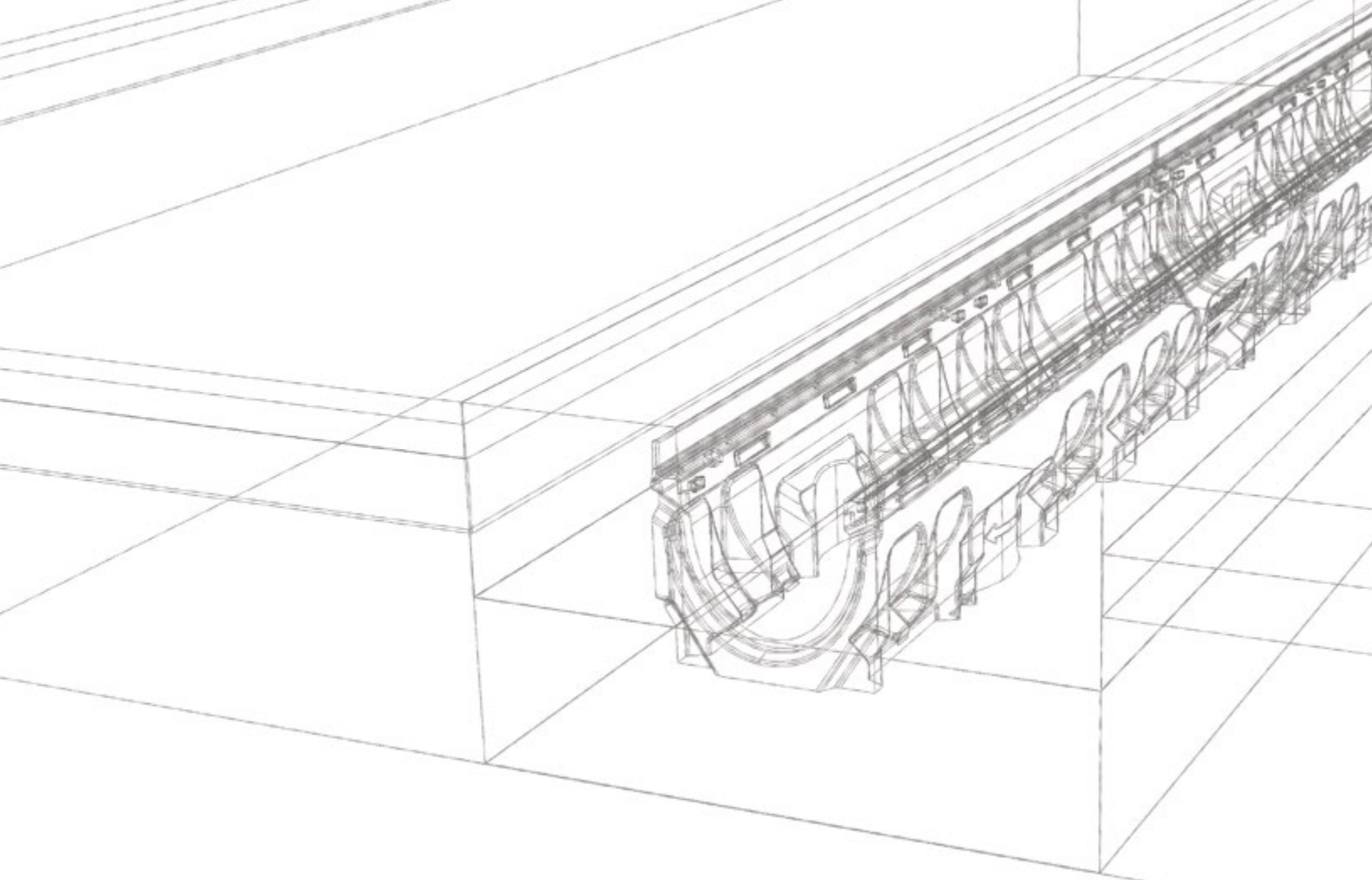
Note 1 : Compatible with galvanized steel grating, perforated galvanized steel grate and cast iron grate (optional).

Note 2 : Two units may be used per grate, but only one is required.

RISER FOR SIDE CONNECTION WELL



Note 1 : Compatible only with the 300 mm (12 in) side connection well.



FILCOTEN[®] INFRA

INSTALLATION GUIDE

Filcoten Infra 4 and 5 mm

INSTALLATION

1. Drainage outlets
2. Installation
3. Installing the vandal-proof lock (optionnal)

Table 1: Trench dimensions

Figure 1: Typical trenches

Figure 2: Positioning the expansion joints

Figure 3: Tees and elbows (optional)

STEP 1

DRILLING THE DRAINAGE OUTLETS

DRILLING THE SIDE CONNEXION WELL OPENING (IF REQUIRED)

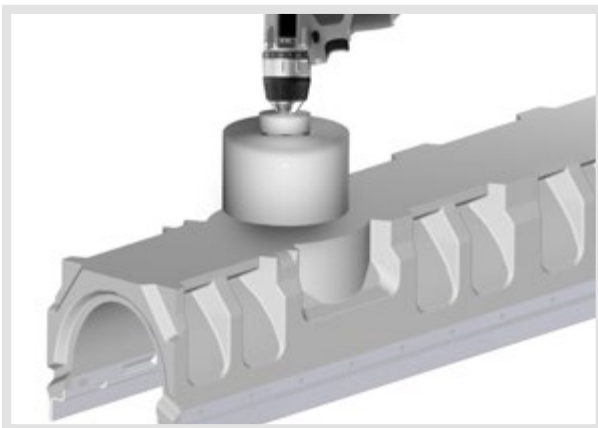


- 1a. Using a concrete drill bit, drill small holes following the guide which is appropriate for your installation.

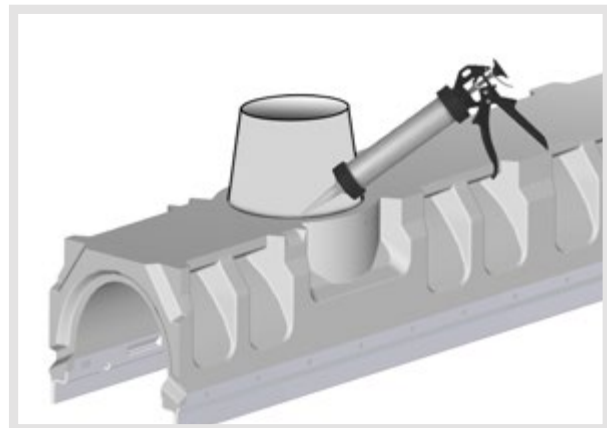


- 1b. Using a cold chisel, clear the side connection well opening.

AND/OR DRILLING THE VERTICAL CHANNEL OUTLET (IF REQUIRED)



- 1a. Using a hole saw, drill an opening in order to install the vertical channel outlet.



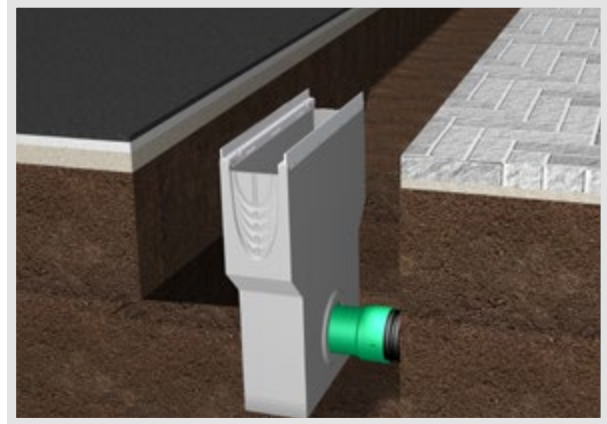
- 1b. Insert the vertical channel outlet above the opening and seal.

STEP 2

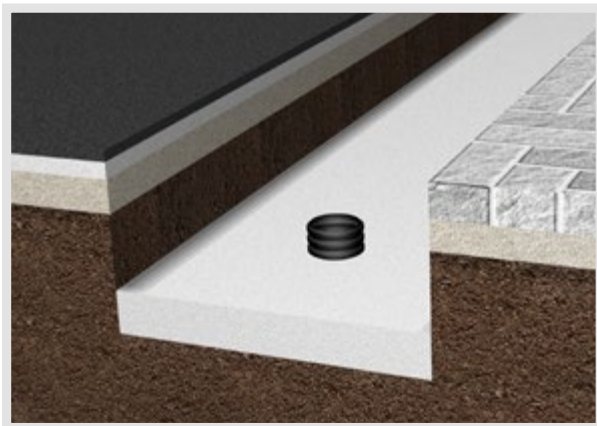
INSTALLATION



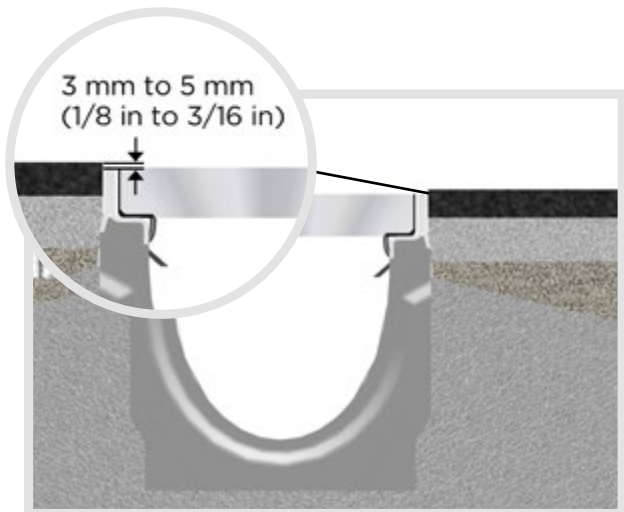
- 2a. Dig the trench in accordance with Table 1.



- 2b. Determine the outlet type: lateral for side connection well, vertical for channel or horizontal for the end of channel with end plate with outlet used.

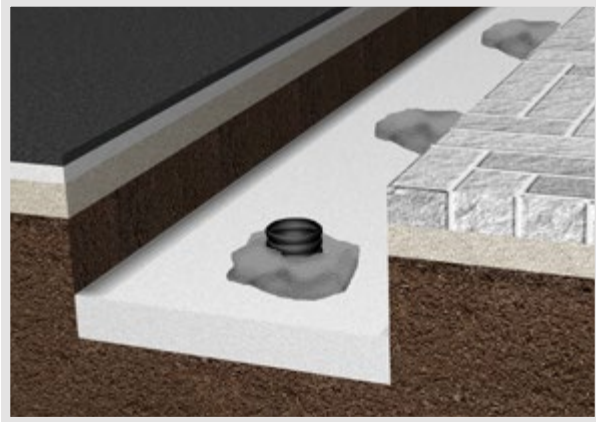


- 2c. Pour a concrete slab if the clearance between the base of the channel and the bottom of the trench exceeds 125 mm (5 in) and let it harden.

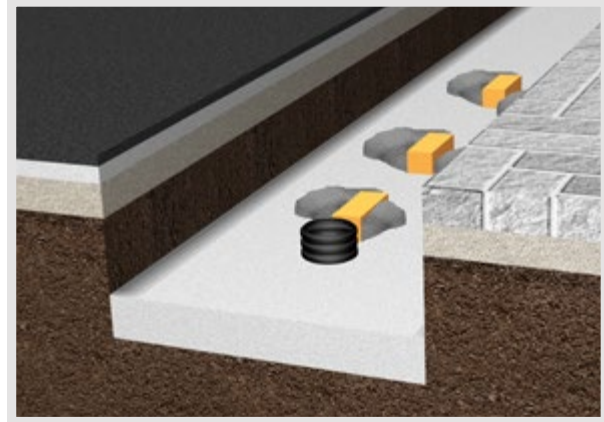


- 2d. Determine the final elevation level of the top of the corner bracket using a chalk line or a laser level. The final elevation level must be 3 mm to 5 mm (1/8 in to 3/16 in) lower than the finished surface.

2e. Preparation for levelling: Two suggested methods

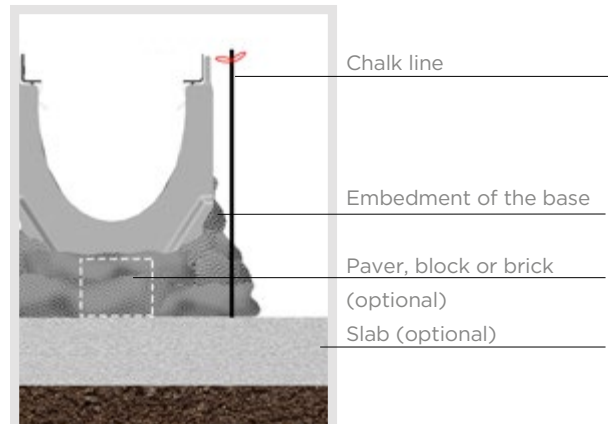


Method #1:
Place clumps of concrete having a consistent texture against the channel joints.

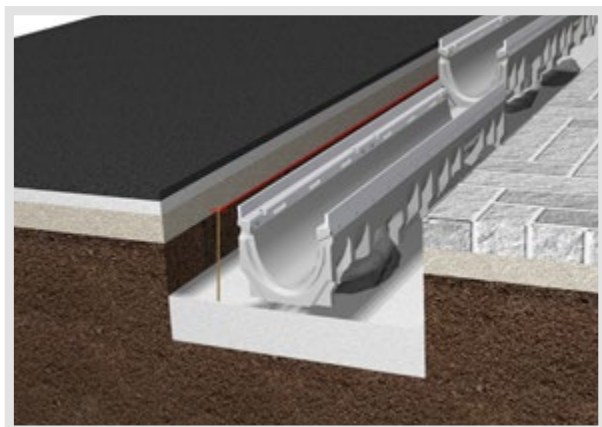


or Method #2:
Add concrete pavers or bricks against the channel joints. Embed these leveling blocks in concrete having a consistent texture.

2f. Installation and levelling



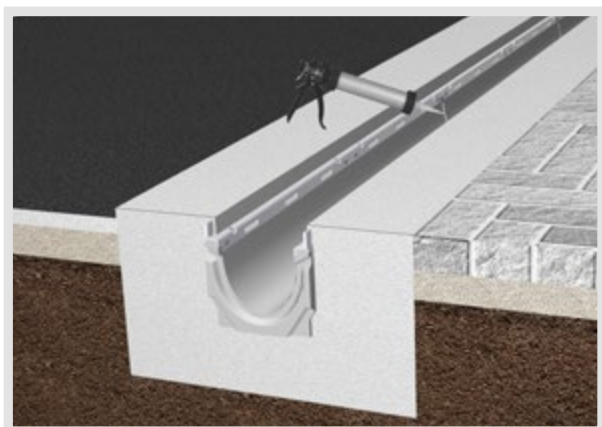
Place the first drainage channel by pushing its base in the concrete clump, until the angle corner reaches the desired level.



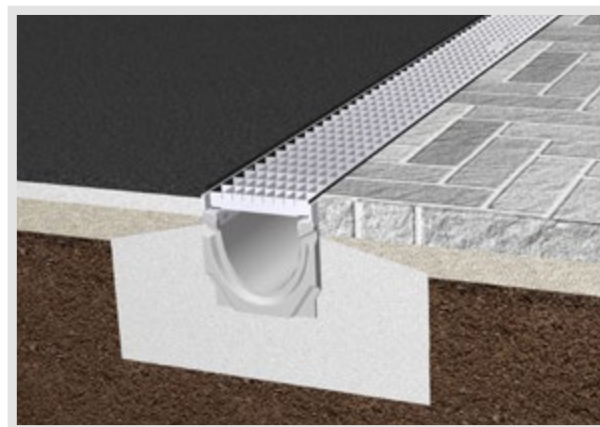
- 2g. Install the other drainage channels. Let the concrete embedment harden at the base of the channels prior to pouring the final concrete.



- 2h. Apply a concrete sealer at the joints between each channel.



- 2i. It is essential that the grates be installed in the drainage channels and covered with a plastic film or protective tape during the concrete pour. Backfill the bottom and sides of the trench, in accordance with Figure 1. Take care to spread the concrete evenly.



- 2j. Unpack the grates and reinstall them. Add the finish coating, where applicable.

STEP 3

INSTALLING THE VANDAL-PROOF LOCK (OPTIONAL)



3a. Insert the vandal-proof lock under the grate.



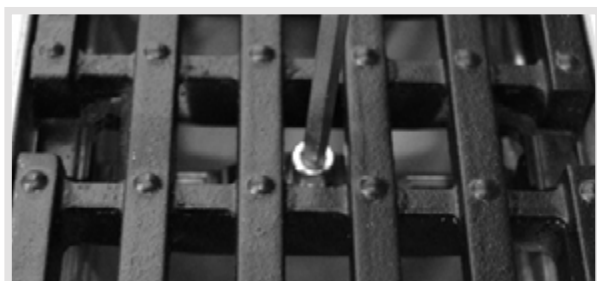
3b. Press on the lock until it fits completely between the clips.



3c. Insert the grate into the channel.



3d. Press the grate until the clips reach the recesses inside the channel.



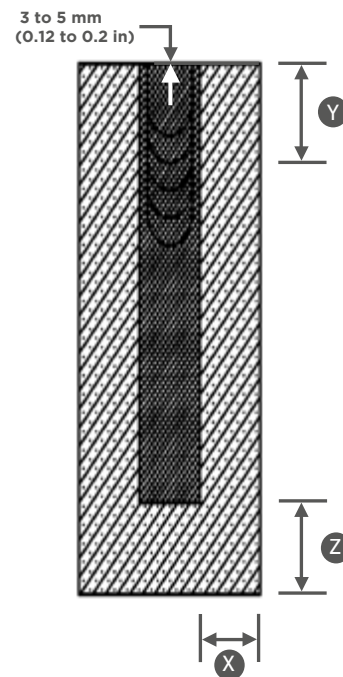
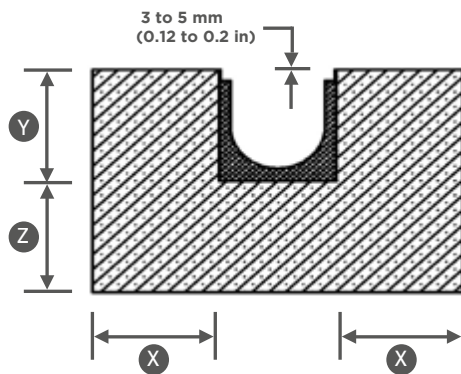
3e. Using a 5-mm Allen wrench, tighten the screw to expand the lock and thus prevent removal of the grate.

TABLE 1

TABLE TRENCH DIMENSIONS ACCORDING TO THE LOAD CLASSES OF THE INSTALLATION*

	Load Classes																	
	A 15 kN			B 125 kN			C 250 kN			D 400 kN			E 600 kN			F 900 kN		
Channel width (mm)	100	150	300	100	150	300	100	150	300	100	150	300	100	150	300	100	150	300
X _{min} (mm)	100	150	200	100	150	200	150	200	250	150	200*	300	200	250	300	200	250	300
Z _{min} (mm)	100	150	200	100	150	200	150	200	250	150	200*	300	200	250	300	200	250	300
Y _{min} (mm)	Total height of the channel																	
Re-bar	Not required															Treillis of 10 m 200 mm x 200 mm		

Note : For channel with cast iron angle brackets, forecast 250 mm (10 in) of concrete instead of 200 mm (8 in).



*The load class of the installation is governed by the lowest component. For example, an installation of a Class F channel with a Class C grate will have a global load Class C.

FIGURE 1
TYPICAL FILCOTEN INFRA TRENCHES 4 mm AND 5 mm

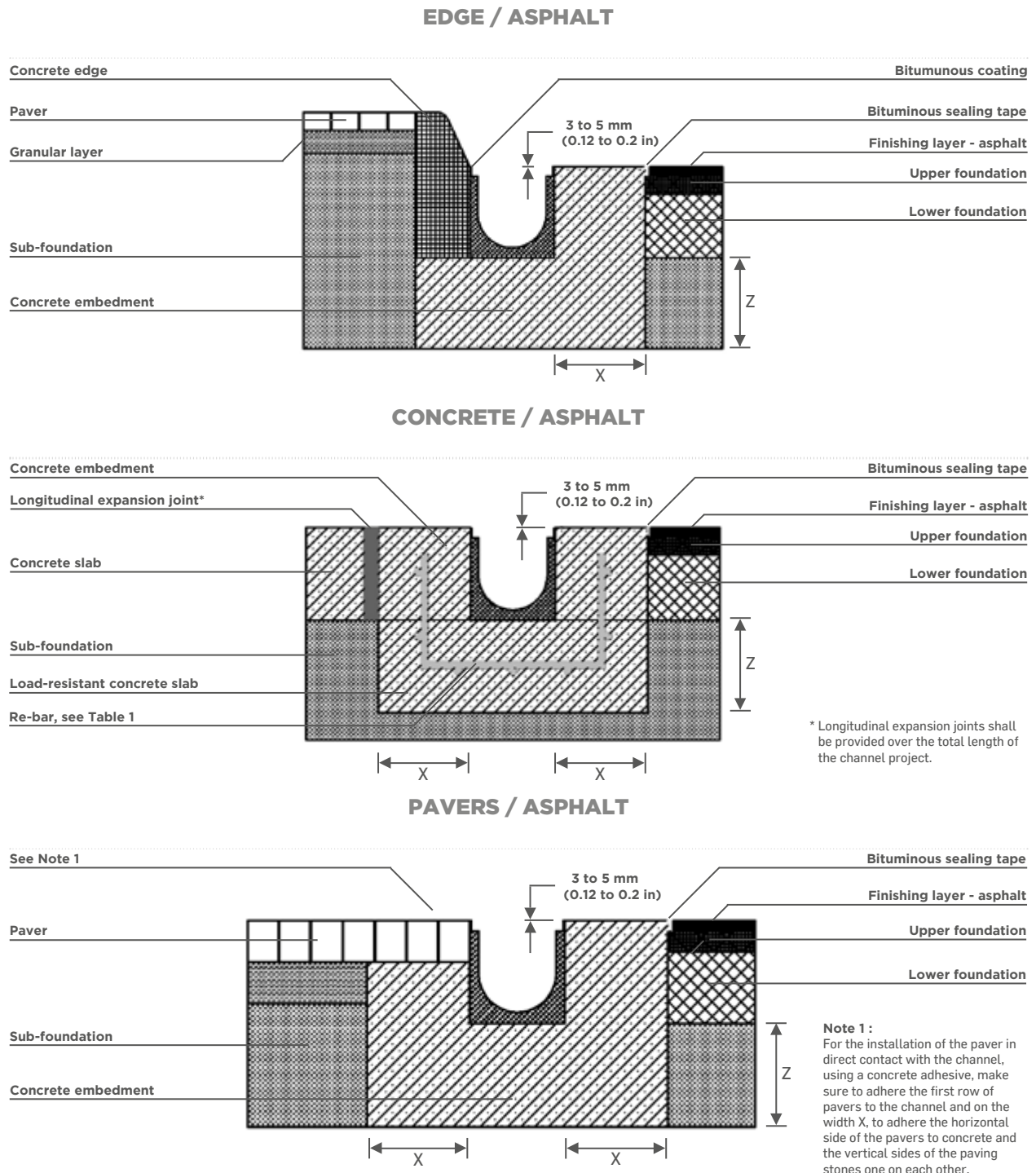
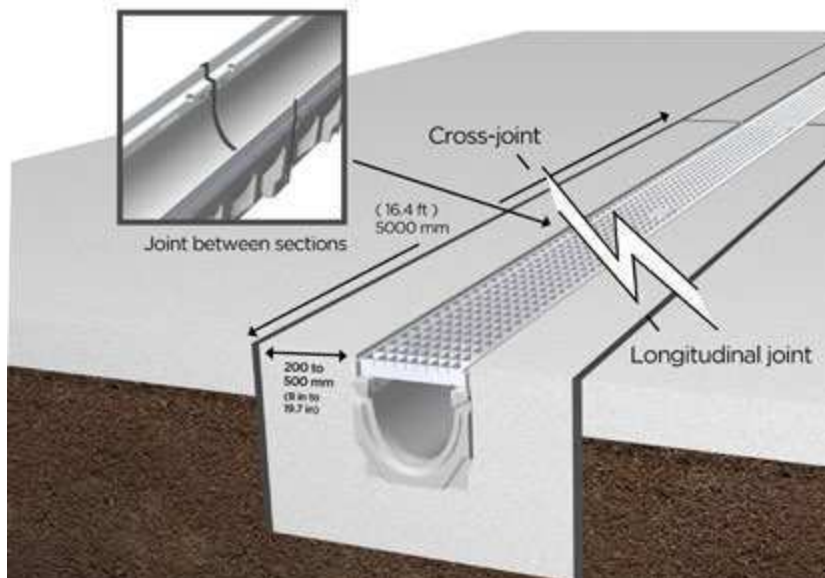


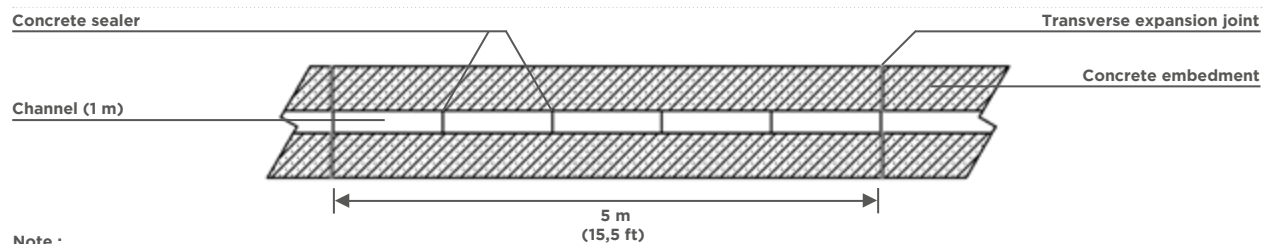
FIGURE 2
POSITIONING THE EXPANSION JOINTS



LONGITUDINAL EXPANSION JOINTS SHALL BE PROVIDED OVER THE TOTAL LENGTH OF THE CHANNEL PROJECT.

THE TRANSVERSE JOINTS SHOULD BE INSTALLED EVERY 5 METERS WHEN THE CONCRETE IS IN DIRECT CONTACT WITH THE CHANNELS.

TRANSVERSE EXPANSION JOINT
INSTALLED WHEN CONCRETE IS IN CONTACT WITH THE CHANNEL



Note :
A transverse expansion joint must be provided every 5 meters at the junction between two channels.

FIGURE 3
TEES AND ELBOWS (OPTIONAL)



ON SITE, IT IS POSSIBLE TO MAKE TEES AND ELBOWS, IF REQUIRED. CONTACT YOUR SOLENO REPRESENTATIVE FOR QUESTIONS REGARDING YOUR PROJECT.

DATA SHEET

TX-70

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Filtration and separation

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	0.9 mm	36 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	445 N	100 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 105 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	190 N	43 lb
	Puncture strength (CBR)	ASTM D6241	1250 N	281 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	1170 kPa	170 psi
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.25 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	2.05 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	75 - 140 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	150.0 m	492.0 ft

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APPLICATIONS: Permanent public roadway
Slab and building foundation
Drainage trench
Retaining walls and embankments

DATA SHEET

TX-80

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Filtration and separation

PRODUCT CERTIFICATION: standard GCTTG 3001-06/2006, using protocol BNQ 7009-910/07 and the single identifier 0140

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness ⁽¹⁾	ASTM D5199	1.0 mm	40 mils
	Mass per unit area ⁽¹⁾	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	470 N	106 lb
	Elongation at break SP	CAN/CGSB-148.1 - No 7.3	56 – 84 %	
	Elongation at break ST	CAN/CGSB-148.1 - No 7.3	60 – 90 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	220 N	49 lb
	Puncture strength (CBR) ⁽¹⁾	ASTM D6241	1510 N	339 lb
HYDRAULIC	Burst strength	CAN/CGSB-4.2 - No 11.1	1350 kPa	196 psi
	Permeability ⁽¹⁾	CAN/CGSB-148.1 - No 4	0.24 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	1.54 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	70 - 130 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours) ⁽¹⁾	ASTM D4355	50 %	
DIMENSIONS	Standard width ⁽¹⁾	-	3.5 m	11.5 ft
	Standard length ⁽¹⁾	-	150.0 m	492.0 ft

Note 1: Property not part of the product certification by the BNQ

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APPLICATIONS: Temporary roadway
Slab and building foundation
Drainage trench
Residential parking areas
Retaining walls, embankments and berms

DATA SHEET

TX-90

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTION: Separation

PRODUCT CERTIFICATION: standard GCTTG 3001-06/2006, using protocol BNQ 7009-910/07
and the single identifier 0141

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness ⁽¹⁾	ASTM D5199	1.0 mm	40 mils
	Mass per unit area ⁽¹⁾	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	555 N	125 lb
	Elongation at break SP	CAN/CGSB-148.1 - No 7.3	55 – 85 %	
	Elongation at break ST	CAN/CGSB-148.1 - No 7.3	60 – 90 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	255 N	57 lb
	Puncture strength (CBR) ⁽¹⁾	ASTM D6241	1650 N	371 lb
HYDRAULIC	Burst strength	CAN/CGSB-4.2 - No 11.1	1585 kPa	230 psi
	Permeability ⁽¹⁾	CAN/CGSB-148.1 - No 4	0.23 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	1.34 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	63 - 117 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours) ⁽¹⁾	ASTM D4355	50 %	
DIMENSIONS	Standard width ⁽¹⁾	-	3.5 m	11.5 ft
	Standard length ⁽¹⁾	-	150.0 m	492.0 ft

Note 1: Property not part of the product certification by the BNQ

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road
Commercial parking areas

Storage and loading areas
Roadway widening
Airport runways
Basin
Residential parking areas

DATA SHEET

TX-110

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTION: Filtration

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	2.3 mm	92 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	670 N	151 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 100 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	325 N	73 lb
	Puncture strength (CBR)	ASTM D6241	2170 N	488 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	2170 kPa	315 psi
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.42 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	1.42 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	70 - 115 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	100.0 m	328.0 ft

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APPLICATIONS: Slab and building foundation
Drainage trench

DATA SHEET

TX-170

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTION: Separation

PRODUCT CERTIFICATION: GCTTG 3001-06/2006 standard, using protocol BNQ 7009-910/07 and the single identifier 0142

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness ⁽¹⁾	ASTM D5199	1.3 mm	52 mils
	Mass per unit area ⁽¹⁾	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	805 N	181 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	60 – 90 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	365 N	82 lb
	Puncture strength (CBR) ⁽¹⁾	ASTM D6241	2450 N	551 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	2275 kPa	330 psi
HYDRAULIC	Permeability ⁽¹⁾	CAN/CGSB-148.1 - No 4	0.20 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.96 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	49 - 91 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours) ⁽¹⁾	ASTM D4355	50 %	
DIMENSIONS	Standard width ⁽¹⁾	-	3.5 m	11.5 ft
	Standard length ⁽¹⁾	-	100.0 m	328.0 ft

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road
Commercial parking areas

Storage and loading areas
Roadway widening
Airport runways
Basins

DATA SHEET

TX-200

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Separation, reinforcement and protection

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	1.7 mm	68 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	875 N	197 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 110 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	360 N	81 lb
	Puncture strength (CBR)	ASTM D6241	2400 N	540 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	2400 kPa	348 psi
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.43 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	1.00 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	50 - 100 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	100.0 m	328.0 ft

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road
Commercial parking areas
Storage and loading areas
Roadway widening

Airport runways
Railway tracks
Basins
Retaining walls and embankments
Bank protection
Dams and dikes

DATA SHEET

TX-200B

PRODUCT DESCRIPTION: Absorbent nonwoven needled geotextile used for curing concrete

FUNCTION: Concrete curing

PRODUCT STANDARD: Meets the 3501 standard of the Quebec Ministry of Transportation (MTQ)
Volume VII, Chapter 3, Concrete and related products, 3.5 Curing materials

RAW MATERIAL: Polypropylene and polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Mass per unit area	ASTM D5261	300 g/m ²	8,8 oz/yd ²
	Standard width	-	3.5 m	11.5 ft
DIMENSIONS	Widths are available with seams	-	7.0 m	23.0 ft
			10.5 m	34.4 ft
	Standard length	-	100.0 m	328.0 ft

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APPLICATIONS: Permanent public roadway
Airport runways
Slab and building foundation

DATA SHEET

TX-300

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Separation, reinforcement and protection

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	2.6 mm	104 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	1050 N	236 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 110 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	460 N	103 lb
	Puncture strength (CBR)	ASTM D6241	2930 N	659 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	2910 kPa	422 psi
	Puncture resistance	ASTM D4833	580 N	130 lb
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.24 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.90 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	50 - 100 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	100.0 m	328.0 ft

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APPLICATIONS: Railway tracks
Basins
Retaining walls and embankments
Bank protection
Dams and dikes

DATA SHEET

TX-400

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Separation, reinforcement and protection

RAW MATERIAL: Polypropylene / polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	2.8 mm	112 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	1200 N	270 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 110 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	530 N	119 lb
	Puncture strength (CBR)	ASTM D6241	3350 N	753 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	3300 kPa	479 psi
	Puncture resistance	ASTM D4833	665 N	149 lb
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.19 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.80 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	50 - 90 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	100.0 m	328.0 ft

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APPLICATIONS: Railway tracks
Basins
Retaining walls and embankments
Bank protection
Dams and dikes

DATA SHEET

TX-800

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Separation, reinforcement and protection

RAW MATERIAL: Polypropylene / polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	3.5 mm	140 mils
	Mass per unit area	ASTM D5261	420 g/m ²	12.39 oz/yd ²
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	1470 N	330 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	70 – 110 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	600 N	135 lb
	Puncture strength (CBR)	ASTM D6241	4000 N	899 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	3500 kPa	508 psi
	Puncture resistance	ASTM D4833	850 N	191 lb
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.25 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.41 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	40 - 110 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	100.0 m	328.0 ft

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APPLICATIONS: Railway tracks
Basins
Retaining walls and embankments
Bank protection
Dams and dikes

DATA SHEET

TX-1600

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Separation, reinforcement and protection

RAW MATERIALS: Polypropylene / polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	5.8 mm	232 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	2500 N	562 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	60 – 120 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	1050 N	236 lb
	Puncture strength (CBR)	ASTM D6241	7000 N	1574 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	7000 kPa	1015 psi
	Puncture resistance	ASTM D4833	1250 N	281 lb
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.17 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.27 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	40 - 70 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	50.0 m	164.0 ft

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APPLICATIONS: Railway tracks
Basins
Bank protection
Dams and dikes

DATA SHEET

TX-2000

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Separation, reinforcement and protection

RAW MATERIALS: Polypropylene / polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	6.5 mm	260 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	3300 N	742 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 110 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	1350 N	303 lb
	Puncture strength (CBR)	ASTM D6241	10000 N	2249 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	10000 kPa	1450 psi
	Puncture resistance	ASTM D4833	1750 N	393 lb
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.15 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.20 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	50 - 80 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	50.0 m	164.0 ft

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APPLICATIONS: Railway tracks
Basins
Bank protection
Dams and dikes

DATA SHEET

TX-F

PRODUCT DESCRIPTION: Nonwoven needled geotextile mainly used in forestry

FUNCTION: Reinforcement

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	1.5 mm	60 mils
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	1000 N	225 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 100 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	350 N	79 lb
	Bursting strength	CAN/CGSB-4.2 - No 11.1	2600 kPa	377 psi
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.22 cm/s	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	50 - 80 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width	-	3.5 m	11.5 ft
	Standard length	-	100.0 m	328.0 ft

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APPLICATION: Temporary roadway

DATA SHEET

ROUTE X II

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTION: Reinforcement

PRODUCT CERTIFICATION: GCTTG 3001-06/2006 standard, using protocol BNQ 7009-910/07 and the single identifier 0138

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness ⁽¹⁾	ASTM D5199	0.8 mm	32 mils
	Mass per unit area ⁽¹⁾	ASTM D5261	225 g/m ²	6.6 oz/yd ²
MECHANICAL	Wide-width strip method testing ⁽¹⁾	ASTM D4595	11 kN/m	62.8 lb/in
	Wide-width strip method at 5 % elongation ⁽¹⁾	ASTM D4595	5 kN/m	28.5 lb/in
	Puncture strength (CBR) ⁽¹⁾	ASTM D6241	1590 N	357 lb
HYDRAULIC	Permeability ⁽¹⁾	CAN/CGSB-148.1 - No 4	0.02 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.05 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	56 - 104 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width ⁽¹⁾	-	3.5 m	11.5 ft
	Standard length ⁽¹⁾	-	100.0 m	328.0 ft

Note 1: Properties not part of the product certification by the BNQ

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APPLICATIONS: Permanent public roadway
Roadway widening

DATA SHEET

ROUTE III-IV

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Filtration and separation

PRODUCT CERTIFICATION: GCTTG 3001-06/2006 standard. using protocol BNQ 7009-910/07
and the single identifier 0133

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness ⁽¹⁾	ASTM D5199	0.9 mm	36 mils
	Mass per unit area ⁽¹⁾	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	420 N	94 lb
	Elongation at break SP	CAN/CGSB-148.1 - No 7.3	60 - 90 %	
	Elongation at break ST	CAN/CGSB-148.1 - No 7.3	64 - 96 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	170 N	38 lb
	Puncture strength (CBR) ⁽¹⁾	ASTM D6241	1250 N	281 lb
	Bursting strength	CAN/CGSB-4.2 - No 11.1	1100 kPa	160 psi
HYDRAULIC	Permeability ⁽¹⁾	CAN/CGSB-148.1 - No 4	0.15 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	1.23 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	77 - 143 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours) ⁽¹⁾	ASTM D4355	50 %	
DIMENSIONS	Standard width ⁽¹⁾	-	3.5 m	11.5 ft
	Standard length ⁽¹⁾	-	150.0 m	492.0 ft

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APPLICATIONS: Permanent public roadway
Roadway widening
Drainage trench
Retaining walls and embankments

DATA SHEET

ROUTE X V

PRODUCT DESCRIPTION: Nonwoven needled geotextile

FUNCTIONS: Separation and protection

PRODUCT CERTIFICATION: GCTTG 3001-06/2006 standard, using protocol BNQ 7009-910/07
and the single identifier: 0144

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness ⁽¹⁾	ASTM D5199	1.7 mm	68 mils
	Mass per unit area ⁽¹⁾	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	1050 N	234 lb
	Elongation at break SP	CAN/CGSB-148.1 - No 7.3	68 - 102 %	
	Elongation at break ST	CAN/CGSB-148.1 - No 7.3	64 - 96 %	
	Puncture strength (CBR) ⁽¹⁾	ASTM D6241	2930 N	659 lb
	Puncture resistance ⁽¹⁾	ASTM D4833	580 kPa	130 psi
HYDRAULIC	Permeability ⁽¹⁾	CAN/CGSB-148.1 - No 4	0.20 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.70 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	49 - 91 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours) ⁽¹⁾	ASTM D4355	50 %	
DIMENSIONS	Standard width ⁽¹⁾	-	3.5 m	11.5 ft
	Standard length ⁽¹⁾	-	100.0 m	328.0 ft

Note 1: Property not part of the product certification by the BNQ

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APPLICATIONS: Basins
Dams and dikes
Bank protection

DATA SHEET

TX-R

PRODUCT DESCRIPTION: Geocomposite combining a nonwoven needled geotextile reinforced with a woven geotextile

FUNCTION: Reinforcement

PRODUCT CERTIFICATION: GCTTG 3001-06/2006 standard, using protocol BNQ 7009-910/07 and the single identifier 0143

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness ⁽¹⁾	ASTM D5199	1.7 mm	68 mils
	Mass per unit area ⁽¹⁾	ASTM D5261	355 g/m ²	10.5 oz/yd ²
	Wide-width strip method testing ⁽¹⁾	ASTM D4595	13.5 kN/m	77 lb/in
MECHANICAL	Wide-width strip method at 5 % elongation ⁽¹⁾	ASTM D4595	6 kN/m	34.2 lb/in
	Tensile strength	CAN/CGSB-148.1 - No 7.3	650 N	146 lb
	Puncture strength (CBR) ⁽¹⁾	ASTM D6241	2800 N	629 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	2600 kPa	377 psi
	Permeability ⁽¹⁾	CAN/CGSB-148.1 - No 4	0.01 cm/s	
HYDRAULIC	Permittivity	CAN/CGSB-148.1 - No 4	0.05 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	40 - 68 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
DIMENSIONS	Standard width ⁽¹⁾	-	3.5 m	11.5 ft
	Standard length ⁽¹⁾	-	100.0 m	328.0 ft

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road
Commercial parking areas

Storage and loading areas
Roadway widening
Airport runways
Railway tracks

DATA SHEET

TX-MP

PRODUCT DESCRIPTION: Nonwoven needled geotextile, laminated with a PVC geomembrane

FUNCTION: Protection

RAW MATERIAL: Polypropylene, polyester and PVC

TECHNICAL DATA TABLE - GEOMEMBRANE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	0.5 mm	20 mils
	Resistance at low temperatures	ASTM D1790	-26 °C	-14.8 °F
	Dimensional stability	ASTM D1204	5 %	
MECHANICAL	Tensile strength	ASTM D882	182 N	41 lb
	Elongation at break	ASTM D882	400 %	
	Tear strength	ASTM D1004	22 N	5 lb

TECHNICAL DATA TABLE - GEOTEXTILE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	4.5 ± 0.5 mm	180 ± 20 mils
	Mass per unit area	ASTM D5261	450 g/m ²	13.2 ± 1.5 oz/yd ²

TECHNICAL DATA TABLE - COMPOSITE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	5.0 ± 0.5 mm	200 ± 20 mils
	Mass per unit area	ASTM D5261	1090 ± 65 g/m ²	32 ± 1.9 oz/yd ²
HYDRAULIC	Transmissivity under 8 kPa at a gradient = 1	ASTM D4716	6 x 10 ⁻⁵ m ² /s	6 x 10 ⁻⁵ m ² /s
PACKAGING	Standard width	-	1.83 m	6.00 ft
	Standard length	-	15.0 or 30.0 m	49.2 ft x 98.4 ft

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APPLICATION: Catch basin and concrete manhole protection

DATA SHEET

DRAINCOTEX

PRODUCT DESCRIPTION: Nonwoven needled geocomposite, designed for vertical drainage

FUNCTION: Drainage

RAW MATERIAL: Polypropylene / polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	7.0 mm	280 mils
	Mass per unit area	ASTM D5261	-	-
HYDRAULIC	Resistance to penetration by water	CAN/CGSB-4.2 - No 26.3	0 – 20 mm	0 – 0.79 in

TECHNICAL DATA TABLE - TRANSMISSIVITY AS A FUNCTION OF COMPRESSION

CONSTRAINT		THICKNESS (± 15 %)		TRANSMISSIVITY (± 15 %)
kPa	psi	mm	mils	
8	1.2	8.00	320	$2.0 \times 10^{-4} \text{ m}^2/\text{sec}$
20	2.9	6.90	276	$1.5 \times 10^{-4} \text{ m}^2/\text{sec}$
50	7.3	5.75	230	$0.9 \times 10^{-4} \text{ m}^2/\text{sec}$
200	29.0	3.40	136	$0.2 \times 10^{-4} \text{ m}^2/\text{sec}$

Note : test method ASTM D4716, gradient = 1

TECHNICAL DATA TABLE - POCKET FILTER HOSE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Mass per unit area	ASTM D5261	$75 \text{ g/m}^2 \pm 15 \%$	$2.2 \text{ oz/yd}^2 \pm 15 \%$
	Permeability	CAN/CGSB-148.1 - No 4	0.22 cm/s	
HYDRAULIC	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	$150 \mu\text{m} \pm 15 \%$	

DATA SHEET

DRAINCOTEX (CONT.)

TECHNICAL DIMENSION TABLE

PIPE DIAMETER (DRAINAGE CORE)		HEIGHT		LENGTH	
mm	in	m	ft	m	ft
100	4	0.45	0.14	30.0	9.1
100	4	0.65	0.20	30.0	9.1
100	4	0.85	0.26	30.0	9.1
150	6	1.00	0.30	30.0	9.1
150	6	1.20	0.37	30.0	9.1

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APPLICATIONS: Permanent public roadway
Roadway widening
Airport runways
Slab and building foundation

DATA SHEET

DRAINATEX

PRODUCT DESCRIPTION: Nonwoven needled geocomposite, designed for horizontal drainage

FUNCTION: Drainage

RAW MATERIAL: Polypropylene / polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	7.0 mm	280 mils
	Mass per unit area	ASTM D5261	-	-
HYDRAULIC	Resistance to penetration by water	CAN/CGSB-4.2 - No 26.3	0 – 20 mm	0 – 0.79 in
DIMENSIONS	Standard width	-	3.4 m	11.2 ft
	Standard length	-	30.0 m	98.4 ft

TECHNICAL DATA TABLE - TRANSMISSIVITY AS A FUNCTION OF COMPRESSION

CONSTRAINT		THICKNESS (± 15 %)		TRANSMISSIVITY (± 15 %)
kPa	psi	mm	mils	
8	1.2	8.00	320	$2.0 \times 10^{-4} \text{ m}^2/\text{sec}$
20	2.9	6.90	276	$1.5 \times 10^{-4} \text{ m}^2/\text{sec}$
50	7.3	5.75	230	$0.9 \times 10^{-4} \text{ m}^2/\text{sec}$
200	29.0	3.40	136	$0.2 \times 10^{-4} \text{ m}^2/\text{sec}$

Note: test method ATSM D4716, gradient =1

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APPLICATIONS: Permanent public roadway
Roadway widening
Airport runways
Slab and building foundation

DATA SHEET

BX-2000

PRODUCT DESCRIPTION: Geogrid extruded in a single layer

FUNCTION: Reinforcement

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Aperture size	-	39 mm x 39 mm (± 1 mm)	1.54 in x 1.54 in (± 0.04 in)
	Carbon black content	ASTM D4218	2 %	
	Average rib thickness	-	1.3 mm	52 mils
MECHANICAL	Tensile strength. at 2 % strain	ASTM D6637	8 kN/m	548 lb/ft
	Tensile strength. at 5 % strain	ASTM D6637	14 kN/m	959 lb/ft
	Ultimate tensile strength	ASTM D6637	19 kN/m	1302 lb/ft
	Junction strength	GRI-GG2	19 kN/m	1302 lb/ft
	Flexural stiffness/rigidity	ASTM D7748	1585 g-cm (SM) - 930 g-cm (ST)	
	Aperture stability ⁽¹⁾	-	3.4 kg-cm/deg @ 20 kg-cm torque	
	Radial stiffness at low strain ⁽²⁾		284.9 kN/m @ 0.5 % strain	
	Multi-axial tension test	ASTM D7748	-	
	- Vessel pressure at rupture		73 kPa	10.6 psi
	- Axisymmetric break resistance strain		8.3 %	
	- Average deflection at rupture		109 mm	4360 mils
PERFORMANCE AND DURABILITY	pH resistance	-	2 -13	
	Ultraviolet stability (500 hours)	ASTM D4355	100 %	
DIMENSIONS	Standard width	-	3.95 m	13.0 ft
	Standard length	-	50.0 m	164.0 ft

NOTE 1: In-plane torsional rigidity, measured by applying a moment of 20 kg-cm, at the central junction of a 225 mm x 225 mm specimen (8.85 in x 8.85 in). *U.S. Army Corps of Engineers Methodology (Kinney, T.C. Aperture Stability Modulus ref 3, 3.1.2000).*

NOTE 2: Radial stiffness is determined from tensile stiffness measured in any in-plane axis from testing in accordance with ASTM D6637-01.

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road
Commercial parking areas

Storage and loading areas
Roadway widening
Airport runways
Railway track

DATA SHEET

BX-3000

PRODUCT DESCRIPTION: Biaxial geogrid, extruded in a single layer

FUNCTION: Strengthening

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Aperture size	-	39 mm x 39 mm (± 1 mm)	1.54 in x 1.54 in (± 0.04 in)
	Carbon black content	ASTM D4218	2 %	
	Average rib thickness	-	2.0 mm	80 mils
MECHANICAL	Tensile strength. at 2 % strain	ASTM D6637	12.0 kN/m	822 lb/ft
	Tensile strength. at 5 % strain	ASTM D6637	21.6 kN/m	1480 lb/ft
	Ultimate tensile strength	ASTM D6637	30.0 kN/m	2056 lb/ft
	Junction strength	GRI-GG2	27.9 kN/m	1912 lb/ft
	Flexural stiffness/rigidity	ASTM D7748	4806 g-cm (MD) - 2619 g-cm (XMD)	
	Aperture stability ⁽¹⁾	-	5.7 kg-cm/deg @ 20 kg-cm torque	
	Radial stiffness at low strain ⁽²⁾	-	384.9 kN/m @ 0.5 % strain	
	Multi-axial tension test	ASTM D7748	-	
	- Vessel pressure at rupture		107.6 kPa	15.6 psi
	- Axisymmetric break resistance strain		7.3 %	
PERFORMANCE AND DURABILITY	- Average deflection at rupture		102 mm	4080 mils
	pH resistance	-	2 - 13	
	Ultraviolet stability (500 hours)	ASTM D4355	100 %	
DIMENSIONS	Standard width	-	3.95 m	13.0 ft
	Standard length	-	50.0 m	164.0 ft

NOTE 1: In-plane torsional rigidity, measured by applying a moment of 20 kg-cm to the central junction of a 225mm x 225mm specimen (8.85 in x 8.85 in). *U.S. Army Corps of Engineers Methodology (Kinney, T.C. Aperture Stability Modulus ref 3, 3.1.2000).*

NOTE 2: Radial stiffness is determined from tensile stiffness measured in any in-plane axis from testing in accordance with ASTM D6637-01.

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road
Commercial parking areas

Storage and loading areas
Roadway widening
Airport runways
Railway tracks

DATA SHEET

EPURTEX

PRODUCT DESCRIPTION: Nonwoven needled textile

FUNCTION: Separation

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	0.6 mm	24 mils
	Mass per unit area	ASTM D5261	75 g/m ² ± 15 %	2.2 oz/yd ² ± 15 %
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	180 N	40 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	55 - 95 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	60 N	13 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	500 kPa	73 psi
HYDRAULIC	Permittivity	CAN/CGSB-148.1 - No 4	6.0 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	180 µm	
PACKAGING	Standard size	-	1.5 m x 50.0 m	4.9 ft x 164.0 ft
			1.5 m x 150.0 m	4.9 ft x 492.0 ft
			3.0 m x 150.0 m	9.8 ft x 492.0 ft

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APPLICATION: Septic fields

DATA SHEET

HIBERTEX

PRODUCT DESCRIPTION: Nonwoven needled textile

FUNCTION: Protection

RAW MATERIAL: Polypropylene and polyester

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	0.68 mm	27 mils
	Mass per unit area	ASTM D5261	150 g/m ² ± 7 %	4.4 oz/yd ² ± 7 %
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	260 N	58 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	50 – 110 %	
	Burst strength (Mullen)	CAN/CGSB-4.2 - No 11.1	760 kPa	110 psi
PACKAGING	Standard size	-	1.1 m x 150.0 m	3.6 ft x 492.0 ft
			1.65 m x 30.0 m	5.4 ft x 98.4 ft
			1.65 m x 150.0 m	5.4 ft x 492.0 ft

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APPLICATION: Winter protection

DATA SHEET

2002W

PRODUCT DESCRIPTION: Woven geotextile

FUNCTION: Separation

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
MECHANICAL	Tensile strength	ASTM D4632	890 N	200 lb
	Elongation at break	ASTM D4632	15 %	
	Tear strength	ASTM D4533	330 N	75 lb
	Puncture strength (CBR)	ASTM D6241	3114 N	700 lb
HYDRAULIC	Permittivity	ASTM D4491	0.05 s ⁻¹	
	Filtration opening size (AOS)	ASTM D4751	0.425 mm	40 US Sieve
	Flow rate	ASTM D4491	204 l/min/m ²	5 gal/min/ft ²
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	70 %	
DIMENSIONS	Standard width	-	3.81 m	12.50 ft
	Standard length	-	131.00 m	432.00 ft

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road

Commercial parking areas
Storage and loading areas
Roadway widening
Airport runways

DATA SHEET

2004W

PRODUCT DESCRIPTION: Woven geotextile

FUNCTION: Separation

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
MECHANICAL	Tensile strength	ASTM D4632	1113 N	250 lb
	Elongation at break	ASTM D4632	12 %	
	Tear strength	ASTM D4533	400 N	90 lb
	Puncture strength (CBR)	ASTM D6241	3338 N	750 lb
HYDRAULIC	Permittivity	ASTM D4491	0.05 s ⁻¹	
	Filtration opening size (AOS)	ASTM D4751	0.425 mm	40 US Sieve
	Flow rate	ASTM D4491	163 l/min/m ²	4 gal/min/ft ²
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	70 %	
DIMENSIONS	Standard width	-	3.81 m	12.50 ft
	Standard length	-	110.0 m	360.00 ft

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road

Commercial parking areas
Storage and loading areas
Roadway widening
Airport runways

DATA SHEET

2006W

PRODUCT DESCRIPTION: Woven geotextile

FUNCTION: Reinforcement

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
MECHANICAL	Tensile strength	ASTM D4632	1402 N	315 lb
	Elongation at break	ASTM D4632	15 %	
	Tear strength	ASTM D4533	503 N	113 lb
	Puncture strength (CBR)	ASTM D6241	4005 N	900 lb
HYDRAULIC	Permittivity	ASTM D4491	0.05 s ⁻¹	
	Filtration opening size (AOS)	ASTM D4751	0.425 mm	40 US Sieve
	Flow rate	ASTM D4491	163 l/min/m ²	4 gal/min/ft ²
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	70 %	
DIMENSIONS	Standard width	-	3.81 m	12.50 ft
	Standard length	-	110.0 m	360.00 ft

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APPLICATIONS: Temporary roadway
Permanent public roadway
Permanent private roadway
Public access road

Commercial parking areas
Storage and loading areas
Roadway widening
Airport runways

DATA SHEET

MICROFAB

PRODUCT DESCRIPTION: Coated woven micro-perforated textile

FUNCTION: Protection

RAW MATERIAL: Polyethylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	200 N	45 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	5 - 20 %	
	Tear strength	CAN/CGSB-4.2 - No 12.2	30 N	7 lb
	Burst strength	CAN/CGSB-4.2 - No 11.1	485 kPa	11 psi
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4 / ASTM D4491	1 (± 0.2) x 10 ⁻³ cm/sec	
	Permittivity	CAN/CGSB-148.1 - No 4 / ASTM D4491	0.02 s ⁻¹	
	Pore openings	ASTM D4751	53 - 85 µm	
	Flow rate	ASTM D4491	49 (± 10) l/min/m ²	1.2 (± 0.25) gal/min/ft ²
PACKAGING	Standard size	-	1.17 m x 30.0 m	3.84 ft x 98.4 ft
			1.17 m x 100.0 m	3.84 ft x 328.0 ft

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APPLICATION: Grow vegetation areas

DATA SHEET

WOOD FIBERS LOGS TO CONTROL EROSION AND GROW VEGETATION

PRODUCT DESCRIPTION: Roll made of shriveled aspen fibers, seed-free

FUNCTION: Control

RAW MATERIAL: Wood fibers

PERFORMANCES:

- Channel erosion: Reduced by a minimum of 50 % in bare soil channels
- pH absorption: At the end, the pH must not exceed 8.3
- Product life: ≤ 24 months

TECHNICAL DATA TABLE

PRODUCT DIAMETER		LENGTH (± 10 %)		WEIGHT (± 10 %)		MESH OPENING (hexagonal shape)	
mm	in	m	ft	kg	lb	mm	in
222	9.0	7.62	25.0	11.35	25	190	0.75
300	12.0	3.05	10.0	9.02	20	250	1.0
500	20.0	3.05	10.0	13.62	30	320	1.3

Note 1: Based on ASTM D7208

Note 2: Based on ASTM D1117, amended

Note 3: Product life may depend on the geographical area, owing to different climatic conditions

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APPLICATIONS : Retaining walls and embankments
Grow vegetation areas
Sediment retention

DATA SHEET

TURBIDITY CURTAIN 400W

PRODUCT DESCRIPTION: Permeable geotextile membrane installed in a body of water to minimize the movement of sediments by creating a confined space. A tubular float made of UV-resistant polyethylene which is installed at the top and a chain ballast inserted into a specially designed sheath at the base of the curtain, maintain the membrane in place in the water.

FUNCTION: Protection

RAW MATERIAL: Polypropylene and polyethylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
MECHANICAL	Tensile strength	ASTM D4632	1400 N	311 lb
	Elongation at break	ASTM D4632	15 %	
	Tear strength	ASTM D4533	533 N	118 lb
	Puncture resistance	ASTM D4833	533 N	118 lb
HYDRAULIC	Filtration opening size (AOS)	ASTM D4751	0.425 mm	40 US Sieve

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APPLICATION: Sediment retention

DATA SHEET

ROUTEXGUARD – TYPE III-IV

PRODUCT DESCRIPTION: Sediment barrier

FUNCTION: Protection

COMPLIANT WITH MTQ STANDARDS: this sediment barrier meets the requirements of *Volume II, Chapter 9, Temporary environmental mitigation measures, 9.4.3.3, Sediment barriers* from the Quebec Ministry of Transportation (MTQ)

A measure allowing for the acquisition of LEED® points from the Canada Green Building Council, Sustainable Sites

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	0.9 mm	36 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	420 N	94 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	64 - 90 %	
	Puncture strength (CBR)	ASTM D6241	1250 N	281 lb
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.15 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	1.23 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	77 - 143 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
PACKAGING	Standard width	-	1.17 m (3.8 pi) with wood poles at every 1.5 m (4.9 ft)	
	Standard length	-	30.0 m	98.4 ft

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APPLICATION: Sediment retention

DATA SHEET

ROUTEXGUARD – TYPE V

PRODUCT DESCRIPTION: Sediment barrier

FUNCTION: Protection

COMPLIANT WITH MTQ STANDARDS: this sediment barrier meets the requirements of *Volume II, Chapter 9, Temporary environmental mitigation measures, 9.4.3.3, Sediment barriers* from the Quebec Ministry of Transportation (MTQ)

A measure allowing for the acquisition of LEED® points from the Canada Green Building Council, Sustainable Sites

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D5199	1.7 mm	68 mils
	Mass per unit area	ASTM D5261	-	-
MECHANICAL	Tensile strength	CAN/CGSB-148.1 - No 7.3	1050 N	236 lb
	Elongation at break	CAN/CGSB-148.1 - No 7.3	68 - 96 %	
	Puncture strength (CBR)	ASTM D6241	2930 N	659 lb
HYDRAULIC	Permeability	CAN/CGSB-148.1 - No 4	0.20 cm/s	
	Permittivity	CAN/CGSB-148.1 - No 4	0.70 s ⁻¹	
	Filtration opening size (FOS)	CAN/CGSB-148.1 - No 10	49 - 91 µm	
PERFORMANCE AND DURABILITY	UV resistance (500 hours)	ASTM D4355	50 %	
PACKAGING	Standard width	-	1.17 m (3.8 ft). with wood poles at every 1.5 m (4.9 ft)	
	Standard length	-	30.0 m	98.4 ft

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APPLICATION: Sediment retention

DATA SHEET

TEXGUARD

PRODUCT DESCRIPTION: Sediment barrier

FUNCTION: Protection

COMPLIANCE: A measure allowing for the acquisition of LEED points® from the Canada Green Building Council, Sustainable Sites

RAW MATERIAL: Polypropylene

TECHNICAL DATA TABLES

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
MECHANICAL	Tensile strength	ASTM D4632	601 N x 556 N	135 lb x 125 lb
	Elongation at break	ASTM D4632	15 %	
	Tear strength	ASTM D4533	267 N	60 lb
	Puncture strength (CBR)	ASTM D6241	1446 N	325 lb
HYDRAULIC	Permittivity	ASTM D4491	0.1 s ⁻¹	
	Filtration opening size (AOS)	ASTM D4751	595 µm	30 US Sieve
	Flow rate	ASTM D4491	408 l/min/m ²	10 gal/min/ft ²
	UV resistance (500 hours)	ASTM D4355	80 %	
PERFORMANCE AND DURABILITY	Standard width	-	0.91 m	3.0 ft
PACKAGING	Standard length	-	30.48 m	100.0 ft
	With stakes at every:	-	2.50 m	8.24 ft

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APPLICATION: Retention of sediments

DATA SHEET

BLANKET FOR EROSION CONTROL - WOOD FIBERS (SINGLE NETTING)

PRODUCT DESCRIPTION: Blanket composed of a mixture of poplar wood of which 80 % of the fibers are 150 mm (6 in) long or more, with its top covered with a degradable polypropylene mesh

FUNCTION: Control

RAW MATERIAL: Poplar wood fibers and polypropylene mesh

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D6525	9.70 mm	0.382 in
	Mass per unit area	ASTM D6475	216 g/m ²	0.40 lb/yd ²
	Light penetration	ASTM D6567	47.4 %	
MECHANICAL	Tensile strength SM	ASTM D6818	1.28 kN/m	87.6 lb/ft
	Tensile strength ST	ASTM D6818	0.44 kN/m	30.0 lb/ft
	Elongation at break SM	ASTM D6818	30.0 %	
	Elongation at break ST	ASTM D6818	25.9 %	
	Swelling	ECTC procedure	52.0 %	
OTHER	Water absorbency	ASTM D1117/ECTC	189.0 %	
	Laboratory testing of rain splash	ASTM D7101	SLR = 7.08 @ 2 in/hr ^(2,3)	
			SLR = 5.57 @ 4 in/hr ^(2,3)	
			SLR = 4.4 @ 6 in/hr ^(2,3)	
	Shear test	ASTM D7207	1.76 lb/ft ² @ 0.5 in of soil loss	
	Germination improvement	ASTM D7322	586.0 %	

TECHNICAL DIMENSIONS TABLE

WIDTH		LENGTH		SURFACE		WEIGHT		MASS PER UNIT AREA (± 10 %)		MESH OPENING	
m	ft	m	ft	m ²	yd ²	kg	lb	kg/m ²	lb/yd ²	mm	in
1.2	4.0	45.7	150.0	55.7	66.7	16.6	36.7	0.3	0.55	25.4 x 50.8	1.0 x 2.0
2.4	8.0	45.7	150.0	111.5	133.3	33.3	73.3	0.3	0.55	25.4 x 50.8	1.0 x 2.0
4.9	16.0	45.7	150.0	223.0	266.7	66.5	146.7	0.3	0.55	25.4 x 50.8	1.0 x 2.0

Note 1: Weight is based on dry fibre. During its manufacture, the reference humidity level for wood fibers is 22 %

Note 2: SLR is the soil leach ratio, as defined by NTPEP/AASHTO

Note 3: The laboratory tests indexes should not be used for design purposes

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DATA SHEET

BLANKET FOR EROSION CONTROL - WOOD FIBERS (SINGLE NETTING) (CONT.)

Wood mats have a rate of soil loss of 0.039 and are generally appropriate for slopes up to 2H: 1V.

They are designed to channel flows up to a stress of 1.5 m/s (5.0 ft/s) and 77 Pa (1.60 lbs/ft²) of shear.

APPLICATIONS: Retaining walls and embankments
Grow vegetation areas

DATA SHEET

BLANKET FOR EROSION CONTROL - WOOD FIBERS (DOUBLE NETTING)

PRODUCT DESCRIPTION: Blanket composed of a mixture of poplar wood of which 80 % of the fibers are 150 mm (6 in) long or more, and all sides covered with a degradable polypropylene mesh

FUNCTION: Control

RAW MATERIAL: Wood fibers and polypropylene mesh

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D6525	10.62 mm	0.418 in
	Mass per unit area	ASTM D6475	309 g/m ²	0.57 lb/yd ²
	Light penetration	ASTM D6567	34.6 %	
MECHANICAL	Tensile strength SM	ASTM D6818	1.85 kN/m	127.0 lb/ft
	Tensile strength ST	ASTM D6818	0.74 kN/m	50.9 lb/ft
	Elongation at break SM	ASTM D6818	28.64 %	
	Elongation at break ST	ASTM D6818	29.84 %	
	Swelling	ECTC procedure	89.0 %	
	Water absorbency	ASTM D1117/ECTC	199.0 %	
OTHER	Laboratory testing of rain splash	ASTM D7101	SLR = 6.84 @ 2 in/hr ^(2,3)	
			SLR = 7.19 @ 4 in/hr ^(2,3)	
			SLR = 7.56 @ 6 in/hr ^(2,3)	
	Shear test	ASTM D7207	2.6 lb/ft' @ 0.5 in of soil loss	
	Germination improvement	ASTM D7322	645.0 %	

TECHNICAL DIMENSIONS TABLE

WIDTH		LENGTH		SURFACE		WEIGHT		MASS PER UNIT AREA (± 10%)		MESH OPENING	
m	ft	m	ft	m ²	yd ²	kg	lb	kg/m ²	lb/yd ²	mm	in
1.2	4.0	34.3 m	112.5	41.8	50.0	16.6	36.5	0.4	0.73	25.4 x 50.8	1.0 x 2.0
2.4	8.0	34.3 m	112.5	83.6	100.0	33.1	73.0	0.4	0.73	25.4 x 50.8	1.0 x 2.0
4.9	16.0	34.3 m	112.5	167.2	200.0	66.2	146.0	0.4	0.73	25.4 x 50.8	1.0 x 2.0

Note 1: Weight is based on dry fibre. During its manufacture, the reference humidity level for wood fibers is 22 %

Note 2: SLR is the soil leach ratio, as defined by NTPEP/AASHTO

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DATA SHEET

BLANKET FOR EROSION CONTROL - WOOD FIBERS (DOUBLE NETTING) (CONT.)

Wood mats have a rate of soil loss of 0.022 and are generally appropriate for slopes up to 1.5H:1V. They are designed to channel flows up to a limit of 2.7 m/s (9.0 ft/s) and 108 Pa (2.25 lbs/ft²) of shear.

APPLICATIONS: Retaining walls and embankments
Grow vegetation areas

DATA SHEET

BLANKET FOR EROSION CONTROL - COIR FIBERS

PRODUCT DESCRIPTION: Coir fibers blanket, whose top and bottom are covered with a polypropylene mesh

FUNCTION: Control

RAW MATERIAL: Coir fibers and polypropylene mesh

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D6525	6.60 mm	0.26 in
	Mass per unit area	ASTM D6475	250 g/m ²	0.40 lb/yd ²
	Resiliency	ASTM D6524	83.0 %	
	Light penetration	ASTM D6567	23.1 %	
MECHANICAL	Tensile strength SM	ASTM D6818	4.57 kN/m	313.2 lb/ft
	Tensile strength ST	ASTM D6818	2.50 kN/m	171.6 lb/ft
	Elongation at break SM	ASTM D6818	29.0 %	
	Elongation at break ST	ASTM D6818	44.0 %	
	Swelling	ECTC procedure	20.3 %	
OTHER	Water absorbency	ASTM D1117/ECTC	234.0 %	
	Laboratory testing of rain splash	ASTM D7101	SLR = 15.05 @ 2 in/hr ^(2,3)	
			SLR = 16.51 @ 4 in/hr ^(2,3)	
			SLR = 18.01 @ 6 in/hr ^(2,3)	
	Shear test	ASTM D7207	2.50 lb/ft @ 0.5 in of soil loss	
PACKAGING	Germination improvement	ASTM D7322	415.0 %	
	Width	-	2.4 m	8.0 ft
	Length	-	34.3 m	112.5 ft
	Surface	-	83.6 m ²	100.0 yd ²
	Weight (± 10 %)	-	22.7 kg/m ²	50.0 lb/yd ²
	Mass per unit area	-	0.27 kg/m ²	0.50 lb/yd ²
	Mesh opening	-	19.1 mm x 19.1 mm	0.75 in x 0.75 in

Note 1: Weight is based on the dry fibre. During its manufacture, the reference humidity level for coir fibers is 20 %

Note 2: SLR is the soil leach ratio, as defined by NTPEP/AASHTO

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DATA SHEET

BLANKET FOR EROSION CONTROL - COIR FIBERS (CONT.)

Coir mats are designed to channel a flow rate of 2.7 m/s (9 ft/s) and 108 Pa (2.25 lbs/ft²) up to the shear stress limit.

Coir mats have a rate of soil loss of 0.10 and are generally appropriate for slopes up to 1H:1V.

APPLICATIONS: Retaining walls and embankments
Grow vegetation areas

DATA SHEET

BLANKET FOR EROSION CONTROL - STRAW FIBERS

PRODUCT DESCRIPTION: Blanket composed of a mixture of straw of which 75 % of fibers are a 100 mm (4 in) long or more, whose top and bottom are covered with a polypropylene mesh

FUNCTION: Control

RAW MATERIAL: Straw fibers and polypropylene mesh

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D6525	7.87 mm	0.31 in
	Mass per unit area	ASTM D6475	233 g/m ²	0.43 lb/yd ²
	Resiliency	ASTM D6524	65.0 %	
	Light penetration	ASTM D6567	21.6 %	
MECHANICAL	Tensile Strength SM	ASTM D6818	1.93 kN/m	132.2 lb/ft
	Tensile Strength ST	ASTM D6818	0.68 kN/m	46.8 lb/ft
	Elongation at break SM	ASTM D6818	28.4 %	
	Elongation at break ST	ASTM D6818	26.5 %	
	Swelling	ECTC procedure	22.0 %	
OTHER	Water absorbency	ASTM D1117/ECTC	447.0 %	
	Laboratory testing of rain splash	ASTM D7101	SLR = 7.53 @ 2 in/hr ^(2,3)	
			SLR = 8.38 @ 4 in/hr ^(2,3)	
			SLR = 9.32 @ 6 in/hr ^(2,3)	
	Shear test	ASTM D7207	1.59 lb/ft ² @ 0.5 in of soil loss	
	Germination improvement	ASTM D7322	476.0 %	

TECHNICAL DIMENSIONS TABLE

WIDTH		LENGTH		SURFACE		WEIGHT (± 10 %)		MASS PER UNIT AREA		MESH OPENING	
m	ft	m	ft	m ²	yd ²	kg/m ²	lb/yd ²	kg/m ²	lb/yd ²	mm	in
2.4	8.0	34.3	112.5	83.6	100.0	22.7	50.0	0.27	0.5	19.1 x 19.1	0.75 x 0.75
4.9	16.0	34.3	112.5	167.2	200.0	45.4	100.0	0.27	0.5	19.1 x 19.1	0.75 x 0.75

Note 1: Weight is based on the dry fibre. During its manufacture, the reference humidity level is 15 % for straw fibers

Note 2: SLR is the soil leach ratio, as defined by NTPEP/AASHTO

Note 3: The laboratory tests indexes should not be used for design purposes

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DATA SHEET

BLANKET FOR EROSION CONTROL - STRAW FIBRES (CONT.)

Straw mats are designed to channel a flow rate of 1.4 m/s (4.5 ft/s) and 74 Pa (1.55 lb/ft²) up to the shear stress limit.

Straw mats have a rate of soil loss of 0.05 and are generally appropriate for slopes up to 3H: 1V.

APPLICATIONS: Retaining walls and embankments
Grow vegetation areas

DATA SHEET

BLANKET FOR EROSION CONTROL - STRAW AND COIR FIBERS

PRODUCT DESCRIPTION: Blanket composed of a mixture of straw and coir fibers, whose top and bottom are covered with a polypropylene mesh

FUNCTION: Control

RAW MATERIAL: Straw (70 %) fibers, coir fibers (30 %) and polypropylene mesh

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	VALUES	
			Metric	Imperial
PHYSICAL	Thickness	ASTM D6525	6.90 mm	0.27 in
	Mass per unit area	ASTM D6475	228 g/m ²	0.42 lb/yd ²
	Resiliency	ASTM D6524	74.0 %	
	Light penetration	ASTM D6567	15.0 %	
MECHANICAL	Tensile strength SM	ASTM D6818	2.97 kN/m	203.5 lb/ft
	Tensile strength ST	ASTM D6818	2.12 kN/m	145.0 lb/ft
	Elongation at break SM	ASTM D6818	27.54 %	
	Elongation at break ST	ASTM D6818	36.52 %	
OTHER	Swelling	ECTC procedure	55.0 %	
	Water absorbency	ASTM D1117/ECTC	415.0 %	
	Laboratory testing of rain splash	ASTM D7101	SLR = 10.7 @ 2 in/hr ^(2,3)	
			SLR = 13.1 @ 4 in/hr ^(2,3)	
			SLR = 16.1 @ 6 in/hr ^(2,3)	
	Shear test	ASTM D7207	2.16 lb/ft ² @ 0.5 in of soil loss	
PACKAGING	Germination improvement	ASTM D7322	345.0 %	
	Width	-	2.4 m	8.0 ft
	Length	-	34.3 m	112.5 ft
	Surface	-	83.6 m ²	100 yd ²
	Weight (± 10 %)	-	22.7 kg/m ²	50.0 lb/yd ²
	Mass per unit area	-	0.27 kg/m ²	0.50 lb/yd ²
	Mesh openings	Polypropylene (black HV) on top	19.1 mm x 19.1 mm	0.75 in x 0.75 in
		Polypropylene (green) underneath	12.7 mm x 12.7 mm	0.50 in x 0.50 in

Note 1: Weight is based on the dry fibre. During its manufacture, the reference humidity levels are 20 % for coir fibers, and 15 % for straw

Note 2: SLR is the soil leach ratio, as defined by NTPEP/AASHTO

Note 3: The laboratory tests indexes should not be used for design purposes

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DATA SHEET

BLANKET FOR EROSION CONTROL - STRAW AND COIR FIBERS (CONT.)

Straw mats are designed to channel a flow rate of 2.6 m/s (8.5 ft/s) and 96 Pa (2.0 lbs/ft²) up to the shear stress limit.

Straw mats have a rate of soil loss of 0.15 and are generally appropriate for slopes up 1.5H:1V.

APPLICATIONS: Retaining walls and embankments
Grow vegetation areas

DATA SHEET

GABION

PRODUCT DESCRIPTION: Rectangular wire mesh baskets that, when filled with stone and set together, form gravity retaining wall systems

FUNCTION: Protection

RAW MATERIAL: galvanized steel, with or without PVC coating

TECHNICAL DATA TABLE - STEEL WIRE

	PROPERTIES	TEST METHOD	VALUES
PHYSICAL	Type of steel	ASTM D461-92	Hardness: Mild steel
	Wire mesh	ASTM A90	259.4 g/m ²
	Salvage wire	ASTM A90	274.6 g/m ²
	Lacing wire	ASTM A90	244.1 g/m ²
MECHANICAL	Tear strength	ASTM A370	< 12 %

Note: Mass per zinc area unit (zinc coating finish 5, class 3)

TECHNICAL DATA TABLE - PVC COATING

	PROPERTIES	TEST METHOD	INITIAL VALUES	LOSS AFTER WEAR ⁽¹⁾
PHYSICAL	Density	ASTM D792	1.3 to 1.34 g/m ³	< 6 %
	Hardness	ASTM D2240	50 - 60	< 10 %
MECHANICAL	Tensile strength	ASTM D412	20.55 Mpa	< 25 %
	Modulus of elasticity at 100 % elongation	ASTM D412	18.6 Mpa	< 25 %
	Elongation	ASTM D412	200 to 280 %	< 25 %
	High temperature exposure	ASTM D1203, ASTM D2287	No effect after 240 hours at 105 °C	
	Abrasion resistance	ASTM D12425, B method	Loss of Weight < 12 %	

Note 1: The accelerated wear tests are test spraying of salt water according to ASTM B117 for 3000 hours and the UV test according to ASTM D1499 and ASTM G152 for 3000 hours at 63 °C.

TECHNICAL DATA TABLE - GABION

PROPERTIES	TEST METHOD	GALVANISED GABIONS	PVC COATED GABIONS ⁽²⁾
Diameter of steel used for wire mesh	Measured according to ASTM A641	3.05 mm (0.12 in)	2.70 mm (0.11 in)
Diameter of steel used for salvage wire		3.88 mm (0.15 in)	3.40 mm (0.13 in)
Diameter of steel used for lacing wire		2.31 m (5.57 ft)	2.31 m (5.57 ft)
Mesh size (10 % tolerance)	The hexagonal mesh is doubly twisted to 83 x 114 mm (3.3 x 4.5 in)		

Note 2: Note that the nominal thickness of the PVC layer is 0.55 mm (minimum of 0.38 mm) for each type of steel wire.

DATA SHEET

GABION (CONT.)

TECHNICAL DATA TABLE - TWISTED WIRE GABION

LENGTH		WIDTH		HEIGHT	
m	ft	m	ft	m	ft
2.0	6.50	1.0	3.25	0.3	1.00
3.0	9.75	1.0	3.25	0.3	1.00
4.0	13.00	1.0	3.25	0.3	1.00
2.0	6.50	1.0	3.25	0.5	1.50
3.0	9.75	1.0	3.25	0.5	1.50
4.0	13.00	1.0	3.25	0.5	1.50
2.0	6.50	1.0	3.25	1.0	3.25
3.0	9.75	1.0	3.25	1.0	3.25
4.0	13.00	1.0	3.25	1.0	3.25

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APPLICATIONS: Retaining walls and embankments
Bank protection

DATA SHEET

GEOCELLS

PRODUCT DESCRIPTION: Cellular confinement system made of light and flexible polyethylene, composed of three-dimensional cells in a honeycomb structure

FUNCTIONS: Reinforcement and protection

RAW MATERIAL: Polyethylene

TECHNICAL DATA TABLE

	PROPERTIES	TEST METHOD	TYPICAL VALUES	
			Metric	Imperial
PHYSICAL	Minimum polymer density	ASTM D1505	0.941 - 0.960 g/cm ³	
	Sheet nominal thickness	ASTM D5199	1.25 mm	50 mils
	Carbon Black Content	ASTM D1603	1.5 %	
	Nominal-expanded cell size (width x length)	-	259 mm x 224 mm	10.2 in x 8.8 in
	Nominal-expanded cell area	-	289 cm ²	44.8 in ²
	Nominal-expanded section size (width x length)	-	2.56 m x 6.52 m	8.4 ft x 21.4 ft
	Nominal-expanded section area (width x length)	-	16.7 m ²	180.0 ft ²
MECHANICAL	Environmental stress crack resistance	ASTM D1693	4000 hours	
	Seam hang strength	-	A 102 mm (4 in) weld joint which supports a load of 72.5 kg (160 lbs) for at least 30 days or a weld joint 102 mm (4 in) that supports a load of 72.5 kg (160 lbs) for at least 7 days, while undergoing changes in temperature from 23 °C (74 °F) to 54 °C (130 °F) over a one-hour cycle	

CELL TECHNICAL DATA TABLE

CELL DEPTH		SEAM PEEL STRENGTH		SECTION WEIGHT		SECTIONS PER PALLET
mm	in	N	lb	kg	lb	
100	4	1420	320	25.9	57.0	50
150	6	2130	480	39.0	86.0	30
200	8	2840	640	51.7	114.0	25

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DATA SHEET

GEOCELLS (CONT.)

APPLICATIONS: Retaining walls and embankments
Public access road

ORDER FORM

CORRUGATED MANHOLE/CATCH BASIN 300 mm (12 in) TO 900 mm (36 in) FOR NON-ROADWAY USE

PROJECT: _____

DATE REQUIRED: _____

CLIENT: _____

TELEPHONE: _____

APPROVED BY: _____

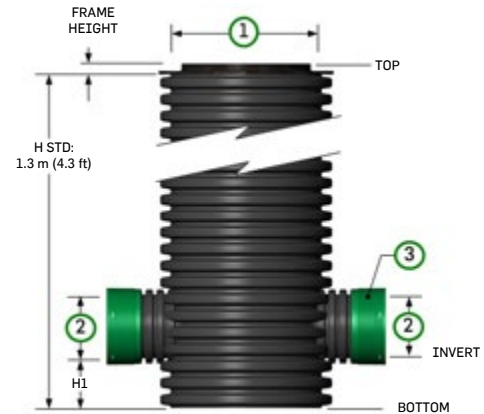
QUANTITY: _____

MANHOLE/CATCH BASIN NUMBER: _____ As on the site plan.

DIMENSIONS:

① Nominal diameter: _____ mm _____ in

Height (H): 1.3 m (4.3 ft) ☐ Other: _____ m _____ ft



OUTLETS AND COUPLERS:

② Outlet diameter mm (in)	H1 (mm)	ANGLE (°)	③ TYPE OF COUPLER			
	Standard: 300 mm (12 in)	Top view	PLAIN	SNAP	GSKT.	PVC
A						
B						
C						
D						

FRAMES and COVERS/GRATES:

☐ HDPE grate

300 mm (12 in)
375 mm (15 in)
450 mm (18 in)



☐ Frame and grate in
S series cast iron

300 mm (12 in)
375 mm (15 in)
450 mm (18 in)
600 mm (24 in)
750 mm (30 in)
900 mm (36 in)



☐ Frame and cover
in S series cast
iron

600 mm (24 in)
750 mm (30 in)
900 mm (36 in)



OPTIONS:

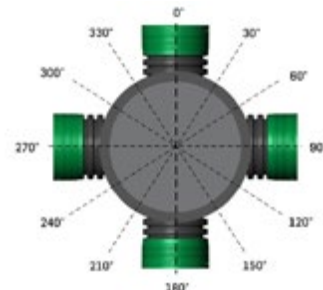
Ladder
(galvanised steel, installed)



Hook or eyelet
(for hooking system)



OUTLET ANGLES:



COMMENTS: _____

ORDER FORM

CORRUGATED MANHOLE/CATCH BASIN 1050 mm (42 in) TO 1200 mm (48 in) FOR NON-ROADWAY USE

PROJECT: _____

DATE REQUIRED: _____

CLIENT: _____

TELEPHONE: _____

APPROVED BY: _____

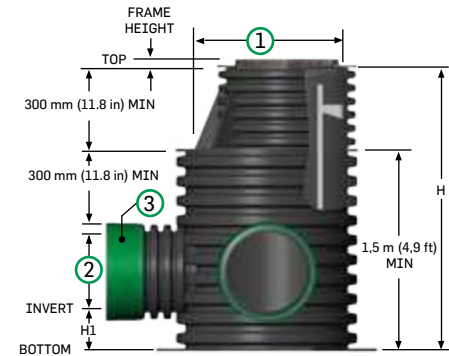
QUANTITY: _____

MANHOLE/CATCH BASIN NUMBER: _____ As on the site plan.

DIMENSIONS:

① Nominal diameter: _____ mm _____ in

Height (H): _____ m _____ ft



OUTLETS AND COUPLERS:

② Outlet diameter mm (in)	H1 (mm)	ANGLE (°)	③ TYPE OF COUPLER			
	Standard: 300 mm (12 in)	Top view	PLAIN	SNAP	GSKT.	PVC
A						
B						
C						
D						

FRAMES and COVERS/GRATES:

☐ Frame and grate in
S series cast iron

900 mm (36 in)



Non-roadway

☐ Frame and cover in
S series cast iron

900 mm (36 in)



Non-roadway

OPTIONS:

Ladder
(galvanised steel, installed)



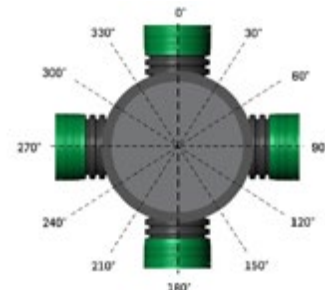
Hook or eyelet
(for hooking system)



Safety platform



OUTLET ANGLES:



COMMENTS: _____

ORDER FORM

CATCH BASIN WITH SMOOTH EXTERIOR WALL 450 mm (18 in) TO 900 mm (36 in) FOR ROADWAY* OR NON-ROADWAY USE

PROJECT: _____

DATE REQUIRED: _____

CLIENT: _____

TELEPHONE: _____

APPROVED BY: _____

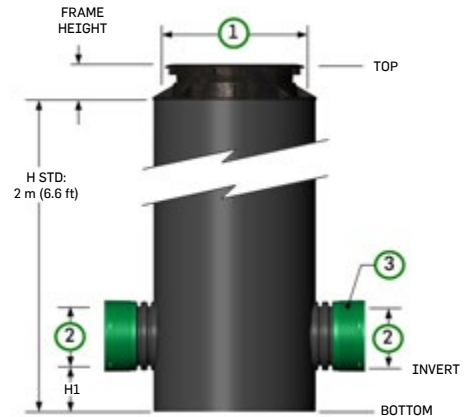
QUANTITY: _____

CATCH BASIN NUMBER: _____ As on the site plan.

DIMENSIONS:

① Nominal diameter: _____ mm _____ in

Height (H): 2 m (6.6 ft) ☐ Other: _____ m _____ ft



OUTLETS AND COUPLERS:

② Outlet diameter mm (in)	H1 (mm)	ANGLE (°)	③ TYPE OF COUPLER			
	Standard: 300 mm (12 in)	Top view	PLAIN	SNAP	GSKT.	PVC
A						
B						
C						
D						

FRAMES AND GRATES:

☐ Adjustable cast iron frame, guide frame and grate

450 mm (18 in) **
600 mm (24 in)
750 mm (30 in)
900 mm (36 in)

**no guide frame



Roadway

☐ Rectangular adjustable cast iron frame, guide frame and grate

355.6 mm x 609.6 mm
(14 in x 24 in)
457.2 mm x 914.4 mm
(18 in x 36 in)



Roadway

☐ Frame and grate in S series cast iron

450 mm (18 in)
600 mm (24 in)
750 mm (30 in)
900 mm (36 in)



Non-roadway

OPTIONS:

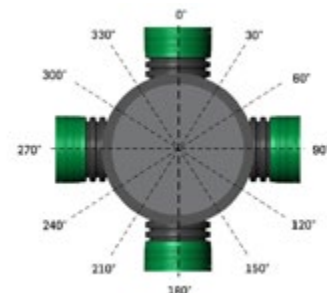
Hook or eyelet
(for hooking system)



Lift eyelets
(anchors not included)



OUTLET ANGLES:



COMMENTS: _____

* Note: When used with a suitable frame and grate.

ORDER FORM

MANHOLE WITH SMOOTH EXTERIOR WALL 600 mm (24 in) TO 900mm (36 in) FOR ROADWAY* OR NON-ROADWAY USE

PROJECT: _____

DATE REQUIRED: _____

CLIENT: _____

TELEPHONE: _____

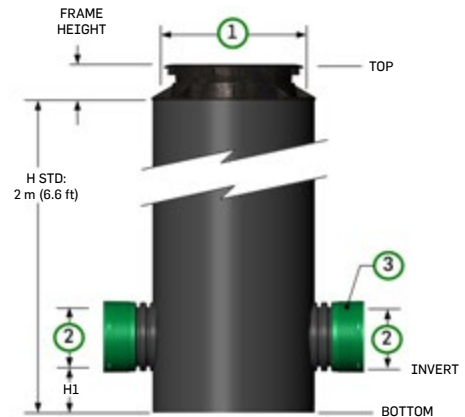
APPROVED BY: _____

QUANTITY: _____

MANHOLE NUMBER: _____ As on the site plan.

DIMENSIONS:

- ① Nominal diameter: _____ mm _____ in
Height (H): 2 m (6.6 ft) ☐ Other: _____ m _____ ft



OUTLETS AND COUPLERS:

② Outlet diameter mm (in)	H1 (mm)	ANGLE (°)	③ TYPE OF COUPLER			
	Standard: 300 mm (12 in)	Top view	PLAIN	SNAP	GSKT.	PVC
A						
B						
C						
D						

FRAME AND COVERS:

- ☐ Adjustable cast iron frame,
guide frame and cover

600 mm (24 in)
750 mm (30 in)
900 mm (36 in)



Roadway

- ☐ Frame and cover in S
series cast iron

600 mm (24 in)
750 mm (30 in)
900 mm (36 in)



Non-roadway

OPTIONS:

Ladder
(galvanised steel, installed)



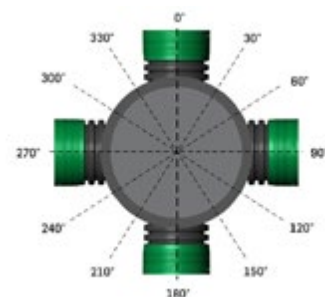
Lift eyelets
(anchors not included)



Safety platform



OUTLET ANGLES:



COMMENTS: _____

* Note: When used with a suitable frame and grate.

ORDER FORM

MANHOLE WITH SMOOTH EXTERIOR WALL 1,050 mm (42 in) TO 3,350 mm (132 in) FOR ROADWAY* OR NON-ROADWAY USE

PROJECT: _____

DATE REQUIRED: _____

CLIENT: _____

TELEPHONE: _____

APPROVED BY: _____

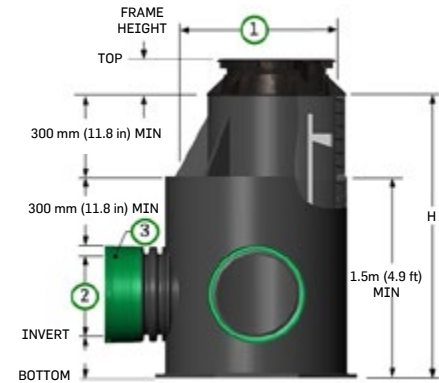
QUANTITY: _____

MANHOLE NUMBER: _____ As on the site plan.

DIMENSIONS:

① Nominal diameter: _____ mm _____ in

Height (H): _____ m _____ ft



OUTLETS AND COUPLERS:

② Outlet diameter mm (in)	H1 (mm)	ANGLE (°)	③ TYPE OF COUPLER			
	Standard: 300 mm (12 in)	Top view	PLAIN	SNAP	GSKT.	PVC
A						
B						
C						
D						

FRAME AND COVERS:

☐ Adjustable cast iron frame,
guide frame and cover

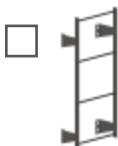
900 mm (36 in)



Roadway

OPTIONS:

Ladder
(galvanised steel, installed)



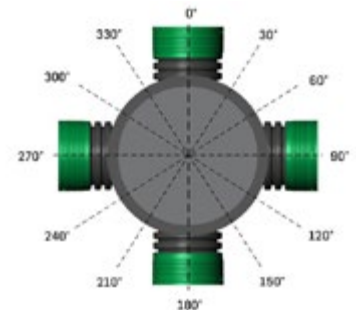
Lift eyelets
(anchors not included)



Safety platform



OUTLET ANGLES:



COMMENTS: _____

* Note: When used with a suitable cover and frame.

ORDER FORM

MANHOLE/CATCH BASIN CHIMNEY TYPE 600 mm (24 in) TO 900 mm (36 in) FOR ROADWAY* OR NON-ROADWAY USE

PROJECT : _____
 DATE REQUIRED : _____
 CLIENT : _____
 TÉLÉPHONE : _____
 APPROVED BY : _____
 QUANTITY : _____
 MANHOLE NUMBER: _____ As on the site plan.

DIMENSIONS:

A) Diameters

Chimney ___ mm ___ in
 Conduit ___ mm ___ in
 Smooth exterior wall (FOR ROADWAY) ☐
 Corrugated wall (NON-ROADWAY) ☐

B) Heights

H: ___ mm ___ in
 H1 (Bottom): ___ mm ___ in

C) Lengths

L totale: ___ mm ___ in
 L1: ___ mm ___ in

D) Angle: _____°

GRATES, COVERS AND FRAMES:

☐ Adjustable cast iron
frame, guide frame
and cover



☐ Frame and cover in
S Series cast iron



Non-Roadway

☐ Frame and grate
in S series cast
iron



Non-Roadway

☐ Adjustable cast iron
frame, guide frame and
grate



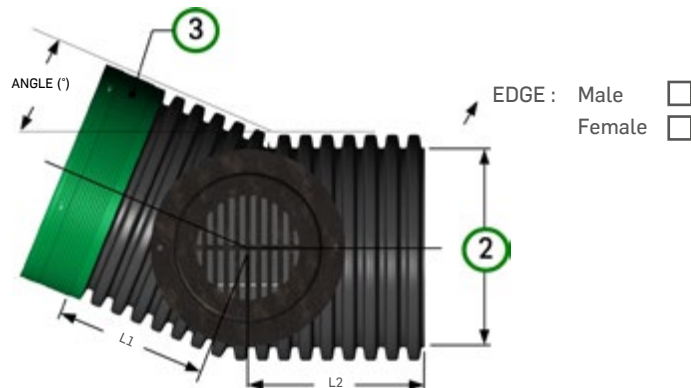
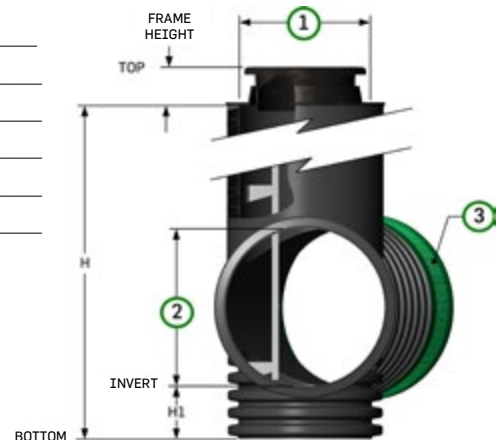
Roadway

OPTIONS :

☐ LADDER
(galvanised steel, installed)



☐ LIFT EYELETS
(anchors not included)



COMMENTS : _____

* Note : When used with a suitable frame and grate.

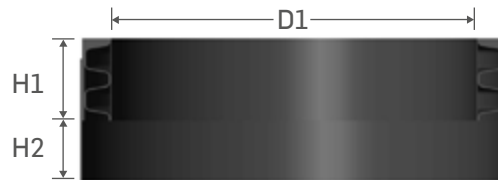
ORDER FORM

STANDARD OR CUSTOM-MADE RISER FOR MANHOLE OR CATCH BASIN WITH SMOOTH EXTERIOR WALL

PROJECT: _____
CLIENT: _____
DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____



Diameter D1	Quantity	Rings	Specify lengths
☐ 525 mm (21 in)		2 ☐	Standard (according to data sheet) ☐
☐ 600 mm (24 in)		3 ☐	or
☐ 750 mm (30 in)			H1: _____ ☐ mm
☐ 900 mm (36 in)			☐ in
			H2: _____ ☐ mm
			☐ in

Note : For a manhole riser, validate the ladder height before ordering.

APPROVAL

ORDER FORM

OUTSIDE OR INSIDE ADAPTER

PROJECT: _____
 DATE REQUIRED: _____
 CLIENT: _____
 TELEPHONE: _____
 APPROVED BY: _____
 QUANTITY: _____

OUTSIDE ADAPTER



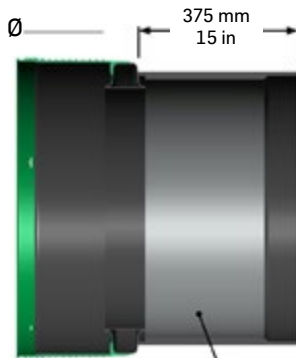
SOLENO PIPE NOMINAL
DIAMETER: Ø _____

TYPE OF COUPLER

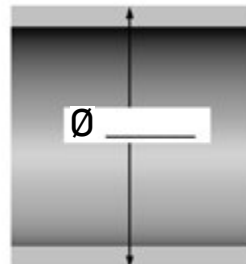
PLAIN ☐

SNAP ☐

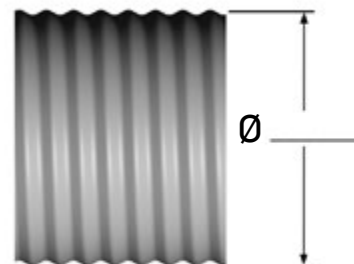
CGI ☐



EXISTING CONCRETE PIPE ☐



EXISTING STEEL PIPE ☐



INSIDE ADAPTER



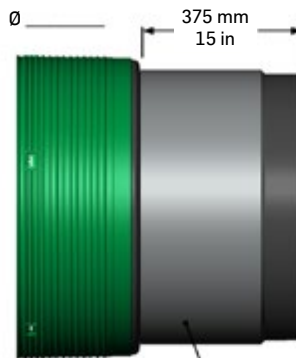
SOLENO PIPE
NOMINAL DIAMETER: Ø _____

TYPE OF COUPLER

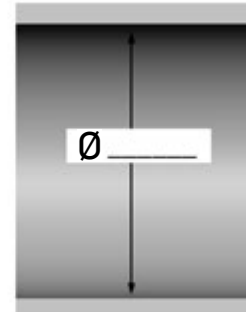
PLAIN ☐

SNAP ☐

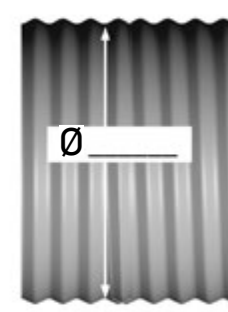
CGI ☐



EXISTING CONCRETE PIPE ☐



EXISTING STEEL PIPE ☐



COMMENTS: _____

ORDER FORM

MONOLITHIC CATCH BASIN INLET/CONNECTION

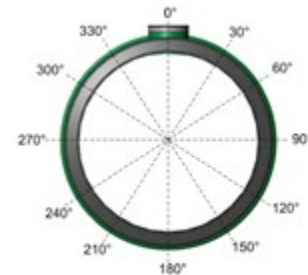
PROJECT: _____

DATE REQUIRED: _____

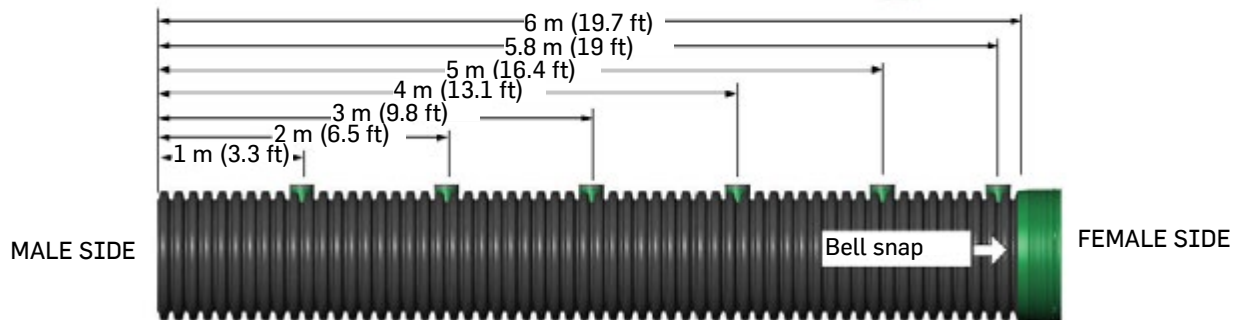
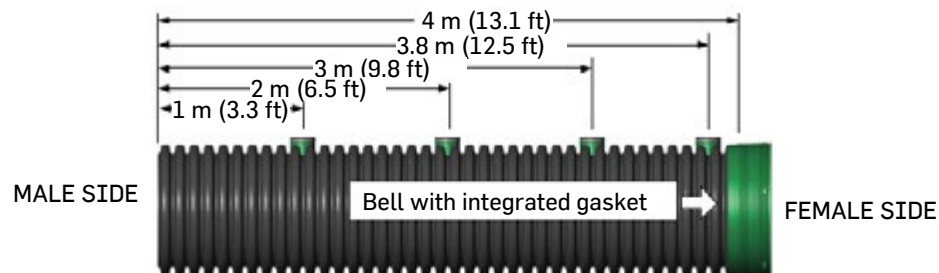
CLIENT: _____

TELEPHONE: _____

APPROVED BY: _____



VIEW FROM MALE SIDE



# IDENTIFICATION	Ø MAIN		Ø FEEDER MAIN (CONNECTIONS)			CONNECTIONS POSITIONING								ANGLE	QTY
	4 m (13.1 ft)	6 m (19.7 ft)				MALE END									
	GSKT.	SNAP	PVC	HDPE SNAP	HDPE GSKT.	1 m (3.3 ft)	2 m (6.5 ft)	3 m (9.8 ft)	3.8 m (12.5 ft)	4 m (13.1 ft)	5 m (16.4 ft)	5.8 m (19 ft)			

COMMENTS: _____

ORDER FORM

AQUA-SWIRL

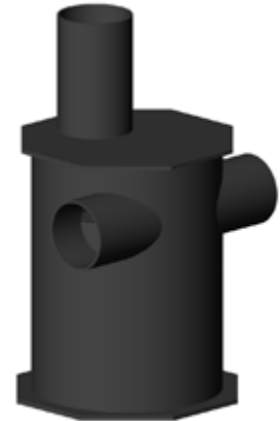
PROJECT : _____
 SITE: _____
 CLIENT: _____
 PHONE | E-MAIL: _____
 DATE REQUIRED: _____
 DESIGNING ENGINEER: _____
 COMPANY AND CONTACT INFORMATION: _____

INSPECTION CONTRACT AND MAINTENANCE REQUIRED: YES ☐ NO ☐

TREATMENT DEVICE SPECIFICATIONS:

Rate of flow to be treated: _____ L/s

Model number (if specified in the plans): AS-X _____



ONLY FOR LÉVIS

Area (ha)	
Runoff Coefficient	

Online or offline configuration: Online ☐ Offline ☐

If offline, how many bypass manholes? _____

Pipe diameter: _____ inlet _____ outlet

Pipe Type: _____ inlet _____ outlet

Invert: _____ inlet _____ outlet

Angle between inlet and outlet lines: _____

Slope of the inlet line: _____ %

Slope of the outlet pipe: _____ %

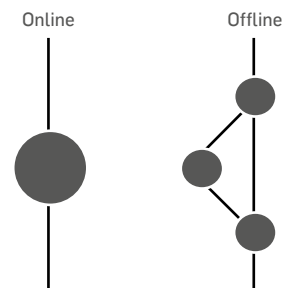
Direction of flow (clockwise or counterclockwise): _____

Elevation of the water table: _____

Final grade (finished ground elevation): _____

Will unit be subjected to traffic loading? YES ☐ NO ☐

NOTES: _____



APPROVAL

ORDER FORM

NON-STANDARD MANHOLE ADAPTER - 300 mm (12 in) to 900 mm (36 in)

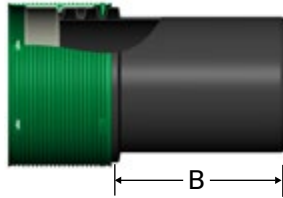
PROJECT : _____

CLIENT: _____

DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____

Type of opening	Specify the diameter SOLFLO MAX (A)	Quantity	Product Drawing	Specify dimension required
PVC	☐ 300 mm (12 in)			B: _____ ☐ mm ☐ in
PVC	☐ 375 mm (15 in) ☐ 450 mm (18 in) ☐ 525 mm (21 in) ☐ 600 mm (24 in)			B: _____ ☐ mm ☐ in
Concrete	☐ 525 mm (21 in) ☐ 600 mm (24 in) ☐ 750 mm (30 in) ☐ 900 mm (36 in)			B: _____ ☐ mm ☐ in

APPROVAL

ORDER FORM

NON-STANDARD MANHOLE ADAPTER - 1050 mm (42 in) to 1200 mm (48 in)

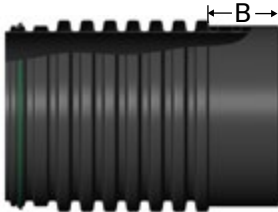
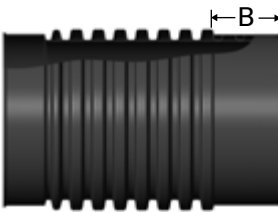
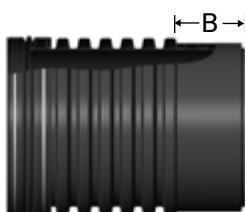
PROJECT: _____

CLIENT: _____

DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____

Type of opening	Specify the diameter SOLFLO MAX (A)	Quantity	Product Drawing	Specify dimension required
Concrete	☐ 1050 mm (42 in) M ☐ 1200 mm (48 in) M			B: _____ ☐ mm ☐ in
Concrete	☐ 1050 mm (42 in) F ☐ 1200 mm (48 in) F			B: _____ ☐ mm ☐ in
Concrete	☐ 1050 mm (42 in) CS ☐ 1200 mm (48 in) CS			B: _____ ☐ mm ☐ in

Legend: M = male outlet with O-ring seal;
 F = female outlet;
 CS = smooth outlet with collar sleeve; both male rings of the pipe to connect must be cut before installation

APPROVAL

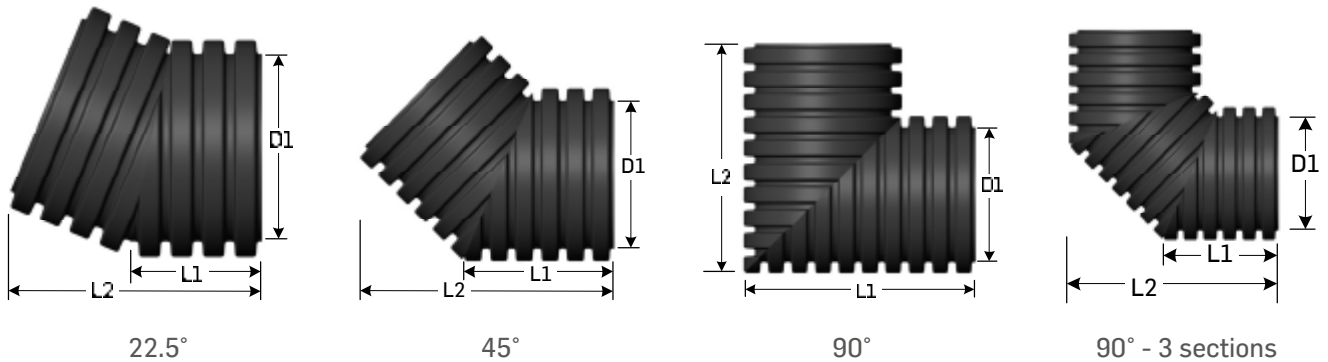
ORDER FORM

STANDARD OR CUSTOM MADE ELBOW

PROJECT: _____
CLIENT: _____
DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____



D1 Diameter	Quantity	Angle	Fitting	Specify lengths
☐ 100 mm (4 in)		☐ 22.5°	Side A	Standard ☐
☐ 150 mm (6 in)		☐ 45°	☐ Smooth	(according to data sheet)
☐ 200 mm (8 in)		☐ 90°	☐ Clip	or
☐ 250 mm (10 in)		☐ 90° - 3 sections	☐ Gasket	L1: _____ ☐ mm
☐ 300 mm (12 in)				☐ in
☐ 375 mm (15 in)				L2: _____ ☐ mm
☐ 450 mm (18 in)			Side B	Not required for 90° ☐ in
☐ 525 mm (21 in)			☐ Smooth	
☐ 600 mm (24 in)			☐ Clip	
☐ 750 mm (30 in)			☐ Gasket	
☐ 900 mm (36 in)				
☐ 1050 mm (42 in)				
☐ 1200 mm (48 in)				

APPROVAL

ORDER FORM

STANDARD OR CUSTOM MADE CROSS AND CROSS-REDUCER

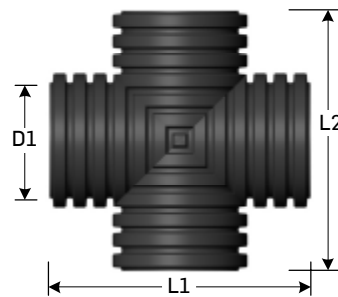
PROJECT: _____

CLIENT: _____

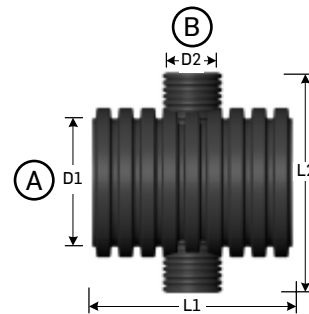
DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____



Cross



Cross-reducer

D1 Diameter	D2 Diameter	Quantity	Fitting	Specify lengths
☐ 100 mm (4 in)	☐ 100 mm (4 in)		Side A	Standard (according to data sheet) ☐
☐ 150 mm (6 in)	☐ 150 mm (6 in)		☐ Smooth	or
☐ 200 mm (8 in)	☐ 200 mm (8 in)		☐ Clip	L1: _____ ☐ mm
☐ 250 mm (10 in)	☐ 250 mm (10 in)		☐ Gasket	☐ in
☐ 300 mm (12 in)	☐ 300 mm (12 in)			L2: _____ ☐ mm
☐ 375 mm (15 in)	☐ 375 mm (15 in)		Side B	☐ in
☐ 450 mm (18 in)	☐ 450 mm (18 in)		☐ Smooth	
☐ 525 mm (21 in)	☐ 525 mm (21 in)		☐ Clip	
☐ 600 mm (24 in)	☐ 600 mm (24 in)		☐ Gasket	
☐ 750 mm (30 in)	☐ 750 mm (30 in)			
☐ 900 mm (36 in)				
☐ 1050 mm (42 in)				
☐ 1200 mm (48 in)				

APPROVAL

ORDER FORM

STANDARD OR CUSTOM MADE REDUCER

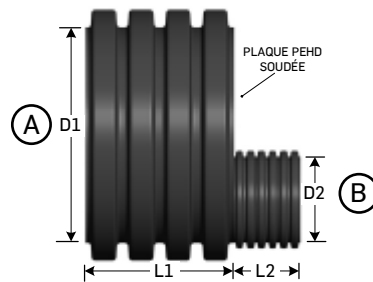
PROJECT: _____

CLIENT: _____

DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____



D1 Diameter	D2 Diameter (If different from D1)	Quantity	Fitting	Specify lengths
☐ 150 mm (6 in)	☐ 100 mm (4 in)		Side A	Standard (according to data sheet) ☐
☐ 200 mm (8 in)	☐ 150 mm (6 in)		☐ Smooth	or
☐ 250 mm (10 in)	☐ 200 mm (8 in)		☐ Clip	L1: _____ ☐ mm
☐ 300 mm (12 in)	☐ 250 mm (10 in)		☐ Gasket	☐ in
☐ 375 mm (15 in)	☐ 300 mm (12 in)			L2: _____ ☐ mm
☐ 450 mm (18 in)	☐ 375 mm (15 in)			☐ in
☐ 525 mm (21 in)	☐ 450 mm (18 in)		Side B	
☐ 600 mm (24 in)	☐ 525 mm (21 in)		☐ Smooth	
☐ 750 mm (30 in)	☐ 600 mm (24 in)		☐ Clip	
☐ 900 mm (36 in)	☐ 750 mm (30 in)		☐ Gasket	

APPROVAL

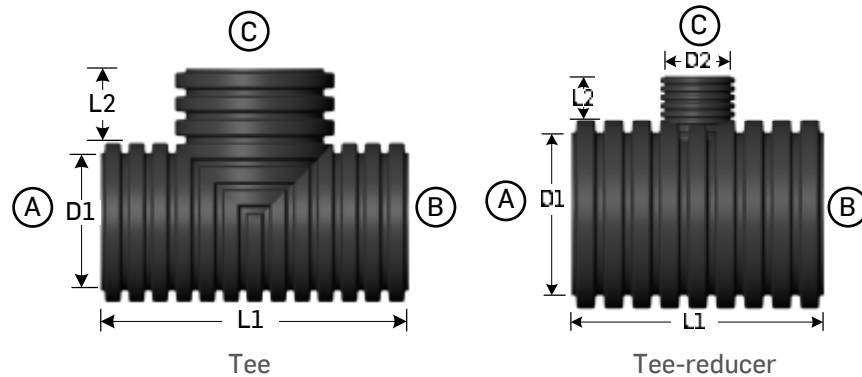
ORDER FORM

STANDARD OR CUSTOM TEE / REDUCER-TEE OR CORRUGATED MANHOLE/CATCH BASIN-TEE

PROJECT: _____
CLIENT: _____
DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____



D1 Diameter	D2 Diameter (If different from D1)	Quantity	Fitting	Specify lengths
☐ 100 mm (4 in)	☐ 100 mm (4 in)		Side A	Standard ☐ (according to data sheet)
☐ 150 mm (6 in)	☐ 150 mm (6 in)		☐ Smooth	or
☐ 200 mm (8 in)	☐ 200 mm (8 in)		☐ Clip	L1: _____ ☐ mm
☐ 250 mm (10 in)	☐ 250 mm (10 in)		☐ Gasket	☐ in
☐ 300 mm (12 in)	☐ 300 mm (12 in)		Side B	L2: _____ ☐ mm
☐ 375 mm (15 in)	☐ 375 mm (15 in)		☐ Smooth	☐ in
☐ 450 mm (18 in)	☐ 450 mm (18 in)		☐ Clip	
☐ 525 mm (21 in)	☐ 525 mm (21 in)		☐ Gasket	
☐ 600 mm (24 in)	☐ 600 mm (24 in)		Side C	
☐ 750 mm (30 in)	☐ 750 mm (30 in)		☐ Smooth	
☐ 900 mm (36 in)			☐ Clip	
☐ 1050 mm (42 in)			☐ Gasket	
☐ 1200 mm (48 in)				

APPROVAL

FRAMES AND GRATES/COVERS:

HDPE grate
300 mm (12 in)
375 mm (15 in)



Frame and grate in
S series cast iron

300 mm (12 in)
375 mm (15 in)
450 mm (18 in)
600 mm (24 in)
750 mm (30 in)
900 mm (36 in)



Frame and cover in
S series cast iron

600 mm (24 in)
750 mm (30 in)
900 mm (36 in)



ORDER FORM

STANDARD OR CUSTOM MADE WYE (Y) AND WYE (Y) REDUCER

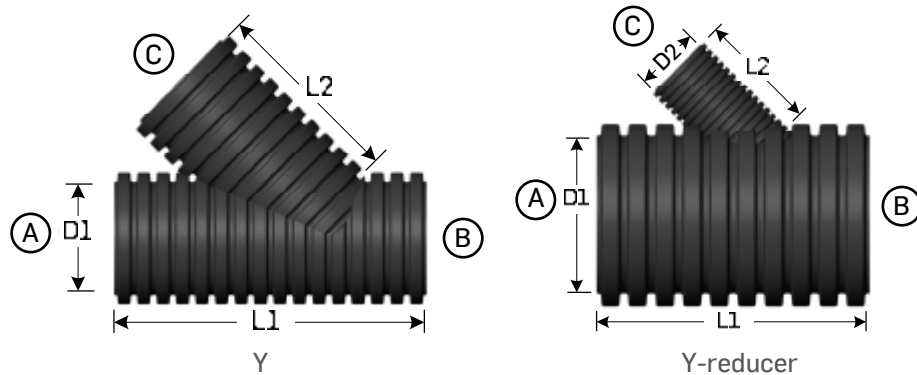
PROJECT: _____

CLIENT: _____

DATE REQUIRED: _____

FOR OFFICE USE

ORDER #: _____



D1 Diameter	D2 Diameter (If different from D1)	Quantity	Fitting	Specify lengths
☐ 100 mm (4 in)	☐ 100 mm (4 in)		Side A	Standard ☐ (according to data sheet)
☐ 150 mm (6 in)	☐ 150 mm (6 in)		☐ Smooth	or
☐ 200 mm (8 in)	☐ 200 mm (8 in)		☐ Clip	L1: _____ ☐ mm
☐ 250 mm (10 in)	☐ 250 mm (10 in)		☐ Gasket	☐ in
☐ 300 mm (12 in)	☐ 300 mm (12 in)			L2: _____ ☐ mm
☐ 375 mm (15 in)	☐ 375 mm (15 in)		Side B	☐ in
☐ 450 mm (18 in)	☐ 450 mm (18 in)		☐ Smooth	
☐ 525 mm (21 in)	☐ 525 mm (21 in)		☐ Clip	
☐ 600 mm (24 in)	☐ 600 mm (24 in)		☐ Gasket	
☐ 750 mm (30 in)	☐ 750 mm (30 in)			
☐ 900 mm (36 in)			Side C	
☐ 1050 mm (42 in)			☐ Smooth	
☐ 1200 mm (48 in)			☐ Clip	
			☐ Gasket	

APPROVAL